

Disorders of Sodium Balance

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Sodium (Na^+) and water balance and their distribution among the various body compartments are essential for the maintenance of fluid homeostasis, particularly intravascular volume. Disturbances of either or both of these components have serious medical consequences, are relatively frequent and are among the most common conditions encountered in clinical practice. In fact, abnormalities of Na^+ and water balance are responsible for, or associated with, a wide spectrum of medical and surgical admissions or complications. The principal disorders of Na^+ balance are manifested clinically as hypovolemia or hypervolemia, whereas disruption in water balance can be diagnosed only in the laboratory as hyponatremia or hypernatremia. Although disorders of Na^+ and water balance are often interrelated, the latter are considered separately in Chapter 15. In this chapter, the physiologic and pathophysiologic features of Na^+ balance are discussed. Because Na^+ is restricted predominantly to the extracellular compartment, this chapter also addresses perturbations of extracellular fluid (ECF) volume homeostasis.

PHYSIOLOGY

Approximately 60% of adult body mass is composed of solute-containing fluids divided into extracellular and intracellular compartments. Because water flows freely across cell membranes in accordance with the prevailing osmotic forces on either side of the membrane, the solute/water ratios in the intracellular fluid (ICF) and ECF are almost equal. However, the solute compositions of the ICF and ECF are quite different, as shown in Fig. 14.1. The principal ECF cation is sodium; other cations are potassium (K^+), calcium, and magnesium. In contrast, potassium is the major ICF cation. The accompanying anions in the ECF are chloride, bicarbonate and plasma proteins (mainly albumin), whereas electroneutrality of the ICF is maintained by phosphate and the negative charges on organic molecules. The difference in

cationic composition of the two compartments is maintained by a pump-leak mechanism consisting of sodium–potassium adenosine triphosphatase ($\text{Na}^+\text{-K}^+\text{-ATPase}$), which operates in concert with sodium and potassium conductance pathways in the cell membrane.

The free movement of water across the membrane ensures that the ECF and ICF osmolalities are the same. However, the intracellular volume is greater because the amount of potassium salts inside the cell is larger than that of sodium salts outside the cell. The movement of water is determined by the “effective osmolality,” or tonicity, of each compartment, so that if tonicity of the ECF rises—for example, as a result of excess Na^+ —water will move from the ICF to ECF to restore tonicity. On the other hand, addition of solute-free water leads to a proportionate decrease in both osmolality and tonicity of all body fluid compartments (see Chapters 10 and 15 for a detailed discussion). The restriction of Na^+ to the ECF compartment by the pump-leak mechanism, in combination with maintenance of the osmotic equilibrium between ECF and ICF, ensures that ECF volume is determined mainly by total body Na^+ content. Departures from this paradigm relate to the identification in recent years of subcutaneous Na^+ pools, which do not participate in movement of water in response to osmotic equilibrium, and therefore do not respond equivalently to ECF volume regulation as described below. Further departures from the classical paradigm that maintenance of osmotic balance in response to salt surfeit is mediated by increased water consumption to balance the salt load, were recently suggested by studies demonstrating that under the influence of glucocorticoid and mineralocorticoid hormone fluctuations, long-term increased sodium intake increased water retention rather than consumption in human subjects. Parallel studies in murine models attributed the water retention to increased medullary urea and urea transport.^{1,2}

To maintain constancy of the ECF and ICF and thereby safeguard hemodynamic stability, even minute changes in

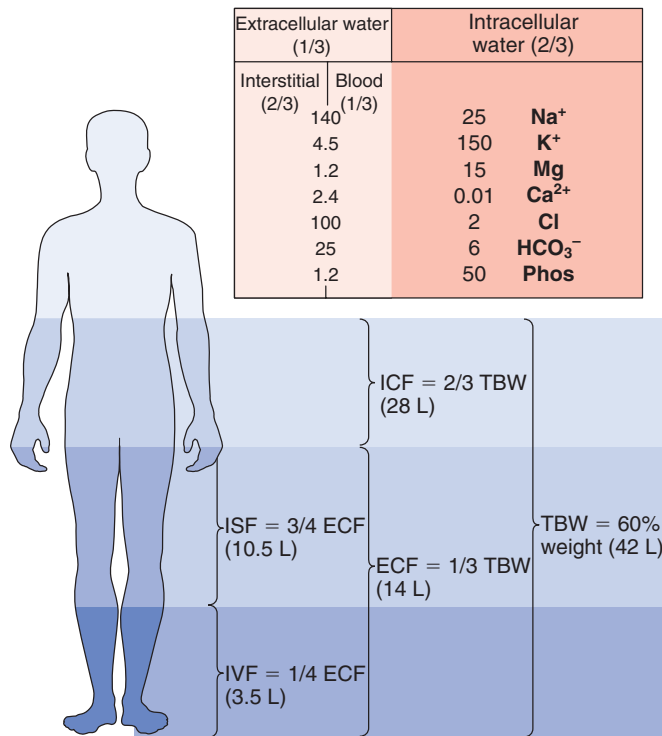


Fig. 14.1 Composition of body fluid compartments. This is a schematic representation of electrolyte composition (*upper panel*) and volumes (*lower panel*) of the body fluid compartments in humans. In the *upper panel*, electrolyte concentrations are in millimoles per liter; intracellular concentrations are typical values obtained from muscle. In the *lower panel*, shaded areas depict the approximate size of each compartment as a function of body weight. In a normally built individual, the total body water content is roughly 60% of body weight. Because adipose tissue has a low concentration of water, the relative water/total body weight ratio is lower in obese individuals. Relative volumes of each compartment are shown as fractions; approximate absolute volumes of the compartments (in liters) in a 70-kg adult are shown in parentheses. ECF, Extracellular fluid; ICF, intracellular fluid; ISF, interstitial fluid; IVF, intravascular fluid; TBW, total body water. (From Verbalis JG. Body water osmolality. In Wilkinson B, Jamison R, editors: *Textbook of nephrology*, London: Chapman & Hall; 1997: pp 89–94. Reproduced with permission of Hodder Arnold.)

these parameters can be detected by a number of sensing mechanisms. These sensory signals lead to activation of neural and hormonal factors, which, in turn, cause appropriate adjustments in urinary Na⁺ and water excretion (Fig. 14.2). Constancy of ECF volume ensures a high degree of circulatory stability, whereas constancy of ICF volume protects against significant brain cell swelling or shrinkage.

SODIUM BALANCE

Na⁺ balance is the difference between intake (diet or supplementary fluids) and output (renal, gastrointestinal, perspiratory, and respiratory). In healthy humans in steady state, dietary intake is closely matched by urinary output of Na⁺. Thus, a person consuming a chronically low-Na⁺ diet (20 mmol/day, or ≈1.2 g of salt/day) excretes, in the steady state, a similar quantity of Na⁺ in the urine (minus extrarenal losses). Conversely, on a high-Na⁺ diet (200 mmol/day, or

12 g of salt/day), approximately 200 mmol of Na⁺ is excreted in the urine. Any perturbation of this balance leads to activation of the sensory and effector mechanisms outlined in the following discussions. In practice, any deviation in ECF volume in relation to its capacitance is sensed and translated, under the influence of neural and hormonal factors, into the appropriate change in Na⁺ excretion, principally through the kidneys but also, to a much lesser degree, through stool and sweat.

According to the traditional two-compartmental model, body sodium balance and partitioning of extracellular fluid volume (ECFV) is based solely on exchangeable and osmotically active Na⁺. For normal functioning of the afferent sensing and efferent effector mechanisms that regulate ECF volume, the integrity of the intravascular and extravascular subcompartments of the ECF is crucial³ (see Fig. 14.2). Although the composition and concentration of small, noncolloid electrolyte solutes in these two subcompartments are approximately equal (slight differences are due to the Gibbs-Donnan effect), the concentration of colloid osmotic particles (mainly albumin and globulin) is higher in the intravascular compartment. The balance between transcapillary hydraulic and colloid osmotic (oncotic) gradients (Starling forces) favors the net transudation of fluid from the intravascular to interstitial compartment. However, this is countered by movement of lymphatic fluid from the interstitial to intravascular compartment via the thoracic duct. The net effect is to restore and maintain the intravascular subcompartment at 25% of the total ECF volume (corresponding to 3.5 L of plasma); the remaining 75% is contained in the interstitial space (equivalent to 10.5 L in a 70-kg man; see Fig. 14.1). The constancy of ECF volume and the appropriate partitioning of the fluid between intravascular and interstitial subcompartments are crucial for maintaining hemodynamic stability. In particular, intravascular volume in relation to overall vascular capacitance is a major determinant of left ventricular filling volume and, hence, cardiac output and mean arterial pressure (MAP).

Recently, the traditional two-compartment model of volume regulation (according to which the intravascular and interstitial spaces are in equilibrium) has been challenged. It now appears that Na⁺ can be bound to and stored on proteoglycans in interstitial sites, where it becomes osmotically inactive; accordingly, a novel mechanism of volume regulation has been elucidated.^{4–11} In rats fed a high-salt diet, this uniquely bound Na⁺ induced a state of subcutaneous interstitial hypertonicity and systemic hypertension.⁸ This hypertonicity is sensed by macrophages,⁵ which then produce vascular endothelial growth factor C (VEGF-C), an angiogenic protein. In turn, VEGF-C stimulates increased numbers and density of lymphatic capillaries. In parallel, macrophages subjected to osmotic stress display activation of a transcription factor, tonicity-responsive enhancer-binding protein (TonEBP). This factor is known to activate osmoprotective genes in other hypertonic environments, such as the renal medulla.¹² Moreover, the VEGF-C promoter contains two TonEBP binding sites, which are upregulated in parallel with VEGF-C. The effect of TonEBP on VEGF-C was shown to be specific, inasmuch as small interfering RNA for TonEBP or deletion of the murine TonEBP gene, but not nonspecific small interfering RNA, inhibited the VEGF-C upregulation and increased blood pressure (BP). Furthermore, macrophage depletion or inhibition of VEGF-C signaling led to

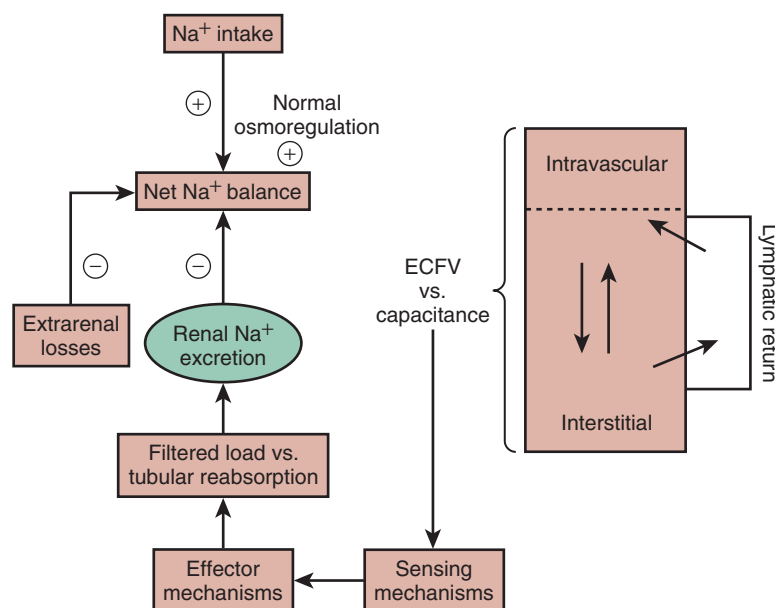


Fig. 14.2 Traditional two-compartment scheme for body sodium balance and partitioning of extracellular fluid volume (ECFV), based on exchangeable and osmotically active Na⁺. In the setting of normal osmoregulation, extracellular Na⁺ content is the primary determinant of ECFV. Overall Na⁺ homeostasis depends on the balance between losses (extrarenal and renal) and intake. Renal Na⁺ excretion is determined by the balance between filtered load and tubule reabsorption. This latter balance is modulated under the influence of effector mechanisms, which, in turn, are responsive to sensing mechanisms that monitor the relationship between ECFV and capacitance. In rats, a high-salt diet leads to interstitial hypertonic Na⁺ accumulation in skin, resulting in increased density and hyperplasia of the lymphatic capillary network.

exacerbation of high-salt diet-induced hypertension.⁵ Also, VEGFR-3, an antibody that blocks the lymph-endothelial VEGF-C receptor, selectively inhibited macrophage-driven increases in cutaneous lymphatic capillary density, led to skin chloride (Cl⁻) accumulation and induced salt-sensitive hypertension. Mice overexpressing soluble VEGFR-3 in epidermal keratinocytes exhibited hypoplastic cutaneous lymph capillaries and increased Na⁺, Cl⁻ and water retention in skin and salt-sensitive hypertension.¹³ A high-salt diet also led to elevated skin osmolality above plasma levels. In addition, in humans with relatively resistant hypertension, elevated levels of VEGF-C were found,⁵ and skin Na⁺ concentration correlated with the presence of left ventricular hypertrophy in chronic kidney disease.¹⁴ These data are consistent with a role for VEGF-C in the redistribution of excess volume to the intravascular space and exacerbation of hypertension.¹⁵

Interestingly, mice skin, but not muscle, arterioles isolated from animals fed a high-salt diet compared with those on a normal salt diet exhibited increased contractile sensitivity to concentrations of angiotensin II (Ang II) $\geq 10^{-10}$ M and to norepinephrine (NE) at 10^{-5} to 10^{-4} M. Finally, a unique study involving astronauts on a simulated Mars expedition, who received diets with fixed salt intake that varied between 6 and 12 g daily, each for 35 days, was recently reported (reviewed in Titze et al.¹⁶). At each level of salt in the diet, the astronauts reached overall equilibrium between intake and output, as measured in 24-hour urine collections, within the expected 6 days. In parallel, there were the expected early changes in body weight, ECF water and inverse relationship with the urine aldosterone level. However, changes in total body Na⁺ only occurred after 7 days and BP reached a new steady state after 3 weeks. Moreover, on the 12-g salt diet, BP continued to rise over a further 4 weeks, with an initial rise and then subsequent fall in body weight and ECF

water. During this period, urine aldosterone levels did not change, whereas total body Na⁺ decreased back to original levels, despite the maintained high salt intake. From these data, it appears that intrinsic rhythms with a periodicity of 30 days or more exist for aldosterone and Na⁺ retention, independent of salt intake. Taken together, all these results clearly demonstrate that the skin contains a hypertonic interstitial fluid compartment in which macrophages exert homeostatic and BP regulatory control by local organization of interstitial electrolyte clearance via TonEBP and VEGF-C/VEGFR-3-mediated modification of cutaneous lymphatic capillary function.¹³ This compartment may be associated with increased vasoreactivity in precapillary arterioles, the major resistance vessel of rat skin, which could increase peripheral resistance and contribute, independently of the kidney, to higher BP in salt-sensitive individuals.^{17,18}

Fig. 14.3 summarizes the novel three-compartment model of Na⁺ balance. The reader is also referred to an excellent recent review of this fascinating subject.¹⁶

EFFECTIVE ARTERIAL BLOOD VOLUME

To understand the mechanisms regulating ECF volume, it is important to appreciate that what is sensed is the effective arterial blood volume (EABV), defined as that part of the ECF in the arterial blood system that effectively perfuses the tissues. In physiologic terms, what is sensed is the threat to arterial pressure induced by the EABV¹⁹ that perfuses the arterial baroreceptors in the carotid sinus and glomerular afferent arterioles. Any change in perfusion pressure (or stretch) at these sites evokes appropriate compensatory responses. EABV is usually correlated with actual ECF volume and is proportional to total body Na⁺. This means that the regulation of Na⁺ balance and the maintenance of EABV are

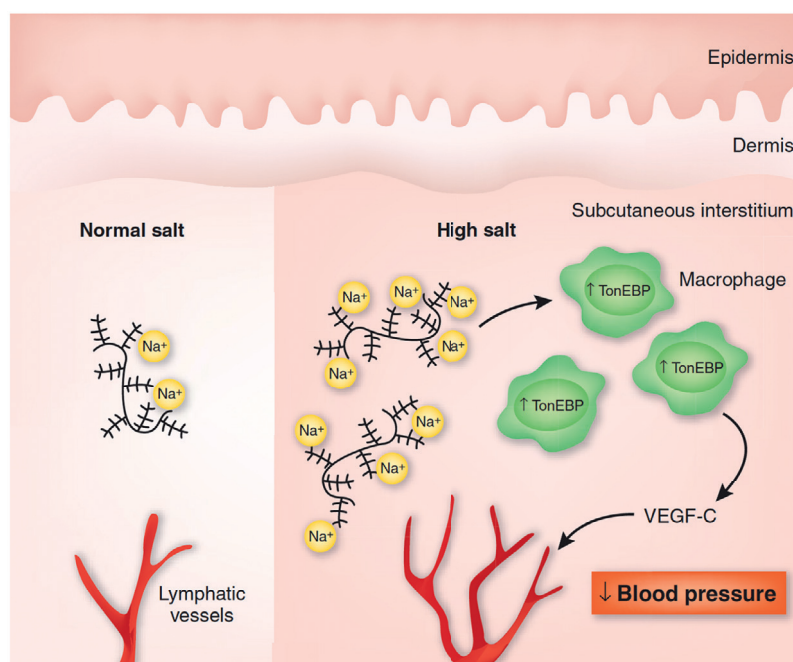


Fig. 14.3 New model of body electrolyte balance and blood pressure homeostasis. Increased subcutaneous Na^+ modulates lymphangiogenesis and blood pressure. After high salt intake, osmotically inactive Na^+ accumulates in the skin interstitium, binding proteoglycans. Macrophages accumulate in the subcutaneous compartment, and increased interstitial tonicity activates tonicity-responsive enhancer-binding protein (TonEBP). TonEBP transactivates the *VEGFC* gene and increases vascular endothelial growth factor C secretion by macrophages. This increases lymph capillary density and attenuates the blood pressure response to high salt. (Modified from Marvar PJ, Gordon FJ, Harrison DG. Blood pressure control: salt gets under your skin. *Nature Medicine*. 2009;15:487–488.)

closely related functions. Na^+ loading generally leads to EABV expansion, whereas loss leads to depletion. However, in some situations, EABV and actual blood volume are not well correlated (see Table 14.5). For example, in heart failure (HF), a primary decrease in cardiac output leads to lowered perfusion pressure of the baroreceptors and reduced EABV is sensed. This leads to renal Na^+ retention and ECF volume expansion. The net result is increased plasma and total ECF volume, in association with increased EABV. The increase in plasma volume is partially appropriate in that intraventricular filling pressure rises leading to increasing myocardial stretching and improved ventricular contractility, thereby raising cardiac output and restoring systemic BP and baroreceptor perfusion. However, this response is also maladaptive in that the elevated intraarterial pressure promotes fluid movement out of the intravascular space into the tissues, which leads to peripheral and pulmonary edema.

In HF, EABV is dependent on cardiac output; in other disease settings, however, these two parameters may be dissociated. For example, in the presence of an arteriovenous fistula cardiac output rises in proportion to the blood flow through the fistula. However, the flow through the fistula shunts blood away from the capillaries perfusing the tissues and, therefore, the EABV does not rise in conjunction with the rise in cardiac output. Similarly, a fall in systemic vascular resistance (SVR)—which, together with cardiac output, is a determinant of BP—leads to reductions in BP and EABV.

Another situation in which cardiac output and EABV change in opposite directions is advanced cirrhosis with ascites. ECF volume expands because of the ascites and plasma volume is increased because of fluid accumulation in the splanchnic circulation, in which the vessels are dilated but flow is sluggish.

Although cardiac output may increase modestly because of arteriovenous shunting, marked peripheral vasodilation leads to a fall in SVR, with reductions in EABV and BP. In the presence of reduced EABV, renal perfusion is impaired; under the influence of hormones, such as renin, NE, and antidiuretic hormone (or arginine vasopressin [AVP])—released in response to the perceived hypovolemia—further Na^+ and water retention ensue (see later section, “Efferent Limb: Effector Mechanisms for Maintaining Effective Arterial Blood Volume”).

To summarize, EABV is an unmeasured index of tissue perfusion that usually, but not always, reflects actual arterial blood volume. Therefore, EABV can be viewed as a functional parameter of organ perfusion.

The diagnostic hallmark of reduced EABV is evidence of renal sodium retention, manifested as a urinary sodium (U_{Na}) less than 15 to 20 mmol/L. This relationship holds true with the following exceptions. If renal Na^+ wasting occurs because of diuretic therapy or intrinsic tubular disease or injury, then U_{Na} is relatively high, despite low EABV. Conversely, the presence of selective renal or glomerular ischemia (e.g., because of bilateral renal artery stenosis or acute glomerular injury) will be misinterpreted as poor renal perfusion and is associated with renal Na^+ retention (low U_{Na}).

REGULATION OF EFFECTIVE ARTERIAL BLOOD VOLUME

Regulation of EABV can be divided into two stages, afferent sensing and efferent effector mechanisms. A number of mechanisms for sensing low EABV exist, all of them primed to stimulate renal Na^+ retention.

AFFERENT LIMB: SENSING OF EFFECTIVE ARTERIAL BLOOD VOLUME

Volume sensors are strategically situated at critical points in the circulation (Table 14.1). Each sensor reflects a specific characteristic of overall circulatory function so that atrial and ventricular sensors sense cardiac filling, arterial sensors respond to cardiac output, and renal, central nervous system (CNS) and gastrointestinal (GI) tract sensors monitor perfusion of the kidneys, brain and gut, respectively. The common mechanism whereby volume is monitored is by physical alterations in the vessel wall, such as stretch or tension. The process of mechanosensing probably is dependent on afferent sensory nerve endings in the vessel wall and activation of endothelial cells. Signal transduction mechanisms in endothelial cells include stretch-activated ion channels, cytoskeleton-associated protein kinases, integrin-cytoskeletal interactions, cytoskeletal-nuclear interactions and generation of reactive oxygen species.^{20,21} In addition, mechanical stretch and tension of blood vessel walls, as well as the frictional forces of the circulation or shear stress, can lead to alterations in gene expression, mediated by specific recognition sites in the upstream promoter elements of responsive genes.^{22,23} These signals induce efferent effector mechanisms that lead to modifications in renal Na⁺ excretion, appropriate to the volume status.

Sensors of Cardiac Filling

Atrial Sensors. The role of the atria in volume regulation in humans was elucidated in experiments involving head-out water immersion (HWI) and exposure to head-down tilt or nonhypotensive lower body negative pressure (LBNP). During HWI, the increased hydrostatic pressure of the water on the lower limbs leads to redistribution of the intravascular fluid from the peripheral to central circulation. The resulting increase in central blood volume causes a rise in cardiac output, which in turn produces a brisk increase in Na⁺ and water excretion in an attempt to restore euvolemia.²⁴ In

contrast, LBNP results in a redistribution of blood to the lower limbs, thereby reducing central venous and cardiac filling pressures without affecting arterial pressure, heart rate, or atrial diameter. The resulting retention of Na⁺ and water occurs without any change in renal plasma flow (RPF).²⁵

A second and, possibly, more dominant, mechanism of HWI-induced Na⁺ and water diuresis involves the external hydrostatic pressure of the water reducing the hydrostatic pressure gradient across the capillary wall in the legs, leading to a net transfer of fluid from the interstitial to intravascular compartment. The resulting hemodilution causes a fall in the colloid osmotic pressure (COP).^{26,27} The combination of hemodilution and central hypervolemia, through atrial stretch, induces neurohumoral changes that bring about the subsequent diuresis and natriuresis.

Neural Pathways. Two types of neural receptors in the atrium have been described, type A and type B. They are thought to be branching ends of small medullated fibers running in the vagus nerve. Only type B receptor activity is increased by atrial filling and stretch.²⁸ The signal is then thought to travel along cranial nerves IX and X to the hypothalamic and medullary centers, where a series of responses is initiated—inhibition of AVP release (left atrial signal),²⁹ a selective decrease in renal but not lumbar sympathetic nerve discharge,^{30,31} and decreased tone in peripheral precapillary and postcapillary resistance vessels. Conversely, reduction in central venous pressure and atrial volume stimulates renal sympathetic nerve activity (RSNA).^{32,33}

The effects just described occur in response to acute atrial stretch, whereas chronic atrial stretch leads to adaptation and downregulation of the neural responses.³³ Cardiac nerves appear to be essential only for the restoration of Na⁺ balance in states of repletion, but not for the renal response to acute volume depletion.³⁴ For example, after human cardiac transplantation, a natural model of cardiac denervation, the expected suppression of the renin-angiotensin-aldosterone (RAAS) system in response to chronic volume expansion is not observed.³⁵

Humoral Pathways. Even after cardiac denervation the natriuresis and diuresis induced by atrial distension is maintained, due to the presence of natriuretic peptides (NP) of cardiac origin.^{36,37} The NP family is comprised of atrial NP (ANP), brain NP (BNP), C-type NP (CNP), *Dendroaspis* NP (DNP), and urodilatin. Although their structures are quite similar, each is encoded by different genes and has distinct, albeit overlapping, functions.^{38–41} The actions of NPs and their interaction with other hormone systems are discussed in detail later (see section, “Efferent Limb: Effector Mechanisms for Maintaining Effective Arterial Blood Volume” and Chapter 11). This section is confined to a discussion of the afferent mechanisms of NP stimulation.

From studies in animals and humans, it is clear that acute increments in atrial stretch or pressure cause a brisk release of ANP. The process involves cleavage of the prohormone, located in preformed stores in atrial granules, to the mature 28-amino acid C-terminus peptide, in a sequence-specific manner, by corin, a transmembrane serine protease.⁴² Release of the hormone appears to occur in two steps, the first a Ca²⁺-sensitive K⁺ channel-dependent release of ANP from myocytes into the intercellular space and then

Table 14.1 Mechanisms for Sensing Regional Changes in Effective Arterial Blood Volume

Sensors of Cardiac Filling

- Atrial
 - Neural pathways
 - Humoral pathways
- Ventricular
- Pulmonary

Sensors of Cardiac Output

- Carotid and aortic baroreceptors

Sensors of Organ Perfusion

- Renal sensors
- CNS sensors
- GI tract sensors
 - Hepatic receptors
 - Guanylin peptides

CNS, Central nervous system; GI, gastrointestinal.

a Ca^{2+} -independent translocation of the hormone into the atrial lumen.⁴³ The afferent mechanism for ANP release is activated by intravascular volume expansion, supine posture, HWI, saline administration, exercise, Ang II, tachycardia and ventricular dysfunction.^{44,45} Conversely, volume depletion induced by Na^+ restriction, furosemide administration or LBNP-mediated reduction in central venous pressure causes a fall in plasma ANP concentration.

In contrast to the effects of acute changes in atrial pressure on ANP release, the role of this peptide in the long-term regulation of plasma volume appears to be minimal. For example, although incremental oral salt loading was associated with correspondingly higher baseline plasma ANP levels, only intravenous (not oral) salt loading led to increased ANP levels.⁴⁶ Moreover, in humans given intravenous or oral salt loads, no correlation could be found between changes in ANP levels and the degree of natriuresis.^{47,48} The contrasting relationships among acute and chronic Na^+ loading, plasma ANP levels and natriuresis have been elegantly demonstrated in ANP gene knockout mice. These mice display a reduced natriuretic response to acute ECFV expansion in comparison with their wild type counterparts. However, no differences in cumulative Na^+ and water excretion were observed between the knockout and wild type mice after a high- or low- Na^+ diet for 1 week. The only difference between the two types of mice was a significant increase in MAP. Further experiments using disruptions of the genes for ANP or its receptor, guanylate cyclase A (GC-A), have shown the importance of this system in the maintenance of normal BP and in modulating cardiac hypertrophy.⁴⁹

In contrast to ANP, the other members of the NP family appear not to be involved in the physiologic regulation of Na^+ excretion.^{38,50}

Ventricular and Pulmonary Sensors. Volume sensors have been found in the ventricles, coronary arteries, main pulmonary artery and bifurcation,⁵¹ and juxtapulmonary capillaries in the interstitium of the lungs,⁵² but not in the intrapulmonary circulation.⁵³ These sensors apparently mediate reflex changes in heart rate and SVR through modulation of the sympathetic nervous system (SNS) and ANP. This also appears to be true for the coronary baroreceptor reflex in anesthetized dogs, by which changes in coronary artery pressure lead to alterations in lumbar and renal sympathetic discharge and a coronary artery response much slower than that of the carotid and aortic baroreceptors.⁵⁴ However, ventricular and pulmonary sensors, at least in dogs, may also detect changes in blood volume through increased left ventricular pressure, which causes a reflex inhibition of plasma renin activity (PRA).^{55,56}

Sensors of Cardiac Output

The sensors described so far are situated in low-pressure sites, where they sense the fullness of the circulation and are probably more important for defending against excessive volume expansion and the consequent cardiac failure. The arterial high-pressure sensors, on the other hand, are geared more toward detecting low cardiac output or SVR, manifested as underfilling of the vascular tree (i.e., EABV depletion) threatening arterial pressure¹⁹, and as signaling the kidneys to retain Na^+ . These high-pressure sensors are found in the aortic arch, carotid sinus, and renal vessels.

Carotid and Aortic Baroreceptors. The carotid baroreceptor has a large content of elastic tissue in the tunica media, which makes the vessel wall highly distensible in response to changes in intraluminal pressure, thereby facilitating transmission of the stimulus intensity to sensory nerve terminals. A rise in MAP induces depolarization of these sensory endings, resulting in action potentials. Transient receptor potential vanilloid receptors may mediate this process.⁵⁷ Afferent signals from the baroreceptors are integrated in the nucleus tractus solitarius (NTS) of the medulla oblongata,⁵⁸ which leads to reflex changes in systemic and renal sympathetic nerve activity and, to a lesser degree, release of AVP. Baroreceptor reflex modulation in the NTS may be mediated by endocannabinoid CB(1) receptor activation, via the 5-hydroxytryptamine type 1A (5-HT1A) receptor, leading to endogenous anandamide release.^{59,60,61} Conversely, hypovolemia-induced activation of NTS (A1) adenosine receptors may serve as a negative feedback regulator of sympathoinhibitory reflexes integrated in the NTS.⁶² An important additional function of the carotid baroreceptors is maintenance of adequate cerebral perfusion. The aortic baroreceptor appears to behave in a way similar to the carotid baroreceptor. Finally, there is evidence in dogs for an interaction between pulmonary arterial and carotid baroreceptor reflexes.⁶³

Sensors of Organ Perfusion

Renal Sensors. Not only is the kidney the major effector target responding to signals that indicate the need for adjustments in Na^+ excretion, but also has a central role in the afferent sensing of volume homeostasis by virtue of the local sympathetic innervation. However, despite considerable knowledge concerning the mechanisms of renal sensing of EABV, the molecular identity and exact cellular location of the renal sensor(s) remains elusive.⁶¹ The integral relationship between afferent and efferent RSNA and the central arterial baroreceptors was highlighted by Kopp and colleagues.⁶⁴ They showed that a high- Na^+ diet increases afferent RSNA, which then decreases efferent RSNA and leads to natriuresis. Using dorsal rhizotomy to induce afferent renal denervation in rats maintained on a high- Na^+ diet, they demonstrated increased MAP that was dependent on impaired arterial baroreflex suppression of efferent RSNA. Animals fed a normal- Na^+ diet displayed no changes in arterial baroreceptor function. Kopp and coworkers concluded that arterial baroreflex function contributes to increased efferent RSNA, which, in the absence of intact afferent RSNA, would eventually lead to Na^+ retention and hypertension. The role of RSNA in Na^+ regulation is discussed further later in this chapter (see section, “[Neural Mechanisms: Renal Nerves and Sympathetic Nervous System](#)”).

An additional level of renal sensing depends on the close anatomic proximity of the sensor and effector limbs to one another: Volume changes may be sensed through alterations in glomerular hemodynamics and renal interstitial pressure. These alterations result simultaneously in adjustments in the physical forces governing tubular Na^+ handling (see section, “[Efferent Limb: Effector Mechanisms for Maintaining Effective Arterial Blood Volume](#)”).

The kidneys have the ability to maintain a constant blood flow and glomerular filtration rate (GFR) at varying arterial pressures. This phenomenon, termed “autoregulation,” operates over a wide range of renal perfusion pressures (RPPs).

Autoregulation of renal blood flow (RBF) occurs through three mechanisms—the myogenic response, tubuloglomerular feedback (TGF) and a third mechanism. In the myogenic response, changes in RPP are sensed by smooth muscle elements that serve as baroreceptors in the afferent glomerular arteriole and dynamically respond by adjusting transmural pressure and tension across the arteriolar wall.⁶⁵ An example of this can be seen in Ang II–infused rats on a high salt intake. These animals show reduced dynamic autoregulation of RBF, an effect mediated, at least in part, by superoxide⁶⁶ and attenuated by endothelial nitric oxide synthase (eNOS)–dependent production of nitric oxide (NO).⁶⁷

The second mechanism, TGF, is operated by the juxtaglomerular apparatus (JGA), comprised of the afferent arteriole and, to a lesser extent, the cells of the macula densa in the early distal tubule.^{65,68} The JGA is also important because of its involvement in the synthesis and release of renin,⁶⁸ which is controlled by three pathways, all driven by EABV status. First, renin release is inversely related to RPP and directly related to intrarenal tissue pressure. When RPP falls below the autoregulatory range, renin release is further enhanced. Second, renin secretion is influenced by solute delivery to the macula densa. Increased NaCl delivery past the macula densa leads to inhibition of renin release, whereas a decrease has the opposite effect. Sensing at the macula densa is mediated by NaCl entry through the $\text{Na}^+\text{-K}^+\text{-2Cl}^-$ cotransporter (NKCC2),^{69,70} which leads to increased intracellular Ca^{2+} , production of prostaglandin E_2 (PGE_2),⁷¹ adenosine,⁷² and, subsequently, renin release. Third, changes in RSNA influence renin release. Renal nerve stimulation increases renin release through direct activation of β -adrenergic receptors on juxtaglomerular cells. This effect is independent of major changes in renal hemodynamics.⁷³ Sympathetic stimulation also affects intrarenal baroreceptor input, composition of the fluid delivered to the macula densa and renal actions of Ang II so that renal nerves may serve primarily to potentiate other regulatory signals.⁷³

The nature of the third mechanism of RBF autoregulation is still unclear, but Seeliger and associates,⁷⁴ using a normotensive Ang II clamp in anesthetized rats, were able to abolish the resetting of autoregulation during incremental shaped RPP changes. Under control conditions, the initial TGF response was dilatory after total occlusions but constrictive after partial occlusions. The initial third mechanism response was a mirror image of TGF; it was constrictive after total occlusions but dilatory after partial occlusions. The angiotensin clamp suppressed TGF and converted the initial third mechanism response after total occlusions into dilation. Seeliger and coworkers concluded: 1. pressure-dependent renin angiotensin system (RAS) stimulation was a major factor behind hypotensive resetting of autoregulation; 2. TGF sensitivity depended strongly on pressure-dependent changes in RAS activity; 3. the third mechanism was modulated, but not mediated, by the RAS; and 4. the third mechanism acted as a counterbalance to TGF.⁷⁴ These findings may be related to feedback between the connecting tubule and glomerulus.⁷⁵ TGF is discussed later (see section, “[Integration of Changes in Glomerular Filtration Rate and Tubular Reabsorption](#)”) and in detail in Chapter 3.

Central Nervous System Sensors. Certain areas in the CNS appear to act as sensors to detect alterations in body salt

balance, at least in rats. Thus, intracerebral injection of hypertonic saline led to reduced RSNA and natriuresis,^{76,77} and administration of Ang II into the cerebral ventricles and changes in dietary Na^+ modulated baroreflex regulation of RSNA. Stimulation of neurons in the paraventricular nucleus and in a region extending to the anteroventral third ventricle led to ANP release, inducing Ang II blockade and inhibition of salt and water intake. Conversely, disruption of these neurons, as well as of the median eminence or neural lobe, led to decreased ANP release and impaired response to volume expansion.⁷⁸ However, the exact nature, mode of operation, and relative importance of this aspect of sensing remains unclear.

Gastrointestinal Tract Sensors. Under normal physiologic conditions, Na^+ and water reach the ECF by absorption in the GI tract. Therefore, it is not surprising that sensing and regulatory mechanisms of ECF volume have been found in the GI tract itself. The evidence for this phenomenon comes from experiments that showed more rapid natriuresis after an oral salt load than after a similar intravenous load. Moreover, infusions of hypertonic saline into the portal vein led to greater natriuresis than similar infusions into the femoral vein. These findings were consistent with the presence of Na^+ -sensing mechanisms in the splanchnic or portal circulation, or both⁷⁹ and are probably important in the pathogenesis of the hepatorenal syndrome (HRS; see later).

Hepatoportal Receptors. The two main neural reflexes, termed the “hepatorenal” and “hepatointestinal reflexes,” originate from receptors in the hepatoportal region. They transduce portal plasma Na^+ concentration into hepatic afferent nerve activity; before a measurable increase in systemic Na^+ concentration occurs, the hepatointestinal reflex attenuates intestinal Na^+ absorption via the vagus nerve and the hepatorenal reflex augments Na^+ excretion both in humans and experimental animals.^{80–82} These reflexes are impaired in the chronic bile duct ligation model of cirrhosis and portal hypertension,⁸³ leading to Na^+ retention, mediated in part by the A_1 adenosine receptor (A1AR)^{84,85} and, possibly, also the NKCC2 cotransporter.⁸⁶ In addition, the hepatic artery shows significant autoregulatory capacity, dilating when perfusion pressure falls and constricting when pressure rises, thereby maintaining hepatic arterial blood flow over a wide range of perfusion pressures. This indicates the presence of a sensor in the hepatic artery, which responds to changes in the contribution of the portal vein to total hepatic blood flow.⁸⁷

In addition to hepatoportal Na^+ -sensing chemoreceptors, the liver also contains mechanoreceptors. Increased intrahepatic hydrostatic pressure, as seen in the Budd-Chiari syndrome,⁸⁸ is associated with enhanced RSNA and renal Na^+ retention in various experimental models,^{89,90} so that hepatic volume-sensing mechanisms probably play a role in renal Na^+ retention (see section, “[Specific Treatments Based on the Pathophysiology of Sodium Retention in Cirrhosis](#)”).

Intestinal Natriuretic Hormones. As described previously, the natriuretic response to a Na^+ load in experimental animals is more rapid when the load is delivered orally than intravenously.⁷³ The different responses are observed without changes in plasma aldosterone,⁹¹ suggesting that the gut produces one or more substances that signal the kidneys to

excrete excess Na^+ . The two main candidate substances are guanylin (guanylin and uroguanylin)^{92,93} and gastrin.⁹⁴

Guanylin is small (15 to 16 amino acids), heat-stable peptides with intramolecular disulfide bridges and are found in mammals, birds, and fish.⁹³ Both guanylin and uroguanylin are synthesized as prepropeptides, primarily in the intestine. The former, produced mainly by the ileum through the proximal colon, circulates as proguanylin; the latter, expressed principally in the jejunum, circulates in its active form.⁹² The two peptides differ in their sensitivity to proteases. Because of a tyrosine residue at the ninth amino acid, guanylin is sensitive to renal inactivation by protease digestion, whereas uroguanylin can be locally activated by the same proteases.⁹³ After an oral salt load, guanylin and uroguanylin released in the intestine lead to increased intestinal secretion of Cl^- , HCO_3^- and water and to inhibition of Na^+ absorption. In the kidneys, Na^+ , K^+ , and water excretion is increased, without any change in RBF or GFR and independently of RAAS, AVP, or ANPs.⁹³ Guanylin signal transduction occurs via binding to and activation of the receptor guanylate cyclase C (GC-C), in the intestinal brush border and increased cGMP, which inhibits Na^+/H^+ exchange and activates protein kinases G II and A. These, in turn, activate the cystic fibrosis transmembrane conductance regulator (CFTR), leading to Cl^- secretion, activation of the $\text{Cl}^-/\text{HCO}_3^-$ exchanger, and HCO_3^- secretion.⁹⁵

The best evidence for a link between the gut and kidneys comes from mice lacking the uroguanylin gene, which display an impaired natriuretic response to oral salt loading but not to intravenous NaCl infusion.⁹⁶ However, because plasma proguanylin levels do not rise but urinary uroguanylin levels do increase after a high-salt meal, locally released peptide by the kidneys may play a role in uroguanylin-associated natriuresis.^{97,95} In the kidneys, both GC-C-dependent and GC-C-independent signaling pathways for guanylin peptides exist, inasmuch as knockout of GC-C in mice does not affect the high-salt diet-induced increase in uroguanylin.⁹³

From experiments on cell lines and isolated tubules, it appears that uroguanylin acts to decrease Na^+ reabsorption in the proximal tubule and principal cells of the cortical collecting duct (CCD).⁹³ Crosstalk between guanylin peptides and ANPs may also occur in the proximal tubule.⁹⁸ In the principal cell, uroguanylin activation of a G protein-coupled receptor results in phospholipase A_2 -dependent inhibition of the renal outer medullary potassium (ROMK) channel, which leads to depolarization and a reduced driving force for Na^+ reabsorption.⁹³ Guanylin may cause cell shrinkage in the inner medullary collecting duct (IMCD), suggestive of water secretion from this segment.⁹³ Together, the experimental data are highly suggestive of a role for uroguanylin as a natriuretic hormone, in response to NaCl absorbed via the GI tract.^{92,93}

More recently, gastrin, secreted from the stomach and duodenum, has been proposed as a second candidate substance mediating natriuresis in response to an oral salt load. Gastrin appears to signal natriuresis via its receptor, cholecystokinin B receptor linked to a dopamine D1-like receptor.⁹⁴ However, despite all the evidence from animal studies, recent work has cast substantial doubt on the relevance of a GI renal signaling axis for Na^+ regulation in humans.⁹⁹

A final point is that although multiple receptors are involved in the regulation of EABV, their functions appear to be redundant. In this regard, cardiac or renal denervation and

Table 14.2 Major Renal Effector Mechanisms for Regulating Effective Arterial Blood Volume

Glomerular Filtration Rate and Tubular Reabsorption

Tubuloglomerular feedback
Glomerulotubular balance
Peritubular capillary Starling forces
Luminal composition
Physical factors beyond proximal tubule
Medullary hemodynamics (pressure natriuresis)

Neural Mechanisms

Sympathetic nervous system
Renal nerves

Humoral Mechanisms

Renin-angiotensin-aldosterone system
Vasopressin
Prostaglandins
Natriuretic peptides
Endothelium-derived factors
 Endothelins
 Nitric oxide
Others (see text)

chronic aldosterone administration in nonhuman primates do not significantly affect the maintenance of Na^+ balance.^{100,101}

EFFERENT LIMB: EFFECTOR MECHANISMS FOR MAINTAINING EFFECTIVE ARTERIAL BLOOD VOLUME

The maintenance of Na^+ homeostasis is achieved by adjustments of renal Na^+ excretion according to the body's needs. The adjustments are made by integrated changes in GFR and tubular reabsorption, so that changes in one component lead to appropriate changes in the other to maintain Na^+ homeostasis. In addition, tubular reabsorption is regulated by local peritubular and luminal factors as well as by neural and humoral mechanisms (Table 14.2).

INTEGRATION OF CHANGES IN GLOMERULAR FILTRATION RATE AND TUBULAR REABSORPTION

In humans, normal GFR leads to the delivery of approximately 24,000 mmol of Na^+ /day to the tubules where more than 99% of the filtrate is reabsorbed. Therefore, even minute changes in the relationship between filtered load and fraction of Na^+ absorbed can profoundly influence net Na^+ balance. However, even marked perturbations in GFR are not necessarily associated with drastic alterations in U_{Na} excretion and overall Na^+ balance is usually preserved. Such preservation results from adjustments in two important protective mechanisms—TGF, in which changes in tubular fluid Na^+ inversely affect GFR, and glomerulotubular balance, whereby changes in tubular flow rate resulting from changes in GFR directly affect tubular reabsorption.^{68,102,103}

Tubuloglomerular Feedback

A remarkable feature of nephron architecture is that after emerging from Bowman's capsule and descending deep into the medulla, each tubule returns to its parent glomerulus.

The functional counterpart of this anatomic relationship is TGF^{104,105} (see also Chapter 3). TGF is constructed as a negative feedback loop in which an increase in NaCl concentration at the macula densa (the point of contact between the specialized tubular cells of the cortical thick ascending limb of Henle, cTALH, adjacent to the extraglomerular mesangium) leads to increases in afferent arteriolar resistance and a consequent fall in the GFR. This, in turn, leads to an increase in proximal reabsorption and a reduction in distal delivery of solute, whereby NaCl delivery to the distal nephron is maintained within narrow limits.¹⁰⁵

The complexities of TGF were initially unraveled by micropuncture, imaging and electrophysiologic techniques in isolated perfused tubule/glomerulus preparations. Subsequently, the signaling mechanisms linking changes in tubular composition with altered glomerular arteriolar tone became evident through experiments in gene-manipulated mice.¹⁰⁵ The primary detection mechanism of TGF is uptake of salt by means of the NKCC2, located in the apical membrane of macula densa cells. The evidence comes from TGF inhibition by inhibitors of the cotransporter, furosemide and bumetanide,¹⁰⁶ and by deletions in mice of the A or B isoform of NKCC2, both of which are expressed in macula densa cells.^{69,105} In fact, complete inactivation of the NKCC2 gene leads to the severe salt-losing phenotype of antenatal Bartter syndrome.¹⁰⁷ Similarly, inhibition or deletion of the ROMK channel in mice abolishes TGF.¹⁰⁵

The next step in the juxtaglomerular cascade is less clear. One possibility is direct coupling of NKCC2-dependent NaCl uptake to the mediation step. Results of studies in the isolated perfused rabbit JGA have indicated that depolarization, alkalinization, and various ionic compositional changes occur after increased NaCl uptake; thus, one or more of these changes could trigger the signal.¹⁰⁸ A second possibility is that signal propagation is the consequence of transcellular NaCl transport and Na⁺-K⁺-ATPase-dependent basolateral extrusion. Experiments using double-knockout mice, in which the α_1 -subunit of Na⁺-K⁺-ATPase was made sensitive and the α_2 -subunit resistant to the pump inhibitor, ouabain, clearly indicate an important role for Na⁺-K⁺-ATPase in supporting TGF and that adenosine triphosphate (ATP) consumption is required for the process.¹⁰⁸

In contrast, other studies have yielded strong evidence that ATP release and degradation, rather than consumption, may be the link connecting NaCl changes in the macula densa with alteration of glomerular arteriolar tone. According to the current working model, after NaCl uptake and transcellular transport, ATP is released from macula densa cells and undergoes stepwise hydrolysis and dephosphorylation by ecto-ATPases and nucleotidases to adenosine diphosphate, adenosine monophosphate and then adenosine. Adenosine, in a paracrine manner, then causes A1AR-dependent afferent arteriolar constriction. Although the evidence for ATP breakdown is as yet incomplete, evidence for adenosine as a mediator of TGF is very strong. For example, isolated perfused mouse afferent arterioles exposed to adenosine display vigorous vasoconstriction, an effect not seen in A1AR-deficient mice.^{109,110} As shown by overexpression¹¹¹ and conditional knockout of the receptor, this A1AR effect is primarily on afferent arteriolar smooth muscle cells, although A1AR effects on extravascular, perhaps mesangial, cells appear to contribute to the TGF response.¹¹² The response is mediated

by inhibitory G protein (G_i)-dependent activation of phospholipase C, release of Ca²⁺ from intracellular stores and subsequent entry of Ca²⁺ through L-type Ca²⁺ channels.^{109,113} Cellular adenosine uptake is likely to be involved in the TGF response because targeted deletion of the type 1 equilibrative nucleoside transporter (ENT1) led to significant attenuation of the response.¹¹⁴ Also, vasodilatory adenosine A₂ receptor is more abundant than the A1AR in the renal vasculature and continuous exogenous application of adenosine to mouse kidneys is indeed vasodilatory.¹¹⁵ However, the generation of adenosine in the confines of the juxtaglomerular interstitium and its exclusive delivery to the afferent arteriole, where A1AR expression predominates, ensures the appropriate response for TGF.

Other factors, both coconstrictors and modulators, appear to be involved in TGF. Ang II is an important cofactor in the vasoconstrictive action of adenosine, since deletions of the Ang II receptor or angiotensin converting enzyme (ACE) in mice were found to abolish TGF. The effect may result from nonresponsiveness to adenosine in the absence of an intact RAS.¹¹¹ By contrast, aldosterone appears to blunt TGF through superoxide-mediated activation of mineralocorticoid receptors on macula densa cells.^{116,117} In turn, superoxide may also be upregulated by Ang II via the NOX2 and NOX4 isoforms of nicotinamide adenine dinucleotide phosphate (NADPH) oxidase.¹¹⁸ The high levels of neuronal NO synthase (nNOS) expression in macula densa cells are thought to counterbalance Ang II-induced efferent arteriolar vasoconstriction and to modulate renin secretion by the JGA.^{119–121} In contrast, chronic absence of functional nNOS in macula densa cells is associated with enhanced vasoconstriction in the subnormal flow range, probably as a result of proportional increases in preglomerular and postglomerular tone. In addition, increased delivery of fluid to the macula densa induces NO release from these cells.¹²⁰

Inhibition of the NO system by nonselective blockers of NOS results in an exaggerated TGF response that leads to even further renal vasoconstriction, Na⁺ and water retention, and arterial hypertension.¹²¹ Also, TGF responses are absent in mice with concurrent deficiencies in nNOS and the A1AR, which implies that nNOS deficiency does not overcome deficient A1AR signaling. Moreover, NO modulation of TGF can be mediated by ecto 5'-nucleotidase, the enzyme responsible for adenosine formation.¹²² Furthermore, NO, via eNOS, modulates the afferent arteriolar myogenic response.⁶⁷ Finally, aldosterone-induced modulation of TGF appears to involve interactions between NO and superoxide.¹²³ Together, these data suggest that A1AR signaling is primary and that nNOS and eNOS and superoxide, play modulatory roles in TGF.

Apart from the RAAS, other hormonal systems and secondary messengers appear to be involved in TGF. For example, stimulation of the glucagon-like peptide 1 receptor leads to an increased GFR and reduced proximal tubular reabsorption.¹²⁴ Moreover, high salt intake-induced activation of AMP-activated protein kinase leads to an enhanced TGF response and increased delivery of Na⁺ to the end of the proximal tubule.⁶⁶ Furthermore, acute saline expansion leads to an increased single-nephron glomerular filtration rate (SNGFR) and distal nephron flow rate, independently of the Ang II receptor.¹²⁵

The afferent arteriolar A1AR may not be the sole mediator of TGF. Activation by adenosine of the low-affinity adenosine

A_{2b} receptor,¹¹⁵ via increased levels of eNOS,¹²⁶ has been shown to dilate mouse cortical efferent receptors. This highly specific effect occurs despite the presence of A1AR in the efferent arteriole. Apparently, therefore, the relative abundance of the various adenosine receptor subtypes in afferent and efferent arterioles ultimately allows fine-tuning of TGF by concerted changes in glomerular vascular tone.¹²⁷ On the other hand, purine receptors do not seem to be involved in TGF.¹²⁸

Connexin 40, which plays a predominant role in the formation of gap junctions in the vasculature, also participates in the autoregulation of RBF by the afferent arteriole and, therefore, in TGF.¹²⁹ Connexin 40 knockout mice displayed impaired steady-state autoregulation to a sudden stepped increase in RPP, likely due to a marked reduction in TGF. Connexin 40–mediated RBF autoregulation occurred by paracrine signaling between tubular cells and afferent arteriolar vascular cells,¹³⁰ independently of NO.¹³¹ Other endogenous modulators of TGF include the eicosanoid 20-hydroxyeicosatetraenoic acid (20-HETE), which modifies the myogenic afferent arteriolar and TGF responses;¹³² also heme oxygenase, via carbon monoxide and cGMP generation, blocks TGF through inhibition of depolarization and Ca^{2+} entry into macula densa cells.^{133,134}

Sex hormones also appear to regulate TGF. Testosterone in rats leads to upregulation of TGF by generation of superoxide dismutase,¹³⁵ whereas enhanced Ang II receptor activity attenuates Ang II–dependent resetting of TGF activity in female rats.¹³⁶ A final point in the complexity of TGF is that there is evidence for three sites in addition to the macula densa that are in contact with the efferent arteriole—the terminal cTALH, the early distal tubule and the connecting tubule. In particular, perimacular cells and oscillatory cells of the early distal tubule may be involved in the intracellular Ca^{2+} signaling required for adenosine-induced afferent vasoconstriction. On the other hand, the effect of the connecting tubule on the afferent arteriolar tone appears to be modulatory in that elevations in luminal NaCl and cellular Na^+ entry via the epithelial sodium channel (ENaC) lead to afferent arteriolar dilation¹³⁷ through the release of prostaglandins (PG) and epoxyeicosatrienoic acids.^{138,139} Moreover, connecting tubule glomerular feedback has been shown to antagonize TGF,¹⁴⁰ at least in the acute setting.

Glomerulotubular Balance

Several factors are involved in glomerulotubular balance (GTB), which describes the ability of proximal tubular reabsorption to adapt proportionally to changes in filtered load.

Peritubular Capillary Starling Forces. Following acute, but probably not chronic, changes in ECFV, alterations in hydraulic and oncotic pressures (Starling forces) in the peritubular capillary seem to play an important role in the regulation of Na^+ and water transport, especially in the proximal nephron. The peritubular capillary network is anatomically connected in series with the capillary bed of cortical glomeruli through the efferent arteriole; thus, changes in the physical determinants of GFR critically influence Starling forces in the peritubular capillaries.

Of importance is that about 10% of glomeruli, mainly those at the corticomedullary junction, are connected in

series to the vasa recta of the medulla. In the proximal tubule—whose peritubular capillaries receive 90% of blood flow from glomeruli—the relationship of hydraulic and oncotic driving forces to the transcapillary fluid flux is given by the Starling equation, as follows:

$$\text{Rate}_{\text{abs}} = K_r [(\pi_c - \pi_i) - (P_c - P_i)]$$

where Rate_{abs} is the absolute rate of reabsorption of proximal tubule reabsorbate by the peritubular capillary, K_r is the capillary reabsorption coefficient (the product of capillary hydraulic conductivity and absorptive surface area), π_c and P_c are the local capillary colloid osmotic (oncotic) and hydraulic pressures, respectively, and π_i and P_i are the corresponding interstitial pressures. Whereas π_i and P_c oppose fluid absorption, π_c and P_i tend to favor uptake of the reabsorbate. As a consequence of the anatomic relationship of the postglomerular efferent arteriole to the peritubular capillary, P_c is significantly lower in the peritubular than in the glomerular capillary. Also, because the peritubular capillary receives blood from the glomerulus, P_c is high at the outset as a result of prior filtration of protein-free fluid. It follows that the greater the GFR in relation to plasma flow rate, the higher the protein concentration in the efferent arteriolar plasma and the lower P_c in the proximal peritubular capillary. Consequently, proximal fluid reabsorption is enhanced (Fig. 14.4). Therefore, in contradistinction to the glomerular and peripheral capillary, the peritubular capillary is characterized by high values of $\pi_c - \pi_i$ that greatly exceed $P_c - P_i$, which results in net reabsorption of fluid. Changes in peritubular capillary Starling forces secondary to reduced RBF, may contribute to Na^+ -retaining and edema-forming states, such as HF (see Fig. 14.4).

From studies using micropuncture, microperfusion, and the isolated perfused tubule model,^{141–145} the role of peritubular forces in the setting of increased ECF volume can be summarized as follows:

1. Acute saline expansion results in dilution of plasma proteins and reduction in efferent arteriolar π_c . SNGFR and peritubular P_c may be increased as well, but the decrease in peritubular π_c by itself results in a decreased net peritubular capillary reabsorptive force and decreased Rate_{abs} . GTB is disrupted because Rate_{abs} falls, despite the tendency for SNGFR to rise, thereby allowing the excess Na^+ to be excreted and plasma volume to be restored.
2. Iso-oncotic plasma infusions tend to raise SNGFR and peritubular P_c but lead to relative constancy of efferent arteriolar π_c . Rate_{abs} may, therefore, decrease slightly, resulting in less disruption of GTB and less natriuresis than that observed with saline expansion.
3. Hyperoncotic expansion usually increases both SNGFR (because of volume expansion) and efferent arteriolar π_c . As a result, Rate_{abs} is enhanced and GTB, therefore, tends to be better preserved than with iso-oncotic plasma or saline expansion.
4. Changes in π_i can directly alter proximal tubular reabsorption, independently of the peritubular capillary bed.

The alterations in proximal peritubular Starling forces that modulate fluid and solute movement across the peritubular basement membrane into the surrounding capillary bed appear to be accompanied by corresponding changes in

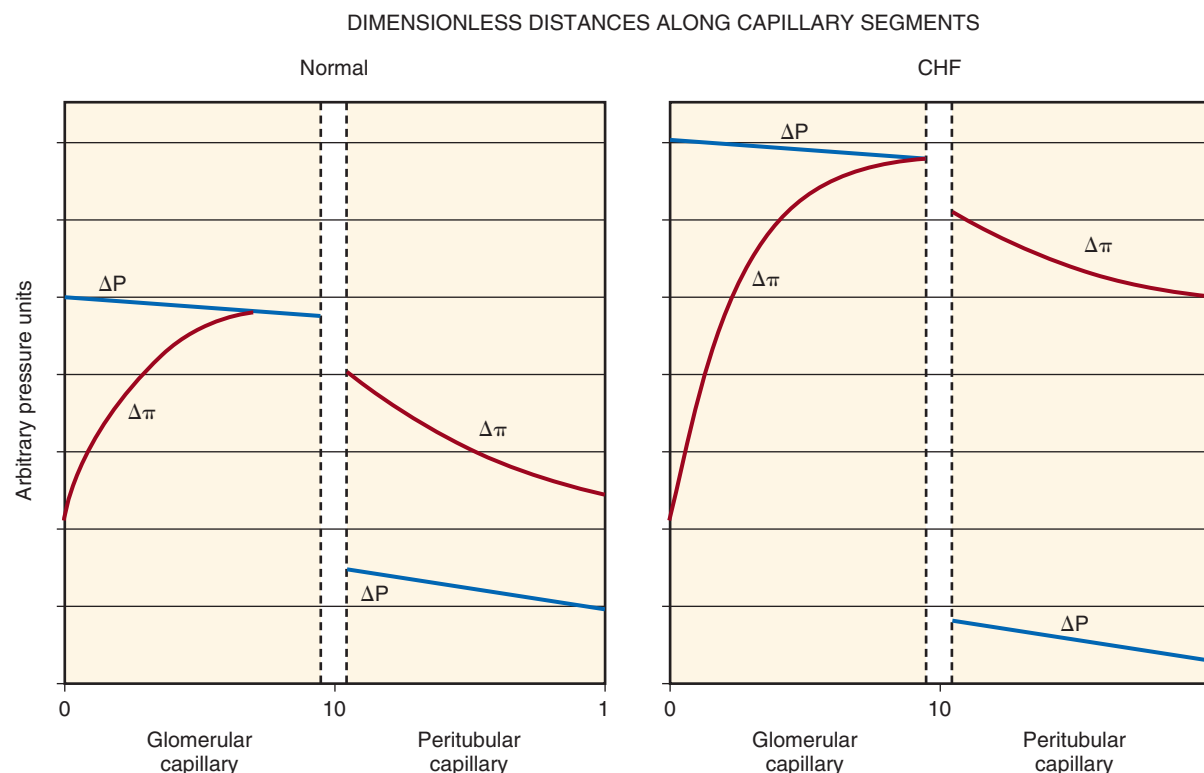


Fig. 14.4 The glomerular and peritubular microcirculations. *Left*, Approximate transcapillary pressure profiles for the glomerular and peritubular capillaries in normal humans. Vessel lengths are given in normalized nondimensional terms, with 0 being the most proximal portion of the capillary bed and 1 the most distal portion. Thus, for the glomerulus, 0 corresponds to the afferent arteriolar end of the capillary bed and 1 corresponds to the efferent arteriolar end. The transcapillary hydraulic pressure difference (ΔP) is relatively constant with distance along the glomerular capillary and the net driving force for ultrafiltration ($\Delta P - \Delta \pi$) diminishes primarily as a consequence of the increase in the opposing COP difference ($\Delta \pi$), the latter resulting from the formation of an essentially protein-free ultrafiltrate. As a result of the drop in pressure along the efferent arteriole, the net driving pressure in the peritubular capillaries ($\Delta P - \Delta \pi$, in which $\Delta \pi$ is the change in transcapillary oncotic pressure) becomes negative, favoring reabsorption. *Right*, Hemodynamic alterations in the renal microcirculation in congestive heart failure (CHF). The fall in renal plasma flow (RPF) rate in heart failure is associated with a compensatory increase in ΔP for the glomerular capillary, which is conducive to a greater than normal rise in the plasma protein concentration and, hence, in $\Delta \pi$ along the glomerular capillary. This increase in $\Delta \pi$ by the distal end of the glomerular capillary also translates to an increase in $\Delta \pi$ in the peritubular capillaries, resulting in an increased net driving pressure for enhanced proximal tubule fluid absorption, believed to take place in heart failure. The increased peritubular capillary absorptive force in heart failure also probably results from the decline in ΔP , a presumed consequence of the rise in renal vascular resistance. (From Humes HD, Gottlieb M, Brenner BM. *The kidney in congestive heart failure: contemporary issues in nephrology*. Vol 1, New York: Churchill Livingstone 1978: pp 51–72.)

structure of the peritubular interstitial compartment. Ultrastructural data from rats have suggested that the peritubular capillary wall is in tight apposition to the tubule basement membrane for about 60% of the tubule basolateral surface. However, over the other 40%, irregularly shaped wide portions of peritubular interstitium intercede between the tubule and peritubular capillaries. Alterations in the physical properties of the interstitial compartment could conceivably modulate net fluid transport in the proximal tubule via changes in P_i . Also, Starling forces in the peritubular capillary are thought to regulate the rate of fluid entry from the peritubular interstitium into the capillary. Any change in this rate of flux could also lead to changes in P_i that secondarily modify proximal tubule solute transport. This formulation could explain why raising P_i (e.g., by infusion of renal vasodilators, renal venous constriction, renal lymph ligation) were associated with a natriuretic response, whereas the opposite effect was seen with renal decapsulation, which lowers P_i (see section, “Medullary Hemodynamics and Interstitial Pressure in the Control of Sodium Excretion: Pressure Natriuresis”).

Because of the relatively high permeability of the proximal tubule, changes in interstitial Starling forces are likely to be transduced mainly through alterations in passive bidirectional paracellular flux through the tight junctions.¹⁴⁶ These changes might be mediated by the claudin family of adhesion molecules present in the tight junctions.^{147–149} Among the 24 known mammalian claudin family members, at least two—claudin-2 and claudin-10—are located in the proximal nephron of the mouse.^{148,150} In particular, claudin-2 is selectively expressed in the proximal nephron.^{151,152} However, the exact role of claudins in the influence of Starling forces on fluid reabsorption remains to be elucidated.¹⁵²

Luminal Composition. Even when the native peritubular environment is kept constant while the rate of perfusion of proximal tubular segments with native tubular fluid is changed, GTB can still be fully expressed.¹⁵³ A transtubular anion gradient, normally present in the late portion of the proximal nephron, is necessary for this flow dependence to occur.¹⁵⁴ Establishment of this anion gradient depends on

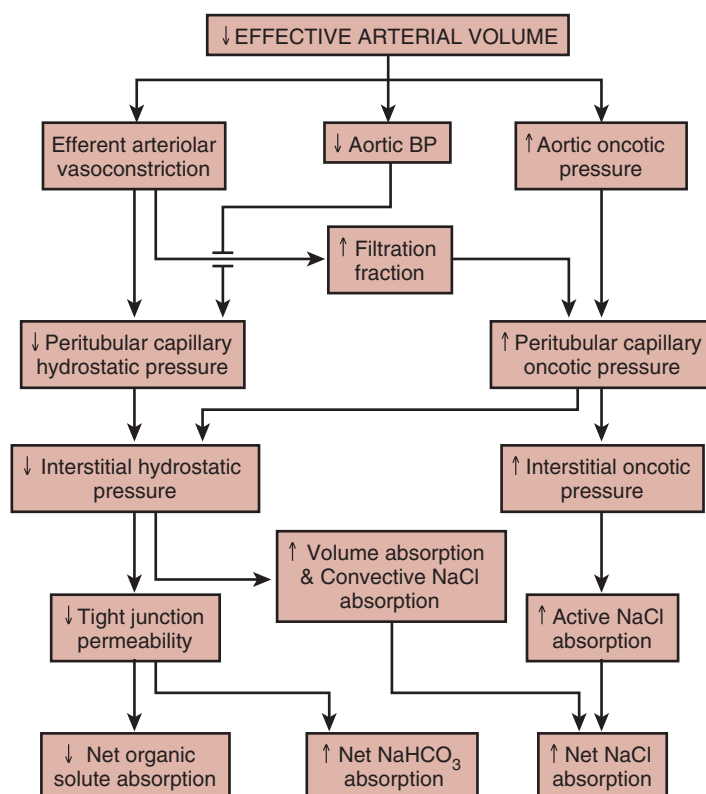


Fig. 14.5 Effects of hemodynamic changes on proximal tubule solute transport. BP, Blood pressure. (From Seldin DW, Preisig PA, Alpern RJ. Regulation of proximal reabsorption by effective arterial blood volume. *Semin Nephrol.* 1991;11:212–219.)

the close coupling of Na^+ transport with the cotransport of glucose, amino acids and other organic solutes. The increased delivery of organic solutes that accompanies increases in GFR, together with the preferential reabsorption of Na^+ with bicarbonate in the early proximal tubule, would lead to increased delivery of both Cl^- and organic solutes to the late proximal tubule. The resulting transtubular anion gradient would then facilitate the “passive” reabsorption of the organic solutes and NaCl in this segment, although overall net reabsorption would be reduced.

In summary, regardless of the exact mechanism, ECFV expansion impairs the integrity of GTB, allowing increased delivery of salt and fluid to more distal parts of the nephron. The major factors acting on the proximal nephron during a decrease in ECF and effective arterial circulating volume are outlined schematically in Fig. 14.5.

Physical Factors Beyond the Proximal Tubule. The final urinary excretion of Na^+ , in response to volume expansion or depletion, can be dissociated from the amount delivered out of the superficial proximal nephron. This is achieved by appropriate modulation of Na^+ and water excretion in the loop of Henle, distal tubule, cortical and papillary collecting ducts. However, direct evidence that these transport processes are mediated by changes in Starling forces per se is lacking (see Jamison et al.¹⁵⁵ for a detailed review of experimental evidence).

In summary, if ECFV is held relatively constant, an increase in GFR leads to little or no increase in salt excretion because of a close coupling between GFR and intrarenal physical forces acting at the peritubular capillary to control Rate_{abs} .

In addition, changes in the filtered load of small organic solutes and perhaps other, as yet uncharacterized, substances in tubular fluid may influence Rate_{abs} . Any changes in the Na^+ load delivered to more distal segments are accompanied by parallel changes in distal reabsorptive rates, thereby ensuring a high overall degree of GTB. Conversely, ECFV expansion leads to large increases in Na^+ excretion, even in the presence of a reduced GFR. Changes in Na^+ reabsorption in the proximal tubule partially account for this natriuresis, but suppression of more distal Na^+ reabsorption is also likely.

Medullary Hemodynamics and Interstitial Pressure in the Control of Sodium Excretion: Pressure Natriuresis. First proposed in the 1960s and elucidated in the 1970s and 1980s,^{156,157} ECFV expansion (or elevation in systemic BP) was shown to elicit an increase in RPP that is transmitted as enhanced medullary plasma flow; this leads to a subsequent loss of medullary hypertonicity, elimination of the medullary osmotic gradient (“medullary washout”), thereby decreasing water reabsorption in the thin descending loop of Henle. This decrease in water reabsorption lowers the Na^+ concentration in the fluid entering the ascending loop of Henle, thus decreasing the transepithelial driving force for salt transport in this nephron segment, particularly in juxtamedullary nephrons. The result is an increase in Na^+ and water excretion, thereby decreasing circulating blood volume and restoring arterial pressure.^{158,159} The phenomenon, known as “pressure natriuresis,”^{158–162} occurred in the absence of changes in total RBF, GFR, or filtered load of Na^+ and seems to be triggered by changes in medullary and papillary circulation.^{156,161,163–165}

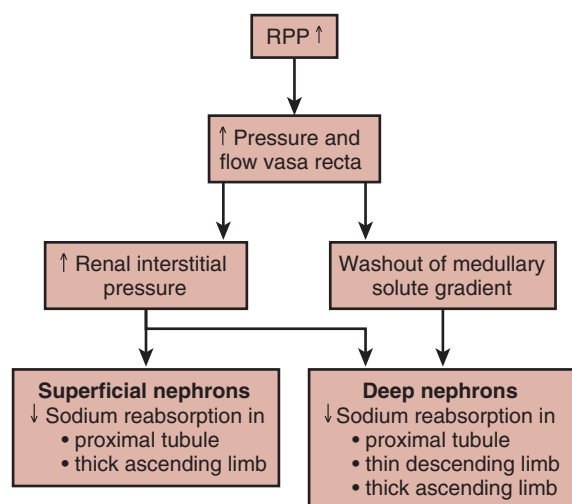


Fig. 14.6 Role of the renal medulla in modulating tubular reabsorption of sodium in response to changes in renal perfusion pressure (RPP). (Modified from Cowley AW, Jr. Role of the renal medulla in volume and arterial pressure regulation. *Am J Physiol.* 1997;273:R1–R15.)

An increase in medullary plasma flow leads not only to medullary washout and a consequent reduction in Na^+ reabsorption in the ascending loop of Henle but also to a rise in P_i . In fact, increasing P_i by ECFV expansion by infusion of renal vasodilatory agents, long-term mineralocorticoid escape, or hilar lymph ligation results in a significant increase in Na^+ excretion.^{166,167} Prevention of the increase in P_i by removal of the renal capsule attenuates, but does not completely block, the natriuretic response to elevations in RPP. Thus as depicted in Fig. 14.6, elevation in RPP is associated with an increase in medullary plasma flow and increased vasa recta capillary pressure, which results in an increase in medullary P_i . This increase of P_i is thought to be transmitted to the renal cortex in the encapsulated kidneys and to provide a signal that inhibits Na^+ reabsorption along the nephron. In that regard, the renal medulla may be viewed as a sensor that detects changes in RPP and initiates pressure natriuresis.

To explain how changes in systemic BP are transmitted to the medulla in the presence of efficient RBF and GFR autoregulation, it has been suggested that shunt pathways connect preglomerular vessels of juxtamedullary nephrons directly to the postglomerular capillaries of the vasa recta.¹⁵⁶ Alternatively, autoregulation of RBF might lead to increased shear stress in the preglomerular vasculature, triggering the release of NO and perhaps cytochrome P450 products of arachidonic acid metabolism (see later), thereby driving the cascade of events that inhibit Na^+ reabsorption.^{168,169} The mechanisms by which changes in P_i decrease tubular Na^+ reabsorption and increase U_{Na} excretion, as well as the nephron sites responding to the alterations in P_i , have not been fully clarified.¹⁶⁷ As noted earlier, elevations in P_i may increase passive backleak or paracellular pathway hydraulic conductivity, with a resultant increase in back flux of Na^+ through this pathway.¹⁶⁶ However, the absolute changes in P_i , in the range of 3 to 8 mm Hg in response to increments of about 50 to 90 mm Hg in RPP, are probably not sufficient to account for the decrease in Na^+ reabsorption, even in the

proximal tubule, the nephron segment with the highest transepithelial hydraulic conductivity.¹⁶⁰ The decreased proximal tubular Na^+ reabsorption may also be explained by redistribution of the apical Na^+/H^+ exchanger from the brush border into intracellular compartments and concomitantly decreased basolateral Na^+/K^+ -ATPase activity in response to increased RPP.¹⁷⁰ The reduction in proximal fluid reabsorption is especially marked in deep nephrons, leading to enhanced delivery to the loop of Henle, inhibition of Na^+ reabsorption in the thin descending limb and reduced blood flow in the vasa recta.¹⁶⁶ Pressure-induced changes in tubular reabsorption may also occur in the ascending loop of Henle, distal tubule, and collecting duct.¹⁷¹

The demonstration of pressure natriuresis in states of volume expansion and renal vasodilation and its significant attenuation in states of volume depletion¹⁶⁶ suggested that changes in P_i may be amplified by hemodynamic, hormonal and paracrine factors.^{156,161,163,166} For example, RAAS inhibition potentiates whereas cyclooxygenase inhibitors attenuate the pressure natriuretic response.^{166,172} Moreover, blockade of the Ang II type 2 receptor allows the same amount of Na^+ to be excreted at lower arterial pressure.¹⁷³ However, RAAS blockade does not completely eliminate pressure natriuresis, which indicates that the RAAS act as modulators and not as mediators of the phenomenon.

Pressure natriuresis is also modulated by endothelium-derived NO and P450 eicosanoids.^{174–176} NO, generated in large amounts in the renal medulla, plays a critical role in the regulation of medullary blood flow and Na^+ excretion.^{174–176} In this regard, inhibition of intrarenal NO production reduces Na^+ excretion and markedly suppresses pressure natriuresis,¹⁷⁷ whereas administration of an NO agonist normalizes the defect in the pressure natriuresis response in Dahl salt-sensitive rats.¹⁷⁸ Similarly, a positive correlation between urinary excretion of nitrites and nitrates (metabolites of NO) and changes in renal arterial pressure or U_{Na} excretion was observed both in dogs^{174,179,180} and rats.¹⁷⁴ Hydrogen peroxide (H_2O_2) has also been invoked in the mediation of RPP-induced changes in outer medullary blood flow and natriuresis. The response appears to be localized to the medullary thick ascending limb of Henle (mTALH), in contrast to the NO effect, which occurs in the vasa recta.¹⁷⁴ Other factors involved in the regulation of medullary blood flow and pressure natriuresis include superoxide, heme oxygenase,^{175,180} and the cytochrome P450 eicosanoid 20-HETE.^{168,169,181,182}

One mechanism by which acute elevations in RPP in the autoregulatory range lead to increased endothelial release of NO and reactive oxygen species could be via increased blood flow velocity and shear stress. Enhanced renal production of these molecules may then increase U_{Na} excretion by acting directly on tubular Na^+ reabsorption or through a renal vasodilatory effect. ATP is another paracrine factor involved in pressure natriuresis by inhibiting salt and water reabsorption. ATP release appears to be mediated by mechanosensitive connexin 30 hemichannels.¹⁸³ The mechanisms of these cellular events have not been fully elucidated, but they may be related directly to changes in P_i or to changes in the intrarenal paracrine agents described previously.

A major assumption of the pressure natriuresis theory is that changes in systemic and RPP mediate the natriuretic response by the kidneys. As noted in comprehensive reviews, acute regulatory changes in renal salt excretion may occur

without measurable elevation in arterial BP.^{48,184–186} In many of these studies, natriuresis was accompanied by a decrease in the activity of the RAAS, without changes in plasma ANP levels.^{48,74,185,186} Thus, whereas increases in arterial BP can drive renal Na⁺ excretion, other pressure-independent control mechanisms must also operate to mediate the natriuresis.⁴⁸

Neural Mechanisms: Renal Nerves and Sympathetic Nervous System

Extensive sympathetic innervation, predominantly adrenergic, occurs at all segments of the renal vasculature and tubule.¹⁸⁷ Only the basolateral membrane separates the nerve endings from the tubular cells. The greatest innervation is found at the level of the afferent arterioles, followed by the efferent arterioles and outer medullary descending vasa recta.¹⁸⁸ However, high-density tubular innervation is found in the ascending limb of the loop of Henle, with the lowest density in the collecting duct, inner medullary vascular elements and papilla.^{189,190} The magnitude of the tubular response to renal nerve activation may thus be proportional to the differential density of innervation.

In accordance with these anatomic observations, stimulation of the renal nerve results in vasoconstriction of afferent and efferent arterioles^{190,191} mediated by activation of postjunctional α_1 -adrenoreceptors.¹⁹² With respect to tubular function, α_1 -adrenergic receptors and most α_2 -adrenergic receptors are localized in the basolateral membranes of the proximal tubule.¹⁹³ In the rat, β_1 -adrenoreceptors have been found in the cTALH.¹⁹⁴ The predominant neurotransmitters in renal sympathetic nerves are noradrenaline and, to a lesser extent, dopamine and acetylcholine.¹⁹⁰ Consistent with their location, changes in RSNA play an important role in controlling body fluid homeostasis and BP.^{73,187,191} Renal sympathetic nerve activity can influence renal function and Na⁺ excretion through several mechanisms: 1. changes in renal and glomerular hemodynamics; 2. effect on renin release from juxtaglomerular cells, with increased formation of Ang II; and 3. direct effect on renal tubular fluid and electrolyte reabsorption.⁷³ Graded direct electrical stimulation of renal nerves produces frequency-dependent changes in RBF and GFR, reabsorption of renal tubular Na⁺ and water and secretion of renin.^{73,187} The lowest frequency (0.5–1.0 Hz) stimulates renin secretion and frequencies of 1.0 to 2.5 Hz increase renal tubule Na⁺ and water reabsorption. Increasing the frequency of stimulation to 2.5 Hz and higher results in decreased RBF and GFR.^{73,191}

The decrease in SNGFR in response to enhanced RSNA results from a combination of increases in afferent and efferent glomerular resistance, as well as decreases in glomerular capillary hydrostatic pressure (ΔP) and ultrafiltration coefficient (Kf).^{187,191} In Munich-Wistar rats, renal nerve stimulation at different frequencies revealed that the effector loci for vasomotor control by renal nerves were in the afferent and efferent arteriole. In addition, although urine flow and Na⁺ excretion declined with renal nerve stimulation, there was no change in absolute proximal fluid reabsorption rate, which suggests that reabsorption is increased in the more distal segments of the nephron.

The SNS also has a role in regulating the renal response to varying Na⁺ loads. In response to isotonic saline volume expansion and furosemide-induced volume contraction, a low-Na⁺ diet resulted in a reduction in right atrial pressure

and an increase in RSNA. Conversely, a high-Na⁺ diet resulted in increased right atrial pressure and a reduction in RSNA.¹⁹¹ Other studies using HWI and left atrial balloon inflation have similarly yielded evidence of the importance of reflex regulation of RSNA.⁷³

Collectively, these studies demonstrated the reciprocal relationship between ECFV and RSNA and the role of central cardiopulmonary mechanoreceptors governing RSNA. Moreover, efferent RSNA is of special significance during conditions of dietary Na⁺ restriction, when the need for renal Na⁺ conservation is maximal. Indeed, when the linkage between the renal SNS and excretory renal function is defective, abnormalities in the regulation of ECF volume and BP may develop.^{187,195} Thus, in euvoletic animals, acute denervation of the kidneys is associated with increased urine flow and Na⁺ excretion, but without alteration in any of the determinants of SNGFR. However, absolute proximal reabsorption was significantly reduced in the absence of changes in peritubular capillary oncotic pressure, hydraulic pressure and renal interstitial pressure. Na⁺ and water reabsorption was also reduced in the loop of Henle and more distal segments.¹⁹¹ In rats with HF, in contrast, denervation ameliorated both renal vasoconstriction, by decreasing afferent and efferent arteriolar resistance and natriuresis, by reduced Na⁺ reabsorption.¹⁹¹

Human studies, using the renal NE spillover technique, have confirmed that a true increase of efferent RSNA and a fall in U_{Na} excretion occurs secondary to Na⁺ restriction, with no change in cardiac NE uptake.¹⁹⁶ Similarly, low-dose NE in salt-replete volunteers resulted in antinatriuresis with a significant decline in lithium (Li⁺) clearance, indicating enhanced proximal tubule reabsorption. The reduced Na⁺ excretion occurred without change in GFR.¹⁹⁷

The cellular mechanisms mediating the tubular actions of NE include stimulation of Na⁺-K⁺-ATPase activity and Na⁺/H⁺ exchange in proximal tubular epithelium.¹⁹¹ This occurs via α_1 -adrenoreceptor stimulation, mediated by phospholipase C, causing an increase in intracellular Ca²⁺ that activates the Ca²⁺ calmodulin-dependent calcineurin phosphatase. Calcineurin converts Na⁺-K⁺-ATPase from its inactive phosphorylated form to its active dephosphorylated form.¹⁹⁸ The stimulatory effect of renal nerves on Na⁺/H⁺ exchange is mediated through the α_2 -adrenoreceptor.¹⁹¹

In addition to the direct action of Na⁺ on epithelial cell transport and renal hemodynamics, interactions of renal nerve input with other effector mechanisms may contribute to the regulation of renal handling of Na⁺. Efferent sympathetic nerve activity influences the rate of renal renin secretion both directly and via the macula densa and vascular baroreceptors.¹⁹¹ The increase in renin secretion is mediated primarily by β_1 -adrenergic receptors located on juxtaglomerular granular cells and is augmented during RPP reduction.¹⁹¹ Intrarenal generation of Ang II facilitates NE release during renal nerve stimulation,¹⁹¹ but the physiologic significance of this remains unclear.¹⁹⁹

Sympathetic activity is also a stimulus for the production and release of renal PG, coupled to adrenergic-mediated renal vasoconstriction.¹⁹¹ Renal vasodilatory PG attenuate the renal hemodynamic vasoconstrictive response to renal adrenergic activation.¹⁹¹ The primary factor responsible for the reduction in glomerular Kf during renal nerve stimulation may be Ang II rather than NE and endogenously produced

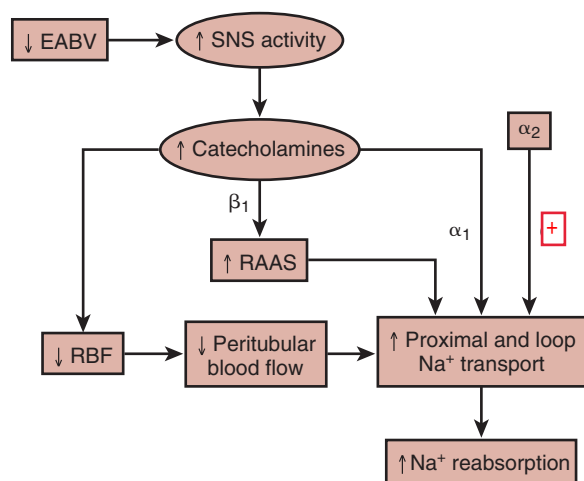


Fig. 14.7 Sympathetic nervous system (SNS)-mediated effects of decreased effective arterial blood volume (EABV) on the kidneys. α_1 , α_2 , β_1 , α_1 , α_2 - and β_1 -Adrenergic receptors, respectively. RAAS, Renin-angiotensin-aldosterone system; RBF, renal blood flow; –, inhibitory effect.

PG neutralize the vasoconstrictive effects of renal nerve stimulation at the intraglomerular rather than arteriolar level.

The renal SNS also interacts with AVP, which exerts a dose-related effect on the arterial baroreflex. Low doses of AVP sensitize the central baroreflex neurons to afferent input, whereas higher doses cause direct excitation of these neurons, resulting in reduced renal sympathetic outflow.¹⁹¹ This response depends on the number of afferent inputs from baroreceptors.²⁰⁷ Conversely, renal nerve stimulation results in elevated plasma AVP levels and arterial pressure in conscious, baroreceptor-intact rats.²⁰⁰ Many studies have demonstrated in normal and pathologic situations that increased RSNA can antagonize the natriuretic/diuretic response to ANP and that removal of the influence of sympathetic activity enhances the natriuretic action of the peptide.¹⁹¹ Conversely, renal denervation in Wistar rats increased ANP receptors and cGMP generation in glomeruli, resulting in increased Kf after ANP infusion.²⁰¹

In summary, renal sympathetic nerves regulate U_{Na} and water excretion by changing renal vascular resistance, by influencing renin release from the juxtaglomerular granular cell and through a direct effect on tubular epithelial cells (Fig. 14.7). These effects may be modulated by other hormonal systems, including ANP, PG, and AVP.

Humoral Mechanisms

Renin-Angiotensin-Aldosterone System. The RAAS plays a central role in the regulation of ECF volume, Na^+ homeostasis and cardiac function.²⁰² The system is activated in situations that compromise hemodynamic stability, such as blood loss, reduced EABV, low Na^+ intake, hypotension and increase in sympathetic nerve activity. The RAAS is comprised of a coordinated hormonal cascade whose synthesis is initiated by the release of renin from the JGA in response to reduced renal perfusion or decrease in arterial pressure.²⁰³ Renin acts on its circulating substrate, angiotensinogen, which is produced and secreted mainly by the liver, but also by the kidneys.²⁰² ACE1, which cleaves Ang I to Ang II, exists in

large amounts in the microvasculature of the lungs but also on endothelial cells of other vascular beds and cell membranes of the proximal nephron brush border, heart, and brain.²⁰² Ang II is the principal effector of the RAAS, although other smaller metabolic products of Ang II also have biologic activities.^{204,205} Nonrenin (cathepsin G, plasminogen-activating factor, tonin) and non-ACE pathways (chymase, cathepsin G) also exist in these tissues and may contribute to tissue Ang II synthesis.²⁰²

In addition to its important function as a circulating hormone, Ang II produced locally acts as a paracrine growth-promoting agent in the cardiovascular system and kidneys.²⁰² For example, proximal tubular epithelial cells, which abundantly express the mRNA for angiotensinogen, synthesize and secrete Ang II into the lumen;²⁰⁶ this leads to approximately 1000 times higher local concentrations in proximal tubular fluid, as well as higher levels in interstitial fluid and medulla than in the plasma.²⁰⁷ Moreover, the mechanisms regulating intrarenal levels of Ang II appear to be dissociated from those controlling the systemic concentrations of the peptide.²⁰⁶

The biologic actions of Ang II are mediated through activation of AT_1 and AT_2 receptors, encoded by different genes residing on different chromosomes.^{208,209} Both receptors are G protein-coupled, seven-transmembrane polypeptides containing approximately 360 amino acids.^{202,210} The AT_1 receptor mediates most of the biologic activities of Ang II, whereas the AT_2 receptor appears to have a vasodilatory and antiproliferative effect.^{204,211} AT_1 is expressed in the vascular poles of glomeruli, JGA and mesangial cells, whereas the quantitatively lower expression of AT_2 is confined to renal arteries and tubular structures.²⁰⁹ In addition to their functional distinction, the two receptor types use different downstream pathways. Stimulation of the AT_1 receptor activates phospholipases A_2 , C and D, resulting in increased cytosolic Ca^{2+} and inositol triphosphate and inhibition of adenylate cyclase. In contrast, activation of the AT_2 receptor results in increases in NO and bradykinin levels, which lead to elevation in cGMP concentrations and vasodilation.²⁰⁸

In addition to being an important source of several components of the RAAS, the kidney is a major target organ for Ang II and aldosterone. The direct effect of Ang II is mediated via AT_1 receptors to induce renal vasoconstriction, stimulation of tubular epithelial Na^+ reabsorption, augmentation of TGF sensitivity, modulation of pressure natriuresis, and stimulation of mitogenic pathways.²⁰² Moreover, circulating levels of Ang II in the picomolar range are highly effective in modulating renal hemodynamic and tubular function, in comparison with the 10- to 100-fold higher concentrations required for its extrarenal effects. Thus the kidneys appear to be uniquely sensitive to the actions of Ang II. Furthermore, the synergistic interactions between the renal vascular and tubular actions of Ang II significantly amplify the influence of Ang II on Na^+ excretion.²⁰⁶ Ang II elicits a dose-dependent decrease in RBF but slightly augments GFR as a result of its preferential vasoconstrictive effect on the efferent arteriole, thereby increasing filtration fraction. In turn, this may further modulate peritubular Starling forces, possibly by decreasing interstitial hydraulic and increasing COP. These peritubular changes eventually lead to enhanced proximal Na^+ and water reabsorption. Of importance, however, is that changes in preglomerular resistance have also been described during

Ang II infusion or blockade.²¹² These may be secondary to changes in systemic arterial pressure (myogenic reflex) or to increased sensitivity of TGF because Ang II does not alter preglomerular resistance when RPP is clamped or adjustments in TGF are prevented.²¹²

In addition, Ang II may affect GFR by reducing Kf, thereby altering the filtered load of Na⁺.²¹³ This effect is likely mediated by mesangial cell contractility and increasing permeability to macromolecules.²¹² Finally, Ang II leads to reduced cortical and medullary blood flow and decreased Na⁺ and water excretion.^{212,214} As noted earlier, changes in medullary blood flow may affect medullary tonicity, which determines the magnitude of passive salt reabsorption in the loop of Henle and may also modulate pressure natriuresis through alterations in renal interstitial pressure.²¹⁴

The other well-characterized renal effect of Ang II is a direct action on proximal tubular epithelial transport, independently of changes in renal or systemic hemodynamics.^{202,215} Ang II exerts a dose-dependent biphasic effect on proximal Na⁺ reabsorption. Peritubular capillary infusion with solutions containing low concentrations of Ang II (10⁻¹² to 10⁻¹⁰ mol) stimulate, whereas higher concentrations (>10⁻⁷ mol) inhibit proximal Na⁺ reabsorption. Addition of ACE inhibitors (ACEIs) or angiotensin II receptor blockers (ARB) directly into the luminal fluid results in a significant decrease in proximal Na⁺ reabsorption, indicative of tonic regulation of proximal tubule transport by endogenous Ang II.²¹⁶

The specific mechanisms by which Ang II influences proximal tubule transport include increases in reabsorption of Na⁺ and HCO₃⁻ by stimulation of the apical Na⁺-H⁺ antiporter, Na⁺/H⁺-exchanger isoform 3 (NHE3), basolateral Na⁺-3HCO₃⁻ symporter and Na⁺-K⁺-ATPase.^{217,218} Thus, Ang II can affect NaCl absorption by two mechanisms:

1. Activation of NHE3 can directly increase NaCl absorption.
2. Increasing the rate of NaHCO₃ absorption can stimulate passive NaCl absorption by increasing the concentration gradient for passive Cl⁻ diffusion.²¹⁹

Na⁺ reabsorption is further promoted by the action of Ang II on NHE3 and Na⁺-K⁺-ATPase in the mTALH.²⁰²

In the early and late portions of the distal tubule, as well as the connecting tubule, Ang II regulates Na⁺ and HCO₃⁻ reabsorption by stimulating NHE3 and the amiloride-sensitive Na⁺ channel.²²⁰⁻²²² Two additional mechanisms may amplify the antinatriuretic effects of Ang II. The first concerns the increased sensitivity of TGF in the presence of Ang II and the second is related to the effect of Ang II on pressure natriuresis. The decrease in distal delivery produced by the action of Ang II on renal hemodynamics and proximal fluid reabsorption could elicit afferent arteriolar vasodilation by means of the TGF mechanism, which, in turn, could antagonize the Ang II-mediated increase in proximal reabsorption. This effect, however, is minimized because Ang II increases the responsiveness of the TGF mechanism, thus maintaining GFR at a lower delivery rate to the macula densa.⁷⁵ The second mechanism by which the antinatriuretic effects of Ang II may be amplified is blunting of the pressure natriuresis mechanism so that higher pressures are needed to induce a given amount of Na⁺ excretion.^{168,202} This shift to the right in the pressure natriuresis curve may be an important

Na⁺-conserving mechanism in situations of elevated arterial pressure.

As already stated, most of the known intrarenal effects of Ang II in the regulation of renal hemodynamics and proximal tubule reabsorption of Na⁺ and HCO₃⁻ are mediated by the AT₁ receptor.²⁰⁹ However, activation of the AT₂ receptor may play a counterregulatory protective role against the AT₁ receptor-mediated antinatriuretic and pressor actions of Ang II.²⁰⁹

Ang I is converted not only to Ang II but also to angiotensin-(1-7) (Ang 1-7)²²³ either directly by the homolog of ACE, ACE2, or indirectly, via angiotensin-(1-9), followed by ACE-mediated conversion to Ang 1-7.²²³ Ang 1-7, through its G protein-coupled receptor, Mas, may play a significant role as a regulator of cardiovascular and renal function by opposing the effects of Ang II, through vasodilation, diuresis, and an antihypertrophic action.²²³ Thus, the RAAS can currently be envisioned as a dual-function system in which the vasoconstrictor/proliferative or vasodilator/antiproliferative actions are driven primarily by the ACE/ACE2 balance. According to this model, an increased ACE/ACE2 activity ratio leads to increased generation of Ang II and increased catabolism of Ang 1-7, which is conducive to vasoconstriction; conversely, a decreased ACE/ACE2 ratio reduces Ang II and increases Ang 1-7 levels, facilitating vasodilation. The additional effect of Ang 1-7/Mas to antagonize the actions of Ang II directly adds a further level of counterregulation.²²³

The final component of the RAAS, aldosterone, is produced via Ang II stimulation of the adrenal cortex and also plays an important physiologic role in the maintenance of ECFV and Na⁺ homeostasis.²²⁴ The primary sites of aldosterone action are the principal cells of the cortical collecting tubule and distal convoluted tubule, in which the hormone promotes the reabsorption of Na⁺ and the secretion of K⁺ and protons.^{224,225} Aldosterone may also enhance electrogenic Na⁺ transport, but not K⁺ secretion, in the IMCD²²⁶ and proximal tubule.²²⁷ Aldosterone exerts its effects by increasing the number of open Na⁺ and K⁺ channels in the luminal membrane and the activity of Na⁺-K⁺-ATPase in the basolateral membrane.²²⁸ The effect of aldosterone on Na⁺ permeability appears to be the primary event because blockade of the ENaC with amiloride prevents the initial increase in Na⁺ permeability and Na⁺-K⁺-ATPase activity.²²⁸ This effect on Na⁺ permeability is mediated by changes in intracellular Ca²⁺, intracellular pH,²²⁹ trafficking via protein kinase D1-phosphatidylinositol 4-kinaseIIIβ trans Golgi signaling,²³⁰ and methylation of channel proteins, thus increasing the mean open probability of ENaC.²²⁹ However, the long-term effect of aldosterone on Na⁺-K⁺-ATPase activity involves de novo protein synthesis, which is regulated at the transcriptional level by serum- and glucocorticoid-induced kinase-I.²²⁹

Aldosterone specifically regulates the α-subunit of ENaC and changes in expression of a variety of genes are important intermediates in this process. Microarray analysis in a mouse IMCD line showed that the most prominent transcript induced acutely by aldosterone was period homolog 1 (Per1), an important component of the circadian clock and that disruption of the Per1 gene led to attenuated expression of mRNA encoding for the α-subunit and increased U_{Na} excretion. mRNA encoded by the α-subunit was also expressed in an apparent circadian pattern that was dramatically altered in mice lacking functional Per1 genes.²³¹ These results imply

that the circadian clock has a role in the control of Na^+ balance and provide molecular insight into how the circadian cycle directly affects Na^+ homeostasis.

The Na^+ -retaining effect of aldosterone in the collecting tubule induces an increase in the transepithelial potential difference, which is conducive to K^+ excretion. In terms of overall body fluid homeostasis, the actions of aldosterone in the defense of ECF result from the net loss of an osmotically active particle confined primarily to the intracellular compartment (K^+) and its replacement with a corresponding particle confined primarily to the ECF (Na^+). The effect of a given circulating level of aldosterone on overall Na^+ excretion depends on the volume of filtrate reaching the collecting duct and the composition of luminal and intracellular fluids. As noted earlier, this delivery of filtrate is, in turn, determined by other effector mechanisms (Ang II, sympathetic nerve activity and peritubular physical forces) acting at more proximal nephron sites.

Na^+ balance can be regulated over a wide range of intake, even in subjects without adrenal glands and despite fixed low or high supplemental doses of mineralocorticoids. Under these circumstances, other effector mechanisms predominate in controlling urinary Na^+ excretion, although often in a setting of altered ECF volume or K^+ concentration. In this regard, how renal Na^+ reabsorption and K^+ excretion are coordinately regulated by aldosterone has long been a puzzle. In states of EABV depletion, aldosterone release stimulated by Ang II induces maximal Na^+ reabsorption without significantly affecting plasma K^+ levels. Conversely, hyperkalemia-induced aldosterone secretion stimulates maximum K^+ excretion without major effects on renal Na^+ handling.

Elegant studies on the intracellular signaling pathways involved in renal Na^+ and K^+ transport have shed light on this puzzle. The key elements in this transport regulation are the Ste20/SPS1-related proline/alanine-rich kinase (SPAK), oxidative stress-related kinase (OSR1) the with-no-lysine kinases (WNKs) and their effectors, the thiazide-sensitive NaCl cotransporter and the K^+ secretory channel ROMK. According to the proposed model, when EABV is reduced or dietary salt intake is low, Ang II, mediated by the AT_1 receptor, leads to phosphorylation of WNK4, which stimulates phosphorylation of SPAK and OSR1. In turn, SPAK and OSR1 phosphorylate the NaCl cotransporter, inducing Na^+ transport and conservation. Simultaneous phosphorylation of the full-length isoform of WNK1, WNK1-L, causes endocytosis of the ROMK channel, thereby enabling K^+ conservation, despite high aldosterone levels. In contrast, in the presence of hyperkalemia or low dietary salt, Ang II levels are low so that WNK4 cannot be activated, SPAK, OSR1 and the NaCl cotransporter are not phosphorylated and NaCl cotransporter trafficking to the apical membrane is inhibited. At the same time, K^+ -induced kidney-specific WNK1 leads to suppression of WNK1-L, which allows ROMK trafficking to the apical membrane and maximal K^+ secretion.²³² For further details, the reader is referred to Chapter 6.

For BP maintenance, systemic vasoconstriction—another major extrarenal action of Ang II—is the appropriate response to perceived ECF volume contraction. As mentioned previously, higher concentrations of Ang II are needed to elicit this response than those governing the antinatriuretic actions of Ang II. Transition from an antinatriuretic to a natriuretic

action of Ang II at high infusion rates is almost entirely due to a concomitant BP rise.²³³

In addition to the adrenal glomerulosa, aldosterone, similar to Ang II, is also produced by the heart and vasculature, exerting powerful mitogenic and fibrogenic effects on blood vessels, independently of regulation of salt and water balance.²³⁴ It directly increases the expression and production of transforming growth factor- β (TGF- β) and thus is involved in the development of glomerulosclerosis, hypertension and cardiac injury/hypertrophy.^{202,224,235}

In summary, Ang II, the principal effector of the RAAS, regulates extracellular volume and renal Na^+ excretion through intrarenal and extrarenal mechanisms. The intrarenal hemodynamic and tubular actions of the peptide and its main extrarenal actions (systemic vasoconstriction and aldosterone release) act in concert to adjust U_{Na} excretion under a variety of circumstances associated with alterations in ECF volume. Many of these mechanisms are synergistic and tend to amplify the overall influence of the RAAS. However, additional counterregulatory mechanisms, induced directly or indirectly by Ang II, provide a buffer against the unopposed actions of the primary components of the RAAS.

Vasopressin. AVP is a nonapeptide hormone, synthesized in the paraventricular and supraoptic nuclei of the hypothalamus and secreted from the posterior pituitary gland into the circulation in response to an increase in plasma osmolality (through osmoreceptor stimulation) or a decrease in EABV and BP (through baroreceptor stimulation).²³⁶ AVP acts through at least three different G protein-coupled receptors. Two of these, $\text{V}_{1\text{A}}$ and V_2 , are abundantly expressed in the cardiovascular system and the kidneys; $\text{V}_{1\text{B}}$ receptors are expressed on the surfaces of corticotropic cells of the anterior pituitary gland, in the pancreas and adrenal medulla. $\text{V}_{1\text{A}}$ and V_2 receptors mediate the two main biologic actions of the hormone, vasoconstriction and increased water reabsorption by the kidneys, respectively. $\text{V}_{1\text{A}}$ and $\text{V}_{1\text{B}}$ receptors operate through the phosphoinositide signaling pathway, causing release of intracellular Ca^{2+} . The $\text{V}_{1\text{A}}$ receptor, found in vascular smooth muscle cells (VSMC), hepatocytes and platelets, mediates vasoconstriction, glycogenolysis and platelet aggregation, respectively. The V_2 receptor, found mainly in renal collecting duct epithelial cells, is linked to the adenylate cyclase/cAMP pathway. A V_3 receptor is not involved in the regulation of ECFV and is not discussed further.²³⁷

Under physiologic conditions, AVP functions primarily to regulate body water content by adjusting reabsorption in the collecting duct according to plasma tonicity. A change in plasma tonicity by as little as 1% causes a parallel change in AVP release. AVP activates V_2 receptors in the basolateral membrane of the principal cells, leading to increased cytosolic cAMP, which stimulates the activity of protein kinase A. The latter triggers a series of phosphorylation events that promotes the translocation of AQP2 from intracellular stores to the apical membrane,²³⁸ as well as the synthesis of AQP2 mRNA and protein;²³⁹ this allows the reabsorption of water from lumen to cells. Water then exits the cell to the hypertonic interstitium via aquaporin-3 and aquaporin-4 channels at the basolateral membrane.²⁴⁰ (See Chapter 10 for further details).

The second major effect of AVP on the collecting duct is to increase the permeability of the IMCD to urea through activation of the urea transporter UT-A1, enabling the accumulation of urea in the interstitium; there, along with Na^+ , it contributes to the hypertonicity of the medullary interstitium, a prerequisite for maximum urine concentration and water reabsorption.²⁴¹ AVP also mediates an increase in Na^+ reabsorption from the mTAL, distal and connecting tubule and collecting duct, by activation of the segment-specific Na^+ transporters; this effect is important for maintenance of the axial corticomedullary osmotic gradient necessary for maximal water reabsorption (see reviews in Kortenoeven et al.²³⁶ and Knepper et al.²⁴²).

In addition, AVP, via the V_{1A} receptor, reduces RBF, especially to the inner medulla, an effect modulated by local release of NO and PG.²⁴³ At higher concentrations, AVP may also decrease total RBF and GFR as part of the generalized vasoconstrictor effect.^{239,244} Experiments in V_{1A} receptor-deficient ($V_{1A}R^{-/-}$) mice have shown decreases in plasma volume, BP, GFR, U_{Na} excretion, AVP-dependent cAMP generation, levels of V_2 receptor and renal AQP2 expression with concomitantly increased urine volume compared with wild type mice; the data imply impaired urinary concentration in $V_{1A}R^{-/-}$ mice. Moreover, plasma renin and Ang II levels were decreased, as was renin expression in granule cells. In addition, the expression of renin stimulators such as nNOS and cyclooxygenase-2 (COX-2) in macula densa cells, where $V_{1A}R$ is specifically expressed, was decreased in $V_{1A}R^{-/-}$ mice. Thus, AVP regulates body fluid homeostasis and GFR through the $V_{1A}R$ in macula densa cells by activating the RAAS and subsequently the V_2 receptor–AQP2 system.^{244,245}

In addition to its renal effects, AVP also regulates extrarenal vascular tone through the V_{1A} receptor. Stimulation of this receptor by AVP results in a potent arteriolar vasoconstriction in various vascular beds, with a significant increase in SVR.²⁴² However, physiologic increases in AVP do not usually cause a significant increase in BP because AVP also potentiates the sinoaortic baroreflexes that subsequently reduce heart rate and cardiac output.²⁴⁶ Nevertheless, at supraphysiologic concentrations, such as those seen when EABV is severely compromised (e.g., in shock or HF), AVP plays an important role in supporting arterial pressure and maintaining adequate perfusion to vital organs such as the brain and myocardium. AVP also has a direct, V_1 receptor-mediated, inotropic effect in the isolated heart, but, in vivo, leads to decreased myocardial function; this effect is attributed to AVP-induced cardioinhibitory reflexes or coronary vasoconstriction.²⁴⁷ Of more importance is that AVP, similar to Ang II and catecholamines, has been shown to stimulate cardiomyocyte hypertrophy and protein synthesis in neonatal rat cardiomyocytes and intact myocardium through a V_1 -dependent mechanism.²⁴⁷ This effect may contribute to the induction of cardiac hypertrophy and remodeling in HF.²⁴⁷

To summarize, regardless of the effects of AVP on Na^+ excretion, the predominant influence of the hormone is indirectly through water accumulation or vasoconstriction. In fact, the vasoconstrictive V_1 receptor effect of AVP overrides the osmotically driven effect in the presence of an ECF volume deficit of 20% or more (see Chapters 10 and 15). Nevertheless, the vasoconstrictive and water-retaining effects of AVP are modulated, respectively, by concomitant

increases in baroreflex-mediated sympathoinhibition or PGE_2 and direct or PGE_2 -mediated suppression of V_2 receptor activation.²⁴⁸

Prostaglandins. Prostaglandins (see also Chapter 16), or cyclooxygenase-derived prostanoids, possess diverse regulatory functions in the kidneys, including hemodynamic, renin secretion, growth response, tubular transport, and immune responses.^{249,250} Two principal isoforms of cyclooxygenase, COX-1 and COX-2, catalyze the synthesis of prostaglandin H_2 (PGH_2) from arachidonic acid, released from membrane phospholipids. PGH_2 is then metabolized to the five major prostanoids— PGE_2 , PGD_2 , $\text{PGF}_2\alpha$, and thromboxane A_2 (TXA_2)—through specific synthases (see also Chapter 16).

Prostanoids are rapidly degraded, so their effect is localized strictly to their site of synthesis, which accounts for their predominant autocrine and paracrine modes of action. Each prostanoid has a location-specific cell surface G protein-coupled receptor that determines the function of the PG in the given cell type.²⁵⁰ The major sites for PG production (and hence for local actions) are the renal arteries, arterioles, and glomeruli in the cortex and interstitial cells in the medulla, with additional contributions from epithelial cells of the cortical and medullary collecting tubules.^{251,252} COX-1 is constitutively and abundantly expressed in the kidneys, especially in the collecting duct but also in medullary interstitial, mesangial, and arteriolar endothelial cells.²⁵⁰ In contrast, COX-2 is inducible, cell type-specific, and prominently expressed in cells of the medullary interstitium, cTALH and macula densa, in which expression is regulated in response to salt intake.²⁵⁰ PGE_2 and PGI_2 are the main products in the cortex of normal kidneys, whereas PGE_2 predominates in the medulla.²⁵⁰ PGF_2 and TXA_2 are also produced in smaller amounts.²⁵⁰ In addition, the metabolism of arachidonic acid by other pathways (e.g., lipoxygenase, epoxygenase) leads to products involved in crosstalk with COX.²⁴⁹

The two major roles for PG in volume homeostasis are modulation of 1. RBF and GFR and 2. tubular handling of salt and water. PGI_2 and PGE_2 have predominantly vasodilating and natriuretic activities, modulate the action of AVP and tend to stimulate renin secretion. TXA_2 causes vasoconstriction, although the importance of the physiologic effects of TXA_2 on the kidneys is still controversial. The end results of the stimulation of renal PG secretion in the kidneys are vasodilation, increased renal perfusion, natriuresis, and facilitation of water excretion.

The cellular targets of vasodilatory PG are the afferent and efferent arteriolar VSMC, glomerular mesangial cells peritubular capillaries, and vasa recta to modulate renal vascular resistance and glomerular function. Intrarenal infusions of PGE_2 and PGI_2 cause vasodilation and increased RBF.²⁵¹ In isolated renal microvessels, PGE_2 and PGE_1 attenuate Ang II-induced afferent and PGI_2 antagonizes Ang II-induced efferent arteriolar vasoconstriction.²⁵³ Similarly, PGE_2 can counteract renal nerve stimulation²⁵⁴ and Ang II-induced contraction of isolated glomeruli and cultured mesangial cells.²⁵⁵ Furthermore, in volume-contracted states, COX-2 expression and PGE_2 release in the macula densa and cTALH dramatically increases in response to decreased luminal Cl^- delivery. In addition to its direct vasodilatory effect on afferent arterioles, PGE_2 leads to increased renin release from the macula densa.²⁵⁰ The resulting rise in Ang II and

consequent efferent arteriolar constriction also ensures maintenance of GFR.

In volume-replete states, the renal vasoconstrictive influences of Ang II and NE are mitigated by their simultaneous stimulation of vasodilatory renal PG, so that RBF and GFR are maintained.²⁵⁶ However, in the setting of heightened vasoconstrictor input from the RAAS, SNS and AVP, as in states of EABV depletion, the vasorelaxant action of PGE₂ and PGI₂ is overwhelmed, with the concomitant risk for the development of acute kidney injury.²⁵⁰ Similarly, when this PG-mediated counterregulatory mechanism is suppressed by nonselective or COX-2-selective inhibitors, the unopposed actions of Ang II and NE can lead to a rapid deterioration in renal function.²⁵⁷ Moreover, COX-2-derived prostanoids also promote natriuresis and stimulate renin secretion.²⁵⁰ Therefore, during states of volume depletion, low Na⁺ intake or the use of loop diuretics, COX-2 inhibitors and nonselective COX inhibitors can cause Na⁺ and K⁺ retention, edema, HF and hypertension.²⁵²

Besides modulating glomerular vasoreactivity, PGE₂-induced renal vasodilation affects U_{Na} excretion, by inducing medullary interstitial solute washout.²⁵¹ The natriuretic response to PGE₂ may also be attenuated by preventing an increase in renal interstitial hydraulic pressure, even in the presence of a persistent increase in RBF.²⁵⁸ In addition, the natriuresis accompanying direct expansion of renal interstitial volume is significantly attenuated by inhibition of PG synthesis.²⁵⁸ PGE₂ also affects U_{Na} excretion by direct effects on epithelial transport processes.²⁵¹ In the mTALH and collecting tubule, PGE₂ causes a decrease in the reabsorption of water, Na⁺ and Cl⁻ which correlates with reduced Na⁺-K⁺-ATPase activity. In contrast, in the distal convoluted tubule, PGE₂ causes increased Na⁺-K⁺-ATPase activity.²⁵⁹ The net effect of PG on tubular Na⁺ handling is probably inhibitory because complete blockade of PG synthesis by indomethacin in rats receiving a normal or salt-loaded diet increases fractional Na⁺ reabsorption and enhances medullary Na⁺-K⁺-ATPase activity.²⁶⁰ In addition, PGE₂ inhibits AVP-stimulated NaCl reabsorption in the mTALH and AVP-stimulated water reabsorption in the collecting duct.^{261,262} Both these effects tend to antagonize the overall hydroosmotic response to AVP. However, the cTALH, which can augment NaCl reabsorption in response to an increased delivered load, is not affected by PG; also, the effects of PG on solute transport in the collecting tubule remain unresolved. Therefore, the contribution of direct epithelial effects of PG to overall Na⁺ excretion is unclear.²⁶¹

Not only does renal COX expression influence salt handling, but also changes in Na⁺ intake affect the renal expression of these enzymes. COX-2 expression in the macula densa and TALH is increased by a low-salt diet, RAAS inhibition and renal hypoperfusion. Conversely, a high-salt diet leads to decreased COX-2, but unchanged COX-1, expression in the renal cortex.^{249,250} In the medulla, whereas a low-salt diet downregulated both COX-1 and COX-2, a high-salt diet enhanced their expression.^{249,250} High osmolarity of the medium of cultured IMCD cells induced COX-2 expression whereas infusion of a selective COX-2 inhibitor into anesthetized dogs on a normal Na⁺ diet reduced U_{Na} excretion and urine flow rate, without affecting renal hemodynamics or systemic BP.²⁵²

Collectively, the differential regulation of COX-2 in the renal cortex and medulla can be integrated into a

physiologically relevant model in which upregulation of COX-2 in the cTALH and macula densa is induced in a volume-contracted or vasoconstrictor state. In the cTALH, the effect is by direct inhibition of Na⁺ excretion, whereas in the macula densa, COX-2 stimulates renin release, which leads to Ang II-mediated Na⁺ retention. In contrast, medullary COX-2 is induced by a high-salt diet, which leads to net Na⁺ excretion.²⁵⁰

Finally, in addition to their hemodynamic and direct epithelial effects, PG may mediate the physiologic responses to other hormonal agents. The intermediacy of PG in renin release responses has already been cited. As another example, some, but not all, of the physiologic effects of bradykinin are mediated by PG production (e.g., inhibition of AVP-stimulated osmotic water permeability in the cortical collecting tubule).²⁶¹ In addition, the renal and systemic actions of Ang II appear to be differentially regulated by COX-1 and COX-2. For example, COX-2 deficiency in mice, induced by inhibitors or gene knockout, dramatically augmented the systemic pressor effect of Ang II, whereas COX-1 deficiency abolished this effect. Similarly, Ang II infusion reduced medullary blood flow in COX-2-deficient, but not COX-1-deficient animals, which suggests that COX-2-dependent vasodilators are synthesized in the renal medulla. Moreover, the diuretic and natriuretic effects of Ang II were absent in COX-2-deficient but remained in COX-1-deficient animals. Thus, COX-1 and COX-2 seem to exert opposite effects on systemic BP and renal function.²⁶³

Natriuretic Peptides. There are four NPs: ANP, BNP, CNP, and DNP.²⁶⁴ Although encoded by different genes, their structures, gene regulation and degradation pathways are very similar, and they exert various actions on renal, cardiac, and vascular tissues.²⁶⁵ ANP, a 28-amino acid peptide, plays an important role in BP and volume homeostasis through its natriuretic/diuretic and vasodilatory responses.^{264,265} BNP has an amino acid sequence similar to that of ANP, with an extended NH₂-terminus. In humans, BNP is produced from pro-brain NP (proBNP), which contains 108 amino acids and, by proteolysis, releases a mature, 32-amino acid molecule and N-terminal fragment into the circulation. Although BNP was originally cloned from the brain, it is now considered a circulating hormone produced mainly in the cardiac ventricles.^{266,267} CNP, produced mostly by endothelial cells, shares the ring structure common to all NP members; however, it lacks the C-terminal tail. DNP is released by the kidney and is a more effective activator of renal functions than ANP.²⁶⁸

The biologic effects of the natriuretic peptides (NPs) are mediated by binding to specific membrane receptors localized to numerous tissues, including the vasculature, renal arteries, glomerular mesangial and epithelial cells, collecting ducts, adrenal zona glomerulosa, and CNS.²⁶⁴ At least three different, subtypes of NP receptors have been identified—NP-A, NP-B and NP-C. NP-A and NP-B, single-transmembrane proteins with molecular weights of approximately 120 to 140 kDa, mediate most of the biologic effects of NPs. Both are coupled to guanylate cyclase (GC) in their intracellular portions.²⁶⁹ After binding to their receptors, all three NP isoforms markedly increase cGMP in target tissues and plasma. Therefore analogs of cGMP or inhibitors of its degradation mimic the vasorelaxant and renal effects of NPs. NP-C (molecular weight, 60 to 70 kDa), the most abundant NP receptor in many key

target organs, is believed to be a clearance receptor because it is not coupled to any known second-messenger system.²⁷⁰

Two additional routes for the removal of NPs should be noted. The first is the enzymatic degradation by neutral endopeptidase (NEP) 24.11, a metalloproteinase located mainly in the lungs and the kidneys.²⁷⁰ Extensive research into this pathway culminated in the development of a specific NEP inhibitor, which leads to enhanced NP activity and improved HF outcomes (see section, “[Specific Treatments Based on the Pathophysiology of Congestive Heart Failure](#)”). The second route involves the negative regulation of ANP by microRNA-425 (miRNA-425). In this context, carriers of the rs5068 minor G allele of the gene encoding ANP, *NPPA*, have ANP levels 50% higher than those with two copies of the major A allele and have a 15% lower risk of hypertension. miRNA-425, expressed in human atria and ventricles, is predicted to bind the sequence spanning rs5068 in the 3′ untranslated region of the A, but not G, allele. Only the A allele was silenced by miRNA-425, whereas possession of the G allele conferred resistance to miR-425. The results raise the possibility that miR-425 antagonists could be used to treat disorders of salt overload, such as hypertension and HF.²⁷¹

Atrial Natriuretic Peptide. Studies in humans and experimental animals have established the role of ANP in the regulation of ECFV and BP by acting on all organs and tissues involved in the homeostasis of Na⁺ and BP ([Table 14.3](#)).²⁶⁷ Therefore,

Table 14.3 Physiologic Actions of the Natriuretic Peptides

Target Organ	Biologic Effects
Kidney	Increased GFR Afferent arteriolar vasodilation Efferent arteriolar vasoconstriction Natriuresis Inhibition of Na ⁺ /H ⁺ exchanger (proximal tubule) Inhibition of Na ⁺ -Cl ⁻ cotransporter (distal tubule) Inhibition of Na ⁺ channels (collecting duct) Diuresis Inhibition of AVP-induced AQP2 incorporation into CD-AM
Cardiac	Reduction in preload, leading to reduced cardiac output Inhibition of cardiac remodeling
Hemodynamic	Vasorelaxation Elevation of capillary hydraulic conductivity Decreased cardiac preload and afterload
Endocrine	Suppression of RAAS Suppression of sympathetic outflow Suppression of AVP Suppression of endothelin
Mitogenesis	Inhibition of mitogenesis in VSMC Inhibition of growth factor-mediated hypertrophy of cardiac fibroblasts

AQP2, Aquaporin 2; AVP, arginine vasopressin; CD-AM, collecting duct apical membrane; GFR, glomerular filtration rate; RAAS, renin-angiotensin-aldosterone system.

it is not surprising that ANP and NH₂-terminal ANP levels are increased in: 1. conditions associated with enhanced atrial pressure; 2. systolic or diastolic cardiac dysfunction; 3. cardiac hypertrophy/remodeling; and 4. severe myocardial infarction.³⁸ In the kidneys, ANP exerts hemodynamic/glomerular effects that increase Na⁺ and water delivery to the tubule and inhibits tubular Na⁺ and water reabsorption, leading to significant natriuresis and diuresis.²⁶⁷

In addition, ANP relaxes vascular smooth muscle and leads to vasodilation by antagonizing the concomitant vasoconstrictive influences of Ang II, endothelin (ET), AVP, and α₁-adrenergic input.²⁶⁷ This vasodilation reduces preload, resulting in a fall in cardiac output.²⁶⁷ ANP also reduces cardiac output by shifting fluid from the intravascular to extravascular compartment, an effect mediated by increased capillary hydraulic conductivity.²⁷² Studies in endothelial-restricted, GC-A knockout mice have found that ANP, through GC-A, enhances albumin permeability in the microcirculation of the skin and skeletal muscle. This effect is mediated by caveolae²⁷³ and is critically involved in the hypovolemic and hypotensive actions of ANP in vivo.²⁷⁴

ANP also inhibits proliferation of cultured mesangial, vascular smooth muscle and endothelial cells.²⁷⁵ ANP causes afferent vasodilation, efferent vasoconstriction and mesangial relaxation, which leads to increased glomerular capillary pressure, GFR and filtration fraction.²⁶⁷ In combination with increased medullary blood flow, these hemodynamic effects enhance diuresis and natriuresis, although the natriuretic effect of ANP infusion does not usually require these changes in glomerular function. In the tubules, ANP inhibits the stimulatory effect of Ang II on the luminal Na⁺/H⁺ exchanger of the proximal tubule, the thiazide-sensitive NaCl cotransporter in the distal tubule, ENaC and AVP-induced AQP2 incorporation into the apical membrane of the collecting duct (see [Table 14.3](#)).²⁶⁷

Brain Natriuretic Peptide. BNP is produced by activated satellite cells in ischemic skeletal muscle or by cardiomyocytes, mainly by the ventricles but also, in small amounts, by the atrium, in response to volume or pressure load, as seen in HF and hypertension. BNP, like ANP, induces natriuretic, endocrine, and hemodynamic responses.²⁶⁷ BNP levels rise with age, more than doubling from age 55 to 64 years to 75 years and older. However, in HF and other chronic volume-expanded conditions much greater elevations occur in BNP levels.²⁷⁶

Studies in animals and humans have demonstrated the natriuretic effects of pharmacologic doses of BNP. The combination with ANP produced no further effect.²⁶⁷ Moreover, like ANP, BNP exerts a hypotensive effect in animals and humans. For example, transgenic mice, overexpressing either the BNP or ANP gene exhibit lifelong hypotension.²⁶⁷ Therefore, it is clear that BNP induces its biologic actions through mechanisms similar to those of ANP.²⁶⁷

This notion is supported by several findings: 1. both ANP and BNP act through the same receptors and induce similar renal, cardiovascular and endocrine actions in association with an increase in cGMP production (see [Table 14.3](#)); and 2. BNP suppresses ACTH-induced aldosterone generation both in cell culture and following infusion. The latter action may be attributed to inhibition of renin secretion in dogs, although apparently not in humans.²⁶⁷ At high, but not low,

doses, BNP caused a profound fall in systolic BP in humans.²⁶⁷ In the clinical setting, the effects of BNP are used routinely to monitor volume overload in HF (see section, “**Specific Treatments Based on the Pathophysiology of Congestive Heart Failure**”).

C-Type Natriuretic Peptide. Although CNP is considered a neurotransmitter in the CNS, considerable amounts are produced by endothelial cells, where it plays a role in the local regulation of vascular tone.²⁷⁰ Smaller amounts of CNP are produced by the kidneys, heart ventricles and intestines²⁷⁰ and can be detected in human plasma.²⁷⁷ CNP is produced in response to hypoxia, cytokines, shear stress, and fibrotic growth factors²⁷⁸ and enhanced expression of CNP mRNA has been reported after volume overload.²⁷⁰ Infusion of CNP decreases BP, cardiac output, urinary volume and Na⁺ excretion; the effects are less pronounced than those of ANP and BNP, despite strong stimulation of cGMP and inhibition of vascular smooth muscle cell proliferation.

All three NPs inhibit the RAAS, although CNP does not induce significant changes in cardiac output, BP or plasma volume in sheep.²⁷⁰ This finding supports the widely-accepted concept that ANP and BNP are the major circulating NPs, whereas CNP is a local regulator of vascular structure and tone.

D-Type Natriuretic Peptide. DNP infused into rabbits increased urine volume and electrolyte excretion. The effects were more pronounced than those of ANP, possibly because of DNP resistance to degradation by endogenous peptidases. DNP, via the NP-A receptor, preferentially induced cGMP production in glomeruli compared with cortical, outer medullary and inner medullary tubules. Thus, DNP may play a regulatory role in the kidney, via specific NP receptors with a GC domain.²⁶⁸

Together, the various biologic actions of NPs lead to reduction of EABV, in response to perceived overfilling of the central intrathoracic circulation. Furthermore, all NPs counteract the adverse effects of the RAAS, suggesting that the two systems act in opposite directions in the regulation of body fluid and cardiovascular homeostasis.

Endothelial-Derived Factors. The endothelium is a major source of factors that regulate vascular tone in health and disease.³⁰⁰ These factors regulate the perfusion pressure of multiple organ systems involved in water and Na⁺ balance, such as the kidneys, heart and vasculature. This section summarizes some concepts regarding actions of ET and NO relevant to volume homeostasis.

Endothelin. The ET system consists of three vasoactive peptides—endothelin 1 (ET-1), endothelin 2 (ET-2) and endothelin 3 (ET-3), which act in a paracrine and autocrine manner.²⁷⁹ Endothelins are synthesized by proteolytic cleavage from specific preproendothelins that are further cleaved to form 37- to 39-amino acid precursors called “big ET”. Big ET is then converted into the biologically active 21-amino acid peptide by a highly specific endothelin-converting enzyme (ECE), a phosphoramidon-sensitive, membrane-bound metalloprotease. There are two main isoforms of ECE, ECE-1 and ECE-2. ECE-1 itself has four isoforms.²⁷⁹ ECE-2 is localized mainly to VSMC and is probably intracellular. Both chymase²⁸⁰

and carboxypeptidase A²⁸¹ are also involved in mature ET production.

Endothelins bind to two distinct receptors, types A and B (ET-A and ET-B).²⁷⁹ The ET-A receptor shows a higher affinity for ET-1 than for ET-2 or ET-3. The ET-B receptor shows equal affinity for all three ETs. ET-A receptors are found mainly on VSMC, on which their activation leads to vasoconstriction through an increase in cytosolic Ca²⁺. ET-B receptors are also found on VSMC, where they can mediate vasoconstriction, but are found predominantly on vascular endothelium, where their activation results in vasodilation through prostacyclin and NO.²⁷⁹ Endothelin is detectable in the plasma of humans and experimental animals and, therefore, can also act as a circulating vasoactive hormone.²⁷⁹

Selective ET-A receptor antagonism is associated with vasodilation and a reduction in BP, whereas selective ET-B antagonism is accompanied by vasoconstriction and a rise in BP.²⁷⁹ These data suggest complementary roles for the ET receptor subtypes in the maintenance of vascular tone. In addition to its vasoconstrictive action, ET has a variety of renal effects.^{282–284} The kidney (mainly the inner medulla) is both a source and target of ET. ET-1, the principal subtype involved in renal functional regulation, is synthesized by vascular endothelial cells, whereas ET-1 and ET-3 are produced by various renal cell types, but at a rate one to two orders of magnitude lower than ET-1.²⁷⁹

In relation to volume homeostasis, ET-1 acts in a paracrine or autocrine manner to regulate: 1. renal and intrarenal blood flow; 2. glomerular hemodynamics; and 3. tubular salt and water transport. Both ET-A and ET-B receptors are present in the glomerulus, renal vessels and tubular epithelial cells, but most ET-B receptors are found in the medulla.²⁸⁵ The renal vasculature is exquisitely sensitive to the vasoconstrictor action of ET-1. Infusion of ET-1 into the renal artery of anesthetized rabbits was found to decrease RBF, GFR, natriuresis, and urine volume.²⁸⁶ ET-1 increases afferent and efferent arteriolar resistance (afferent > efferent), which results in reduced glomerular plasma flow. In addition, Kf is reduced because of mesangial cell contraction, resulting in a diminished SNGFR.

The profound reduction of RBF and concomitant lesser reduction in GFR should result in a rise in filtration fraction, but this is not evident in all models. Infusion of ET-1 for 8 days into conscious dogs increased plasma levels of ET by two- to threefold and resulted in increased renal vascular resistance and decreased GFR and RBF.²⁸⁷ Interestingly, the effect of ET on regional intrarenal blood flow is not homogeneous. As measured by laser Doppler flowmetry, administration of ET-1 to rats produced sustained cortical vasoconstriction but only transient medullary vasodilation.²⁸⁸ These results are in line with the medullary predominance of ET-B receptors and the high density of ET-A-binding sites in the cortex.²⁷⁹

The effect of ET on Na⁺ and water excretion varies, depending on the dose and source of ET. Systemic infusion of ET in high doses results in profound antinatriuresis and antidiuresis, apparently secondary to a decrease in GFR and RBF. However, in low doses, or when produced locally in tubular epithelial cells, ET decreases salt and water reabsorption, consistent with tubular ET-1 target sites.²⁸⁹ Also, administration of big ET has been shown to cause natriuresis, supporting the notion of direct autocrine inhibition of ET on tubular salt reabsorption.²⁹⁰

The natriuretic and diuretic actions of big ET-1 are significantly reduced by ET-B–specific blockade.²⁹¹ Furthermore, ET-B knockout rats have salt-sensitive hypertension reversed by luminal ENaC blockade with amiloride, by a Ca^{2+} -dependent effect, which suggests that collecting duct ET-B tonically inhibits ENaC activity.²⁹² Similarly, mice with collecting duct–specific knockout of the ET-1 gene have impaired Na^+ excretion in response to Na^+ load and develop hypertension with a high salt intake.²⁷⁹ These mice also have heightened sensitivity to AVP and reduced ability to excrete an acute water load. These findings are in line with observations that ET-B mediates the inhibitory effects of ET-1 on Na^+ and water transport in the collecting duct and TALH.²⁸³ Thus if vascular and mesangial ET exert a greater physiologic effect than tubule-derived ET, then RBF is diminished and net fluid retention occurs. Whereas if the tubule-derived ET effect predominates, salt and water excretion are increased.

Renal endothelin production is regulated differently than in the vasculature. Whereas vascular (and mesangial) ET generation is controlled by thrombin, Ang II and TGF- β , tubular ET production seems to particularly depend on medullary tonicity. For example, a high-salt diet, by raising medullary tonicity, stimulates ET-1 release, which in turn leads to increased eNOS (NOS3) expression and natriuresis.²⁹³ (see “Nitric Oxide” section, later). The signaling mechanisms for these phenomena, as well other renal actions of ET-1, continue to be a subject of intensive research and the interested reader is referred to a recent review summarizing the current state of knowledge.²⁷⁹

Nitric Oxide. Nitric oxide (NO) is a diffusible gaseous molecule produced from L-arginine by the enzyme NOS, which exists in three distinct isoforms—nNOS (NOS1), inducible NOS (iNOS, or NOS2) and eNOS (NOS3).²⁹¹ NOS is expressed in renal vascular endothelial cells (mainly eNOS), tubular epithelial and mesangial cells and macula densa (mainly nNOS). There is controversy regarding iNOS expression in normal kidneys, but upregulation is clearly seen in pathologic conditions such as ischemia-reperfusion injury.²⁹⁴

Selective NOS inhibitors and NOS knockout mice have been used to elucidate the role of NOS isoforms in the regulation of renal function.²⁹⁴ The action of NO is mediated by activation of a soluble GC, thereby increasing intracellular levels of cGMP.²⁹⁵ In the kidneys, the physiologic roles of NO include the regulation of glomerular hemodynamics, attenuation of TGF, mediation of pressure natriuresis, maintenance of medullary perfusion, inhibition of tubular Na^+ reabsorption and modulation of RSNA.^{291,296} Renal NOS activity is regulated by several factors, such as Ang II (see earlier section, “Tubuloglomerular Feedback”) and salt intake.¹²⁰

The role of NO in the regulation of renal hemodynamics and excretory function is best illustrated by the fact that inhibition of intrarenal NO production results in increased BP and impaired renal function.²⁹⁴ Infusion of the NOS inhibitor, *N*-monomethyl-L-arginine (L-NMMA), into one kidney in anesthetized dogs resulted in a dose-dependent decrease in urinary cGMP levels, decreases in RBF and GFR, Na^+ and water retention in the ipsilateral compared with the contralateral kidney.²⁹⁷ In addition, acute NO blockade amplified the renal vasoconstrictive action of Ang II in isolated microperfused rabbit afferent arterioles and in conscious

rats.^{120,294} Also L-NMMA–induced vasoconstriction led to decreased RBF and KF, effects reversed by RAAS blockade. Thus, a major effect of NO is to counterbalance the efferent vasoconstrictive action of Ang II as well as modulate renin secretion by the JGA (see section, “Tubuloglomerular Feedback”).

The involvement of NO in promoting diuresis and natriuresis in normal and increased salt intake/volume-expanded states is well characterized.²⁹¹ In conscious dogs on a normal Na^+ diet, NO inhibition induced a significant decrease in natriuresis and diuresis with no change in arterial pressure. On a high- Na^+ diet, treatment with the NO inhibitor, *N*^G-nitro-L-arginine methyl ester (L-NAME), increased arterial pressure and cumulative Na^+ balance.²⁹⁸ Similarly, rats on a high salt intake for 2 weeks had increased Na^+ excretion and proportionally elevated urinary concentrations of the NO metabolites, NO_2 and NO_3 . The increase in urinary NO metabolites is attributed to enhanced expression of all three NOS isoforms in the renal medulla.¹²¹

NO exerts a vasodilatory effect on the renal medullary circulation and promotes Na^+ excretion.¹⁶⁴ Consistent with these data are the high levels of eNOS in the renal medulla and the inhibitory effect of NO on collecting duct Na^+ -K⁺-ATPase.²⁹⁸ In contrast, in salt-sensitive hypertension, NOS activity (mainly nNOS) is significantly lower in salt-sensitive than salt-resistant rats maintained on a high-salt diet.^{299,300} Also, L-arginine-induced NO production prevented the development of salt-induced hypertension in Dahl salt-sensitive rats.³⁰¹ These findings suggest that nNOS plays an important role in Na^+ handling and decreased nNOS activity may be involved in the mechanism of salt-sensitive hypertension.

The involvement of NO in salt retention and subsequent hypertension could result from an inadequate direct effect on proximal and distal tubular Na^+ reabsorption. However, attenuated NO inhibition of renin secretion and TGF may also contribute. In this context, macula densa-derived NO could blunt the TGF-mediated vasoconstriction during high salt intake in salt-resistant but not salt-sensitive rats.³⁰² As noted, the medullary and other effects of NO occur in response to local ET production.²⁹¹ For example, the inhibition of NOS by L-NAME or the highly selective ET-B antagonist A-192621 abolished the diuretic and natriuretic effects of big ET-1 in anesthetized rat kidneys.³⁰³ In addition, ET-1 acutely activated eNOS in the isolated mTALH and nNOS in isolated IMCD cells, via ET-B activation, an effect dependent on increased NOS protein, but not mRNA, expression.²⁹¹ These data suggest that nNOS and eNOS activation occur by posttranscriptional pathways. NO also reduces Cl^- absorption in CCD through an ENaC-dependent mechanism.³⁰⁴

Activation of eNOS, by a paracrine effect in the IMCD, where the highest renal NOS activity is found,³⁰⁵ is also associated with inhibition of Na^+ reabsorption in the mTALH through phosphatidylinositol-3-kinase (PI3K)–stimulated Akt activity, leading to eNOS phosphorylation at Ser1177.³⁰⁵ However, the functional corollary of nNOS activation in the IMCD remains to be determined. NO also reduces paracellular Na^+ reabsorption in the mTALH, and the magnitude of this effect is equal to that due to NO inhibition of transcellular transport.³⁰⁶ A further action of NO is inhibition of AVP-enhanced Na^+ reabsorption and hydroosmotic water permeability of the CCD.³⁰⁷ A mouse model of specific collecting duct NOS1 gene deletion should be a valuable tool to study

the signaling mechanisms involved in the NO effects on AVP-enhanced Na^+ reabsorption, as well as salt-dependent BP mechanisms.²⁹⁶ The role of NO in pressure natriuresis and RSNA has been discussed in the relevant sections.

Kinins. The kallikrein-kinin system (KKS) is a complex cascade responsible for the generation and release of vasoactive kinins. The active peptides bradykinin (BK) and kallidin are formed from precursors (kininogens) that are cleaved by tissue and circulatory kinin-forming enzymes.²²⁹ Kinins are produced by many cell types and can be detected in urine, saliva, sweat, interstitial fluid and, rarely, venous blood. Circulating BK is almost undetectable because of rapid metabolism by kininases, particularly kininase II/ACE1. The renal KKS can produce local BK concentrations much higher than those in blood. In the kidney, bradykinin is metabolized by NEP.³⁰⁸

Kinins play an important role in hemodynamic and excretory processes through their G protein-coupled receptors, BK-B₁ and BK-B₂. BK-B₂ receptors mediate most of the actions of kinins and are located mainly in the kidneys, although they are also detectable in the heart, lungs, brain, uterus and testes.³⁰⁸ Activation of BK-B₂ receptors results in vasodilation, probably through an NO- or arachidonic acid metabolite-dependent mechanism.³⁰⁹ BK selectively increases medullary perfusion, especially to the inner layer, via activation of NO and Ca^{2+} -activated K^+ channels.³¹⁰ BK also has multiple effects on the cardiovascular system, particularly vasodilation and plasma extravasation.³⁰⁹

In the kidney, kinins induce diuresis and natriuresis through activation of BK-B₂ receptors, increased RBF and secondary inhibition of Na^+ and water reabsorption via ENaC in the distal nephron.²²⁹ Unlike many vasodilators, BK increases RBF without significantly affecting GFR or proximal tubular Na^+ reabsorption.

Studies with transgenic animals have elaborated the physiologic role of kinins and the interaction between the KKS and the RAAS.³⁰⁹ In the kidney, Na^+ depletion-induced Ang II, acting via the AT₂ receptor, stimulates a vasodilator cascade of BK, NO and cGMP.²⁹⁷ In the absence of the AT₂ receptor, pressor and antinatriuretic hypersensitivity to Ang II is associated with BK and NO deficiency.²⁹⁷ Furthermore, renal kinins are involved in pressure natriuresis.³¹¹ BK also mediates Ang 1–7-mediated diuresis and natriuresis, as shown in rats transgenic for the kallikrein gene.³¹² Because ACE is involved in kinin degradation, ACE inhibitors not only attenuate Ang II formation but may also lead to kinin accumulation. The latter effect may be responsible in part for the beneficial effects of ACE inhibitors in HF, but also for their troublesome side effect of cough.³¹³ In summary, the KKS seems to play a pivotal role as a counterregulatory modulator of vasoconstrictor and Na^+ -retaining mechanisms.

Adrenomedullin. Human adrenomedullin (AM) is a 52-amino acid peptide, discovered in 1993 in extracts of human pheochromocytoma cells and approximately 30% homologous in structure with calcitonin gene-related peptide and amylin.³¹⁴ AM is produced from a 185-amino acid preprohormone that contains a unique NH_2 -terminus 20-amino acid sequence, “proadrenomedullin NH_2 -terminal 20 peptide,” with biologic activity similar to that of AM. AM mRNA is expressed in endothelial cells, glomeruli, distal and medullary

collecting tubules. Synthesis and secretion of AM are stimulated by Ang II, NE, ET, bradykinin, and shear stress.³¹⁴ AM acts through a 395-amino acid G protein-like-coupled receptor, belonging to the calcitonin receptor-like receptor and a family of receptor activity-modifying proteins.³¹⁴ Receptor activation in VSMC increases intracellular cAMP and calcium-activated potassium channel activity,³¹⁴ leading to prolonged dose-dependent generalized vasodilation and hypotension, accompanied by increases in heart rate and cardiac output caused by positive inotropic effects.³¹⁴ The vasodilating effect of AM is also stimulated by calcium-dependent NO synthesis in endothelial cells.³¹⁴

In addition to its hypotensive action, AM increases RBF through preglomerular and postglomerular arteriolar vasodilation,^{315,316} accompanied by dose-dependent diuresis and natriuresis.^{316,317} These effects result from a decrease in tubular Na^+ reabsorption, despite the AM-induced hyperfiltration³¹⁵ and may be mediated partially by locally released NO^{318,319} and PG.³²⁰ In addition, NEP inhibition potentiates exogenous AM-induced natriuresis without affecting GFR.³²¹ Like NPs, AM suppresses aldosterone secretion in response to Ang II and high potassium levels.³¹⁷ Furthermore, in cultured VSMC, AM inhibits ET production induced by various stimuli.³¹⁷ In the hypothalamus, AM inhibits AVP secretion, which may also contribute to its diuretic and natriuretic actions.³¹⁷

Two other members of the AM family, AM-2 (intermedin) and AM-5, have cardiovascular effects similar to those of AM-1. AM-2, but not AM-5, also has renal effects similar to those of AM-1.^{322,323}

Together, these findings show that AM may be involved in the physiologic control of vascular tone and cardiac function; in the kidney, AM may modulate sodium and water excretion in a paracrine fashion.³²⁴

Urotensin. Urotensin II (UT II) is a highly-conserved peptide that binds to the human orphan G protein-coupled receptor GPR14, or UT II receptor. The parent peptide, prepro-UT II, is widely expressed in human tissues, including the kidney where it can be detected in tubular epithelial cells, especially of the distal tubule, and endothelial cells of renal capillaries.³²⁵ The C-terminus of the prohormone is cleaved to produce UT II, an 11-amino acid residue peptide. Human UT II includes a cyclic hexapeptide sequence fundamental for the action of the peptide. Substantial local UT II has been demonstrated in the heart, liver, and kidneys.³²⁶

Infusion of UT II leads to local forearm vasoconstriction, no effect, or cutaneous vasodilation, according to species variation, site and modality of injection, dose, vascular bed and experimental model.³²⁷ The vasoconstrictive action is probably direct, whereas the vasodilatory response is likely mediated by factors such as cyclooxygenase products and NO.

The involvement of the UT II system in the regulation of renal function in mammals is unclear and the data are as contradictory as those for vascular tone. In normal rats, intravenous boluses in the nanomolar range caused minor reductions in GFR and no effect on Na^+ excretion.³²⁵ However, bolus injections in the picomolar range produced a dose-dependent decrease in GFR associated with reduced urine flow and Na^+ excretion.³²⁸ In contrast, continuous infusion of UT II in the picomolar range elicited increases

in GFR and NO-dependent diuresis and natriuresis.³²⁵ Thus, the effect of UT II on renal function seems dependent on the mode of administration and experimental condition.

The variability in renal and vascular responses to UT II may also depend on regulation at the receptor level. The binding density of UT II is correlated with vasoconstrictor response in rats and small changes in receptor density may result in pathophysiologic effects. Under normal conditions, most UT II receptors are already occupied by UT II. Changes in unoccupied receptor reserve—perhaps in response to alterations in UT II levels in experimental models or in disease states—might explain, at least in part, the observed variability in renal and vascular actions.³²⁵

Selective UT II receptor antagonists have been developed and, in normal rats, can increase GFR, urine flow and Na⁺ excretion.³²⁹ However, in light of the complex renal effects of UT II, it is unlikely that these antagonists will have a role in the management of sodium disorders.³²⁵

Digitalis-Like Factors. The existence of endogenous digitalis-like factors was hypothesized in the 1960s and initially reported in the late 1970s.³³⁰ Among these factors, also known as endogenous cardiotonic steroids, two have been characterized extensively in humans, cardenolide (or ouabain) and bufadienolide (marinobufagenin). The main site of synthesis of these compounds is the adrenal cortex.³³¹ Cardiotonic steroids act by inhibiting Na⁺-K⁺-ATPase, leading to attenuation of tubular Na⁺ transport and increased vascular resistance via enhanced cytosolic Ca²⁺ in VSMC.³³⁰ The latter mechanism has been implicated in the pathogenesis of hypertension.³³⁰

Neuropeptide Y. Neuropeptide Y (NPY), a 36-residue peptide, is a sympathetic cotransmitter stored and released together with NE by adrenergic nerve terminals of the SNS.^{332,333} Although originally isolated from the brain and highly expressed in the CNS, the peptide exhibits a wide spectrum of biologic activities in the cardiovascular system, GI tract and kidneys through multiple G_{i/o} protein-coupled receptors.^{333,334} Experimentally, NPY can reduce RBF and increase renal vascular resistance without significantly affecting GFR in various species, including humans.³³³ Similarly, the peptide may exert a natriuretic or antinatriuretic action, depending on experimental conditions and species studied.³³³ However, overall, NPY appears to have no significant role in the physiological regulation of sodium.

Apelin. Apelin is the endogenous ligand of the APJ receptor, a G protein-coupled receptor involved in water homeostasis, regulation of cardiovascular tone and cardiac contractility.³³⁵ Apelin and its receptor are widely expressed in the CNS, endothelial cells (systemic and renal), VSMC of glomerular arterioles and, to a lesser extent, in other nephron segments.³³⁶

APJ activation leads to inhibition of cAMP production and activation of the Na⁺/H⁺ exchanger type 1 (NHE1). Through the former pathway, apelin enhances vascular dilation via induction of eNOS, whereas NHE1 activation in cardiomyocytes leads to a dose-dependent increase in myocardial contractility.³³⁶ With regard to the renal effects of apelin, direct injection into the hypothalamus of lactating rats inhibited AVP release and reduced circulating AVP. Conversely,

water deprivation led to increased systemic AVP and decreased apelin levels.³³⁵ Moreover, APJ expression in the rat collecting duct occurs near the vasopressin V₂ receptor and apelin directly counteracts the V₂ receptor-mediated antidiuretic effect of AVP.³³⁷ Thus, AVP and apelin have a reciprocal relationship in controlling water diuresis.

With respect to vascular tone, the apelin/APJ system acts as a counterregulator of the RAS. For example, intravenous apelin caused an NO-dependent fall in arterial pressure, whereas apelin receptor knockout mice displayed an enhanced vasopressor response to systemic Ang II.³³⁶ Intravenous injection of apelin also induced vasorelaxation of Ang II-precontracted efferent and afferent arterioles, as well as a significant diuresis.³³⁸ Activation of endothelial apelin receptors caused release of NO, which inhibited the Ang II-induced rise in intracellular Ca²⁺ levels.³³⁶ Furthermore, apelin had a direct receptor-mediated vasoconstrictive effect on vascular smooth muscle.³³⁶ These results indicate that apelin has complex effects on the preglomerular and postglomerular microvasculature regulating renal hemodynamics.

Glucagon-Like Peptide-1

The incretin hormone glucagon-like peptide-1 (GLP-1) is released from the gut in response to fat or carbohydrate and contributes to negative feedback control of blood glucose by stimulating insulin secretion, inhibiting glucagon and slowing gastric emptying. GLP-1 receptors (GLP-1Rs) are also expressed in the proximal tubule and the GLP-1 agonist, exenatide, has natriuretic effects by inhibiting NHE-3, thereby reducing proximal tubular Na⁺ reabsorption; this effect may be mediated by Ang II inhibition.³³⁹ Exenatide also increased SNGFR by 33% to 50%, doubled early distal flow rate and increased urine flow rate sixfold without altering GTB, TGF responsiveness, or the tonic influence of TGF. This implies that exenatide is a proximal diuretic and a renal vasodilator.³³⁹ Because the natural agonist for the GLP-1 receptor is regulated by intake of fat and carbohydrate, but not by salt or fluid, the control of salt excretion by the GLP-1R system departs from the usual negative-feedback paradigm for regulating salt balance.³⁴⁰

Novel Factors

A novel intrarenal paracrine mechanism for sodium regulation in mice was recently described. Early studies showed that changes in dietary acid-base load could reverse the direction of apical transport of the tricarboxylic acid intermediate, α -ketoglutarate (α -KG), in the proximal tubule and loop of Henle from reabsorption following an acid load to secretion following a base load.³⁴¹ Work on isolated microperfused CCDs from *Oxgr1*^{-/-} mice indicated that the concentration of α -KG is sensed by the α -KG receptor, OXGR1, expressed in type B and non-A, non-B intercalated cells of the connecting tubule and CCD. Addition of 1 mM α -KG to the tubular lumen strongly stimulated Cl-dependent HCO₃⁻ secretion and electroneutral NaCl reabsorption in tubules of wild type but not *Oxgr1*^{-/-} mice. *Oxgr1*^{-/-} mice also displayed significantly increased ENaC activity, without changes in plasma aldosterone.³⁴¹ In contrast, α -KG inhibited amiloride-sensitive sodium reabsorption in principal cells independently of OXGR1 activation. This effect is possibly related to increased ATP production, inducing autocrine activation of P2Y₂ receptors and, thereby, inhibition of ENaC.³⁴² It appears that

receptor-dependent and receptor-independent effects of α -KG converge to compensate for an alkalosis-induced decrease in proximal tubular reabsorption of NaCl by favoring NaCl reabsorption over Na^+/K^+ exchange along the connecting tubule and CCD.³⁴¹ Taken together, the data indicate that α -KG acts as a paracrine mediator involved in the functional coordination of proximal and distal parts of the tubule in the adaptive regulation of HCO_3^- secretion and NaCl reabsorption in the presence of acid-base disturbances.³⁴¹

Members of the epidermal growth factor (EGF) family have been shown to be important for maintaining trans-epithelial Na^+ transport. For example, a high salt diet was shown to decrease cortical EGF levels promoting ENaC-mediated Na^+ reabsorption in the collecting duct and the development of hypertension. Conversely, intravenous EGF decreased ENaC activity, prevented the development of hypertension and attenuated glomerular and renal tubular damage in the Dahl salt-sensitive rat.³⁴³ The inhibitory effect of EGF on ENaC-dependent Na^+ absorption appears to be mediated via the H-Ras/c-Raf, MEK/ERK signaling pathway, and Cav-1 is an essential component of this EGF-activated signaling mechanism.³⁴⁴ The physiological implications of these observations are eagerly awaited.

The role of obesity in the pathogenesis of hypertension and renal dysfunction has led to the exploration of appetite-related hormones in salt and water retention. In this context, the orexigenic hormone, ghrelin, secreted by the stomach, has been shown to stimulate Na^+ absorption through cAMP-dependent trafficking of ENaC in the CCD.³⁴⁵ The effect appears to be mediated via ghrelin receptor upregulation of Sirt-1 and was also shown in rats fed a low-salt diet.³⁴⁶ These data indicate that the ghrelin-Sirt1 system may participate in regulating sodium reabsorption in the distal nephron, but its physiologic and pathologic roles in sodium homeostasis remain to be clarified.³⁴⁷

Another novel development is the growing interest in the circadian rhythmicity of many basic physiologic functions. These functional rhythms are driven, in part, by the circadian clock, a ubiquitous molecular mechanism allowing cells and tissues to anticipate and prepare for regular environmental events. This clock has been shown to play a role in the regulation and maintenance of RPF, GFR, tubular reabsorption, and secretion of Na^+ , Cl^- and K^+ .³⁴⁸ Studies in clock-deficient mice have identified the 20-HETE synthesis pathway as one of the clock's principal renal targets to mediate Na^+ excretion.³⁴⁹ By exerting dynamic control over renal sodium handling, the circadian clock could affect BP, at least in part.

Rhythmicity of salt regulation seems to occur not only at the circadian level but also on a longer periodic basis (so-called infradian rhythms). In a fascinating study on men involved in space flight simulations, Rakova and colleagues³⁵⁰ have shown that even on fixed salt diets (6, 9, or 12 g/day), daily Na^+ excretion exhibited aldosterone-dependent, weekly (circaseptan) rhythms, resulting in periodic Na^+ storage. Changes in total-body Na^+ (± 200 to 400 mmol) exhibited monthly or longer period lengths, without parallel changes in body weight and extracellular water. These changes were directly related to urinary aldosterone excretion and inversely to urinary cortisol, suggesting rhythmic hormonal control. These findings suggest the existence of rhythmic Na^+ excretory and retention patterns independent of BP or body water and irrespective of salt intake.³⁵⁰

SODIUM BALANCE DISORDERS

HYPVOLEMIA

DEFINITION

Hypovolemia is the condition in which the volume of the ECF compartment is reduced in relation to its capacitance. As noted, the reduction may be absolute or relative. In states of absolute hypovolemia, Na^+ balance is truly negative, reflecting past or ongoing losses. Hypovolemia is described as “relative” when there is no Na^+ deficit but the capacitance of the ECF compartment is increased. In this situation of reduced EABV, the ECF intravascular and extravascular (interstitial) compartments may vary in the same or opposite directions. ICF volume, reflected by measurements of plasma Na^+ or osmolality, may or may not be concomitantly disturbed; thus, hypovolemia may be classified as normonatremic, hyponatremic or hypernatremic.

ETIOLOGY

The causes of hypovolemia are summarized in Table 14.4. Absolute and relative hypovolemia, in turn, can have extrarenal or renal causes. Absolute hypovolemia results from massive blood loss or fluid loss from the skin, GI or respiratory

Table 14.4 Causes of Absolute and Relative Hypovolemia

Absolute

Extrarenal

Gastrointestinal fluid loss
Bleeding
Skin fluid loss
Respiratory fluid loss
Extracorporeal ultrafiltration

Renal

Diuretics
Obstructive uropathy/postobstructive diuresis
Hormone deficiency
Hypoadosteronism
Adrenal insufficiency
 Na^+ wasting tubulopathies
Genetic
Acquired tubulointerstitial disease

Relative

Extrarenal

Edematous states
Heart failure
Cirrhosis
Generalized vasodilation
Sepsis
Drugs
Pregnancy
Third-space loss

Renal

Severe nephrotic syndrome

system, or kidneys. Relative hypovolemia results from states of vasodilation, generalized edema or third-space loss. In both absolute and relative hypovolemia, the perceived reduction in EABV prompts the compensatory hemodynamic changes and renal responses described earlier (see section, “Physiology”).

PATHOPHYSIOLOGY

Absolute Hypovolemia

Extrarenal. The commonest causes of absolute hypovolemia are persistent diarrhea, vomiting and massive bleeding, either gastrointestinal or as a result of trauma. The reduction in ECF volume is isotonic inasmuch as there is a proportionate loss of water and plasma. The consequent fall in systemic BP leads to compensatory tachycardia and vasoconstriction and the ensuing altered transcapillary Starling hydraulic forces enable a shift of fluid from the interstitial to intravascular compartment. In addition, the neural and hormonal responses to hypovolemia (see section, “Physiology”) result in renal Na^+ and water retention, with the aim of restoring intravascular volume and hemodynamic stability.

Similar compensatory mechanisms become activated after fluid losses from the skin, GI system and respiratory system. Because of the large surface area of the skin, large amounts of fluid can be lost from this tissue as a result of burns or excessive perspiration. Severe burns allow the loss of large volumes of plasma and interstitial fluid and can lead rapidly to profound hypovolemia. Without medical intervention, hemoconcentration and hypoalbuminemia supervene. As occurs after massive bleeding, the fluid loss is isotonic, so plasma Na^+ concentration and osmolality remain normal. In contrast, excessive sweating, induced by exertion in a hot environment, leads to hypotonic fluid loss as a result of the relatively low Na^+ concentration (20 to 50 mmol/L) in sweat. The resulting hypovolemia may therefore be accompanied by hypernatremia and hyperosmolality and the type of fluid replacement must be tailored accordingly (see Chapter 15).

In addition to oral intake, the GI tract is characterized by the entry of approximately 7 L of isotonic fluid, the overwhelming majority of which is reabsorbed in the large intestine. Hence, in normal conditions, fecal fluid loss is minimal. However, in the presence of pathologic conditions, such as vomiting, diarrhea, colostomy and ileostomy secretions, especially those caused by infection, considerable or even massive fluid loss may occur. The ionic composition, osmolality and pH of secretions vary according to the part of the GI tract involved; therefore, the resulting hypovolemia is associated with a large spectrum of electrolyte and acid–base abnormalities (see Chapters 16 and 17 for further discussion).

In contrast to the massive losses that can occur from the skin and GI system, fluid loss from the respiratory tract—as occurs in febrile states and in patients who receive mechanical ventilation with inadequate humidification—is usually modest and hypovolemia ensues only in the presence of accompanying causes. Finally, a special situation in which hypovolemia can occur is after excessive ultrafiltration in dialysis patients (see Chapter 63).

Renal. Even when GFR is markedly impaired, the amount of filtered Na^+ far exceeds the dietary intake, and all but 1%

of the filtered load is reabsorbed. However, if one or more of the tubular reabsorptive mechanisms is impaired, serious Na^+ deficit and absolute volume depletion can occur. The causes of absolute renal Na^+ losses include pharmacologic agents and renal structural, endocrine, and systemic disorders (see Table 14.4). All diuretics used to treat hypervolemic states may induce hypovolemia if administered in excess or inappropriately. Particularly, the powerful loop diuretics, furosemide, bumetanide, torsemide, and ethacrynic acid, are often given in combination with diuretics acting on other tubular segments (e.g., thiazides, aldosterone antagonists, distal ENaC blockers, and carbonic anhydrase inhibitors). Patients receiving these combinations need to be carefully monitored and fluid balance scrupulously adjusted to prevent hypovolemia. Patients commonly at risk are those with HF or underlying hypertension who develop intercurrent infections.

In patients with hypertension, diuretic treatment appreciably increases the risk of volume depletion. Osmotic diuretics, endogenous or exogenous, may also reduce tubular Na^+ reabsorption. Endogenous agents include urea, the principal molecule involved in the polyuric recovery phase of acute kidney injury and postobstructive diuresis, and glucose in hyperglycemia. In patients with increased intracranial pressure, exogenous agents, such as mannitol or glycerol, may be used to induce translocation of fluid from the ICF to the ECF compartment and decrease brain swelling. The resulting polyuria may be associated with electrolyte and acid–base disturbances, the nature of which depends on the complex interplay between fluid intake and intercompartmental fluid shifts.

Na^+ reabsorption may also be disrupted in inherited and acquired tubular disorders. Inherited disorders of the proximal tubules (e.g., Fanconi syndrome) and distal tubules (e.g., Bartter and Gitelman syndromes) may lead to salt-wasting states in association with other electrolyte or acid–base disturbances. Acquired disorders of Na^+ reabsorption may be acute, as in nonoliguric acute kidney injury, the period immediately after renal transplantation, the polyuric recovery phase of acute kidney injury and postobstructive diuresis (see relevant chapters for further details), or they may be chronic as a result of tubulointerstitial diseases with a propensity for salt wasting. Chronic kidney disease stages 3 to 5 of any cause is associated with heightened vulnerability to Na^+ losses because the ability to match tubular reabsorption with the sum of filtered load minus dietary intake is impaired.

In addition to intrinsic tubular disorders, endocrine and other systemic disturbances may lead to impaired Na^+ reabsorption. The principal endocrine causes are mineralocorticoid deficiency and resistance states. A controversial cause is the systemic disturbance known as cerebral salt wasting (CSW). In this condition, salt wasting is thought to occur in response to an as yet unidentified factor released in the setting of acute head injury or intracranial hemorrhage.^{351,352} CSW is usually diagnosed because of concomitant hyponatremia and signs of volume depletion in contrast to the normovolemia characteristic of the syndrome of inappropriate antidiuresis.³⁵³ However, CSW remains an enigmatic and not universally accepted clinical entity.³⁵⁴

An underappreciated, but not uncommon, clinical setting for renal Na^+ loss is after the administration of large volumes

of intravenous saline to patients over several days after surgery or after trauma. In this situation, tubular reabsorption of Na^+ is downregulated. If intravenous fluids are stopped before full reabsorptive capacity is restored, volume depletion may ensue. The phenomenon can be minimized by graded reduction in the infusion rate, which allows Na^+ reabsorptive pathways to be restored gradually.

In the context of volume depletion, diabetes insipidus should be mentioned. However, because this results from a deficiency of or tubular resistance to AVP, water loss is the main consequence and the impact on ECF volume is only minor. AVP-related disorders are considered in Chapter 15.

Relative Hypovolemia

Extrarenal. As outlined previously, the principal causes of relative hypovolemia are edematous states, vasodilation and third-space loss (see Table 14.4). Vasodilation may be physiologic, as in normal pregnancy, or induced by drugs (hypotensive agents, such as hydralazine or minoxidil, that cause arteriolar vasodilation), or it may occur in sepsis during the phase of peripheral vasodilation and consequent low SVR.³⁵⁵

Edematous states in which the EABV and, hence, tissue perfusion, are reduced include HF, decompensated cirrhosis with ascites and nephrotic syndrome. In severe HF, low cardiac output and resulting low systemic BP lead to a fall in RPP. As in absolute hypovolemia, the kidneys respond by retaining Na^+ . Because the increased venous return cannot raise the cardiac output, a vicious cycle is created in which edema is further exacerbated and the persistently reduced cardiac output leads to further Na^+ retention. In decompensated cirrhosis, splanchnic venous pooling leads to decreased venous return, a consequent fall in cardiac output and compensatory renal Na^+ retention. The pathophysiology of edematous states is discussed later (see section, “**Hypervolemia**”). Third-space loss occurs when fluid is sequestered into compartments not normally perfused with fluids, as in states of GI obstruction, after trauma, burns, or in pancreatitis, peritonitis, or malignant ascites. The result is that, even though total body Na^+ is markedly increased, the EABV is severely reduced.

Renal. Approximately 10% of patients with the nephrotic syndrome—especially children with minimal change disease, but also any patient with a serum albumin level lower than 2 g/dL—manifest clinical signs of hypovolemia. The low plasma oncotic pressure is conducive to movement of fluid from the ECF compartment to the interstitial space, thereby leading to reduced EABV.³⁵⁶

CLINICAL MANIFESTATIONS

The clinical manifestations of hypovolemia depend on the magnitude and rate of volume loss, solute composition of the net fluid loss (i.e., the difference between input and output) and vascular and renal responses. The clinical features are related to the underlying pathophysiologic process, hemodynamic consequences and electrolyte and acid-base disturbances accompanying the renal response to hypovolemia. A detailed history usually reveals the cause of volume depletion (bleeding, vomiting, diarrhea, polyuria, diaphoresis, medications).

The symptoms and physical signs of hypovolemia appear only when intravascular volume is decreased by 5% to 15%

and are often related to tissue hypoperfusion. Symptoms include generalized weakness, muscle cramps, and postural light-headedness. Thirst is prominent if concomitant hypertonicity is present (hypertonic hypovolemia). Physical signs are related to the hemodynamic consequences of hypovolemia and include tachycardia, hypotension (postural, absolute or relative to the usual BP) and low central or jugular venous pressure. Elevated jugular venous pressure, however, does not rule out hypovolemia, because of the possible confounding effects of underlying HF or lung disease. When volume depletion exceeds 10% to 20%, circulatory collapse is liable to occur, with severe supine hypotension, peripheral cyanosis, cold extremities and impaired consciousness, extending even to coma. This is especially likely if fluid loss is rapid or occurs against a background of comorbid conditions. When the source of volume loss is extrarenal, oliguria also occurs. The traditional signs—reduced skin turgor, sunken eyes, and dry mucous membranes—are inconstant findings and their absence does not rule out hypovolemia. Reduction in the EABV, as manifested by relative hypotension, may also be observed in generalized edematous states, even though there is an overall excess of Na^+ and water; however, this excess is maldistributed between the extracellular and interstitial spaces.

DIAGNOSIS

The diagnosis of hypovolemia is based essentially on the clinical findings. Nevertheless, when these are equivocal, various laboratory parameters may be helpful for confirming the diagnosis or for elucidating other changes that may be associated with volume depletion.

Laboratory Findings

Hemoglobin and Plasma Albumin. Hemoglobin may decrease if significant bleeding has occurred or is ongoing, but the change, which is caused by hemodilution owing to fluid translocation from the interstitial to intravascular compartment, may take up to 24 hours. Therefore, stable hemoglobin does not rule out significant bleeding. Moreover, the adaptive response of hemodilution may moderate the severity of hemodynamic compromise and resulting physical signs. In hypovolemic situations that do not arise from bleeding, hemoconcentration is often, but not universally, seen, inasmuch as underlying anemia of chronic disease may mask the differential loss of plasma.

Hemoconcentration may also be manifested as a rise in plasma albumin concentration if albumin-free fluid is lost from the skin, GI tract, or kidneys. On the other hand, when albumin is lost, either in parallel with other extracellular fluids (as in proteinuria, hepatic disease, protein-losing enteropathy, or catabolic states) or in protein-rich fluid (third-space sequestration, burns), significant hypoalbuminemia is observed.

Plasma Na^+ Concentration. This may be low, normal, or high, depending on the solute composition of the fluid lost and the replacement solution administered by the patient or physician. For example, the hypovolemic stimulus for AVP release may lead to preferential water retention and hyponatremia, especially if hypotonic replacement fluid is used. In contrast, the fluid content of diarrhea may be hypotonic or hypertonic, resulting in hyponatremia or hypernatremia,

respectively. The plasma Na^+ concentration reflects the tonicity of plasma and provides no direct information about volume status, which is a clinical diagnosis.

Plasma K^+ and Acid-Base Parameters. These can also change in hypovolemic conditions. After vomiting and after some forms of diarrhea, loss of K^+ and Cl^- may lead to alkalosis. More often, the principal anion lost in diarrhea is bicarbonate, which leads to hyperchloremic (nonanion gap) acidosis. When diuretics or Bartter and Gitelman syndromes (the inherited tubulopathies; see Chapter 44) are the cause of hypovolemia, hypokalemic alkalosis is again typically seen. On the other hand, U_{Na} loss that occurs in adrenal insufficiency or due to aldosterone hyporesponsiveness is accompanied by a tendency for hyperkalemia and metabolic acidosis. Finally, when hypovolemia is sufficiently severe to impair tissue perfusion, high anion gap acidosis caused by lactic acid accumulation may be observed.

Blood Urea and Creatinine Levels. These frequently rise in hypovolemic states and reflect impaired renal perfusion. If tubular integrity is preserved, then the rise in urea levels is typically disproportionate to that of creatinine, so-called prerenal azotemia (see Chapter 28). This results mainly from AVP-enhanced urea reabsorption in the medullary collecting duct (MCD), but also from augmented proximal tubular reabsorption due to increased filtration fraction.³⁵⁷ In critically ill patients, an increased urea generation rate (from exogenous or endogenous protein catabolism) or low creatinine generation rate due to muscle wasting may lead to an erroneous diagnosis of prerenal azotemia.³⁵⁸ In the presence of severe hypovolemia, acute kidney injury may ensue, leading to loss of the differential rise in urea level. Proportional rises in urea and creatinine are also observed when hypovolemia occurs against a background of underlying renal functional impairment, as in chronic kidney disease, stages 3 to 5.

Urine Biochemical Parameters. In hypovolemia due to extrarenal fluid losses, the intact kidney will respond to hypoperfusion by enhanced tubular reabsorption of Na^+ and water. The ensuing oliguria will be characterized by urine-specific gravity greater than 1.020, Na^+ concentration less than 10 mmol/L and osmolality greater than 400 mOsm/kg. When urine Na^+ concentration is 20 to 40 mmol/L, the finding of a fractional

excretion of Na^+ $\frac{\text{urine } \text{Na}^+ \times \text{plasma creatinine}}{\text{plasma } \text{Na}^+ \times \text{urine creatinine}} \times 100$ of

less than 1%, in the presence of oliguria, may be helpful. However, in a patient on previous diuretic therapy, especially with loop diuretics, these indices may merely reflect U_{Na} losses. In that case, fractional excretion of urea of less than 30% to 35% may help in the diagnosis of hypovolemia, although the specificity of this test is rather low.^{359,360}

Other Laboratory Parameters. When hypovolemia occurs in the presence of arterial vasodilation, as observed in sepsis, some, but not all, of the clinical manifestations of hypovolemia are observed. Thus, tachycardia and hypotension are usually present, but the extremities are warm, suggesting that perfusion is maintained. This finding is misleading because vital organs, particularly the brain and kidneys, are underperfused

as a result of the hypotension. The presence of lactic acidosis helps establish the correct diagnosis.

TREATMENT

Absolute Hypovolemia

General Principles. The goals of treatment of hypovolemia are to restore normal hemodynamic status and tissue perfusion. These goals are achieved by reversal of the clinical symptoms and signs, described previously. Treatment can be divided into three stages: 1. initial replacement of the immediate fluid deficit; 2. maintenance of the restored ECF volume in the presence of ongoing losses; and 3. treatment of the underlying cause, whenever possible. The main strategies to be addressed by the clinician are the route, volume, rate of administration, and composition of the replacement and maintenance fluids. These are liable to change according to the patient's response.

In general, when hypovolemia is associated with a significant hemodynamic disturbance, intravenous rehydration is required. (The use of oral electrolyte solutions in the management of infants and children is discussed in Chapter 73.) The volume of fluid and rate of administration should be determined on the basis of the urgency of the threat to circulatory integrity, adequacy of the clinical response, and underlying cardiac function. Older patients are especially vulnerable to aggressive fluid challenge and careful monitoring is required, particularly to prevent acute left ventricular failure and pulmonary edema from overzealous correction.

Sometimes the clinical signs do not point unequivocally to the diagnosis of hypovolemia, even though the history is strongly suggestive. Invasive monitoring of central venous and pulmonary venous pressures has not been shown to improve outcomes in this situation^{361,362}; monitoring preload by stroke volume variation may improve outcomes, at least after major abdominal surgery.³⁶³ However, in case of doubt, a diagnostic fluid challenge should be performed. If the patient improves clinically, BP and urine output increase and no overt signs of HF appear over the succeeding 6 to 12 hours, then the diagnosis is substantiated, and fluid therapy can be cautiously continued. Conversely, if overt signs of fluid overload appear, the fluid challenge can be stopped, and diuretic therapy reinstituted.

The initial calculations for replacing the fluid deficit are based on hemodynamic status. These deficits are notoriously difficult to calculate; therefore, good clinical judgment is necessary for successful management. Patients with life-threatening circulatory collapse and hypovolemic shock require rapid intravenous replacement through the cannula with the widest bore possible. Replacement should continue until BP and tissue perfusion are restored. In the second stage, the rate of fluid replacement should be reduced to maintain BP and tissue perfusion. In older patients and those with underlying cardiac dysfunction, the risk of overrapid correction and precipitating pulmonary edema is heightened; therefore slower treatment is preferable, to allow gradual filling of the ECF volume rather than causing pulmonary edema and the threat of mechanical ventilation associated with adverse outcomes.³⁶⁴

Composition of Replacement Fluids. The composition of replacement fluid may also affect outcomes. The two main

categories of replacement solution are crystalloid and colloid solutions. Crystalloid solutions are based largely on NaCl of varying tonicity or dextrose. Isotonic (0.9%) saline, containing 154 mmol of Na^+ /L, is the mainstay of volume replacement therapy being confined to the ECF compartment in the absence of deviations in Na^+ concentration. One L of isotonic saline increases plasma volume by approximately 300 mL; the rest is distributed to the interstitial compartment. In contrast, 1 L of 5% dextrose in water (D_5W), which is also isosmotic (277 mOsm/L), is eventually distributed throughout all the body fluid compartments so that only 10% to 15% (100 to 150 mL) remains in the ECF. Therefore, D_5W should not be used for volume replacement.

Administration of 1 L of .45% saline (77 mmol of Na^+ /L) in D_5W is equivalent to giving 500 mL of isotonic saline and the same volume of solute-free water. The distribution of the solute-free compartment throughout all the fluid compartments would result in plasma dilution and reduction in the plasma Na^+ . Therefore, this solution should be reserved for the management of hypernatremic hypovolemia. Even in that situation, it must be remembered that volume replacement is less efficient than with isotonic saline and, early in the treatment course, may cause plasma tonicity to fall too rapidly.

When hypovolemia is accompanied by severe metabolic acidosis (pH <7.10; plasma HCO_3^- <10 mmol/L), bicarbonate supplementation may be indicated. (For a discussion of bicarbonate balance, see Chapter 16.) Because this anion is manufactured as 8.4% sodium bicarbonate (1000 mmol/L) for use in cardiac resuscitation, appropriate dilution is required for the treatment of acidosis associated with hypovolemia. Two convenient methods are suggested. Either 75 mL (75 mmol) of 8.4% NaHCO_3^- can be added to 1 L of .45% saline, or 150 mL of concentrated bicarbonate can be added to 1 L of D_5W . Although the latter is hypertonic in the short term, it is unlikely to be harmful.

In the presence of accompanying hypokalemia, especially if metabolic alkalosis is also present, volume replacement solutions should be supplemented with K^+ . Commercially available 1-L solutions of isotonic saline supplemented with 10 or 20 mmol of KCl make this option safe and convenient. (For details, see Chapter 17.) On the other hand, newer, commercially available crystalloid solutions containing lactate (converted by the liver to bicarbonate) and low concentrations of KCl may offer advantages over isotonic saline. In a recent large prospective observational study performed in the intensive care unit setting, two periods were compared; in the control period, all patients received isotonic saline as fluid replacement, whereas during the intervention period, Hartmann solution (lactate-containing), Plasma-Lyte 148 (a balanced salt solution), or chloride-poor 20% albumin solution was administered. The chloride-poor solutions were associated with a significantly lower risk of subsequent acute kidney injury, even after adjustment for covariates.³⁶⁵ Clearly, these provocative results indicate the need for randomized controlled trials comparing chloride-rich with more balanced salt solutions for fluid resuscitation.^{366,367}

Colloid solutions include plasma, albumin and high-molecular-weight carbohydrate molecules, such as hydroxyethyl starch and dextrans, at concentrations that exert COPs equal to or greater than that of plasma. Because the transcapillary barrier is impermeable to these large molecules, in theory

they expand the intravascular compartment more rapidly and efficiently than crystalloid solutions. Colloid solutions may be useful in the management of burns and severe trauma when plasma protein losses are substantial and rapid plasma expansion with relatively small volumes is efficacious. However, when capillary permeability is increased, as in states of multiorgan failure or the systemic inflammatory response syndrome, colloid administration is ineffective. Moreover, randomized controlled studies, in which crystalloid solutions were compared with colloid solutions, have shown no survival benefit and even harm with some colloid solutions, particularly hydroxyethyl starch.³⁶⁷ Therefore, the much cheaper and more readily available crystalloid solutions should remain the mainstay of therapy.

Relative Hypovolemia

Treatment of relative hypovolemia is more difficult than that of absolute hypovolemia because there is no real fluid deficit. If the relative hypovolemia is caused by peripheral vasodilation, as in sepsis, it may be necessary to administer cautiously a crystalloid solution, such as isotonic saline, to maintain ECF volume until the SVR and venous capacitance return to normal; the excess volume administered can then be excreted by the kidneys. When vasodilation is more severe, vasoconstrictor agents may be needed to maintain systemic BP. In severe HF, advanced cirrhosis with portal hypertension and severe nephrotic syndrome, when EABV is low but there is an overall excess of Na^+ and water, treatment may be extremely challenging. Crystalloid solution will, likely, lead to worsening interstitial edema without significantly affecting EABV. In these situations, prognosis is determined by whether the underlying condition can be reversed.

HYPERVOLEMIA

DEFINITION

Hypervolemia occurs when the volume of the ECF compartment is expanded relative to its capacitance. Normally, increments in Na^+ intake are matched by corresponding changes in Na^+ excretion as elaborated earlier (see section, “Physiology”). However, in the approximately 20% of the population who are salt sensitive, the upward shift in ECF volume induced by high salt intake leads to a persistent rise in systemic arterial pressure, albeit without other signs of fluid retention (see Chapter 46). Here, the discussion is confined to hypervolemia, in which Na^+ retention is ongoing and inappropriate for the prevailing ECF volume, with the appearance of clinical signs of volume overload.

ETIOLOGY

Hypervolemia may result from either primary renal Na^+ retention or can be secondary to disease in other major organs (Table 14.5).

Primary Renal Na^+ Retention

This can be subclassified as caused by intrinsic kidney disease or primary mineralocorticoid excess. Of the primary renal diseases causing Na^+ retention, oliguric renal failure limits the ability to excrete Na^+ and water and affected patients are at risk for rapidly developing ECF volume overload (see Chapter 28). In contrast, in chronic kidney disease, renal tubular adaptation to salt intake is usually efficient until late

Table 14.5 Causes of Renal Sodium Retention**Primary**

Oliguric acute kidney injury
 Chronic kidney disease
 Glomerular disease
 Severe bilateral renal artery stenosis
 Na⁺-retaining tubulopathies (genetic)
 Mineralocorticoid excess

Secondary

Heart failure
 Cirrhosis
 Idiopathic edema

stage 4 and stage 5. However, in some primary glomerular diseases, especially in the presence of nephrotic range proteinuria, significant Na⁺ retention may occur, even when GFR is close to normal (see section, “**Pathophysiology**,” and Chapter 30). Primary mineralocorticoid excess leads to transient Na⁺ retention. However, because of “mineralocorticoid escape,” the dominant clinical feature is hypertension (see Chapters 12 and 46).

Secondary Renal Na⁺ Retention

This occurs in low- and high-output cardiac failure with systolic and/or diastolic dysfunction. Nephrotic syndrome and hepatic cirrhosis with portal hypertension are also accompanied by renal Na⁺ retention. In this chapter, only HF and cirrhosis are considered. Nephrotic syndrome is discussed in Chapter 28.

PATHOPHYSIOLOGY

Primary renal Na⁺ retention is caused by disruption of normal renal function. In contrast, secondary renal Na⁺ retention occurs because of reduced EABV in the presence of total ECF volume expansion or in response to factors secreted by the heart or liver that signal the kidneys to retain Na⁺ (Fig. 14.8). In secondary Na⁺ retention, the renal effector mechanisms that normally operate to conserve Na⁺ and protect against a Na⁺ deficit are exaggerated and maintained, despite subtle or overt ECF volume expansion. The pathophysiology of hypervolemia involves local mechanisms of edema formation and stimulation of renal Na⁺ retention by reduced EABV either directly or indirectly, via abnormalities of the afferent volume sensing mechanisms.

Local Mechanisms of Edema Formation

Peripheral interstitial fluid accumulation, which is common to all conditions causing hypervolemia, results from disruption of the normal balance of transcapillary Starling forces. Transcapillary fluid and solute transport consists of both convective and diffusive flow. Bulk water movement occurs via convective transport induced by hydraulic and osmotic pressure gradients. Capillary hydraulic pressure (P_c) is under the influence of several factors, including systemic arterial and venous BPs, local blood flow and precapillary and postcapillary resistance. Systemic arterial BP, in turn, is determined by cardiac output, intravascular volume and SVR; systemic venous pressure is determined by right atrial pressure, intravascular volume and venous capacitance. Na⁺

balance is a key determinant of these latter hemodynamic parameters. Also, massive accumulation of fluid in the peripheral interstitial compartment (anasarca) can itself diminish venous compliance and, thereby, alter overall cardiovascular performance.³⁶⁸

The balance of Starling forces prevailing at the arteriolar end of the capillary ($\Delta P > \Delta \pi$, in which $\Delta \pi$ is the change in transcapillary oncotic pressure) favors net filtration of fluid into the interstitium. Net outward movement of fluid along the length of the capillary is associated with an axial decrease in P_c and an increase in π_c . Nevertheless, the local ΔP continues to exceed the opposing $\Delta \pi$ throughout the length of the capillary bed in several tissues; thus, filtration occurs along its entire length.³⁶⁹ In such capillary beds, a substantial volume of filtered fluid must, therefore, return to the circulation via lymphatic vessels. Hence, to minimize edema formation, the lymphatic vessels must be able to expand and proliferate and lymphatic flow increase in response to increased interstitial fluid formation.

Several other mechanisms for minimizing edema formation have been identified. First, precapillary vasoconstriction tends to lower P_c and diminish the filtering surface area in a given capillary bed. Indeed, in the absence of appropriate regulation of the microcirculatory myogenic reflex, as occurs with some Ca²⁺ channel blockers, excessive precapillary vasodilation may lead to lower extremity interstitial edema.³⁷⁰ Second, increased net filtration itself is associated with dissipation of P_c , dilution of interstitial fluid protein concentration and a corresponding rise in intracapillary plasma protein concentration. The resulting change in the balance of Starling forces will tend to mitigate further interstitial fluid accumulation.³⁷¹ Finally, interstitial fluid hydraulic pressure (P_i) is normally subatmospheric; however, even small increases in interstitial fluid volume tend to augment P_i , again opposing further transudation of fluid into the interstitial space.³⁷² The appearance of generalized edema in association with expansion of the ECF volume therefore implies the presence of one or more disturbances in microcirculatory hemodynamics—increased venous pressure transmitted to the capillary, unfavorable adjustments in precapillary and postcapillary resistances, and/or inadequacy of lymphatic flow for draining the interstitial and replenishing the intravascular compartment.

For the clinical detection of generalized edema, the volume of accumulated interstitial fluid required (>2 to 3 L) necessitates expansion of ECF volume and, hence, body exchangeable Na⁺ content. Since continued net accumulation of interstitial fluid without renal Na⁺ retention might result in serious intravascular volume contraction and cessation of interstitial fluid formation, generalized edema must indicate substantial renal Na⁺ retention.

Systemic Factors Stimulating Renal Sodium Retention

Reduced Effective Arterial Blood Volume. Renal Na⁺ (and water retention) in edematous disorders occurs due to reduced EABV, despite an increase in total blood and ECF volumes and normal intrinsic renal function.¹⁹ If the underlying stimulus for hypervolemia is removed, as dramatically seen after heart³⁷³ or liver transplantation,³⁷⁴ Na⁺ excretion is restored to normal. Conversely, when kidneys from patients with end-stage liver disease are transplanted into patients with normal liver function, Na⁺ retention no longer occurs.³⁷⁴

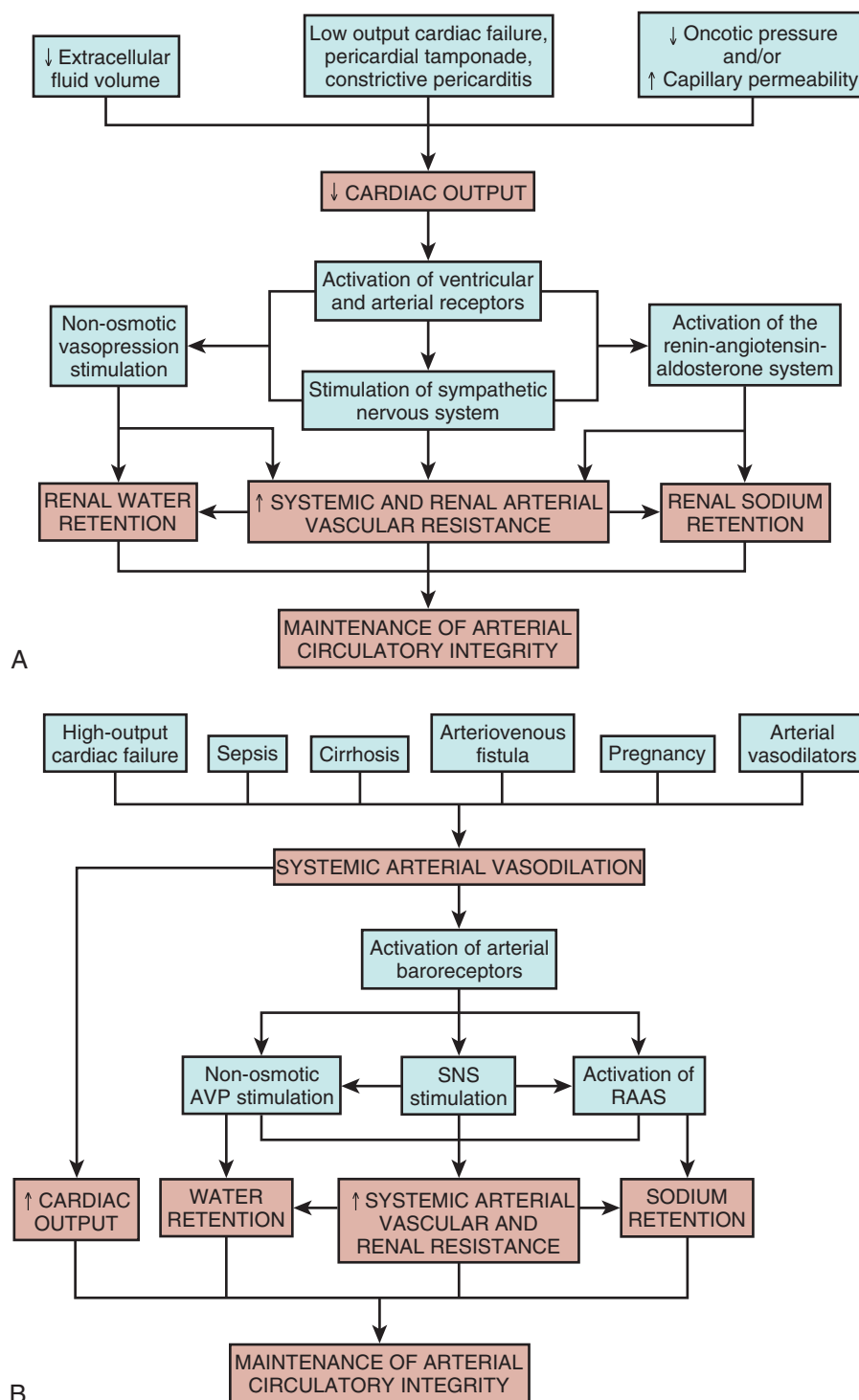


Fig. 14.8 Sensing mechanisms that initiate and maintain renal sodium and water retention in various clinical conditions in which arterial underfilling, with resultant neurohumoral activation and renal sodium and water retention, is caused by a decrease in cardiac output (A) and by systemic arterial vasodilation (B). In addition to activating the neurohumoral axis, adrenergic stimulation causes renal vasoconstriction and enhances sodium and fluid transport by the proximal tubule epithelium. (From Schrier RW. Decreased effective blood volume in edematous disorders: what does this mean? *J Am Soc Nephrol.* 2007;18:2028–2031.)

Because 85% of blood circulates in the venous compartment, expansion of that compartment leads to overall ECF volume excess that could occur concurrently with arterial underfilling. The latter could result from low cardiac output, peripheral arterial vasodilation, or a combination of the two.

In turn, low cardiac output could result from true ECF volume depletion (see earlier discussion), cardiac failure, or decreased π_c , with or without increased capillary permeability. All these stimuli would cause activation of ventricular and arterial sensors. Similarly, conditions such as high-output cardiac

failure, sepsis, cirrhosis and normal pregnancy lead to peripheral arterial vasodilation and activation of arterial baroreceptors. Activation of these afferent mechanisms would then induce the neurohumoral mechanisms that result in renal Na^+ and water retention (see Fig. 14.8).³⁷⁵

Although the mechanisms leading to Na^+ retention in HF and cirrhosis are similar, specific differences between the two conditions have been observed and are discussed separately in the following sections.

Renal Sodium Retention in Heart Failure

Abnormalities of Sensing Mechanisms in Heart Failure. Both the cardiopulmonary and baroreceptor reflexes are blunted in HF, so that they cannot exert an adequate tonic inhibitory effect on sympathetic outflow.³⁷⁶ The resulting SNS activation triggers renal Na^+ retention, as already described. A variety of models of HF have shown marked attenuation of atrial receptor firing and loss of nerve ending arborization in HF.³⁷⁷ Similarly, altering central cardiac filling pressures in response to postural stimuli (e.g., head-up tilt, LBNP) in HF patients, in contrast to normal subjects, usually do not demonstrate significant alterations in limb blood flow, circulating catecholamines, AVP, or renin activity.^{378,379} This diminished reflex responsiveness is proportionate to the severity of ventricular dysfunction.

Arterial baroreceptor reflex impairment has been observed in HF. High baseline values of muscle sympathetic activity were found in patients with HF who failed to respond to activation and deactivation of arterial baroreceptors by infusion of phenylephrine and Na^+ nitroprusside, respectively.³⁷⁷ Carotid and aortic baroreceptor function were also depressed in experimental models of HF.³⁷⁷ These changes were associated with upward resetting of receptor threshold and a reduced range of pressures over which the receptors functioned.

Multiple abnormalities have been described in cardiopulmonary and arterial baroreceptor control of RSNA in HF. Thus, rats with coronary ligation displayed an increased basal level of efferent RSNA that failed to decrease normally during volume expansion.^{73,254} Similarly, in sinoaortic denervated dogs with pacing-induced HF, or following left atrial baroreceptor stimulation, the cardiopulmonary baroreflex control of efferent RSNA was markedly attenuated.³⁸⁰

The abnormal regulation of efferent RSNA was caused by impaired function of aortic and cardiopulmonary baroreflexes; the latter defect was functionally more important.²⁵⁴ Mechanisms implicated in the pathogenesis of these abnormal baroreflexes include loss of compliance in the dilated hearts, gross changes in receptor structure and augmented $\text{Na}^+\text{-K}^+\text{-ATPase}$ activity in the baroreceptor membranes.³⁷⁷ Increased activity of Ang II through the AT_1 receptor also contributes to depressed baroreflex sensitivity. Thus RAAS inhibition in rats or rabbits with HF significantly improved arterial baroreflex control of RSNA or heart rate, respectively.^{377,380} This effect of Ang II could also be blocked by central α_1 -adrenoreceptor stimulation.³⁸¹

More recent studies have indicated that Ang II in the paraventricular nucleus potentiates—and AT_1 receptor antisense mRNA normalizes—the enhanced cardiac sympathetic afferent reflex in rats with chronic HF.³⁷⁷ AT_1 receptors in the nucleus tractus solitarius are thought to mediate the interaction between the baroreflex and cardiac sympathetic

afferent reflex.³⁸² Consistent with this notion, Ang II generation is enhanced and its degradation reduced in central sympathoregulatory neurons, as shown by upregulation of ACE1 and downregulation of ACE2.³⁸³ AT_2 receptors in the rostral ventrolateral medulla inhibited sympathetic outflow, an effect mediated at least partly by an arachidonic acid metabolic pathway.³⁸⁴ These studies indicated that a downregulation in the AT_2 receptor was a contributory factor in the sympathetic neural excitation in HF.³⁷⁷

Together, these data provide evidence of the role of high endogenous levels of Ang II, acting through the AT_1 receptor in concert with downregulation of the AT_2 receptor, in the impaired baroreflex sensitivity observed in HF, both in the afferent limb of the reflex arch and at more central sites. The central effect may be mediated through a central α_1 -adrenoreceptor. The blunted cardiopulmonary and arterial baroreceptor sensitivity in HF may also lead to an increase in AVP release and renin secretion.³⁷⁷

The disturbances in sensing mechanisms that initiate and maintain renal Na^+ retention in HF are summarized in Fig. 14.8A. As indicated, a decrease in cardiac output or a diversion of systemic blood flow diminishes the blood flow to the critical sites of the arterial circuit with pressure- and flow-sensing capabilities. The responses to diminished blood flow culminate in renal Na^+ retention, mediated by the effector mechanisms. An increase in systemic venous pressure promotes the transudation of fluid from the intravascular to interstitial compartment by increasing the peripheral transcapillary ΔP . These processes augment the perceived loss of volume and flow in the arterial circuit. In addition, distortion of the pressure-volume relationships as a result of chronic dilation in the cardiac atria attenuates the normal natriuretic response to central venous congestion. This attenuation is manifested predominantly as a diminished neural suppressive response to atrial stretch, which results in increased sympathetic nerve activity and augmented release of renin and AVP.

Abnormalities of Effector Mechanisms in Heart Failure. The adaptive changes in the efferent limb of the volume control system in HF are generally similar to those seen in states of true Na^+ depletion. These include adjustments in glomerular hemodynamics and tubular transport, brought about by alterations in the neural, humoral and paracrine systems. However, in contrast to true Na^+ depletion, HF is also associated with activation of vasodilatory/natriuretic agents, which tend to oppose the effects of the vasoconstrictor/antinatriuretic systems. The final effect on urinary Na^+ excretion is determined by the dynamic balance among these antagonistic effector systems.

Alterations in Glomerular Hemodynamics

HF is characterized not only by increased renal vascular resistance and reduced GFR, but also by an even greater reduction in RPF, so that the filtration fraction is increased.³⁸⁵ As shown in rat models of HF, these changes seem to result from diminished Kf and elevated afferent and efferent arteriolar resistances. The rise in filtration fraction is probably caused by a disproportionate increase in efferent arteriolar resistance.³⁸⁵

In Fig. 14.9, a comparison of the glomerular capillary hemodynamic profile in the normal versus the HF state is illustrated on the left graph of each panel. First, ΔP declines

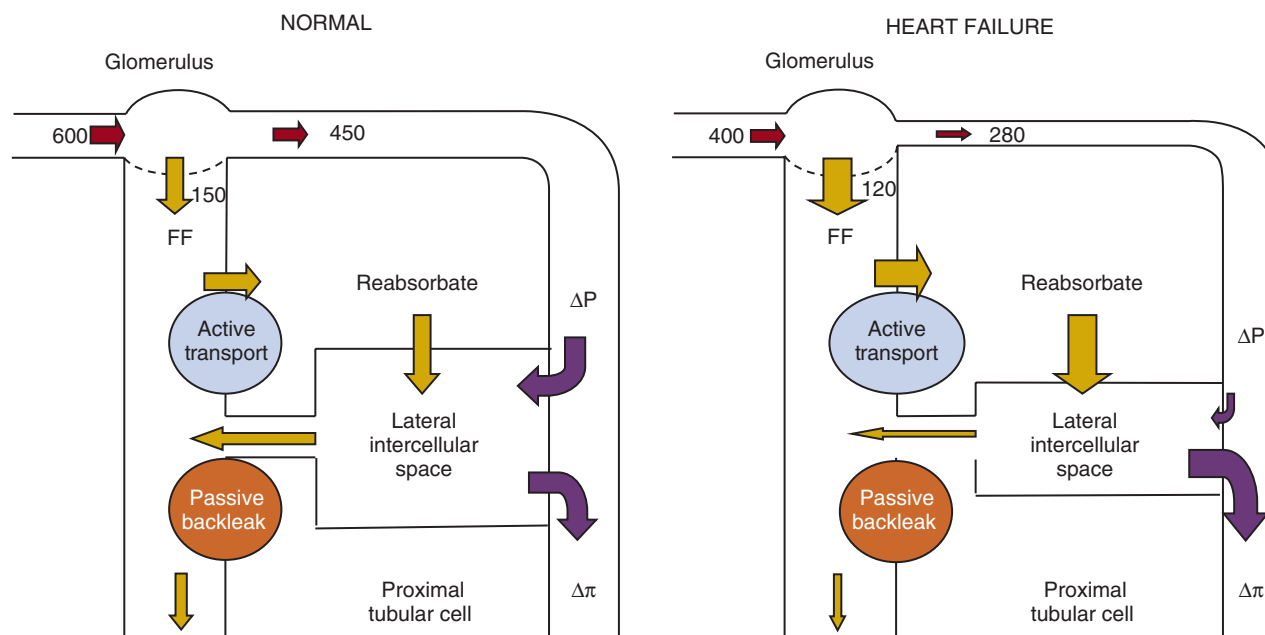


Fig. 14.9 Peritubular control of proximal tubule fluid reabsorption. Fluid reabsorption in the normal state (*left*) and in patients with heart failure (*right*) is shown. Increased postglomerular arteriolar resistance in heart failure is depicted as narrowing. The thickness and font size of the *block arrows* depict relative magnitude of effect. The increase in filtration fraction (FF) in heart failure causes $\Delta\pi$ to rise. The increase in renal vascular resistance in heart failure is believed to reduce ΔP . Both the increase in $\Delta\pi$ and the fall in ΔP enhance peritubular capillary uptake of proximal reabsorbate and thus increase absolute Na^+ reabsorption by the proximal tubule. Numbers and *red block arrows* depict blood flow in preglomerular and postglomerular capillaries; ΔP and $\Delta\pi$ are the transcapillary hydraulic and oncotic pressure differences across the peritubular capillary, respectively; *yellow block arrows* indicate transtubular transport; *purple block arrows* represent the effect of peritubular capillary Starling forces on uptake of proximal reabsorbate. (Modified from Humes HD, Gottlieb M, Brenner BM. *The kidney in congestive heart failure: contemporary issues in nephrology*. Vol 1. New York; Churchill Livingstone; 1978, pp 51–72.)

along the length of the glomerular capillary in normal and HF states, but much more so in HF because of the increased efferent arteriolar resistance. Second, $\Delta\pi$ increases over the length of the glomerular capillary in both states as fluid is filtered into Bowman's space, but again to a greater extent in HF because of the increased filtration fraction. As outlined below (see section, "[Renin-Angiotensin-Aldosterone System](#)"), the preferential increase in efferent arteriolar resistance is mediated principally by Ang II and is critical for the preservation of GFR in the presence of reduced RPF. Because of the intense efferent arteriolar vasoconstriction, further compensation is not possible if RPP falls as a result of systemic hypotension, causing a sharp decline in GFR. This phenomenon is dramatically illustrated by HF patients whose Ang II drive is removed by RAAS inhibitors, particularly those with preexisting renal failure, massive diuretic treatment and limited cardiac reserve.³⁸⁵ In these patients, BP may fall below the level necessary to maintain renal perfusion.

Enhanced Tubular Reabsorption of Sodium. A direct consequence of the glomerular hemodynamic alterations and augmented single-nephron filtration fraction is an increase in the fractional reabsorption of filtered Na^+ in the proximal tubule. In [Fig. 14.9](#), the peritubular capillary hemodynamic profile of the normal state is compared with that of HF on the right graph of each panel. In HF, in comparison with the normal state, the average value of $\Delta\pi$ along the peritubular capillary is increased and that of ΔP is decreased. These values favor fluid movement into the capillary and may also

help reduce paracellular backleak of fluid into the tubule, promoting overall net reabsorption.

The peritubular control of proximal fluid reabsorption in normal and HF states is illustrated schematically in [Fig. 14.9](#). A critical mediator of the enhanced tubular reabsorption of Na^+ is Ang II, which, by increasing efferent arteriolar resistance, increases the filtration fraction and augments proximal epithelial transporter activities directly, thereby amplifying the overall increase in proximal Na^+ reabsorption. This is clearly illustrated by the favorable effects of RAAS blockers in HF to modulate single-nephron filtration fraction and normalize proximal peritubular capillary Starling forces and Na^+ reabsorption.³⁸⁵

Enhanced reabsorption of Na^+ in HF has been shown in the loop of Henle, probably due to altered renal hemodynamics, as in the proximal tubule.¹⁵⁶ In the distal tubule and collecting duct, elevated Ang II and aldosterone levels, respectively, enhance activities of the NaCl cotransporter and ENaC.³⁸⁶

Neurohumoral Mediators. The primary vasoconstrictor/antinatriuretic (and antidiuretic) systems mediating Na^+ and water retention in HF include the RAAS, SNS, AVP, and ETs. The antagonistic vasodilator/natriuretic substances include NO, PG, AM, UT II, and NPY. The development of positive Na^+ balance and edema in HF occurs at the point when the vasoconstrictor/antinatriuretic forces predominate ([Fig. 14.10](#)). The dominant activity of Na^+ -retaining systems in HF is clinically important, as it is associated with globally-impaired

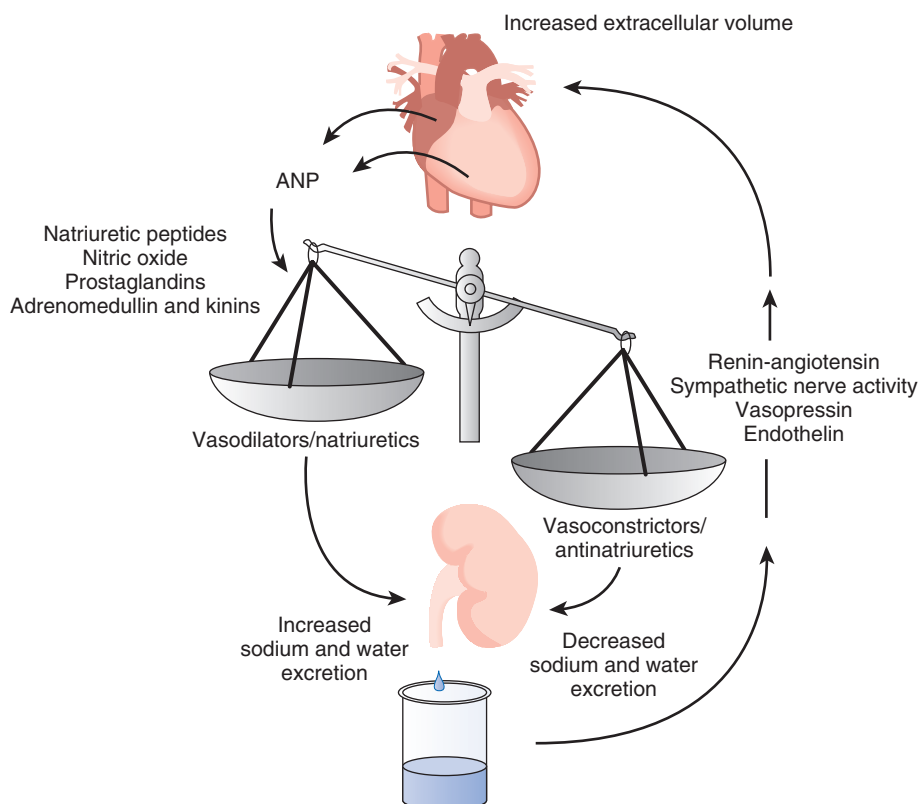


Fig. 14.10 Efferent limb of extracellular fluid volume control in heart failure. Volume homeostasis in heart failure is determined by the balance between natriuretic and antinatriuretic forces. In decompensated heart failure, enhanced activities of the Na^+ -retaining systems overwhelm the effects of the vasodilatory/natriuretic systems, which leads to a net reduction in Na^+ excretion and an increase in ECF volume. ANP, Atrial natriuretic peptide. (Modified from Winaver J, Hoffman A, Abassi Z, et al. Does the heart's hormone, ANP, help in congestive heart failure? *News Physiol Sci.* 1995;10:247–253.)

renal function, a strong predictor of mortality³⁸⁵; moreover, reversal of neurohumoral impairment is associated with improved outcomes.³⁸⁵

VASOCONSTRICTOR/ANTINATRIURETIC (ANTIDIURETIC) SYSTEMS

Renin-Angiotensin-Aldosterone System

The activity of the RAAS is enhanced in most patients with HF in correlation with the severity of cardiac dysfunction²³⁴ and provides a prognostic index for HF patients. Initially, RAAS activation is beneficial by inducing direct systemic vasoconstriction and activating other neurohormonal systems such as AVP, which contribute to maintaining adequate intravascular volume.²³⁴ However, numerous studies in patients and in experimental models of HF have established that continued activation of the RAAS leads to maladaptive myocardial remodeling²³⁴ and progression of cardiovascular and renal dysfunction.³⁶⁴

The kidneys in particular are highly sensitive to the action of Ang II, and a decrease in RPF and SNGFR, as well as elevations in efferent arteriolar resistance and filtration fraction, are observed in both clinical and experimental HF. These changes are completely reversed by ACE inhibitors, as well as a low-salt diet.³⁸⁷

Activation of Ang II in response to the decreased pumping capacity of the failing myocardium also promotes systemic

vasoconstriction and mesangial cell contraction.²⁰² In addition, Ang II reduces renal cortical circulation in rats with HF and increases tubular Na^+ reabsorption directly and by augmenting aldosterone release.²⁰²

Local RAAS in the heart and kidney is also important in maintaining Na^+ retention in HF. The phenomenon explains the presence of positive Na^+ balance as well as the maintained efficacy of RAAS inhibition in chronic HF, in the absence of elevated systemic levels of the component hormones.³⁸⁸ In general, it appears that systemic RAAS activation is most pronounced in acute decompensated HF, whereas local renal RAAS activation may dominate in chronic stable HF.

In the heart, local RAAS activation has a number of effects. In addition to the mechanical stress exerted on the myocardium due to systemic Ang II-mediated increased afterload, pressure overload activates local Ang II production as a result of upregulation of angiotensinogen and tissue ACE.³⁸⁹ Local Ang II acts through AT_1 in a paracrine/autocrine manner, leading to cell swelling and cardiac hypertrophy, remodeling and fibrosis (mediated by $\text{TGF-}\beta$) and reduced coronary flow, hallmarks of severe HF.³⁸⁸ These observations explain the improved cardiac function, prolonged survival, prevention of end-organ damage, and prevention or regression of cardiac hypertrophy in HF treated with RAAS inhibitors.^{203,389} In addition, these drugs may improve endothelial dysfunction, vascular remodeling and potentiation of the vasodilatory effects of kinins.³⁰⁹

Like Ang II, aldosterone is produced locally by and acts directly on the myocardium in HF, inducing structural remodeling of the interstitial collagen matrix.³⁹⁰ These adverse effects of aldosterone were elegantly illustrated using eplerenone, a specific aldosterone antagonist, which prevented progressive left ventricular systolic and diastolic dysfunction by reducing interstitial fibrosis, cardiomyocyte hypertrophy and left ventricular chamber sphericity in dogs with HF. Similarly, eplerenone attenuated ventricular remodeling and reactive (but not reparative) fibrosis after myocardial infarction in rats.^{391,392} These findings have been translated into the now routine clinical use of aldosterone antagonists in HF (see section, “[Specific Treatments Based on the Pathophysiology of Heart Failure](#)”).³⁹³

As noted, in addition to its renal and cardiovascular hemodynamic effects, the RAAS is involved directly in the exaggerated tubular Na⁺ reabsorption in HF. Ang II, produced systemically and locally, directly stimulates proximal tubular Na⁺ reabsorption.³⁹⁴ In contrast, in the cortical and MCD, enhanced Na⁺ reabsorption is mediated largely by aldosterone, as outlined previously. The pivotal role of aldosterone in HF is amply illustrated by elevated plasma and urine levels and the natriuretic effects of aldosterone antagonists in HF, despite further activation of other antinatriuretic systems.³⁹⁵

The importance of RAAS action in the Na⁺ retention of HF varies with the stage and severity of disease and, in more severe HF, positive Na⁺ balance is associated with blunted renal and hemodynamic responses to ANP; this response to ANP is restored by RAAS inhibition (for further details, see later section, “[Natriuretic Peptides](#)”).³⁹⁶ Also, despite low plasma osmolality, patients with HF display increased thirst, probably because of the high Ang II concentrations, which stimulate thirst center cells in the hypothalamus.³⁹⁷ This phenomenon may contribute to the positive water balance and hyponatremia often seen in advanced HF (see section, “[Arginine Vasopressin](#)”).

Sympathetic Nervous System

As mentioned earlier, patients with HF experience progressive activation of the SNS with declining cardiac function^{234,376} and the adverse influence of sympathetic overactivity on the progression and outcome of patients with HF is abundantly clear.^{398,399} Thus, plasma NE levels are frequently elevated and correlate with increased neural traffic. SNS activity is also significantly correlated with intracardiac pressures, cardiac hypertrophy and left ventricular ejection fraction (LVEF).^{398,399} Activation of the SNS not only precedes the appearance of congestive symptoms but also is preferentially directed toward the heart and kidneys, as seen in patients with mild HF who have higher NE levels in the coronary sinus than in the renal veins.³⁹⁹ In early HF, increased SNS activity ameliorates the hemodynamic abnormalities, including hypoperfusion, diminished plasma volume and impaired cardiac function, via vasoconstriction and avid Na⁺ reabsorption.³⁷⁶ However, chronic SNS activation induces several long-term adverse myocardial effects, including apoptosis and hypertrophy, with overall reduction in cardiac contractility. Some of these effects may be mediated by RAAS activation which, in turn, can augment sympathetic activity and create a vicious cycle.²³⁴

Basal sympathetic outflow to the kidneys is significantly increased in patients with HF³⁷⁶ and increased efferent RSNA

contributes to the increased renal vasoconstriction, avid Na⁺ and water retention, renin secretion, and attenuation of the renal actions of ANP.²⁶⁴ In rats with experimental HF caused by coronary artery ligation, renal denervation resulted in increased RPF and SNGFR and decreased afferent and efferent arteriolar resistance.³⁹⁸ In the same model, the decrease in RSNA in response to an acute saline load was less than that of control rats.²⁵⁴ Conversely, bilateral renal denervation restored the natriuretic response to volume expansion. Similarly, in dogs with low cardiac output induced by vena caval constriction, administration of a ganglion blocker resulted in a marked increase in Na⁺ excretion.³⁹⁸ Also, in dogs with high-output HF induced by aortocaval fistula, total postprandial urinary Na⁺ excretion was approximately twofold higher in dogs with renal denervation than in those with intact nerves.³⁹⁸ In line with these observations, administration of the α -adrenoreceptor blocker dibenamine to patients with HF caused an increase in fractional Na⁺ excretion, without a change in RPF or GFR. Treatment with ibopamine, an oral dopamine analog, resulted in vasodilation and positive inotropic and diuretic effects in these patients.⁴⁰⁰ Moreover, for a given degree of cardiac dysfunction, the concentration of NE is significantly higher in patients with abnormal than in those with preserved renal function.⁴⁰¹ These findings suggest that the association between renal function and prognosis in patients with HF is linked by both systemic and CNS neurohormonal activation.

RSNA may also affect renal hemodynamics and Na⁺ excretion in HF by an antagonistic interaction with ANP. On the one hand, ANP has sympathoinhibitory effects;^{402,403} on the other, the SNS-induced salt and water retention in HF may reduce renal responsiveness to ANP. For example, the blunted diuretic/natriuretic response to ANP in rats with HF could be restored by prior renal denervation⁴⁰⁴ or clonidine,⁴⁰⁵ a centrally acting α_2 -adrenoreceptor agonist, which decreases RSNA in HF. These examples illustrate the complexity of interactions between the SNS and other humoral factors involved in the pathogenesis of Na⁺ retention in HF.

In summary, the SNS plays an important role in the regulation of Na⁺ excretion and glomerular hemodynamics in HF, either by a direct renal action or by a complex interplay between the SNS itself and other neurohumoral mechanisms that act on the glomeruli and renal tubules. The recent introduction of renal denervation as a potential therapeutic treatment for HF should facilitate further elucidation of these neurohumoral interactions.

Vasopressin

Numerous studies have demonstrated elevated plasma levels of AVP in HF, mostly in advanced HF with hyponatremia, but also in asymptomatic patients with left ventricular dysfunction.⁴⁰⁶ These high levels are related to nonosmotic factors such as attenuated left atrial compliance, hypotension and RAAS activation, and are reversed by RAAS inhibition or α -blockade (prazosin).⁴⁰⁷

The high circulating levels of AVP adversely affect the kidneys and cardiovascular system. In fact, raised levels of the C-terminal portion of the AVP prohormone (copeptin) at the time of diagnosis of acute decompensated HF are highly predictive of 1-year mortality.⁴⁰⁸ The prognostic power of an increased copeptin level in HF is similar to that of BNP levels (see section, “[Brain Natriuretic Peptide](#)”). The most

recognized renal effect of AVP in HF is the development of hyponatremia, especially in advanced stages of the disease, which most probably results from impaired solute-free water excretion, independent of plasma osmolality. In accordance with this notion, animal models of HF have demonstrated increased collecting duct expression of AQP2.⁴⁰⁹ In addition, administration of specific V_2 receptor antagonists (VRAs) is associated with improvement in plasma Na^+ levels in animals and patients with hyponatremia.^{410,411} The improvement is associated with correction of the impaired urinary dilution in response to acute water load,⁴¹² increased plasma osmolality and downregulation of renal AQP2 expression, but with no effect on RBF, GFR or Na^+ excretion.⁴¹³

The adverse effects of AVP on cardiac function⁴¹⁴ occur through its V_{1A} receptor to increase SVR (i.e., cardiac afterload), as well as by V_2 -receptor-mediated water retention, which leads to systemic and pulmonary congestion (increased preload). In addition, AVP, through its V_{1A} receptor, causes a direct rise in cardiomyocyte intracellular Ca^{2+} and activation of mitogen-activated kinases and protein kinase C. These signaling mechanisms appear to mediate the observed cardiac remodeling, dilation and hypertrophy. The remodeling might be further exacerbated by the abnormalities in preload and afterload.

In summary, the data suggest that AVP is involved in the pathogenesis of water retention and hyponatremia that characterize HF; and that AVP receptor antagonists result in remarkable diuresis in experimental and clinical models of HF. Treatment of HF with VRA will be discussed further (see section, “Specific Treatments Based on the Pathophysiology of Heart Failure”).

Endothelin

ET-1 is involved in both the development and progression of HF as well as in the associated reduced renal function by inducing renal remodeling, interstitial fibrosis, glomerulosclerosis, hypoperfusion, hypofiltration and positive salt and water balance.²⁷⁹ The pathophysiologic role of ET-1 in HF is supported by two major lines of evidence: 1. The ET system is activated in HF; and 2. ET-1 receptor antagonists modify this pathophysiologic process.²⁷⁹ The first line of evidence is based on elevated plasma ET-1 and big ET-1 concentrations in clinical HF and experimental models of HF; these levels correlate with hemodynamic severity and symptoms.⁴¹⁵ Also, the degree of pulmonary hypertension was the strongest predictor of plasma ET-1 level in patients with HF.⁴¹⁶ Moreover, plasma levels of big ET and ET-1 are especially high in patients with moderate to severe HF and are independent markers of mortality and morbidity.⁴¹⁵ The increase in plasma ET-1 levels may be the result of enhanced synthesis in the lungs, heart and circulation by stimuli such as Ang II and thrombin or decreased pulmonary clearance.⁴¹⁶ In parallel to ET-1 levels, ET-A receptors are upregulated, whereas ET-B receptors are downregulated in the failing human heart.⁴¹⁷

A cause-and-effect relationship between these hemodynamic abnormalities and ET-1 in HF was demonstrated using selective and highly specific ET receptor antagonists.⁴¹⁶ In this regard, acute administration of the mixed ET-A/ET-B receptor antagonists, bosentan and tezosentan significantly improved renal cortical perfusion, reversed the profoundly increased renal vascular resistance and increased RBF and Na^+ excretion in rats with severe decompensated HF.⁴¹⁸ In addition, chronic

blockade of ET-A by selective or dual ET-A/ET-B receptor antagonists attenuated the magnitude of Na^+ retention and prevented the decline in GFR in experimental HF.⁴¹⁹ These data are in line with earlier observations that rats with decompensated HF, as compared with normal rats, displayed severely blunted cortical vasoconstriction but significantly prolonged medullary vasodilation in response to ET-1 infusion. These effects could have resulted from activation of vasodilatory systems such as PG and NO, as exemplified by higher medullary immunoreactive eNOS levels in rats with HF than controls.⁴²⁰ Taken together, the data indicate a role for ET in the pathogenesis of renal cortical vasoconstriction and Na^+ retention in HF.

VASODILATORY/NATRIURETIC SYSTEMS

Natriuretic Peptides

In decompensated HF, renal Na^+ and water retention occur, despite ECF volume expansion, even when the NP system is activated. Many clinical and experimental studies have implicated both ANP and BNP in the pathophysiology of the deranged cardiorenal axis in HF.

Atrial Natriuretic Peptide. Plasma levels of ANP and NH_2 -terminal ANP are frequently elevated in HF and are correlated positively with the severity of cardiac failure, elevated atrial pressure and left ventricular dysfunction.²⁶⁴ Hence, circulating ANP level was proposed as a diagnostic marker of cardiac dysfunction and as a predictor of survival in HF.⁴⁴ However, in this context, ANP has since been superseded by BNP. There is also evidence that midregional (MR) proANP may perform similarly to BNP as a biomarker of ADHF (see section, “Brain Natriuretic Peptide”).⁴⁷³

The high plasma ANP levels are attributed to increased production rather than to decreased clearance. Although volume-induced atrial stretch is the main source for the elevated circulating ANP levels in HF, enhanced synthesis and release of the hormone by ventricular tissue in response to Ang II and ET also contribute to the elevated levels.²⁶⁴ Despite the high levels, patients and experimental animals with HF retain salt and water because renal responsiveness to NPs is attenuated.⁴²¹ However, infusion of ANP to patients with HF does lead to hemodynamic improvement and inhibition of activated neurohumoral systems. These data are in line with findings that ANP is a weak counterregulator of the vasoconstriction mediated by the SNS, RAAS, and AVP.⁴²² However, despite the blunted renal response to ANP in HF, elimination of ANP production by atrial appendectomy in dogs with HF aggravated the activation of the vasoconstrictive factors and resulted in marked Na^+ and water retention.⁴²³ These data suggest that ANP plays a critical role in suppressing Na^+ -retaining systems and as an important adaptive or compensatory mechanism aimed at reducing pulmonary vascular resistance and hypervolemia.

Brain Natriuretic Peptide. Plasma levels of BNP and N-terminal (NT)-proBNP are elevated in severe HF in proportion to the degree of myocardial systolic and diastolic dysfunction and New York Heart Association (NYHA) classification.^{264,424} The extreme elevation of plasma BNP in severe HF stems mainly from increased synthesis by the hypertrophied ventricular tissue, but the atria also contribute.^{264,424}

Although echocardiography remains the gold standard for the evaluation of left ventricular dysfunction, plasma levels of BNP and NT-proBNP are reliable markers and, in fact, superior to ANP and NT-proANP for the diagnosis and prognosis of HF.²⁷⁶ NT-proBNP in particular has high sensitivity, specificity and negative predictive value in patients with an ejection fraction less than 35%. Similar high predictive values are found in patients with concomitant left ventricular hypertrophy, either in the absence of or after myocardial infarction.²⁷⁶ The added presence of renal dysfunction appears to enhance these predictive values,^{425,426} and graded increases in mortality throughout each quartile of BNP levels have been shown in several clinical trials.²⁷⁶ In addition, elevated plasma BNP (or NT-proBNP) levels and LVEF lower than 40% are complementary independent predictors of death, HF and new myocardial infarction at 3 years after a first infarction. Moreover, risk stratification with the combination of LVEF lower than 40% and high levels of NT-proBNP is substantially better than that provided by either alone.⁴²⁷ However, even though BNP levels tend to be lower in patients with preserved LVEF than in HF patients with reduced LVEF, the prognosis in patients with preserved LVEF is as poor as in those with reduced LVEF for a given BNP level.⁴²⁷

In asymptomatic patients with preserved LVEF, elevated BNP levels are correlated with diastolic abnormalities on Doppler studies. Conversely, a reduction in BNP levels with treatment is associated with a reduction in left ventricular filling pressures, lower readmission rates and a better prognosis; thus, monitoring of BNP levels may provide valuable information regarding treatment efficacy and expected patient outcomes.⁴²⁸

Another diagnostic role for BNP is in the distinction of dyspnea caused by HF from that caused by noncardiac diseases. Using specific cut points, NT-proBNP levels are highly sensitive and specific for the diagnosis of acute HF. Levels lower than 300 pg/mL rule out acute HF, with a negative predictive value of 99%. An increased level of NT-proBNP is the strongest independent predictor of a final diagnosis of acute HF. NT-proBNP testing alone was superior to clinical judgment, the National Health and Nutrition Examination score and Framingham clinical parameters alone for diagnosing acute HF; NT-proBNP plus clinical judgment was superior to NT-proBNP or clinical judgment alone.²⁷⁶

There is also evidence, albeit of rather low quality, that circulating BNP and NT-proBNP levels can be useful as a guide to therapeutic efficacy of drugs typically prescribed in HF, including RAAS inhibitors, diuretics, digitalis and β -blockers.^{429,430} Also, BNP, but not NT-proBNP, levels at 24 and 48 hours after admission for acute decompensated HF (ADHF) predicted both 30-day and 1-year mortality. Pre-discharge levels of both peptides were predictive of 30-day and 1-year mortality but not 1-year readmission due to HF.⁴³¹ In contrast, BNP levels were not helpful in reducing length of hospital stay and costs.⁴²⁹

Together, these findings suggest that a simple and rapid determination of plasma levels of BNP or NT-proBNP in HF patients, together with clinical and echocardiographic measures, can be used to assess cardiac dysfunction, serve as a diagnostic and prognostic marker and possibly assist in titrating relevant therapy.²⁷⁶ However, it should be emphasized that plasma NP levels are affected by age, salt intake, gender, obesity, hemodynamic status and renal function leading to

considerable overlap among diagnostic groups.^{276,432} Measurement of a panel may enhance the ability of biomarkers to distinguish between cardiac and noncardiac causes of dyspnea.⁴³³

C-Type Natriuretic Peptide. Like those of ANP and BNP, plasma CNP levels are increased in HF and are directly correlated with NYHA classification, levels of BNP, ET-1, and AM and with pulmonary capillary wedge pressure, ejection fraction, and left ventricular end-diastolic diameter.⁴³⁴ CNP is synthesized mainly in the kidney,²⁷⁸ but it is also processed by the myocardium. Overexpression of CNP in the myocardium during HF may be involved in counteracting cardiac remodeling.⁴³⁴ On the other hand, renal CNP secretion is blunted in HF.⁴³⁵ In contrast to the diminished physiologic responses to ANP and BNP in animals with HF, CNP elicited twice as much sGC activity as ANP, due to dramatic reductions in NP receptor A (NPR-A) but not NPR-B activity.⁴³⁴ These findings imply a significant role for NPR-B-mediated NP activity in HF and may explain the modest effects of the NPR-A-selective nesiritide (BNP) treatment in HF.⁴³⁶

Overall, current evidence points to a role of CNP either in the peripheral vascular compensatory response in HF or in mitigating the cardiac remodeling characteristic of HF. Elaboration of the exact role of CNP in HF appears crucial for the design of more effective NP analogs than those currently available for the management of HF.

Overall Relationship Between Natriuretic and Antinatriuretic Factors in Heart Failure

The maintenance of Na^+ balance in the initial compensated phase of HF is at least in part due to the elevated ANP and BNP levels.²⁶⁴ This notion is supported by the findings that, in experimental HF, inhibition of NP receptors by specific antibodies increased renal vascular resistance and decreased GFR, RBF, urine flow, Na^+ excretion and RAAS activation.⁴³⁷ In addition, NPs inhibited the Ang II-induced systemic vasoconstriction, proximal tubule Na^+ reabsorption, and secretion of aldosterone and ET.⁴³⁸

In view of the remarkable activation and ability of NPs to counter the vasoconstrictor/antinatriuretic neurohormonal effects, why then do salt and water retention occur in overt HF? Several mechanisms could explain this paradox:

1. Appearance of abnormal circulating peptides and inadequate secretory reserves in comparison with the degree of HF. Using an extremely sensitive mass spectrometry-based method, altered processing of proBNP1-108 and/or BNP1-32 has been demonstrated, resulting in very low levels of BNP1-32, despite markedly elevated levels of immunoreactive (i.e., total) BNP.⁴³⁹ Moreover, proBNP1-108 has a lower affinity for the GC-A receptor, which would reduce effector function of BNP.⁴⁴⁰
2. Decreased availability of NPs by downregulation of corin⁴⁴¹ or upregulation of NEP and clearance receptors.⁴³⁷ With respect to corin, lower plasma levels have been observed in HF patients in parallel with elevated levels of proANP. On the other hand, circulating cGMP levels are elevated in HF, implying enhanced activity of NPs. These apparently contradictory findings could be reconciled by the demonstration of low intracardiac corin levels in an experimental model of HF caused by dilated cardiomyopathy. Moreover,

transfection of the gene encoding for corin into these animals led to a reduction in cardiac fibrosis, improvement in contractility and reduced mortality.⁴⁴¹ Regarding clearance receptors, there is no convincing evidence to date of upregulation in the renal tissue of HF animals or patients, although increased abundance of clearance receptors for NPs in platelets of patients with advanced HF has been reported.⁴⁴² In contrast, enhanced expression and activity of NEP in experimental HF is well documented²⁶⁴ and NEP inhibitors improve the vascular and renal response to NPs in HF (see section, “[Specific Treatments Based on the Pathophysiology of Heart Failure](#)”).

3. Activation of vasoconstrictor/antinatriuretic factors and renal hyporesponsiveness to ANP. Renal resistance to ANP may be present, even in the early presymptomatic stage of the disease and progresses proportionately as HF worsens.⁴³⁷ In advanced HF, when RPF is markedly impaired, the ability of NPs to antagonize the renal effects of the maximally activated RAAS is limited.²⁶⁴ The mechanisms underlying the attenuated renal effects of ANP in HF include Ang II-induced afferent and efferent vasoconstriction, mesangial cell contraction, activation of cGMP phosphodiesterases that attenuate the accumulation of the second messenger of NPs in target organs and stimulation of Na⁺-H⁺-exchanger and Na⁺ channels in the proximal tubule and collecting duct, respectively.⁴⁴³
4. Activation of the SNS also can overwhelm the renal effects of ANP. As described earlier, overactivity of the SNS leads to vasoconstriction of the peripheral circulation and of the afferent and efferent arterioles, which causes reduction of RPF and GFR. These actions, together with the direct stimulatory effects of SNS on Na⁺ reabsorption in the proximal tubule and loop of Henle, contribute to the attenuated renal responsiveness to ANP in HF. Moreover, the SNS-induced renal hypoperfusion/hyponatremia stimulates renin secretion, thereby aggravating the positive Na⁺ and water balance. In rat models of HF, the natriuretic responses to ANP were increased after sympathetic inhibition by low-dose clonidine⁴⁰⁵ or bilateral renal denervation.⁴⁴⁴ The beneficial effects of renal denervation could be attributed to upregulation of NP receptors and cGMP production.²⁰¹

In summary, the development of renal hyporesponsiveness to NPs is paralleled closely by overreactivity of the RAAS and SNS and represents a critical point in the development of positive salt balance and edema formation in advanced HF.

Nitric Oxide

NO is implicated in the increased vascular resistance and impaired endothelium-dependent vascular responses characteristic of HF.^{445,446} The impaired activity is mediated by reduced shear stress associated with the decreased cardiac output, downregulation or uncoupling of eNOS, decreased availability of the NO precursor L-arginine caused by increased activity of arginase and increased levels of the endogenous NOS inhibitor asymmetric dimethyl arginine (ADMA). In addition, inactivation of NO by superoxide ion and alteration of the redox state of sGC through oxidative stress lead to reduced levels of the NO-sensitive form of sGC and, thereby, of its second messenger cGMP.^{445,446} Oxidative stress may be further exacerbated by overactivity of counterregulatory

neurohumoral systems, such as the RAAS and the release of proinflammatory messengers.^{445,446}

Altered activity of the NO-sGC-cGMP system also underlies the regional vasomotor dysregulation of the renal circulation in HF and Ang II may be involved in mediating the impaired NO-dependent renal vasodilation.⁴⁴⁷ The resulting imbalance between NO and excessive activation of the RAAS and ET systems could explain some of the beneficial effects of RAAS inhibition.⁴⁴⁸ Support for this imbalance concept came from a model of experimental HF in rats overexpressing eNOS in the renal medulla and, to a lesser extent, in the cortex.⁴²⁰ This eNOS might play a role in the preservation of intact medullary perfusion and could attenuate the severe cortical vasoconstriction. Accumulation of ADMA as exemplified by elevated plasma levels in normotensive HF could also account for the impaired renal hemodynamics in HF. In fact, in a multiple regression analysis, ADMA levels independently predicted reduced RBF.⁴⁴⁹

Locally generated NO by the myocardium is also believed to modulate cardiac function and, thereby, lead to the impaired renal function in HF.^{445,446} Alterations in the expression of cardiac NOS isoforms in HF are complex and the functional consequences of these changes depend on a balance among various factors, including disruption of the unique subcellular localization of each isoform and nitroso-redox imbalance.^{445,446}

In summary, endothelium-dependent vasodilation is attenuated in various vascular beds in HF. This attenuation may occur as a result of decreased NO levels and downregulation or inhibition of downstream NO signal transduction pathways. These effects may occur directly or via counterregulatory vasoconstrictor neurohumoral mechanisms.

Prostaglandins

PG play an important role in maintaining renal function in the setting of the impaired RBF in HF. Renal hypoperfusion, directly or by RAAS activation, stimulates the release of PG that exert a vasodilatory effect, predominantly on the afferent arteriole, and promote Na⁺ excretion by inhibiting reabsorption in the TALH and the MCD.^{450,451} Evidence for the compensatory role of PG in experimental and clinical HF comes from two sources. First, plasma levels of PGE₂, PGE₂ metabolites, and 6-keto-PGF₁ were higher in HF patients than in normal subjects.⁴⁵² Moreover, in both experimental and human HF there is a direct relationship between PRA/Ang II and plasma and urinary PGE₂ and PGI₂ metabolite concentrations.⁴⁵³ This correlation probably reflects both Ang II-induced stimulation of PG synthesis and PG-mediated increased renin release. A similar counterregulatory role of PG regarding the other vasoconstrictors (e.g., catecholamines, AVP) may also be inferred.

The second approach, which established the protective role of renal and vascular PG in HF, was by using nonsteroidal antiinflammatory drugs (NSAIDs) to inhibit PG synthesis. In various experimental models of HF, this maneuver was associated with elevated urinary PGE₂ excretion, increase in body weight and renal vascular resistance with resultant decrease in RBF, mainly due to afferent arteriolar constriction.^{452,454} Urine flow rate declined significantly and serum creatinine and urea rose.⁴⁵⁴ Similarly, in patients with HF and hyponatremia, in whom extreme activation of the SNS and RAAS occurred, significant decreases in RBF and GFR

accompanied by reduced urinary Na^+ excretion, followed NSAID treatment.^{452,455} These effects were prevented by intravenous PGE_2 . Moreover, pretreatment with indomethacin attenuated the captopril-induced increase in RBF.⁴⁵⁵ Thus ACEI-associated improvement in renal hemodynamics is mediated in part by increased PG synthesis.

Selective COX-2 inhibitors also lead to a significant worsening of chronic HF and renal function, especially in older patients taking diuretics.^{456–458} These deleterious effects are predictable given the relative abundance of COX-2 in renal tissue and, to a lesser extent, in the myocardium of HF patients.^{450,459}

In summary, HF can be viewed as a PG-dependent state, in which elevated Ang II levels and enhanced RSNA stimulate renal synthesis of PGE_2 and PGI_2 , to counteract the vasoconstrictor neurohumoral stimuli and maintain GFR and RBF. Both COX-2 and nonselective COX inhibitors should be avoided in HF, as they leave the vasoconstrictor systems unopposed, leading to hypoperfusion, hypofiltration, and Na^+ and water retention.⁴⁵⁰

Adrenomedullin

AM seems to play a role in the pathophysiology of HF. HF patients have up to fivefold elevations in plasma levels of AM, in proportion to the severity of cardiac, hemodynamic, and neurohumoral derangements, including pulmonary arterial and capillary wedge pressure, NE, ANP, BNP levels, and PRA.^{317,460} Plasma levels of AM decreased with effective anti-HF treatment, such as carvedilol.⁴⁶¹ High levels of midregional proAM are also strong predictors of mortality in HF.^{462–464} The origin of the increased circulating AM appears to be the failing ventricular and, to a lesser extent, atrial myocardium.^{461,465}

Not only cardiac but also renal AM levels are significantly increased in some, although not all, experimental models of HF.^{466,467} The renal upregulation is consistent with the favorable acute and more prolonged (4 days) effects of AM on creatinine clearance, Na^+ and water excretion, as well as on the hemodynamic abnormalities of experimental HF.⁴⁶¹ In contrast, acute administration of AM to HF patients increased forearm blood flow but less so than in normal subjects. Stroke index and dilation of resistance arteries were increased, and plasma aldosterone reduced, but Na^+ and water excretion were unaffected.⁴⁶¹ Collectively, the data suggest that AM acts to balance the elevation in SVR and volume expansion in HF.³¹⁷

Because the favorable effects of AM alone are rather modest, combination therapy with other vasodilatory/natriuretic substances has been attempted. Combinations with BNP, ACEIs, NEP inhibitors and epinephrine resulted in hemodynamic and renal benefits greater than those achieved by each agent alone.^{461,468} A small long-term clinical trial of combined ANP and AM in acute decompensated HF demonstrated a significant increase in cardiac output, reductions in MAP, pulmonary arterial pressure, systemic and pulmonary vascular resistance without changing heart rate. In addition, levels of aldosterone, BNP and free radical metabolites fell and Na^+ and water excretion rose.⁴⁶⁹

With the advent of NEP inhibitors, which enhance the activities of vasodilatory/natriuretic peptides, including AM, HF outcomes have been significantly improved. (See section “Specific treatments based on the pathophysiology of HF”.)

Urotensin

A role for UT II and its receptor, GPR14, in the pathogenesis of HF has been suggested. First, some, but not all, studies revealed that plasma levels of UT II are elevated in patients with HF in correlation with levels of other markers, such as NT-proBNP and ET-1.⁴⁷⁰ Second, strong myocardial expression of UT II in end-stage HF correlates with the degree of cardiac impairment.⁴⁷⁰ The upregulated UT II in HF may also have a role in the regulation of renal function in HF. In rat models of HF, UT II acted primarily as a renal vasodilator, apparently by an NO-dependent mechanism.³²⁵ RPF and GFR, but not urinary Na^+ excretion, were also increased. On the other hand, UT II in control rats led to intense renal vasoconstriction, a fall in GFR and Na^+ retention.³²⁵ In light of the contradictory effects of UT II in different conditions, the clinical application of these data will be challenging.

Neuropeptides

Because NPY co-localizes and is released with adrenergic neurotransmitters, high circulating NE levels in HF are accompanied by excessive co-release of NPY and plasma levels are correlated with disease severity in HF patients.³³⁴ In contrast, local myocardial levels, like those of NE, were lower than normal in association with decreased Y1 and increased Y2 receptor expression.³³⁴ Because Y1 receptor activation is associated with cardiomyocyte hypertrophy and Y2 receptor activation with angiogenesis, the data in this model suggest that NPY may simultaneously attenuate the maladaptive cardiac remodeling observed in HF and stimulate angiogenesis in the ischemic heart.³³⁴ Similar patterns of receptor expression change were observed in the kidneys and these were proportional to the degree of renal failure and Na^+ retention.³³⁴ In contrast, administration of NPY in experimental models of HF led to diuresis and natriuresis, probably by increasing ANP release and inhibiting the RAAS.⁴⁷¹ Therefore, in HF, the higher circulating levels, together with the reduced tissue levels of NPY, could be a counterregulatory mechanism to modulate the vasoconstrictive and Na^+ retaining, as well as the cardiac remodeling, effects of the RAAS and SNS. In addition, the downregulation of Y1 receptors, by reducing vasoconstriction, could contribute to reduced coronary and renal vascular resistance. However, once the stage of decompensated HF is reached, the RAAS and SNS effects likely dominate, thereby overwhelming any favorable effects of NPY.

Levels of other neuropeptides, such as catestatin, may be elevated in HF and have been investigated as potential biomarkers, but did not improve diagnostic accuracy over BNP.⁴⁷¹ In summary, laboratory data on neuropeptides in HF have not translated into clinical application.

Apelin

The expression of apelin and its receptor in the kidney and heart and the involvement of the system in the maintenance of water balance suggested a potential role in HF. Circulating levels rise in early HF but decline in later stages of the disease.^{472,473} However, this decline correlates poorly with severity of HF, making apelin not useful as a biomarker of HF progression.⁴⁷⁴

The fact that activation of the apelin receptor induces aquaresis, vasodilation and a positive inotropic effect suggested the receptor as a potential therapeutic target in HF. Along these lines, acute IV injection of apelin to

rats with HF following induced myocardial function led to improved systolic and diastolic function. Moreover, more chronic infusion (3 weeks) decreased Ang II–induced cardiac fibrosis and remodeling.⁴⁷² In HF patients, acute intravenous apelin increased cardiac output, reduced BP and vascular resistance.⁴⁷² No data are yet available on the direct renal effects of apelin in HF, although, by reducing AVP levels and improving the renal microcirculation, apelin might increase aquaresis. In addition, the favorable effects on cardiac function are likely to increase renal perfusion and hence promote diuresis.⁴⁷⁵ Stable apelin analogs are currently in development.⁴⁷⁶

Peroxisome Proliferator-Activated Receptors

Peroxisome proliferator-activated receptors (PPARs) are nutrient-sensing nuclear transcription factors, of which PPAR γ is of special interest in the context of Na⁺ and water retention because of its ligands, the thiazolidinediones (TZD). TZD, by increasing insulin sensitivity, are used for the management of type 2 diabetes mellitus. TZD also decrease circulating free fatty acids and triglycerides, lower BP, reduce levels of inflammatory markers and reduce atherosclerosis. Moreover, they have a beneficial effect on cardiac remodeling in myocardial ischemia.⁴⁷⁷ However, a troubling side effect of TZD is fluid retention mainly resulting from PPAR γ -induced Na⁺ reabsorption mediated by increased ENaC expression in collecting duct epithelium.⁴⁷⁸ However, TZD may also augment proximal tubular Na⁺ reabsorption by upregulation of apical NEH3, basolateral Na⁺-HCO₃⁻ cotransporter, and Na⁺-K⁺-ATPase. These effects are mediated by PPAR γ -induced nongenomic transactivation of the epidermal growth factor receptor and downstream extracellular signal-regulated kinases.⁴⁷⁹ Moreover, by reducing SVR, TZD might lead to higher capillary perfusion pressures and fluid extravasation⁴⁷⁸; TZD are also potent VEGF inducers, leading to increased vascular permeability. In clinical terms, the Na⁺-retaining effect of TZD translates into an increased incidence of HF⁴⁷⁸ and, therefore, they are contraindicated in advanced HF.

Because of the Na⁺-retaining and fluid-retaining effects, as well as other concerns related to increased cardiovascular events on the one hand and favorable effects on the myocardium on the other, the exact role of TZD in HF remains a hotly debated subject.⁴⁸⁰

In summary, alterations in the efferent limb of volume regulation in HF include both enhanced activities of vasoconstrictor/Na⁺-retaining systems and counterregulatory vasodilatory/natriuretic systems. The magnitude of Na⁺ excretion by the kidneys and, therefore, the disturbance in volume homeostasis in HF are largely determined by the balance between these antagonistic systems. In the early stages of HF, the vasodilatory/natriuretic systems are important in the maintenance of circulatory and renal function. However, with the progression of HF, the balance shifts toward dysfunction of the vasodilatory/natriuretic systems and enhanced activation of the vasoconstrictor/antinatriuretic systems. The net result is renal circulatory and tubular alterations that result in avid retention of salt and water, and edema formation.

Renal Sodium Retention in Cirrhosis With Portal Hypertension. Avid Na⁺ and water retention commonly occurs in cirrhosis with portal hypertension, leading eventually to ascites,

a major cause of morbidity and mortality, with the occurrence of spontaneous bacterial peritonitis, variceal bleeding, and development of the HRS.^{481,482} As in HF, the pathogenesis of renal Na⁺ and water retention in cirrhosis is related to extrarenal regulation of renal Na⁺ and water handling.

A *sine qua non* for the Na⁺ and water retention in cirrhosis is the development of intrasinusoidal portal hypertension, with values of portal pressure above 12 mm Hg generally being required. In contrast, presinusoidal hypertension alone, as observed in portal vein thrombosis, is not associated with fluid retention. The hallmark of fluid retention in cirrhosis is peripheral arterial vasodilation, in association with renal vasoconstriction. In the early stages of cirrhosis, vasodilation occurs in the splanchnic vascular bed, with arterial pressure maintained through increases in plasma volume and cardiac output, leading to the so-called “hyperdynamic circulation” (overfilling). At this stage, renal Na⁺ and water retention is already evident and aids in the maintenance of EABV.⁴⁸³ However, as cirrhosis progresses, vasodilation in the systemic and pulmonary circulations becomes prominent and cardiac output can no longer compensate for the progressive decrease in SVR.⁴⁸⁴ The resulting relative arterial underfilling⁴⁸³ and reduced EABV leads to unloading of the arterial high-pressure baroreceptors and other volume receptors, in turn, stimulating the classical compensatory neurohumoral response. This response manifests itself as renal, brachial, femoral, and cerebral vasoconstriction and further Na⁺ and fluid retention.⁴⁸⁵

Peripheral Arterial Vasodilation. The initial trigger for splanchnic arterial vasodilation is hepatic tissue damage itself, which leads to venous outflow obstruction, reduced portal venous and increased hepatic arterial blood flow. Moreover, the lower the portal venous flow, the higher the hepatic arterial flow (Fig. 14.11A). These changes lead to increased intrahepatic vascular resistance and sinusoidal pressure.⁴⁸³ Increased hepatic resistance to portal flow causes the gradual development of portal hypertension, collateral vein formation and shunting of blood to the systemic circulation. As portal hypertension develops, local production of vasodilators—mainly NO, but also carbon monoxide, glucagon, prostacyclin, AM, and endogenous opiates—increases, leading to splanchnic vasodilation.⁴⁸⁵ Other contributing factors to splanchnic vasodilation include intestinal bacterial translocation, proinflammatory cytokines and mesenteric angiogenesis.^{486,487}

The decreases in SVR associated with low arterial BP and high cardiac output account for the well-known clinical manifestations of the hyperdynamic circulation commonly seen in patients with cirrhosis. These include warm extremities, cutaneous vascular spiders, wide pulse pressure, capillary pulsations in the nail bed⁴⁸⁸ and pulmonary vasodilation, associated with the hepatopulmonary syndrome.⁴⁸⁹

Abnormalities of Sensing Mechanisms in Cirrhosis

Nitric Oxide. The NO system is integrally involved in the pathogenesis of the hyperdynamic circulation and Na⁺ and water retention in cirrhosis, as well as in hepatic encephalopathy, hepatopulmonary syndrome, and cirrhotic cardiomyopathy.⁴⁹⁰ NO is produced in excess by the vasculature of different animal models of portal hypertension, as well as in cirrhotic patients.⁴⁹⁰ In animal models, the increased production of NO can be detected at the onset of Na⁺ retention

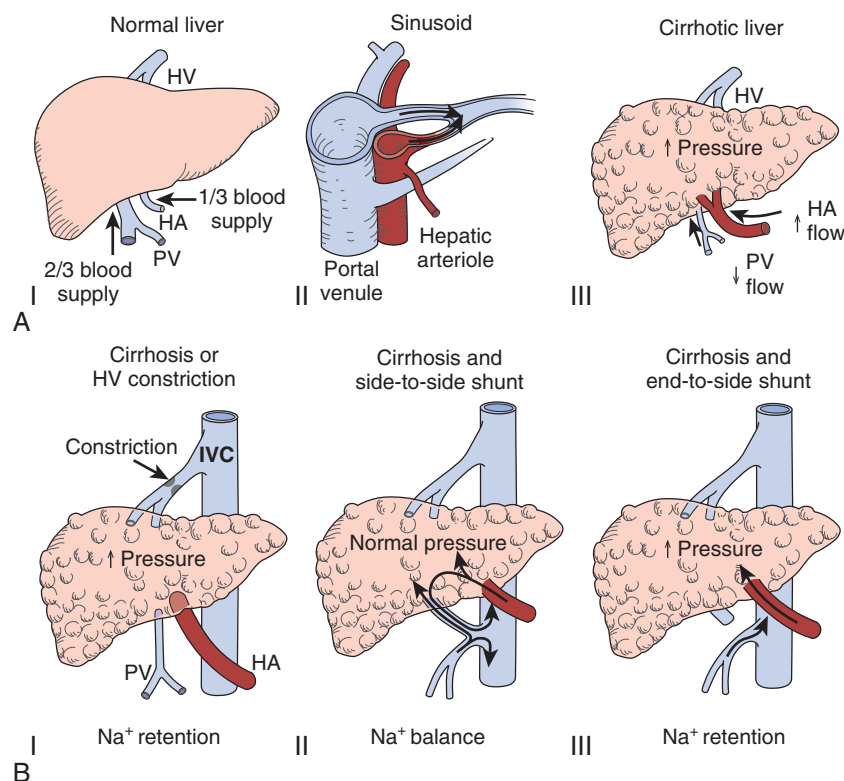


Fig. 14.11 Characteristics of hepatic blood flow. A, Hepatic circulation. I, The normal liver receives two thirds of its blood flow from the portal vein (PV) and the remaining third from the hepatic artery (HA). II, Both the portal venules and hepatic arterioles drain into hepatic sinusoids, but the exact arrangement that allows forward flow of the mixed venous and arterial blood remains unclear. III, Cirrhosis increases intrahepatic vascular resistance and sinusoidal pressure. In addition, PV flow is markedly decreased, and HA flow is unchanged or increased. B, Hepatic vascular hemodynamics and sodium balance. I, Cirrhosis or restriction of HV flow increases intrahepatic vascular resistance and sinusoidal pressure, markedly decreasing PV flow and increasing HA flow. Changes in the physical forces or in the composition of the hepatic blood trigger Na^+ retention and edema formation. II, Insertion of a side-to-side portocaval shunt decreases sinusoidal pressure and maintains mixing of PV and HA blood, irrigating the liver. Under these conditions and despite cirrhosis, there is no Na^+ retention. III, Insertion of an end-to-side portocaval shunt only partially decreases the elevated sinusoidal pressure and prevents mixing of PV and HA blood supplies, inasmuch as the PV blood is diverted to the inferior vena cava (IVC). Under these conditions and, despite normalization of PV pressure, Na^+ retention continues unabated. (Modified from Oliver JA, Verna EC. Afferent mechanisms of sodium retention in cirrhosis and HRS. *Kidney Int.* 2010;77:669–680.)

and before the appearance of ascites and NO has been implicated in the impaired vascular responsiveness to vasoconstrictors.⁴⁹¹ Moreover, removal of the vascular endothelial layer abolishes the difference in vascular reactivity between cirrhotic and control vessels.⁴⁸⁸

Inhibition of NOS has beneficial effects in experimental models of cirrhosis and in cirrhotic patients. By reducing the high NO production to control levels, the hyperdynamic circulation in cirrhotic rats with ascites was corrected and accompanied by a marked increase in Na^+ and water excretion and regression of ascites. Concomitant decreases in PRA, aldosterone and vasopressin concentrations were also observed.^{492,493} In cirrhotic patients, the vascular hyporesponsiveness of the forearm circulation to NE could be reversed by NOS inhibition.⁴⁹⁴ Inhibition of NO production also corrected the hypotension and hyperdynamic circulation, led to improved renal function and Na^+ excretion and to a decrease in plasma NE levels in these patients. However, in patients with established ascites, NOS inhibition did not improve renal function.⁴⁹⁵

The main enzymatic isoform responsible for the increased systemic vascular NO generation in cirrhosis appears to be

eNOS in the systemic and splanchnic circulations.⁴⁹⁰ Upregulation of eNOS appears, at least in part, to be caused by increased shear stress as a result of portal venous hypertension with increased splanchnic blood flow.⁴⁹⁰ Increased NO release, as well as eNOS upregulation, in the superior mesenteric arteries was found to precede the development of the hyperdynamic splanchnic circulation.⁴⁹⁰ In accord with this concept, upregulation of hepatic eNOS (or nNOS) expression in rats with experimental cirrhosis was associated with a decrease in portal hypertension.⁴⁹⁶ However, mice with targeted deletion of eNOS alone or combined deletions of eNOS and iNOS, can still develop a hyperdynamic circulation in association with portal hypertension.⁴⁹³ This suggests that activation of other vasodilatory agents such as PGI_2 , endothelium-derived hyperpolarizing factor, carbon monoxide, and AM may participate in the pathogenesis of the hyperdynamic circulation in experimental cirrhosis.⁴⁹¹

In addition to eNOS, other isoforms may be involved in the generation of the hyperdynamic circulation and fluid retention in experimental cirrhosis. Increased expression of nNOS in mesenteric nerves may compensate partially for eNOS deficiency in eNOS knockout mice and reduce

intrahepatic venous resistance and portal hypertension.⁴⁹⁶ Also, splanchnic vasodilation is modestly promoted, possibly by modulating neurogenic NE release.⁴⁹⁶ In contrast, the role of iNOS remains controversial; some researchers have shown increased iNOS in the superior mesenteric arteries of animals with experimental biliary cirrhosis but not in other forms of experimental cirrhosis.⁴⁹⁶ Specific iNOS inhibition led to peripheral vasoconstriction, but had no effect on portal hypertension.⁴⁹⁶ iNOS is primarily regulated at the transcription level by many proinflammatory factors, principally nuclear factor-kappaB (NF- κ B), which could be induced by endotoxin, from translocated intestinal bacteria. Interestingly, there is also an interaction between eNOS and iNOS in the vasculature in cirrhosis. Overexpression of eNOS in large arteries results in systemic hypotension and increased blood flow. These effects can be abrogated by activated iNOS in the small splanchnic vessels.⁴⁹⁶ Thus, overall, available data indicate a predominant role for eNOS deficiency, with possible modulation by both nNOS and iNOS.

In marked contrast to the increased NO generation in the splanchnic and systemic circulation, NO production and endothelial function in the intrahepatic microcirculation are impaired in cirrhotic rats.⁴⁹⁶ The resulting paradoxical increase in intrahepatic vascular resistance is likely to result from contraction of myofibroblasts and stellate cells and mechanical distortion of the vasculature by fibrosis.⁴⁹⁶ A second mechanism for the increased intrahepatic vascular resistance could be that the locally decreased NO production shifts the balance in favor of local vasoconstrictors (ET, leukotrienes, TXA₂, Ang II).⁴⁹⁷ The increased vascular resistance may also play a role in the pathogenesis of intrahepatic thrombosis and collagen synthesis in cirrhosis.⁴⁹⁷

Several cellular mechanisms have been implicated in the upregulation of splanchnic eNOS and in the downregulation of intrahepatic eNOS. Elevation in shear stress as a result of the hyperdynamic circulation and portal hypertension has already been mentioned and is generally consistent with this well-documented mechanism for upregulating eNOS gene transcription. However, eNOS activity is not only regulated transcriptionally, but also posttranscriptionally, by tetrahydrobiopterin (THB₄) and direct phosphorylation of eNOS protein.⁴⁹⁶ Furthermore, circulating endotoxins may increase the enzymatic production of THB₄, thereby enhancing mesenteric vascular eNOS activity.⁴⁹⁶

Potential contributors to intrahepatic eNOS downregulation include interactions with caveolin, calmodulin, heat shock protein 90, eNOS trafficking inducer,⁴⁹⁶ disorders of GC activity,⁴⁹⁸ and increased levels of the NO inhibitor, ADMA.⁴⁹⁸ In fact, ADMA levels correlate with the severity of portal hypertension during hepatic inflammation and levels are higher in patients with decompensated than compensated cirrhosis.⁴⁹⁹ Raised ADMA levels have been linked to reduced activity of dimethylarginine dimethylaminohydrolases (DDAHs) that metabolize ADMA to citrulline.⁵⁰⁰ Similarly, patients with alcoholic cirrhosis and superimposed alcoholic hepatitis have higher plasma and tissue levels of ADMA, higher portal venous pressures and decreased DDAH expression.⁴⁹⁹ However, attempts at pharmacological or genetic upregulation of DDAH to reduce ADMA levels and increase NO in experimental cirrhosis have not translated into improved management of decompensated portal hypertension.⁵⁰¹

In the final analysis, the relative importance of the various mechanisms involved in the reduced intrahepatic and increased splanchnic and systemic NOS activity in cirrhosis remains to be determined.

Endocannabinoids. Endogenous cannabinoids are lipid-signaling molecules that mimic the activity of Δ^9 -tetrahydrocannabinol, the main psychotropic constituent of marijuana. *N*-arachidonylethanolamide (anandamide) and 2-arachidonoylglycerol are the most widely studied endocannabinoids that bind the two specific receptors, CB₁ and CB₂. Anandamide also interacts with the vanilloid receptor.⁵⁰²

In animal models of cirrhosis, both CB₁ and CB₂ receptors and endocannabinoid production are greatly upregulated and anandamide caused a dose-dependent increase in intrahepatic vascular resistance, especially in the isolated perfused cirrhotic liver. The effect appeared to be mediated by CB₁ receptor-enhanced production of COX-derived vasoconstrictive eicosanoids. Also, the CB₁ receptor antagonist, rimonabant, in cirrhotic rats reversed arterial hypotension, increased hepatic vascular resistance, decreased mesenteric arterial blood flow and portal venous pressure, and prevented ascites formation. The reduction in splanchnic blood flow was enhanced by the vanilloid receptor antagonist, capsazepine. These findings indicate that the transient receptor potential vanilloid type 1 protein and the CB₁ receptor have a dual role in the splanchnic vasodilation characteristic of cirrhosis.⁵⁰²

Endotoxin was found to be a major stimulus for endocannabinoid generation in monocytes and platelets of cirrhotic animals. This pathway could operate in patients with advanced cirrhosis in whom elevated circulating endotoxin levels are frequently found. Endocannabinoid production could then trigger splanchnic and peripheral vasodilation, arterial hypotension, and intrahepatic vasoconstriction through activation of CB₁ receptors in the vascular wall and perivascular nerves.⁵⁰² The potentially favorable effect of CB₁ receptor blockade on Na⁺ excretion opens the possibility of pharmacologic modification of human HRS.

In summary, afferent sensing of volume in cirrhosis is characterized by increased intrahepatic vascular resistance and sinusoidal pressure, decreased portal venous blood flow, and increased hepatic arterial flow. Either changes in intrahepatic physical forces or in the composition of the mixed intrahepatic blood could initiate abnormal Na⁺ retention and edema formation (see Fig. 14.11B). Side-to-side portocaval shunt (currently performed by transjugular intrahepatic portovenous shunt insertion) prevents (if inserted before induction of cirrhosis) or corrects (if inserted after induction of cirrhosis) renal Na⁺ retention. This outcome could result from decreases in sinusoidal pressure or maintenance of the mixing of portal venous and hepatic arterial blood perfusing the liver. In contrast, it diverts blood to the inferior vena cava (IVC), but only partially decreases sinusoidal pressure and prevents mixing of portal venous and arterial hepatic blood supplies. Although portal venous pressure is normalized, Na⁺ retention continues unabated (see Fig. 14.11B). Consequently, end-to-side shunting is no longer used clinically.

Afferent Sensing of Intrahepatic Hypertension. Available data are most consistent with the view that the putative EABV sensor(s) in the hepatic circulation is pathologically activated in cirrhosis, failing to respond to the expanded ECF volume.⁸⁷

The sensing mechanisms likely respond specifically to elevated hepatic venous pressure with increased hepatic afferent nerve activity. The relays for these impulses consist of two autonomic nerve plexuses, surrounding the hepatic artery and portal vein, respectively.⁵⁰³ These neural networks connect hepatic venous congestion to enhanced renal and cardiopulmonary sympathetic activity.

Occlusion of the IVC at the diaphragm was associated with increases in hepatic, portal and renal venous pressures and resulted in markedly increased hepatic afferent nerve traffic and renal and cardiopulmonary sympathetic efferent nerve activity. Section of the anterior hepatic nerves eliminated the reflex increase in renal efferent nerve activity⁵⁰³ and hepatic denervation in dogs with IVC constriction increased urinary Na⁺ excretion.⁸⁹ This effect of hepatic denervation was shown to be mediated by the adenosine A1 receptor in cirrhotic rats.⁵⁰⁴

Apart from the adenosine-mediated hepatorenal reflex, other currently undefined humoral pathways could provide an anatomic or physiologic basis for the primary effects of alterations in intrahepatic hemodynamics on renal function. Despite the wealth of information on hepatic volume sensing, the molecular identity, cellular location of the sensor, and what is sensed remain elusive.

Arterial Underfilling. Several mechanisms have been proposed to account for the development of relative hypovolemia. The first is disruption in normal Starling relationships governing fluid movement in the hepatic sinusoids. Unlike other capillaries, these are highly permeable to plasma proteins. As a result, partitioning of ECF between the intravascular (intrasinusoidal) and interstitial (space of Disse) and lymphatic compartments of the liver is determined predominantly by the ΔP along the length of the hepatic sinusoids. Obstruction of hepatic venous outflow promotes enhanced efflux of a protein-rich filtrate into the space of Disse and results in augmented hepatic lymph formation.^{486,505} In parallel, vastly increased hepatic lymph formation is accompanied by increased flow through the thoracic duct.⁵⁰⁶ When the rate of enhanced hepatic lymph formation exceeds the capacity for return to the intravascular compartment via the thoracic duct, hepatic lymph accumulates as ascites and the intravascular compartment is further compromised. As liver disease progresses, fibrosis around Kupffer cells lining the sinusoids renders the sinusoids less permeable to serum proteins. Under such circumstances, termed “capillarization of sinusoids,” a decrease in oncotic pressure also promotes transudation of ECF within the hepatic lymph space, as in other vascular beds.⁵⁰⁷

Another consequence of intrahepatic hypertension is transmission of elevated intrasinusoidal pressures to the portal vein. This leads to expansion of the splanchnic venous system, collateral vein formation and portosystemic shunting, resulting in increased vascular capacitance and diversion of blood flow from the arterial circuit.⁵⁰⁸ Not only splanchnic but also systemic vasodilation occurs and this has been attributed to refractoriness to the vasoconstrictive effects of hormones such as Ang II and catecholamines, although the mechanism remains unknown.⁵⁰⁹ Along with diminished hepatic reticuloendothelial cell function, portosystemic shunting allows various products of intestinal metabolism and absorption to bypass the liver and escape hepatic elimination. Among these products, endotoxins are thought to contribute to perturba-

tions in renal function in cirrhosis, either due to intestinal bacterial translocation, stimulating the release of proinflammatory cytokines (e.g., tumor necrosis factor- α [TNF- α] and interleukin-6), secondary to the hemodynamic consequences of endotoxemia, or through direct renal effects.⁴⁸⁶

Levels of conjugated bilirubin and bile acids may become elevated as a result of intrahepatic cholestasis or extrahepatic biliary obstruction. Bile acids directly decrease proximal tubular reabsorption of Na⁺, tending to promote natriuresis.⁵¹⁰ This diuretic effect might contribute to the underfilling state in advanced cirrhosis.^{510,511}

Hypoalbuminemia in advanced cirrhosis, either as a result of decreased synthesis by the liver or dilution caused by ECF volume expansion, could also contribute to the development of hypovolemia by diminishing COP in the systemic capillaries and hepatic sinusoids.⁵⁰⁷ In addition, tense ascites might reduce venous return (preload) to the heart, leading to reduced cardiac output and diminished arterial BP.⁴⁸⁴

Other factors that may also adversely affect cardiac performance include diminished β -adrenergic receptor signal transduction, cardiomyocyte cellular plasma membrane dysfunction, and increased activity or levels of cardiodepressants, such as cytokines, endocannabinoids, and NO. Although the cardiac dysfunction, termed “cirrhotic cardiomyopathy,” usually is clinically mild or silent, overt HF can be precipitated by stresses such as liver transplantation or transjugular intrahepatic portosystemic shunt (TIPS) insertion.⁴⁸⁴ Finally, intravascular volume depletion in cirrhotic patients may be aggravated by vomiting, occult variceal bleeding and excessive diuretic use, leading to cardiovascular collapse.

Table 14.6 summarizes the various causative factors contributing to underfilling of the circulation in patients with advanced liver disease. In summary, the early stages of compensated cirrhosis are characterized by increased plasma volume, which frequently antedates ascites formation.⁵¹² However, as cirrhosis progresses, EABV decreases, leading to increased neurohumoral activity (RAAS, SNS, and AVP) and severe Na⁺ and water retention.

Abnormalities of Effector Mechanisms in Cirrhosis. The efferent limb of volume regulation in cirrhosis is similar to that in HF, consisting of adjustments in glomerular hemodynamics and tubular transport mediated by vasoconstrictor/antinatriuretic forces (RAAS, SNS, AVP, and ET) and

Table 14.6 Factors Causing Underfilling of the Circulation in Cirrhosis

Peripheral vasodilation and blunted vasoconstrictor response to reflex, chemical and hormonal influences
Arteriovenous shunts, particularly in portal circulation
Increased vascular capacity of portal and systemic circulation
Hypoalbuminemia
Impaired left ventricular function, so-called cirrhotic cardiomyopathy
Diminished venous return secondary to advanced tense ascites
Occult gastrointestinal bleeding from ulcers, gastritis, or varices
Volume losses caused by vomiting and excessive use of diuretics

counterbalanced by vasodilator/natriuretic systems (NPs and PG). Tilting the balance toward Na^+ retaining forces leads to renal Na^+ and water retention, as in HF.⁴⁸⁵

VASOCONSTRICTOR AND ANTINATRIURETIC (ANTIDIURETIC) SYSTEMS

Renin-Angiotensin-Aldosterone System

As in HF, the RAAS plays a central role in mediating renal Na^+ retention in cirrhosis. Although positive Na^+ balance may already be evident in the preascitic phase of the disease, PRA and aldosterone levels remain within the normal range or may even be depressed at this stage.⁵¹³ With progression of the disease, RAAS activation increases in parallel and aldosterone levels are inversely correlated with renal Na^+ excretion in preascitic cirrhotic patients, particularly in the upright position.⁵¹³ Moreover, treatment with the ARB, losartan, at a dosage that did not affect systemic and renal hemodynamics or GFR, was associated with a significant natriuresis,⁵¹⁴ likely due to inhibition of the local intrarenal RAAS.^{513,514} Indeed, activation of the intrarenal RAAS may precede systemic activation.⁵¹⁵ In addition, losartan caused a decrease in portal venous pressure in cirrhotic patients with portal hypertension.⁵¹⁶ The postural-induced RAAS activation and the beneficial effects of low-dose losartan treatment in preascitic cirrhosis may be explained by splanchnic venous compartmentalization of the expanded blood volume on standing and translocation toward the central and arterial circulatory beds during recumbency.⁵¹³

In contrast, in Na^+ -retaining cirrhotic patients with ascites, Ang II inhibition, even at low doses, resulted in decreased GFR and Na^+ excretion.⁵¹⁷ At this stage of the disease, RAAS activation serves to maintain arterial pressure and adequate circulation. Therefore, RAAS blockade may lead to a profound decrease in RPP. This scenario might be important in the pathogenesis of the HRS, which is regularly preceded by Na^+ retention and may be precipitated by a hypovolemic insult. Abnormalities of the renal circulation in HRS include marked diminution of RPF with renal cortical ischemia and increased renal vascular resistance, abnormalities consistent with the known actions of Ang II on the renal microcirculation.⁵¹⁸ In this regard, RAAS activation correlates with worsening hepatic hemodynamics and decreased survival in patients with cirrhosis.⁵¹⁹ Therefore ACEIs and ARBs should be avoided in patients with cirrhosis and ascites.

Evolving knowledge on the ACE2, Ang 1–7, Mas receptor pathway has shed new light on the role of the RAAS in the pathogenesis of Na^+ retention in cirrhosis. In this regard, exogenous Ang 1–7 elicited a marked NO-dependent vasodilatory effect on the Ang-II-evoked vasoconstrictive response in the portal vein of isolated perfused cirrhotic rat liver.^{520,521} The data raise the possibility of reducing intrahepatic resistance and portal pressure by targeted upregulation of the alternate RAAS pathway in the liver.⁵²²

Sympathetic Nervous System

Activation of the SNS is characteristic of cirrhosis and ascites.⁵²³ Circulating NE levels, as well as urinary excretion of catecholamines and their metabolites, are elevated in patients with cirrhosis and usually are correlated with the severity of the disease. Moreover, high levels of plasma NE in patients with decompensated cirrhosis are predictive of increased mortal-

ity.⁵²³ The increased NE levels stem from enhanced SNS activity, rather than reduced dissipation, with nerve terminal spillover from hepatic, cardiac, renal, muscular, and cutaneous innervation.⁵²⁴ Elevated plasma NE levels were correlated closely with Na^+ and water retention in cirrhotic patients.⁵²⁵ In addition, increased efferent renal sympathetic tone, perhaps due to defective arterial and cardiopulmonary baroreflex control, was observed in experimental cirrhosis.^{526,527} This scenario could explain why volume expansion does not suppress enhanced RSNA in cirrhosis.

Concomitant with the increase in NE release, cardiovascular responsiveness to reflex autonomic stimulation may be impaired in patients with cirrhosis.⁵²⁵ This impairment could be explained partially by increased occupancy of endogenous catecholamine receptors, downregulation of adrenergic receptors, or a defect in postreceptor signaling.⁵²⁴ Excessive NO-dependent vasodilation alone could, in fact, account for the vascular hyporesponsiveness in cirrhosis.^{528,529} Also, enhanced release of NPY may be a compensatory mechanism to counteract splanchnic vasodilation by restoring the vasoconstrictor efficacy of endogenous catecholamines.⁵²⁴

The increase in RSNA and plasma NE levels could contribute to the antinatriuresis of cirrhosis by decreasing total RBF, or its intrarenal distribution, or by acting directly on the tubular epithelium to enhance Na^+ reabsorption. Patients with compensated cirrhosis may have decreased RBF and, as the disease progresses, RBF tends to decline further, concomitantly with increased sympathetic activity.⁵²³ Indeed, SNS activation in cirrhotic patients is associated with a rightward and downward shift of the RBF-RPP autoregulatory curve such that RBF becomes critically dependent on RPP. This phenomenon was found to contribute to the development of the HRS. Furthermore, insertion of TIPS to reduce portal venous pressure in patients with HRS leads to a fall in plasma NE levels and to an upward shift in the RBF-RPP curve.⁵³⁰

Reflex activation of the splenic afferent and renal sympathetic nerves also controls renal microvascular tone. In portal hypertension, the splenorenal reflex-mediated reduction in renal vascular conductance exacerbates Na^+ and water retention and may eventually contribute to renal dysfunction. Also increased splenic venous outflow pressure resulting from, but independent of, portal hypertension, reflexly activates adrenergic-angiotensinergic and vasodilator mesenteric nerves, and the RAAS. Finally, the spleen itself may be the source of a vasoactive factor.^{531,532}

The centrality of SNS overactivity in cirrhosis has been illustrated by the finding that in patients with cirrhosis and increased SNS activity, addition of clonidine or guanfacine to diuretic treatment induces an earlier and enhanced diuretic response, with fewer complications.^{533,534} In advanced cirrhosis, increased SNS activity parallels increases in RAAS and AVP activities.⁵³⁴ This marked neurohumoral activation probably reflects a shift toward decompensation, characterized by a severe decrease in EABV depletion.⁵³⁵ Overall, the three pressor systems might be activated by the same mechanisms and operate in concert to counteract the low arterial BP and decrease in EABV.⁴⁸³

Arginine Vasopressin

Impaired water excretion as a result of nonosmotic release of AVP secondary to decreased EABV is frequent in advanced cirrhosis, leading to water retention with hyponatremia.⁵³⁶

Affected patients also have higher PRA and aldosterone levels and lower urinary Na^+ excretion.⁴⁸³ In rats with experimental cirrhosis, plasma levels of AVP were elevated in association with overexpression of hypothalamic AVP mRNA and diminished pituitary AVP content.⁵³⁷ Concomitantly increased expression of AQP2, the AVP-regulated water channel in the collecting duct, was significantly diminished by the AVP receptor antagonist, terlipressin, indicating the important role of AQP2 in the water retention associated with hepatic cirrhosis.⁵³⁸

As noted earlier, AVP supports arterial BP through its action on the V_1 receptors on VSMC, whereas the V_2 receptor is responsible for water transport in the collecting duct.⁵³⁹ The availability of selective blockers of these receptors has provided clear evidence for the dual roles of AVP in pathogenesis of cirrhosis.^{412,539} Thus, administration of a V_2 receptor antagonist to cirrhotic patients, as well as to rats with experimental cirrhosis, increases urine volume, decreases urine osmolality, and corrects hyponatremia^{412,539} (see further discussion in section, “[Specific Treatments Based on the Pathophysiology of Sodium Retention in Cirrhosis](#)”).

AVP also increases the synthesis of the vasodilatory PGE_2 and PGI_2 in renal and other vascular beds, as well as in the collecting duct. This increase may offset the vasoconstrictive, as well as the hydroosmotic effect of AVP in cirrhosis.⁵⁴⁰

Endothelin

Levels of ET-1 and big ET-1 in plasma, splanchnic, and renal venous beds are markedly elevated in patients with cirrhosis and ascites, as well as in the HRS.^{541,542} Levels correlate positively with portal venous pressure and cardiac output and inversely with central blood volume.⁵⁴² The rise in ET-1 is accompanied by a reduction in ET-3 levels and the consequently elevated ET-1/ET-3 ratio is associated with a poor outcome of portal hypertension.⁵⁴³ In animal models of cirrhosis with portal hypertension, ET-A receptor activation and attenuated ET-B receptor repression on the portal vein has been reported.⁵⁴⁴ ET-B receptor blockade led to sinusoidal constriction and hepatotoxicity,⁵⁴⁵ whereas the dual ET-A and B receptor blocker, tezosentan, had no effect on hepatic blood flow.⁵⁴⁶

In humans with HRS, ET-1 and big ET-1 levels were significantly reduced in portal and renal veins 1 to 2 months after TIPS insertion, with a parallel increase in creatinine clearance and urinary Na^+ excretion.⁵⁴¹ Similar improvements have been observed within 1 week after successful orthotopic liver transplantation.⁵⁴⁷ Conversely, temporary occlusion of TIPS by angioplasty balloon inflation led to a transient increase in portal venous pressure, increased plasma ET-1, marked reduction of RPF, and increased intrarenal generation of ET-1.⁵⁴⁸

The importance of the intrarenal ET system has been demonstrated in a rat model of HRS.⁵⁴⁹ Plasma ET-1 increased twofold after the onset of liver and renal failure and the ET-A receptor was upregulated in the renal cortex. Bosentan, a nonselective ET receptor antagonist, prevented the development of renal failure when given before or 24 hours after onset of liver injury.⁵⁴⁹

Increased intrahepatic production of ET probably also contributes to the development of portal (and pulmonary) hypertension in cirrhosis through contraction of the stellate cells and a concomitant decrease in sinusoidal blood flow.⁵⁵⁰

To summarize, the hemodynamic changes in cirrhosis with refractory ascites could be related to local ET-1 production by the splanchnic and renal vascular beds. After TIPS and orthotopic liver transplantation, there are improvements in both ET and other vasoconstrictive factors (e.g., RAAS and vasopressin).⁵⁵¹ Therefore, the contribution of the intrarenal ET system relative to other vasoconstrictor hormones in the pathogenesis of the HRS remains speculative.

Apelin

The possible involvement of apelin in the pathogenesis of cirrhosis was suggested by the raised plasma levels⁵⁵² and enhanced expression of its receptor in proliferated arterial capillaries directly connected with sinusoids.^{553,554} In addition, an apelin receptor antagonist led to a reduction in the raised cardiac index, reversal of the increased total peripheral resistance, and improvement in Na^+ and water excretion in rats with experimental cirrhosis.⁵⁵² However, to date no therapeutic role of apelin antagonism in the management of severe HRS has been elucidated, possibly owing to the complex effects of apelin on glomerular hemodynamics.³³⁸

VASODILATORS/NATRIURETICS

Apart from their role in the hyperdynamic circulation characteristic of advanced cirrhosis, vasodilators play an important part in the pathogenesis of renal Na^+ retention. The principal vasodilators involved are NPs and PGs.

Natriuretic Peptides

Atrial Natriuretic Peptide. In recent years, measurements of BNP and NT-proBNP have largely superseded ANP as a biomarker of cirrhosis and portal hypertension. Nevertheless, the role of NPs in the pathogenesis of HRS was largely elucidated through studies on ANP and these are summarized here. Plasma ANP is elevated in cirrhosis at all stages, irrespective of the EABV.^{555,556} In the preascitic stage, increased plasma ANP may be important for the maintenance of Na^+ homeostasis, but with progression of the disease, resistance to the natriuretic action of the peptide develops.^{555,556} The high levels of ANP mostly reflect increased cardiac release rather than impaired clearance.⁵⁵⁷ The stimulus for increased cardiac ANP synthesis and release in early cirrhosis is likely increased left atrial size caused by overfilling of the circulation, secondary to intrahepatic hypertension-related renal Na^+ retention.⁵⁵⁸

In addition to elevated ANP, preascitic patients also had significantly elevated left and right pulmonary volumes, despite normal BP, PRA, aldosterone, and NE levels.⁵⁵⁹ High Na^+ intake in these patients resulted in weight gain and positive Na^+ balance for 3 weeks, followed by a return to normal Na^+ balance, thereby preventing fluid retention and the development of ascites.⁵⁶⁰ The factors responsible for maintaining relatively high levels of ANP during the later stages of cirrhosis, in association with arterial underfilling, may be related to a futile cycle of mutual interactions between vasoconstrictor/ Na^+ -retaining and vasodilatory/natriuretic forces. The fact that ANP levels do not increase further as patients proceed from early to late decompensated stages of cirrhosis would be consistent with this explanation. Furthermore, infusion of Ang II mimicked the nonresponder state by causing patients with cirrhosis, who still responded to ANP,

to become unresponsive.⁵⁶¹ This Ang II effect occurred at proximal (decreased distal delivery of Na⁺) and distal nephron sites to abrogate ANP-induced natriuresis and was reversible. The importance of distal solute delivery was confirmed using mannitol, which also resulted in an improved natriuretic response to ANP in responders but not nonresponders.^{562,563} ANP resistance was also ameliorated by endopeptidase inhibitors, renal sympathetic denervation, peritoneovenous shunting and orthotopic liver transplantation.^{564–567}

To summarize, ANP resistance is best explained by an effect of decreased delivery of Na⁺ to ANP-responsive distal nephron sites (glomerulotubular imbalance caused by abnormal systemic hemodynamics and activation of the RAAS) combined with an overriding effect of more powerful antinatriuretic factors to overcome the natriuretic action of ANP in the medullary collecting tubule.⁵⁶⁸ The latter effect could result from decreased delivery as well as permissive paracrine/autocrine cofactors, such as PGs and kinins.

Brain Natriuretic Peptide and C-Type Natriuretic Peptide. BNP levels are also elevated in cirrhosis with ascites and, like ANP, its natriuretic effect is blunted in these patients.^{569–572} Plasma BNP levels may be correlated with cardiac dysfunction⁵⁷⁰ and severity of disease and may be of prognostic value in the progression of cirrhosis.^{572,573} Plasma CNP levels in cirrhotic preascitic patients, although normal, were directly correlated with natriuresis and urine volume⁵⁷⁴ and inversely correlated with arterial compliance but not SVR.⁵⁷⁵ These data suggested that compensatory downregulation of CNP occurs in cirrhosis when vasodilation persists and that regulation of large and small arteries by CNP may differ.

In contrast with the preascitic stage, patients with more advanced disease and impaired renal function had lower plasma and higher urinary CNP levels than those with intact renal function. Moreover, urinary CNP was correlated inversely with urinary Na⁺ excretion. In patients with refractory ascites or HRS treated with terlipressin infusion or TIPS (see later section, “[Specific Treatments Based on the Pathophysiology of Sodium Retention in Cirrhosis](#)”), urinary CNP declined and urinary Na⁺ excretion increased 1 week later.^{574,665} Thus CNP may have a significant role in renal Na⁺ handling in cirrhosis.

Finally, *Dendroaspis* NP levels were found to be increased in cirrhotic patients with ascites, but not in those without and levels were correlated with disease severity.⁵⁷⁶ The significance of these findings remains unknown.

Prostaglandins

As noted, PG modulate the hydroosmotic effect of AVP and protect RPF and GFR when the activity of endogenous vasoconstrictor systems is increased. These properties of PG appear to be critical in decompensated cirrhosis with ascites but no renal failure. Such patients excrete more vasodilatory PG than healthy subjects, suggesting that renal production of PG is increased.⁵⁷⁷ Similarly, in experimental cirrhosis, there is increased synthesis and activity of renal and vascular PG as well as upregulation of COX-2.^{577,578} PGE₂ upregulation in the thick ascending limb of preascitic cirrhotic rats is mediated by downregulation of calcium-sensing receptors (CaSRs). This maneuver resulted in increased expression of the NKCC2, increased Na⁺ reabsorption in this segment, and augmented free water reabsorption in the collecting duct. The effects were reversed by the CaSR agonist poly-L-arginine.⁵⁷⁹

Not surprisingly, nonselective COX inhibitors resulted in a significant decrease in GFR and RPF in cirrhotic patients, with or without ascites. The decrement in renal hemodynamics varied directly with the degree of Na⁺ retention and neurohumoral activation, so that patients with high PRA and NE levels were particularly sensitive to these adverse effects.^{577,578} These negative effects of COX-inhibition appear to be solely COX-1-dependent because selective COX-2 antagonists spare renal function in both human and experimental cirrhosis, even with ascites.^{577,578} The favorable renal effect of selective COX-2 antagonists may be indirect and related to hepatic upregulation of COX-2 in that celecoxib can ameliorate portal hypertension by hepatic anti-angiogenic and antifibrotic actions.⁵⁸⁰

In contrast with nonazotemic patients with cirrhosis and ascites, patients with HRS have reduced renal synthesis of vasodilatory PG.⁵⁸¹ However, treatment with intravenous PGE₂ or its oral analogue, misoprostol, did not improve renal function in HRS patients.⁵⁸² This PGE₂ resistance may be related to overwhelming neurohumoral vasoconstrictive/antinatriuretic effects and may be crucial in the pathogenesis of HRS.⁵⁷⁷

Integrated View of the Pathogenesis of Sodium Retention in Cirrhosis

Portal hypertension leads to increased intestinal permeability, bacterial translocation, endotoxemia and exposure to bacterial DNA. In turn, hepatic NO and PG increase, leading to splanchnic, then systemic vasodilation, with increased cardiac output, decreased SVR, and arterial pressure (“hyperdynamic circulation”). The resulting imbalance between vascular capacity and plasma volume induces baroreceptor and subsequent neurohumoral activation (RAAS, SNS) leading to systemic and renal vasoconstriction, Na⁺ retention, and maintained circulation (compensated cirrhosis). In addition, a yet elusive hepatic-derived factor may also act directly to induce renal Na⁺ retention (overflow theory). If there is enough compensatory increase in systemic natriuretic factors and renal synthesis of PG and NO, renal function is also maintained. However, as cirrhosis advances and portal hypertension worsens, features of the HRS appear: EABV falls (underfilling), vasoconstrictive/antinatriuretic factors dominate over vasodilatory/natriuretic factors, GFR declines, and Na⁺ retention is further exacerbated leading to edema and ascites. Also, at this stage nonosmotic release of AVP becomes prominent, leading to impaired water excretion and hyponatremia.

CLINICAL MANIFESTATIONS OF HYPERVOLEMIA

Apart from the clinical manifestations of the underlying disease, the symptoms and signs of hypervolemia per se depend on the amount and relative distribution of fluid between the intravascular and interstitial spaces. Arterial volume overload is manifested as hypertension, whereas venous overload is characterized by raised jugular venous pressure (JVP). Interstitial fluid accumulation appears as peripheral edema, effusions in the pleural or peritoneal cavity (ascites) or in the alveolar space (pulmonary edema), or a combination of these. If cardiac and hepatic functions are normal and transcapillary Starling forces intact, the excess volume is distributed proportionately throughout the ECF compartments. In this situation, the signs of hypervolemia

will be hypertension and raised JVP. Peripheral edema appears only when interstitial volume overload exceeds 3 L, usually due to ongoing renal Na^+ retention.

When cardiac systolic function is impaired, as a result of myocardial, valvular, or pericardial disease, pulmonary and systemic venous hypertension predominate, and systemic BP may be low as a result of disproportionate fluid accumulation in the venous rather than the arterial circulation. Disruption in transcapillary Starling forces, as found in advanced cardiac and hepatic disease, may lead to fluid transudation into the pleural and peritoneal spaces, manifested as pleural effusions and ascites, respectively.

As already mentioned, advanced cirrhosis or fulminant hepatic failure lead to ascites and oliguric renal failure in the absence of diuretics, nephrotoxins, shock, and without evidence of significant intrarenal pathology. This is referred to as hepatorenal syndrome (HRS). Two subtypes of HRS have been defined. Type 1 is characterized by a rapid decline in renal function (doubling of serum creatinine level to >2.5 mg/dL, or 50% reduction in creatinine clearance to <20 mL/min) over a 2-week period. Typically, an acute precipitating factor can be identified. Type 2 develops spontaneously and progressively over months (serum creatinine level >1.5 mg/dL, or creatinine clearance <40 mL/min). More recently the International Club of Ascites has adopted the AKIN criteria for AKI, a rise of 0.3 mg/dL and/or 50% or more rise from baseline for the diagnosis of HRS. This more sensitive definition allows the possibility for earlier treatment of HRS but has yet to be tested in terms of prognostic significance.^{483,535} HRS is discussed in detail in Chapter 28.

DIAGNOSIS

The diagnosis of hypervolemia is usually evident from the clinical history and physical examination. Any combination of peripheral edema, raised JVP, pulmonary crepitations, and pleural effusions is likely to be diagnostic. In the presence of these findings, high BP would suggest renal failure as the cause of hypervolemia, whereas relatively low BP would point to severe HF or advanced cirrhosis as the precipitating factor. In more enigmatic cases, in which dyspnea is the sole complaint and clinical findings are minimal, measurement of plasma biomarkers such as BNP, NTproBNP, and MR-proANP may help distinguish between cardiac and pulmonary causes of the dyspnea.^{583,584}

Simple laboratory tests may aid in confirming the clinical diagnosis. An elevated cardiac troponin level is consistent with myocardial damage and high levels may be observed in acute decompensated HF.^{585,586} Transaminase levels may be raised in hepatic disease and hypoalbuminemia would be consistent with hepatic cirrhosis or nephrotic-range proteinuria caused by glomerular disease. The latter, of course, is confirmed by appropriate urine testing.

When BP is low, evidence of prerenal azotemia (increased ratio of blood urea nitrogen [BUN] to creatinine) may be found, as in advanced cardiac or hepatic failure (cardiorenal syndrome and HRS, respectively); intrinsic renal failure—proportionate increases in BUN and creatinine—may also ensue (see Chapter 28 for detailed discussion). Low EABV in the presence of hypervolemia is confirmed by a low urine Na^+ concentration or low fractional excretion of Na^+ , indicative of secondary renal Na^+ retention.

TREATMENT

Therapy for volume overload can be divided into management of the volume overload itself and prevention or minimization of its recurrence and associated morbidity and mortality. The critical first step is recognition and treatment of the underlying cause of hypervolemia. Thus, when the EABV is significantly reduced, as in cardiac and hepatic failure (or severe nephrotic syndrome), hemodynamic parameters should be optimized. Otherwise, therapy to induce negative Na^+ balance is associated with an enhanced risk for worsening hemodynamic compromise.

Once the EABV is restored, negative Na^+ balance can be induced by dietary Na^+ restriction, diuretics, and extracorporeal ultrafiltration. The degree of hypervolemia and the clinical urgency for Na^+ removal determine which modality should be used. Therefore, in a patient with life-threatening pulmonary edema, immediate intravenous loop diuretics are indicated; if high doses of these drugs do not induce significant diuresis, then extracorporeal ultrafiltration may be life-saving. At the other extreme, a hypertensive patient with mild volume overload and preserved renal function may require only dietary salt restriction and a thiazide diuretic.

Once acute hypervolemia has been controlled, therapy is directed toward the prevention or minimization of further acute episodes and improvement in prognosis. In addition to maintenance diuretic treatment, several strategies, based on the pathophysiologic process of Na^+ retention, are available clinically or are under experimental development.

Sodium Restriction

Until recently, the prevailing belief was that effective management of hypervolemia of any cause must include Na^+ restriction. Without this intervention, the success of diuretic therapy was thought to be limited because the relative hypovolemia induced by diuretics led to compensatory Na^+ retention, increased diuretic dosage, further reduction in EABV and still more renal Na^+ retention. However, this view was challenged by a randomized controlled trial in patients with acute decompensated HF. In this study, no difference in weight loss or clinical stability was observed at 3 days between the group restricted to 800 mg sodium/day and those with a more liberal sodium intake. Moreover, those on the severely sodium-restricted diet were significantly thirstier than their less restricted counterparts.⁵⁸⁷ In light of these new data, a reasonable goal is to restrict Na^+ intake to 50 to 80 mmol (approximately 3 to 5 g of salt/day).⁵⁸⁸ Because of the generally poor palatability of salt-restricted diets, salt substitutes may be used; however, these preparations usually contain high concentrations of potassium and must be used with caution by patients with renal impairment or those taking potassium-retaining RAAS antagonists.

In hospitalized patients, extra attention must be paid to amounts and types of intravenous fluids administered. A frequent scenario encountered by nephrologists consulted in internal medicine departments is a patient receiving intravenous saline together with high-dose diuretics. The usual rationale offered for this combination is that the saline will expand the intravascular volume while the diuretic will mobilize excess interstitial volume. This logic has no sound physiologic or therapeutic basis, since both modalities operate principally on the intravascular space. Moreover,

the combination of saline and furosemide in HF is associated with adverse outcomes.⁵⁸⁹ Furthermore, water restriction is also inappropriate, except in the presence of accompanying hyponatremia (plasma Na^+ <135 mmol/L). On the other hand, intravenous infusion of small volumes of hypertonic saline during diuretic dosing and liberalizing dietary salt intake while continuing to limit water consumption may improve fluid removal in HF patients. Furthermore, less deterioration in renal function, shorter hospitalizations, reduced readmission rates, and even reductions in mortality have been observed.^{590,591}

Diuretics

Diuretics are classified according to their sites of action along the nephron and are discussed in detail in Chapter 50. The reader is also referred to a comprehensive review.⁵⁹² Diuretics are described briefly here in relation to the treatment of hypervolemia.

Proximal Tubule Diuretics. The prototype proximal tubular diuretic is acetazolamide, a carbonic anhydrase inhibitor that inhibits the reabsorption of sodium bicarbonate. Prolonged use may cause hyperchloremic metabolic acidosis and the drug is usually used in the management of chronic glaucoma rather than for reducing volume overload. Another proximally acting diuretic is the thiazide-like metolazone, which also inhibits the NaCl cotransporter in the distal tubule. The proximal action of metolazone may be associated with phosphate loss greater than that seen with traditional thiazides.⁵⁹³ In general, metolazone is used as an adjunct to loop diuretics in resistant HF.⁵⁹⁴ Mannitol, which also inhibits proximal tubular reabsorption,⁵⁹⁵ can be used in combination with furosemide for the management of acute decompensated HF.⁵⁹⁶

Loop Diuretics. This group includes the most powerful diuretics, furosemide, bumetanide, torsemide, and ethacrynic acid. They act by inhibiting transport via the NKCC2 in the apical membrane of the thick ascending limb of the loop of Henle (see Chapter 6).⁵⁹² They are used for the treatment of severe hypervolemia and hypertension, especially in stages 4 and 5 of chronic kidney disease. Loop diuretics may lead to hypokalemia, intravascular volume depletion, and worsening prerenal azotemia, especially in older patients and those with reduced EABV. They are also hypercalciuric.⁵⁹⁷

Distal Tubule Diuretics. Diuretics in this segment block the apical NaCl cotransporter and include hydrochlorothiazide, chlorthalidone and metolazone (see also earlier section, “**Proximal Tubule Diuretics**”). They are typically used as adjuncts to loop diuretics in resistant HF, particularly metolazone. Inhibition of Na^+ reabsorption by diuretics in the proximal tubule (except for carbonic anhydrase inhibitors), loop of Henle, and distal tubule leads to increased solute delivery to the collecting duct. Consequently, potassium and proton secretion are enhanced, which may lead to hypokalemia and metabolic alkalosis.⁵⁹⁸ Thiazides are also hypocalciuric.⁵⁹⁷

Collecting Duct Diuretics. Collecting duct (K^+ -sparing) diuretics operate by competing with aldosterone for occupation of the mineralocorticoid receptor (spironolactone, eplerenone) or by direct inhibition of the ENaC (amiloride and triamterene).⁵⁹² As their alternative name implies,

important side effects of this group are hyperkalemia and metabolic acidosis, resulting from concomitant suppression of K^+ and proton secretion. Therefore, they are widely used in combination with thiazide and loop diuretics to minimize hypokalemia. Aldosterone antagonists are especially useful in the management of disorders characterized by secondary hyperaldosteronism, such as cirrhosis with ascites. Moreover, these drugs are cardioprotective and renoprotective via non-epithelial mineralocorticoid receptor blockade (see sections, “**Pathophysiology**” and “**Specific Treatments Based on the Pathophysiology of Heart Failure**” in this chapter; also see Chapter 12).

Diuretic Resistance. As noted, when Na^+ retention is severe and resistant to conventional doses of loop diuretics, combinations of diuretics acting at different nephron sites may produce effective natriuresis. Another method for overcoming diuretic resistance is the administration of a bolus dose of loop diuretic to yield a high plasma level, followed by high-dose continuous infusion. Alternately, high doses given intermittently may be successful in reversing diuretic resistance. There appears to be no difference between continuous infusion and repeated bolus administration of high or low doses of furosemide in improvement of shortness of breath or volume of diuresis at 72 hours after admission for acute decompensated HF.⁵⁹⁹

Whichever method is used to treat diuretic resistant hypervolemia, plasma Na^+ , K^+ , Mg^{2+} , Ca^{2+} , phosphate, BUN, and creatinine should be monitored carefully, and any deviations appropriately corrected. Rarer side effects of diuretics include cutaneous allergic reactions, acute interstitial nephritis (see Chapter 28), pancreatitis, and, rarely, blood dyscrasias.⁶⁰⁰

Extracorporeal Ultrafiltration

On occasion, extreme resistance to diuretics occurs, often accompanied by renal functional impairment. In such cases, effective removal of volume excess may be achieved by ultrafiltration using hemofiltration, hemodialysis, peritoneal dialysis (see Chapter 65), or small devices designed for isolated ultrafiltration (UF).⁵⁹⁹ Chronic ambulatory peritoneal dialysis may also reduce hospitalization rates in patients with resistant HF who are not candidates for surgical intervention.⁶⁰¹ However, a randomized controlled trial comparing intensive diuretic therapy with UF for the management of acute decompensated HF with worsened renal function (CARRESS-HF) was halted early because of lack of benefit of UF and an excess of early and late (60 days) adverse events.⁶⁰² Moreover, UF was inferior to diuretic therapy in terms of the bivariate primary endpoint of body weight and rise in serum creatinine level at 96 hours after commencement of therapy. Therefore, UF should currently be reserved for those patients with unequivocal diuretic resistance and as part of an algorithmic stepped approach to HF management.⁵⁹⁹

SPECIFIC TREATMENTS BASED ON THE PATHOPHYSIOLOGY OF HEART FAILURE

Because the clinical situation of an HF patient at any given time depends on the delicate balance between vasoconstrictor/antinatriuretic and vasodilator/natriuretic factors, any treatment that can tip the balance in favor of the latter should be efficacious. Thus, treatment is aimed at

either pharmacologically increasing natriuretic or reducing antinatriuretic mechanisms. The principal approaches include reducing RAAS or SNS activity or increasing NP activity.

INHIBITION OF THE RENIN-ANGIOTENSIN-ALDOSTERONE SYSTEM

Based on the maladaptive actions of locally produced or circulatory Ang II, numerous studies have shown that ACE inhibition and ARBs improve renal function, cardiac performance and life expectancy of HF patients.^{234,603} A small decline in GFR is occasionally observed as a result of blockade of Ang II–induced preferential efferent arteriolar constriction, which leads to a sharp fall in glomerular capillary pressure, but this is usually not clinically significant. Because patients with HF cannot overcome the Na⁺-retaining action of aldosterone, blockade of the latter by spironolactone or eplerenone induces substantial natriuresis in these patients.⁶⁰³

Overall, the effect of RAAS inhibition on renal function in HF depends on a multiplicity of interacting factors. On the one hand, RBF may improve as a result of lower efferent arteriolar resistance. Systemic vasodilation may be associated with a rise in cardiac output. Under such circumstances, reversal of the hemodynamically mediated effects of Ang II on Na⁺ reabsorption would promote natriuresis. Moreover, RAAS inhibition could facilitate the action of NPs to improve GFR and enhance Na⁺ excretion. On the other hand, Ang II–induced elevation of the single-nephron filtration fraction facilitates preservation of GFR in the presence of diminished RPF. In patients with precarious renal hemodynamics, a fall in systemic arterial pressure below the autoregulatory range, combined with removal of the Ang II effect on glomerular hemodynamics, may cause severe deterioration of renal function. The net result depends on the integrated sum of these physiologic effects, which, in turn, depends on the severity of HF (Table 14.7).

Table 14.7 Renal Effects of Renin-Angiotensin-Aldosterone System Inhibition in Heart Failure

Factors Favoring Improvement in Renal Function

Maintenance of Na⁺ balance
Reduction in diuretic dosage
Increase in Na⁺ intake
Mean arterial pressure >80 mm Hg
Minimal neurohumoral activation
Intact counterregulatory mechanisms

Factors Favoring Deterioration in Renal Function

Evidence of Na⁺ depletion or poor renal perfusion
Large doses of diuretics
Increased urea/creatinine ratio
Mean arterial pressure <80 mm Hg
Evidence of maximal neurohumoral activation
AVP-induced hyponatremia
Interruption of counterregulatory mechanisms
Coadministration of prostaglandin inhibitors
Adrenergic dysfunction (e.g., diabetes mellitus)

AVP, Arginine vasopressin.

In addition to promoting Na⁺ retention, the RAAS contributes to vascular and cardiac remodeling by inducing perivascular and interstitial fibrosis in HF.^{234,604} In accordance with this mechanism, the addition of small doses of aldosterone inhibitors to standard therapy, including ACEIs or ARBs, substantially reduces mortality rate and morbidity in HF patients.^{234,603} In contrast, combinations of ACEI and ARB are not useful and, often, are detrimental, owing to increased risk of hypotension, AKI, and hyperkalemia.⁶⁰⁴ However, the combination of RAAS inhibitor, aldosterone inhibitor β -blocker, and low-dose loop diuretics is associated with a better prognosis than chronic use of increasing loop-diuretic dose alone.⁶⁰⁵

Finally, all patients receiving one or more RAAS inhibitors require careful follow-up for detection and management of hyperkalemia, particularly those with renal dysfunction.⁶⁰⁴

β -BLOCKADE

Insofar as β -blockade is now the standard of care in the management of HF, this review would not be complete without mention of these drugs. However, because their effect in HF is not directly related to Na⁺ and water, this important therapy is not elaborated further in this chapter. The reader is referred to a recent meta-analysis for up-to-date information.⁶⁰⁶

NITRIC OXIDE DONOR AND REACTIVE OXYGEN SPECIES/PEROXYNITRATE SCAVENGERS

Because NO signaling is disrupted in HF, achieving NO balance by NO donors or selective NOS inhibitors could be important for correcting the pathophysiologic process of HF.⁶⁰⁷ In this regard, the beneficial effects of combined isosorbide dinitrate (NO donor) and hydralazine (reactive oxygen species and peroxynitrite scavenger) therapy, particularly in African-American patients, are well-established.⁶⁰⁸ Newer methods for enhancing NO activity include NOS and GC stimulators and inhibitors of GC breakdown.⁶⁰⁷ Relaxin-mediated and waon-mediated NOS stimulation, in particular, are currently being tested in a phase 3 clinical trial. Relaxin mediates NO production both immediately via stimulation of eNOS and chronically via increased expression of iNOS. Treatment of acute HF with serelaxin, recombinant relaxin-2, was associated with dyspnea relief and reduced 180-day but had no effect on hospital readmission. Serelaxin treatment was well tolerated and safe.⁶⁰⁹ Further trials are in the pipeline.

Waon or “soothing warm” therapy, which causes vasodilation via eNOS stimulation, is a thermal treatment developed in Japan and shown in a recent clinical trial to improve NYHA class and exercise tolerance in chronic HF.⁶¹⁰ Waon also enhanced the favorable actions of RAAS inhibition and reduced SNS activity.⁶¹¹

An additional strategy exploits upregulation of nNOS in the paraventricular nucleus and a reduction in sympathetic outflow. This observation raised the potential for nNOS upregulation by ACE2 therapy in HF.^{612,613}

ENDOTHELIN ANTAGONISTS

Acute ET nonselective antagonists decrease vascular resistance and increase cardiac index and cardiac output in HF. However,

both short- and long-term clinical trials with ET-A receptor antagonists have demonstrated no benefits, but serious adverse events, including fluid retention, increased need for hospitalization, hepatic dysfunction, and mortality in HF patients.⁶¹⁴ These disappointing results may be explained by the observation that ET-A receptor antagonism in experimental HF further activates the RAAS in association with sustained Na⁺ retention.²⁷⁹ Moreover, the increased cardiac, pulmonary, and renal ET-1 production in HF, together with the marked vasoconstrictor and mitogenic properties of the molecule, suggest that ET-1 exacerbates Na⁺ retention directly and indirectly by aggravating renal and cardiac functions, respectively.²⁷⁹ In addition, nonspecific ET-B receptor blockade could have exacerbated cardiac remodeling. Given the antiproliferative effect of ET-B1–receptor activation and the detrimental vasoconstrictor effect of ET-B2 stimulation, combined ET-B1 stimulation and ET-B2 blockade was recently proposed for control of the remodeling associated with postischemic HF.⁴¹⁷ Concomitant ET-A blockade would help to prevent the undesired effects of unopposed ET-1 activity and exploit the established therapeutic effects of ET-A antagonists.⁴¹⁷ In this regard, a recent report that dual ET-A/ET-B receptor inhibition improves echocardiographic parameters in a mouse model of HF with preserved ejection fraction (HfPEF) is encouraging.⁶¹⁵ The favorable effects occurred by abrogating adverse cardiac remodeling and reducing stiffness. Additional studies on the role of dual ET-1 receptor antagonists in patients with HfPEF are eagerly awaited.

NATRIURETIC PEPTIDES

As noted previously, circulating NP levels are elevated in HF in proportion to disease severity, but renal actions of NPs are attenuated in severe HF. Nevertheless, elimination of NP action using NPR-A blockers or surgical removal of the atrium disrupted renal function and cardiac performance in experimental HF.⁶¹⁶ Conversely, increasing circulating levels of NPs (BNP > ANP) by intravenous administration improved general clinical status,⁶¹⁷ reduced pulmonary arterial and capillary wedge pressures, right atrial pressure and SVR, with improved cardiac output, systemic BP and diuresis. In parallel, plasma NE and aldosterone levels were suppressed.⁶¹⁸ However, these beneficial effects have not translated into clinical efficacy, with several controlled studies showing minimal natriuretic effects of BNP (nesiritide) compared with placebo.²⁶⁴ Furthermore, serious adverse events of nesiritide treatment were seen, including dose-related hypotension, worsening renal function, and possible increased risk of death.²⁶⁴ Finally, in a randomized controlled trial comparing nesiritide or low-dose dopamine as add-on therapy to intravenous furosemide in patients with HF and renal dysfunction, no improvement in decongestion or renal function was observed with either combination therapy over furosemide alone.⁶¹⁹ Similarly, ularitide, the pharmacological equivalent of the NP, urodilatin, as add-on therapy did not affect a clinical composite endpoint or reduce long-term cardiovascular mortality, despite significant reductions in NT-proBNP levels (TRUE-AHF trial). Therefore, the therapeutic role of NP in HF remains unclear and novel approaches are being explored. These include strategies to convert the prohormone into its active form might be efficacious, CNP analogs and designer

chimeric peptides such as CD-NP, consisting of CNP and part of DNP.⁶²⁰

NEUTRAL ENDOPEPTIDASE INHIBITORS AND VASOPEPTIDASE INHIBITORS

Correcting the imbalance between the RAAS and NP systems can also be achieved by inhibiting the enzymatic degradation of NPs by NEP. In both experimental models and clinical trials NEP inhibitors enhanced plasma ANP and BNP levels in association with vasodilation, natriuresis, diuresis, reduced cardiac preload and afterload.⁴³⁹ However, initial favorable hemodynamic, neurohormonal, and renal effects of NEP inhibitors in HF patients^{621,622} were not confirmed in later studies. These showed enhanced RAAS activation, increased ET-1 levels and attenuated renal and hemodynamic improvement.⁴³⁹ On the basis of these findings, combined RAAS and NEP (vasopeptidase) inhibitors were designed and, indeed, initially shown to have more favorable hemodynamic and renal effects, such as preserved GFR, than each treatment alone.⁴³⁹

However, randomized clinical trials indicated that neither vasopeptidase inhibitors nor NEP inhibitors as add-on therapy to RAAS inhibition were more effective than RAAS inhibitors alone in the treatment of HF. Furthermore, the combination was associated with a significantly increased incidence of severe angioedema.⁴³⁹ Possible reasons for the failure of NEP inhibitors include disproportionate increase in RAAS and ET activity over time, the development of tolerance with chronic treatment and downregulation of NP receptors in response to degradation of NEP inhibitors. The increased incidence of angioedema with vasopeptidase inhibition may result from excessive accumulation of bradykinin or inhibition of aminopeptidase P.⁴³⁹

The challenge of preventing angioedema led to the development of combined ARB-NEP inhibition since ARBs do not disrupt bradykinin metabolism. Three trials in HF with reduced EF, IMPRESS, OVERTURE and PARADIGM-HF, compared combined NEP/RAS inhibition with RAS inhibition alone and reported clinical outcomes. The pooled hazard ratio (HR) for all-cause death or HF hospitalization and all-cause mortality was significantly reduced in patients receiving combined NEP/RAS inhibition in all three trials. Combined NEP/RAS inhibition compared with ACE inhibition was associated with more hypotension, but less renal dysfunction and hyperkalaemia.⁶²³ These results pave the way for a paradigm shift in the management of severe HF.⁶²⁴

VASOPRESSIN RECEPTOR ANTAGONISTS

AVP receptor antagonists (vaptans) are small, orally active, nonpeptide molecules that lack agonist effects and display high affinity for and specificity to their corresponding receptors.^{412,625} Highly selective and potent antagonists for the V_{1A}, V₂, and V_{1B} receptor subtypes and mixed V_{1A}/V₂ receptor antagonists are now available.⁴¹² In clinical trials and experimental models of compensated and decompensated HF with hyponatremia, V₂ receptor-specific vaptans produced hemodynamic improvement, with transient decrease in SVR and increased cardiac output. Water diuresis caused decreases in body weight and edema, hyponatremia was corrected, and RBF and renal function stabilized.^{412,625} Dyspnea lessened in some but not all patients. Both tolvaptan, a V₂ selective

receptor antagonist, and conivaptan, a nonselective V_{1A}/V_2 receptor antagonist (both US Food and Drug Administration [FDA]-approved for use in HF) induced these favorable effects after treatment for up to 60 days. Similar results were reported for tolvaptan in stable class II or III HF.^{626,627} Positive effects were observed, regardless of whether LVEF was less or greater than 40%.^{412,625} In patients with LVEF less than 40%, a single dose of conivaptan reduced pulmonary capillary wedge and right atrial pressure in comparison with placebo; cardiac index, pulmonary arterial pressure, systemic, and pulmonary vascular resistance, systemic arterial pressure and heart rate were unaffected. Urine output rose and osmolality fell significantly.^{628,629} However, functional capacity, exercise tolerance, and overall quality of life were unchanged.⁶²⁷

The effect of vaptans on survival is unclear. In one study, mortality rate two months after discharge was halved in patients treated with tolvaptan who had an increase in serum Na^+ of 2 mmol/L or more, compared with those with no increase in serum Na^+ .⁶³⁰ In contrast, in the other trial, tolvaptan given for 60 days had no effect on survival after 9 months of follow-up.⁶²⁷ Lixivaptan, another V_2 selective antagonist, was associated with an excessive mortality rate, despite favorable clinical and laboratory effects similar to those of tolvaptan.⁶³¹ Hence, the FDA did not approve lixivaptan for use in HF.⁶³²

Tolvaptan did not adversely affect cardiac remodeling or LVEF after 1 year of treatment in patients receiving optimal, background therapies for HF (RAAS and β -blockers).⁶³³ In contrast, in a murine model of HF, tolvaptan was associated with increased remodeling and mortality, which was prevented by concomitant furosemide therapy.⁶³⁴ Therefore, vigilance will need to be exercised in the long-term use of tolvaptan in HF.

In view of the impressive diuretic effect of vaptans, there is considerable interest in a potential loop diuretic-sparing effect, as mentioned previously. To summarize results to date, tolvaptan compared with furosemide alone or in combination, was not associated with improvement in dyspnea 24 hours after admission for acute HF, with or without preserved LVEF, despite greater diuresis and weight loss.^{635–638} Tolvaptan was associated with better preserved renal function and serum electrolytes.^{635–637} A Japanese multicenter clinical effectiveness of tolvaptan in patients with acute decompensated heart failure and renal failure (AQUAMARINE) study, in which tolvaptan is being compared with standard furosemide therapy in HF patients with renal impairment is ongoing (University Medical Information Network [UMIN] clinical trial registry number, UMIN000007109).⁶³⁹

The adverse effects of vaptans appear to be relatively few and, overall, minor. Thirst and dry mouth are not unexpected; hypokalemia occurs in less than 10% of recipients similar to loop diuretics. In the largest study to date, involving more than 4000 patients, there was a small but significant increase in strokes but a small but significant reduction in myocardial infarctions.⁴⁰⁶

In summary, AVP receptor antagonists appear to have important short-term benefits in the treatment of advanced HF. However, the question of improvement in longer term prognosis, especially the high mortality among HF patients already receiving optimal treatment with RAAS inhibitors ($\pm$ NEP inhibition) and β -blockers remains open.⁶⁴⁰

SODIUM-GLUCOSE TRANSPORTER-2 INHIBITORS

Sodium-glucose transporter-2 (SGLT-2) inhibitors decrease proximal tubular Na^+ and Cl^- reabsorption, leading to a reset of TGF. This induces plasma volume contraction without activation of the SNS, decreases glomerular hyperfiltration leading to better long-term renal preservation, and improves diuretic and natriuretic responses to other diuretics. Moreover, SGLT-2 inhibitors might improve the efficiency of myocardial energetics by offering β -hydroxybutyrate as an attractive fuel for oxidation and increase hematocrit improving oxygen transport. Finally, decreased vascular stiffness and improved endothelial function are observed with the use of SGLT-2 inhibitors in diabetes. These multiple nonglycemic effects make SGLT-2 inhibitors preferred glucose-lowering drugs in diabetic patients with HF and raise the possibility for use in HF without diabetes.⁶⁴¹

SPECIFIC TREATMENTS BASED ON THE PATHOPHYSIOLOGY OF SODIUM RETENTION IN CIRRHOSIS

The prognosis of type 1 HRS is dismal. The mortality rate, which rises with increased AKI stage, is as high as 80% in the first 2 weeks and only 10% of patients survive longer than 3 months. Therefore specific aggressive therapy in these patients is usually indicated in preparation for liver transplantation.^{483,535} Type 2 HRS has a better prognosis. Median survival is approximately 6 months.^{483,535} Aggressive management for HRS includes pharmacologic therapy, TIPS insertion, renal replacement therapy (RRT), and liver transplantation. The management of precipitating factors is discussed elsewhere.

PHARMACOLOGIC TREATMENT

The goals of pharmacologic therapy are to reverse the functional renal failure and serve as a bridge to liver transplantation. Earlier attempts to reverse renal vasoconstriction with low-dose dopamine, the PGE₁ analog misoprostol, RAAS and ET inhibitors were unsuccessful and even detrimental.^{483,485,519} Therefore current treatment is based on reversal of splanchnic and systemic arterial vasodilation, AVP-induced water retention and hyponatremia.

SYSTEMIC VASOCONSTRICTORS

Three groups of vasoconstrictors have been studied—vasopressin V_1 receptor analogs, a somatostatin analog, and α -adrenergic agonists.^{483,535}

Vasopressin V_1 Receptor Analogs

These agents cause marked vasoconstriction through their action on V_1 receptors in the arterial smooth muscle wall. They are used for the management of acute variceal bleeding in cirrhosis and portal hypertension. Ornipressin infusion in combination with volume expansion led to a remarkable increase in RPE, GFR, and Na^+ excretion in almost 50% of treated patients. However, significant ischemic adverse effects in 30% of patients led to the abandonment of this drug.⁶⁴²

Terlipressin, without significant adverse ischemic reactions, is now the vasoconstrictor drug of choice. Terlipressin and albumin in type 1 HRS is associated with a significant improvement in GFR, increase in arterial pressure, near-normalization of neurohumoral levels, and reduction of serum creatinine level in 40% to 50% of cases.^{483,535,643} Response rates to terlipressin in type 2 HRS are better than those in type 1, with more than 80% survival at 3 months.^{481,538} Despite relapses in 50% of cases, reintroduction of therapy produces a further response.^{481,538}

The results of several randomized controlled trials, 2 meta-analyses and several Cochrane reviews, have shown that the combination of terlipressin and albumin is clearly superior to albumin or terlipressin alone in improving renal function and survival in both types 1 and 2 HRS.^{644–647} No difference in efficacy between boluses and infusions was found, although the latter was associated with fewer adverse effects.⁶⁴⁸ The optimum duration of terlipressin therapy is not clear, but it is usually given until serum creatinine levels decrease to less than 1.5 mg/dL or for a maximum of 15 days. Whether extending therapy beyond 15 days will add any benefit is unknown. A key prognostic target after combined terlipressin-albumin therapy may be a rise in MAP of 10 mm Hg or more; these patients required less dialysis and were more likely to attain liver transplantation than those with smaller responses in MAP. This response was associated with better short-term, long-term, and transplant-free survival.⁶⁴⁹ Moreover, pretransplantation normalization of renal function by terlipressin resulted in similar posttransplantation outcomes to those of patients with normal renal function.⁶⁵⁰

In patients with HRS undergoing living related donor liver transplantation, terlipressin intra- and postoperatively is associated with significant increases in MAP, SVR, and renal function. Significantly decreased heart rate, cardiac output, hepatic and renal arterial resistive indices, portal venous blood flow, and use of vasoconstrictor drugs were seen during reperfusion.^{651,652} To summarize, terlipressin and albumin infusion may be appropriate only for patients awaiting or already undergoing liver transplantation and a favorable hemodynamic response is associated with a better overall prognosis both before and after liver transplantation.

Unfortunately, terlipressin is currently unavailable in the United States and Canada and the results of a phase 3 clinical trial, comparing terlipressin with placebo (mannitol) in these countries, are, therefore, eagerly awaited (CONFIRM study, [ClinicalTrials.gov](https://clinicaltrials.gov) NCT 02770716).

Somatostatin Analogs and α -Adrenergic Agonists

Octreotide is an α -adrenergic agonist that inhibits the release of glucagon and other vasodilator peptides. Octreotide with albumin infusion or midodrine had no effect on renal function in HRS. However, both agents in combination with albumin infusion led to a significant improvement in renal function and survival in types 1 and 2 HRS in comparison with controls.^{653–655} However, no effect of this combination on outcomes of subsequent liver transplantation was observed.⁶⁵⁶ Attempts to prevent relapse of type 2 HRS with midodrine after terlipressin-induced improvement were also unsuccessful.⁶⁵⁷ Moreover, terlipressin plus albumin was significantly more effective than octreotide, midodrine and albumin, in terms of improved renal function, complete reversal of HRS and

90-day survival.⁶⁵⁸ Finally, NE was compared with terlipressin in the treatment of type 1 HRS in three randomized controlled trials. Responses to both agents were similar in terms of MAP and renal function. Cumulative survival and adverse event rates were not significantly different between the two drugs. NE is less expensive than terlipressin, but, unlike terlipressin, NE administration requires cardiac monitoring. Therefore, total costs might be similar for the two therapies.^{483,535}

VASOPRESSIN V₂ RECEPTOR ANTAGONISTS

As noted, hyponatremia, caused by persistent nonosmotic AVP-induced water retention, is often seen in HRS and is a marker of poor prognosis.⁶⁵⁹ Therefore, attaining a water diuresis and reversing hyponatremia using V₂ receptor antagonists are potentially important therapeutic goals. Several controlled studies, using lixivaptan, tolvaptan, and satavaptan have shown modest improvements in serum Na⁺, but without effect on mortality or adverse events.^{659–661} Satavaptan in combination with diuretics was associated with a higher mortality rate than either agent alone, leading to early termination of one study.⁶⁶² The specific role of satavaptan in the increased mortality was uncertain, given that most deaths were due to complications of cirrhosis. Overall, the effects of vaptans in HRS appear to be modest; this may be explained by avid proximal tubular solute reabsorption leading to reduced distal delivery or by V₂ receptor-independent pathways of water retention.⁴¹⁰

TRANSJUGULAR INTRAHEPATIC PORTOSYSTEMIC SHUNT

The efficacy of TIPS in the reduction of portal venous pressure in patients with cirrhosis and refractory ascites with type 1 or 2 HRS has been demonstrated in several small cohort and controlled studies.^{481,663} Significant improvement in renal hemodynamics, GFR, and vasoconstrictive neurohumoral factors were observed in most patients.⁵¹⁹ Liver transplant-free survival also was markedly improved⁶⁶³ and might be better than that reported for terlipressin and albumin infusion.⁶⁶⁴ Some patients who were dialysis dependent were able to discontinue this treatment after TIPS insertion. Moreover, liver transplantation was performed in two patients when the medical condition that precluded transplantation had resolved after TIPS insertion.⁶⁶⁵ However, TIPS seems to increase wait-time for liver transplantation and days in hospital after transplantation, although without effect on 30-day mortality.⁶⁶⁶

TIPS appears to exert its favorable effects by reducing sinusoidal hypertension with suppression of the putative hepatorenal reflex (see earlier), improvement of EABV by shunting portal venous blood into the systemic circulation, or amelioration of cardiac dysfunction.⁵⁵¹

TIPS is currently recommended in the setting of refractory ascites, particularly in type-2 HRS, but is contraindicated in severe liver failure, severe intrinsic renal failure (serum creatinine >3 mg/dL), HF, severe porto-pulmonary hypertension, recurrent or persistent severe hepatic encephalopathy despite adequate treatment, and uncontrolled sepsis.⁶⁶⁷

RENAL AND LIVER REPLACEMENT THERAPY

The benefits of conventional hemodialysis and continuous RRT, in terms of prolonging survival, are dubious,⁶⁶⁸ and

morbidity resulting from these therapies is high.⁶⁶⁹ In oliguric patients awaiting liver transplantation who do not respond to vasoconstrictors or TIPS and who develop diuretic-resistant volume overload, hyperkalemia, or intractable metabolic acidosis, RRT may be considered. In view of the dismal prognosis of HRS, especially type 1, RRT should probably be used solely as a bridge to transplantation.

In contrast with conventional RRT, albumin-dialysis offers the potential advantage of removing albumin-bound, water-soluble vasoactive agents, toxins, and proinflammatory cytokines. Molecules relevant to the pathogenesis of advanced cirrhosis include bile acids, TNF- α , interleukin-6, and NO.⁴⁸⁵ Three albumin dialysis systems, molecular adsorbent recirculating system (MARS), fractionated plasma separation, adsorption and hemodialysis (Prometheus system), and single-pass albumin dialysis, have been examined in randomized trials for supportive treatment of liver failure. These therapies enable partial recovery of hepatic function and provide simultaneous RRT. They lead to a decrease in renal vascular resistance and improvement in the splenic resistance index, a parameter related to portal resistance. The hemodynamic effects are thought to be mediated by clearance of vasoactive substances.⁶⁷⁰ Bilirubin levels, encephalopathy grade, serum creatinine, and serum Na⁺ all improved.⁶⁷¹ In a recent meta-analysis, albumin dialysis achieved a significant net decrease in serum total bilirubin level relative to standard medical therapy but not in serum ammonia or bile acids. Albumin dialysis achieved an improvement in hepatic encephalopathy relative to standard medical therapy but had no effect on survival. Bearing in mind the limited reporting of adverse events, no major safety concerns were detected. Extracorporeal nonbiologic liver support system should at present probably be considered only as a bridge to transplantation.⁶⁷²

LIVER TRANSPLANTATION

Liver transplantation is the treatment of choice for HRS because it offers a cure for the liver disease and renal dysfunction. Large case series indicate that survival is significantly lower in transplant recipients with HRS (types 1 and 2) both immediately postoperatively and long-term than in those without the syndrome (3-year survival rates of 60% vs. 70% to 80%) but may be improved by the bridging therapies described previously.^{673,674}

With respect to renal function after transplantation in patients with HRS, GFR decreases in the first month owing to the stress of surgery, infections, immunosuppressive therapy,

and other factors. Despite the prompt correction of hemodynamic and neurohumoral parameters, GFR recovers incompletely (30 to 40 mL/min at 1 to 2 months) and often remains impaired over the long term. Although dialysis requirement in the first month is greater in patients with HRS than without HRS (35% vs. 5%), duration of dialysis pretransplantation does not influence renal recovery posttransplantation. Predictors of renal recovery included younger recipient and donor, nonalcoholic liver disease, and low posttransplantation serum bilirubin.⁶⁷⁵ Overall, the rate of posttransplantation reversal of HRS has been estimated to be no greater than 58%.

Unsurprisingly, the 5-year risk of ESRD posttransplantation in patients with HRS is significantly higher than in those without HRS. Moreover, combined simultaneous liver and kidney transplantation (SLKT) apparently offers no greater benefit over liver transplantation alone with respect to early posttransplantation kidney function. However, better 5-year survival rates are more likely after SLKT than after liver transplantation alone in patients with pretransplantation serum creatinine levels greater than 2.2 mg/dL, irrespective of dialysis requirement.⁶⁷⁴ More studies are needed to enable a rational decision about who should receive SLKT, rather than liver transplants alone. At present, SLKT should probably be reserved for patients who are dialysis dependent for 8 weeks or more pretransplantation.^{485,674,676}

The introduction of Model of End-stage Liver Disease (MELD) scores for the allocation of livers has increased the number of transplantations in patients with impaired renal function, but more SLKTs are also being performed.⁶⁷⁷ On the other hand, a favorable response to vasoactive therapy reduces the MELD score and may lead to a paradoxical delay in liver transplantation.⁶⁷⁸ Therefore, only pretreatment MELD scores should be used to predict potential outcomes of liver transplantation in HRS.⁴⁸¹ A further paradox is that patients with type 2 HRS have lower MELD scores than those with type 1 HRS, resulting in the former being ascribed a lower priority for transplantation and a longer time on the waiting list. This delay is associated with higher mortality, especially in the presence of hyponatremia and persistent ascites.⁶⁷⁷ Therefore, the criteria for donor allocation need to be modified to incorporate these factors into the final score for prioritization.⁴⁸¹

For a general summary of all aspects of liver transplantation, the reader is referred to several excellent recent reviews.^{679,680}



Complete reference list available at [ExpertConsult.com](https://www.expertconsult.com).

REFERENCES

- Rakova N, et al. Increased salt consumption induces body water conservation and decreases fluid intake. *J Clin Invest*. 2017;127:1932–1943.
- Kitada K, et al. High salt intake reprioritizes osmolyte and energy metabolism for body fluid conservation. *J Clin Invest*. 2017;127:1944–1959.
- Aukland K, Nicolaysen G. Interstitial fluid volume: local regulatory mechanisms. *Physiol Rev*. 1981;61:556–643.
- Ziomer A, et al. Sodium-, potassium-, chloride-, and bicarbonate-related effects on blood pressure and electrolyte homeostasis in deoxycorticosterone acetate-treated rats. *Am J Physiol Renal Physiol*. 2008;295:F1752–F1763.
- Machnik A, et al. Macrophages regulate salt-dependent volume and blood pressure by a vascular endothelial growth factor-C-dependent buffering mechanism. *Nat Med*. 2009;15:545–552.
- Schafflhuber M, et al. Mobilization of osmotically inactive Na⁺ by growth and by dietary salt restriction in rats. *Am J Physiol Renal Physiol*. 2007;292:F1490–F1500.
- Titze J. Water-free Na⁺ retention: interaction with hypertension and tissue hydration. *Blood Purif*. 2008;26:95–99.
- Titze J, et al. Osmotically inactive skin Na⁺ storage in rats. *Am J Physiol Renal Physiol*. 2003;285:F1108–F1117.
- Titze J, et al. Extrarenal Na⁺ balance, volume, and blood pressure homeostasis in intact and ovariectomized deoxycorticosterone-acetate salt rats. *Hypertension*. 2006;47:1101–1107.
- Titze J, Ritz E. Salt and its effect on blood pressure and target organ damage: new pieces in an old puzzle. *J Nephrol*. 2009;22:177–189.
- Titze J, et al. Glycosaminoglycan polymerization may enable osmotically inactive Na⁺ storage in the skin. *Am J Physiol Heart Circ Physiol*. 2004;287:H203–H208.
- Go WY, Liu X, Roti MA, et al. NFAT5/TonEBP mutant mice define osmotic stress as a critical feature of the lymphoid microenvironment. *Proc Natl Acad Sci USA*. 2004;101:10673–10678.
- Wiig H, et al. Immune cells control skin lymphatic electrolyte homeostasis and blood pressure. *J Clin Invest*. 2013;123:2803–2815.
- Schneider MP, et al. Skin sodium concentration correlates with left ventricular hypertrophy in CKD. *J Am Soc Nephrol*. 2017.
- Marvar PJ, Gordon FJ, Harrison DG. Blood pressure control: salt gets under your skin. *Nat Med*. 2009;15:487–488.
- Titze J, et al. Spooky sodium balance. *Kidney Int*. 2013.
- Helle F, Karlsen TV, Tenstad O, et al. High-salt diet increases hormonal sensitivity in skin pre-capillary resistance vessels. *Acta Physiol (Oxf)*. 2013;207:577–581.
- Kurtz TW, et al. An alternative hypothesis to the widely held view that renal excretion of sodium accounts for resistance to salt-induced hypertension. *Kidney Int*. 2016;90:965–973.
- Anand IS. Cardiorenal syndrome: a cardiologist's perspective of pathophysiology. *Clin J Am Soc Nephrol*. 2013;8:1800–1807.
- Ali MH, Mungai PT, Schumacker PT. Stretch-induced phosphorylation of focal adhesion kinase in endothelial cells: role of mitochondrial oxidants. *Am J Physiol Lung Cell Mol Physiol*. 2006;291:L38–L45.
- Ali MH, Schumacker PT. Endothelial responses to mechanical stress: where is the mechanosensor? *Crit Care Med*. 2002;30:S198–S206.
- Gaucher C, et al. In vitro impact of physiological shear stress on endothelial cells gene expression profile. *Clin Hemorheol Microcirc*. 2007;37:99–107.
- Li YS, Haga JH, Chien S. Molecular basis of the effects of shear stress on vascular endothelial cells. *J Biomech*. 2005;38:1949–1971.
- Epstein M. Renal effects of head-out water immersion in humans: a 15-year update. *Physiol Rev*. 1992;72:563–621.
- Miller JA, Floras JS, Skorecki KL, et al. Renal and humoral responses to sustained cardiopulmonary baroreceptor deactivation in humans. *Am J Physiol*. 1991;260:R642–R648.
- Cowley AW Jr, Skelton MM. Dominance of colloid osmotic pressure in renal excretion after isotonic volume expansion. *Am J Physiol*. 1991;261:H1214–H1225.
- Johansen LB, Pump B, Warberg J, et al. Preventing hemodilution abolishes natriuresis of water immersion in humans. *Am J Physiol*. 1998;275:R879–R888.
- Kappagoda CT, Linden RJ, Sivananthan N. The nature of the atrial receptors responsible for a reflex increase in heart rate in the dog. *J Physiol*. 1979;291:393–412.
- Quail AW, Woods RL, Korner PI. Cardiac and arterial baroreceptor influences in release of vasopressin and renin during hemorrhage. *Am J Physiol*. 1987;252:H1120–H1126.
- Dibona GF, Sawin LL. Renal nerve activity in conscious rats during volume expansion and depletion. *Am J Physiol*. 1985;248:F15–F23.
- Myers BD, et al. Role of cardiac atria in the human renal response to changing plasma volume. *Am J Physiol*. 1988;254:F562–F573.
- Wurzner G, et al. Renal and neurohormonal responses to increasing levels of lower body negative pressure in men. *Kidney Int*. 2001;60:1469–1476.
- Tidgren B, Hjemdahl P, Theodorsson E, et al. Renal responses to lower body negative pressure in humans. *Am J Physiol*. 1990;259:F573–F579.
- Kaczmarczyk G, Schmidt E. Sodium homeostasis in conscious dogs after chronic cardiac denervation. *Am J Physiol*. 1990;258:F805–F811.
- Braith RW, Mills RM Jr, Wilcox CS, et al. Breakdown of blood pressure and body fluid homeostasis in heart transplant recipients. *J Am Coll Cardiol*. 1996;27:375–383.
- de Bold AJ, Borenstein HB, Veress AT, et al. A rapid and potent natriuretic response to intravenous injection of atrial myocardial extract in rats. *Life Sci*. 1981;28:89–94.
- de Bold AJ, de Bold ML. Determinants of natriuretic peptide production by the heart: basic and clinical implications. *J Invest Med*. 2005;53:371–377.
- Rubattu S, Sciarretta S, Valenti V, et al. Natriuretic peptides: an update on bioactivity, potential therapeutic use, and implication in cardiovascular diseases. *Am J Hypertens*. 2008;21:733–741.
- Potter LR, Yoder AR, Flora DR, et al. Natriuretic peptides: their structures, receptors, physiologic functions and therapeutic applications. *Handb Exp Pharmacol*. 2009;341–366.
- Piechota M, Banach M, Jacon A, et al. Natriuretic peptides in cardiovascular diseases. *Cell Mol Biol Lett*. 2008;13:155–181.
- Woodard GE, Rosado JA. Natriuretic peptides in vascular physiology and pathology. *Int Rev Cell Mol Biol*. 2008;268:59–93.
- Wu Q, Xu-Cai YO, Chen S, et al. Corin: new insights into the natriuretic peptide system. *Kidney Int*. 2009;75:142–146.
- Cho KW, Kim SH, Seul KH, et al. Effect of extracellular calcium depletion on the two-step ANP secretion in perfused rabbit atria. *Regul Pept*. 1994;52:129–137.
- Espinosa EA, Richards AM, Yandle TG, et al. Natriuretic hormones. *Endocrinol Metab Clin North Am*. 1995;24:481–509.
- Levin ER, Gardner DG, Samson WK. Natriuretic peptides. *N Engl J Med*. 1998;339:321–328.
- Singer DR, et al. Contrasting endocrine responses to acute oral compared with intravenous sodium loading in normal humans. *Am J Physiol*. 1998;274:F111–F119.
- Andersen LJ, Andersen JL, Pump B, et al. Natriuresis induced by mild hypernatremia in humans. *Am J Physiol Regul Integr Comp Physiol*. 2002;282:R1754–R1761.
- Bie P, Wamberg S, Kjolby M. Volume natriuresis vs. pressure natriuresis. *Acta Physiol Scand*. 2004;161:495–503.
- John SW, et al. Blood pressure and fluid-electrolyte balance in mice with reduced or absent ANP. *Am J Physiol*. 1996;271:R109–R114.
- Kuhn M. Cardiac and intestinal natriuretic peptides: insights from genetically modified mice. *Peptides*. 2005;26:1078–1085.
- Doe CP, et al. Reflex vascular responses in the anesthetized dog to large rapid changes in carotid sinus pressure. *Am J Physiol*. 1998;275:H1169–H1177.
- Paintal AS. Vagal sensory receptors and their reflex effects. *Physiol Rev*. 1973;53:159–227.
- McMahon NC, Drinkhill MJ, Hainsworth R. Absence of early resetting of coronary baroreceptors in anesthetized dogs. *J Physiol*. 1998;513(Pt 2):543–549.
- Tavi P, Laine M, Weckström M, et al. Cardiac mechanotransduction: from sensing to disease and treatment. *Trends Pharmacol Sci*. 2001;22:254–260.
- Bekheirnia MR, Schrier RW. Pathophysiology of water and sodium retention: edematous states with normal kidney function. *Curr Opin Pharmacol*. 2006;6:202–207.
- Kockskamper J, et al. The slow force response to stretch in atrial and ventricular myocardium from human heart: functional relevance and subcellular mechanisms. *Prog Biophys Mol Biol*. 2008;97:250–267.
- Sun H, Li DP, Chen SR, et al. Sensing of blood pressure increase by transient receptor potential vanilloid 1 receptors on baroreceptors. *J Pharmacol Exp Ther*. 2009;331:851–859.

58. Andresen MC, Doyle MW, Jin YH, et al. Cellular mechanisms of baroreceptor integration at the nucleus tractus solitarius. *Ann N Y Acad Sci.* 2001;940:132–141.
59. Alves FH, et al. Cannabidiol injected into the bed nucleus of the stria terminalis modulates baroreflex activity through 5-HT1A receptors. *Pharmacol Res.* 2010;62:228–236.
60. Ferreira-Junior NC, Fedoce AG, Alves FH, et al. Medial prefrontal cortex endocannabinoid system modulates baroreflex activity through CB(1) receptors. *Am J Physiol Regul Integr Comp Physiol.* 2012;302:R876–R885.
61. Gomez RA, Sequeira Lopez ML. Who and where is the renal baroreceptor?: the connexin hypothesis. *Kidney Int.* 2009;75:460–462.
62. Ichinose TK, Minic Z, Li C, et al. Activation of NTS A(1) adenosine receptors inhibits regional sympathetic responses evoked by activation of cardiopulmonary chemoreflex. *Am J Physiol Regul Integr Comp Physiol.* 2012;303:R539–R550.
63. Moore JP, Hainsworth R, Drinkhill MJ. Reflexes from pulmonary arterial baroreceptors in dogs: interaction with carotid sinus baroreceptors. *J Physiol.* 2011;589:4041–4052.
64. Kopp UC, Jones SY, Dibona GF. Afferent renal denervation impairs baroreflex control of efferent renal sympathetic nerve activity. *Am J Physiol Regul Integr Comp Physiol.* 2008;295:R1882–R1890.
65. Navar LG. Integrating multiple paracrine regulators of renal microvascular dynamics. *Am J Physiol.* 1998;274:F433–F444.
66. Saeed A, Dibona GF, Marcussen N, et al. High-NaCl intake impairs dynamic autoregulation of renal blood flow in ANG II-infused rats. *Am J Physiol Regul Integr Comp Physiol.* 2010;299:R1142–R1149.
67. Dautzenberg M, Keilhoff G, Just A. Modulation of the myogenic response in renal blood flow autoregulation by NO depends on endothelial nitric oxide synthase (eNOS), but not neuronal or inducible NOS. *J Physiol.* 2011;589:4731–4744.
68. Schnermann J, Homer W. Smith award lecture. The juxtaglomerular apparatus: from anatomical peculiarity to physiological relevance. *J Am Soc Nephrol.* 2003;14:1681–1694.
69. Oppermann M, et al. Macula densa control of renin secretion and preglomerular resistance in mice with selective deletion of the B isoform of the Na,K,2Cl co-transporter. *J Am Soc Nephrol.* 2006;17:2143–2152.
70. Castrop H, et al. Contribution of the basolateral isoform of the Na-K-2Cl cotransporter (NKCC1/BSC2) to renin secretion. *Am J Physiol Renal Physiol.* 2005;289:F1185–F1192.
71. Komlosi P, Fintha A, Bell PD. Current mechanisms of macula densa cell signalling. *Acta Physiol Scand.* 2004;181:463–469.
72. Oppermann M, et al. Tubuloglomerular feedback and renin secretion in NTPDase1/CD39-deficient mice. *Am J Physiol Renal Physiol.* 2008;294:F965–F970.
73. Dibona GF. Physiology in perspective: the wisdom of the body. Neural control of the kidney. *Am J Physiol Regul Integr Comp Physiol.* 2005;289:R633–R641.
74. Secliger E, et al. The renin-angiotensin system and the third mechanism of renal blood flow autoregulation. *Am J Physiol Renal Physiol.* 2009;296:F1334–F1345.
75. Komlosi P, Bell PD, Zhang ZR. Tubuloglomerular feedback mechanisms in nephron segments beyond the macula densa. *Curr Opin Nephrol Hypertens.* 2009;18:57–62.
76. Bolanos L, Colina I, Purroy A. Intracerebroventricular infusion of hypertonic NaCl increases urinary CGMP in healthy and cirrhotic rats. *Arch Physiol Biochem.* 1999;107:323–333.
77. Hansell P, Isaksson B, Sjoquist M. Renal dopamine and noradrenaline excretion during CNS-induced natriuresis in spontaneously hypertensive rats: influence of dietary sodium. *Acta Physiol Scand.* 2000;168:257–266.
78. Zhu GQ, Gao L, Patel KP, et al. ANG II in the paraventricular nucleus potentiates the cardiac sympathetic afferent reflex in rats with heart failure. *J Appl Physiol.* 2004;97:1746–1754.
79. Carey RM. Evidence for a splanchnic sodium input monitor regulating renal sodium excretion in man. Lack of dependence upon aldosterone. *Circ Res.* 1978;43:19–23.
80. Morita H, et al. Effects of portal infusion of hypertonic solution on jejunal electrolyte transport in anesthetized dogs. *Am J Physiol.* 1990;259:R1289–R1294.
81. Morita H, Matsuda T, Furuya F, et al. Hepatorenal reflex plays an important role in natriuresis after high-NaCl food intake in conscious dogs. *Circ Res.* 1993;72:552–559.
82. Morita H, Matsuda T, Tanaka K, et al. Role of hepatic receptors in controlling body fluid homeostasis. *Jpn J Physiol.* 1995;45:355–368.
83. Matsuda T, Morita H, Hosomi H, et al. Response of renal nerve activity to high NaCl food intake in dogs with chronic bile duct ligation. *Hepatology.* 1996;23:303–309.
84. Ming Z, Lauth WW. Intrahepatic adenosine-mediated activation of hepatorenal reflex is via A1 receptors in rats. *Can J Physiol Pharmacol.* 2006;84:1177–1184.
85. Ming Z, Fan YJ, Yang X, et al. Contribution of hepatic adenosine A1 receptors to renal dysfunction associated with acute liver injury in rats. *Hepatology.* 2006;44:813–822.
86. Morita H, Fujiki N, Hagiike M, et al. Functional evidence for involvement of bumetanide-sensitive Na+K+2Cl cotransport in the hepatoportal Na+ receptor of the Sprague-Dawley rat. *Neurosci Lett.* 1999;264:65–68.
87. Oliver JA, Verna EC. Afferent mechanisms of sodium retention in cirrhosis and hepatorenal syndrome. *Kidney Int.* 2010;77:669–680.
88. Blendis L. Budd-Chiari Syndrome: a clinical model of the hepatorenal reflex? *Gastroenterology.* 2006;131:671–672.
89. Levy M, Wexler MJ. Sodium excretion in dogs with low-grade caval constriction: role of hepatic nerves. *Am J Physiol.* 1987;253:F672–F678.
90. Koyama S, Kanai K, Aibiki M, et al. Reflex increase in renal nerve activity during acutely altered portal venous pressure. *J Auton Nerv Syst.* 1988;23:55–62.
91. Thomas L, Kumar R. Control of renal solute excretion by enteric signals and mediators. *J Am Soc Nephrol.* 2008;19:207–212.
92. Sindic A, Schlatter E. Cellular effects of guanylin and uroguanylin. *J Am Soc Nephrol.* 2006;17:607–616.
93. Sindic A, Schlatter E. Renal electrolyte effects of guanylin and uroguanylin. *Curr Opin Nephrol Hypertens.* 2007;16:10–15.
94. Jose PA, Yang Z, Zeng C, et al. The importance of the gastrosplanchnic axis in the control of body sodium homeostasis. *Exp Physiol.* 2016;101:465–470.
95. Fukae H, et al. Changes in urinary levels and renal expression of uroguanylin on low or high salt diets in rats. *Nephron.* 2002;92:373–378.
96. Lorenz JN, et al. Uroguanylin knockout mice have increased blood pressure and impaired natriuretic response to enteral NaCl load. *J Clin Invest.* 2003;112:1244–1254.
97. Kinoshita H, et al. Urine and plasma levels of uroguanylin and its molecular forms in renal diseases. *Kidney Int.* 1997;52:1028–1034.
98. Santos-Neto MS, Carvalho AF, Monteiro HS, et al. Interaction of atrial natriuretic peptide, urodilatin, guanylin and uroguanylin in the isolated perfused rat kidney. *Regul Pept.* 2006;136:14–22.
99. Preston RA, et al. Sodium challenge does not support an acute gastrointestinal-renal natriuretic signaling axis in humans. *Kidney Int.* 2012;82:1313–1320.
100. Peterson TV, Chase NL, Gray DK. Renal effects of volume expansion in the renal-denervated nonhuman primate. *Am J Physiol.* 1984;247:H960–H966.
101. Peterson TV, Jones CE. Renal responses of the cardiac-denervated nonhuman primate to blood volume expansion. *Circ Res.* 1983;53:24–32.
102. Schnermann J. Juxtaglomerular cell complex in the regulation of renal salt excretion. *Am J Physiol.* 1998;274:R263–R279.
103. Thomson SC, Blantz RC. Glomerulotubular balance, tubuloglomerular feedback, and salt homeostasis. *J Am Soc Nephrol.* 2008;19:2272–2275.
104. Guyton AC, Langston JB, Navar G. Theory for renal autoregulation by feedback at the juxtaglomerular apparatus. *Circ Res.* 1964;15:SUPPL–97.
105. Schnermann J, Briggs JP. Tubuloglomerular feedback: mechanistic insights from gene-manipulated mice. *Kidney Int.* 2008;74:418–426.
106. Castrop H. Mediators of tubuloglomerular feedback regulation of glomerular filtration: ATP and adenosine. *Acta Physiol (Oxf).* 2007;189:3–14.
107. Takahashi N, et al. Uncompensated polyuria in a mouse model of Bartter's syndrome. *Proc Natl Acad Sci USA.* 2000;97:5434–5439.
108. Bell PD, Lapointe JY, Peti-Peterdi J. Macula densa cell signaling. *Annu Rev Physiol.* 2003;65:481–500.
109. Hansen PB, Castrop H, Briggs J, et al. Adenosine induces vasoconstriction through Gi-dependent activation of phospholipase C in isolated perfused afferent arterioles of mice. *J Am Soc Nephrol.* 2003;14:2457–2465.
110. Lai EY, et al. Contribution of adenosine receptors in the control of arteriolar tone and adenosine-angiotensin II interaction. *Kidney Int.* 2006;70:690–698.
111. Gao X, et al. Adenosine A(1)-receptor deficiency diminishes afferent arteriolar and blood pressure responses during nitric oxide

- inhibition and angiotensin II treatment. *Am J Physiol Regul Integr Comp Physiol*. 2011;301:R1669–R1681.
112. Li L, et al. Renal afferent arteriolar and tubuloglomerular feedback reactivity in mice with conditional deletions of adenosine 1 receptors. *Am J Physiol Renal Physiol*. 2012;303:F1166–F1175.
 113. Hansen PB, Friis UG, Uhrenholt TR, et al. Intracellular signalling pathways in the vasoconstrictor response of mouse afferent arterioles to adenosine. *Acta Physiol (Oxf)*. 2007;191:89–97.
 114. Li L, et al. Tubuloglomerular feedback and renal function in mice with targeted deletion of the type 1 equilibrative nucleoside transporter. *Am J Physiol Renal Physiol*. 2013;304:F382–F389.
 115. Al-Mashhadi RH, Skott O, Vanhouette PM, et al. Activation of A(2) adenosine receptors dilates cortical efferent arterioles in mouse. *Kidney Int*. 2009;75:793–799.
 116. Fu Y, et al. Aldosterone blunts tubuloglomerular feedback by activating macula densa mineralocorticoid receptors. *Hypertension*. 2012;59:599–606.
 117. Zhu X, et al. Aldosterone stimulates superoxide production in macula densa cells. *Am J Physiol Renal Physiol*. 2011;301:F529–F535.
 118. Sedeek M, Nasrallah R, Touyz RM, et al. NADPH oxidases, reactive oxygen species, and the kidney: friend and foe. *J Am Soc Nephrol*. 2013;24:1512–1518.
 119. Blantz RC, et al. The complex role of nitric oxide in the regulation of glomerular ultrafiltration. *Kidney Int*. 2002;61:782–785.
 120. Patzak A, Persson AE. Angiotensin II-nitric oxide interaction in the kidney. *Curr Opin Nephrol Hypertens*. 2007;16:46–51.
 121. Herrera M, Garvin JL. Recent advances in the regulation of nitric oxide in the kidney. *Hypertension*. 2005;45:1062–1067.
 122. Satriano J, et al. Regulation of ecto-5'-nucleotidase by NaCl and nitric oxide: potential roles in tubuloglomerular feedback and adaptation. *Am J Physiol Renal Physiol*. 2006;291:F1078–F1082.
 123. Zhang Q, et al. Interaction between nitric oxide and superoxide in the macula densa in aldosterone-induced alterations of tubuloglomerular feedback. *Am J Physiol Renal Physiol*. 2013;304:F326–F332.
 124. Thomson SC, Kashkouli A, Singh P. Glucagon-like peptide-1 receptor stimulation increases GFR and suppresses proximal reabsorption in the rat. *Am J Physiol Renal Physiol*. 2013;304:F137–F144.
 125. Blantz RC, Singh P, Deng A, et al. Acute saline expansion increases nephron filtration and distal flow rate but maintains tubuloglomerular feedback responsiveness: role of adenosine A(1) receptors. *Am J Physiol Renal Physiol*. 2012;303:F405–F411.
 126. Carlstrom M, Wilcox CS, Welch WJ. Adenosine A2A receptor activation attenuates tubuloglomerular feedback responses by stimulation of endothelial nitric oxide synthase. *Am J Physiol Renal Physiol*. 2011;300:F457–F464.
 127. Bell TD, Welch WJ. Regulation of renal arteriolar tone by adenosine: novel role for type 2 receptors. *Kidney Int*. 2009;75:769–771.
 128. Schnermann J. Maintained tubuloglomerular feedback responses during acute inhibition of P2 purinergic receptors in mice. *Am J Physiol Renal Physiol*. 2011;300:F339–F344.
 129. Hanner F, Sorensen CM, Holstein-Rathlou NH, et al. Connexins and the kidney. *Am J Physiol Regul Integr Comp Physiol*. 2010;298:R1143–R1155.
 130. Arendshorst WJ. Connexin 40 mediates tubuloglomerular feedback paracrine signaling by coupling tubular and vascular cells in the renal juxtaglomerular apparatus. *Am J Physiol Renal Physiol*. 2012;303:F1409–F1411.
 131. Just A, et al. Connexin 40 mediates the tubuloglomerular feedback contribution to renal blood flow autoregulation. *J Am Soc Nephrol*. 2009;20:1577–1585.
 132. Ge Y, et al. Endogenously produced 20-HETE modulates myogenic and TGF response in microperfused afferent arterioles. *Prostaglandins Other Lipid Mediat*. 2013;102:103:42–48. doi:10.1016/j.prostaglandins.2013.03.001. Epub 2013 Mar 14.
 133. Ren Y, D'Ambrosio MA, Garvin JL, et al. Mechanism of inhibition of tubuloglomerular feedback by CO and cGMP. *Hypertension*. 2013.
 134. Ren Y, et al. Mechanisms of carbon monoxide attenuation of tubuloglomerular feedback. *Hypertension*. 2012;59:1139–1144.
 135. Fu Y, et al. Testosterone enhances tubuloglomerular feedback by increasing superoxide production in the macula densa. *Am J Physiol Regul Integr Comp Physiol*. 2013.
 136. Brown RD, et al. Sex differences in the pressor and tubuloglomerular feedback response to angiotensin II. *Hypertension*. 2012;59:129–135.
 137. Guan Z, Pollock JS, Cook AK, et al. Effect of epithelial sodium channel blockade on the myogenic response of rat juxtaglomerular afferent arterioles. *Hypertension*. 2009;54:1062–1069.
 138. Ren Y, Garvin JL, Liu R, et al. Crosstalk between the connecting tubule and the afferent arteriole regulates renal microcirculation. *Kidney Int*. 2007;71:1116–1121.
 139. Ren Y, D'Ambrosio MA, Garvin JL, et al. Possible mediators of connecting tubule glomerular feedback. *Hypertension*. 2009;53:319–323.
 140. Wang H, D'Ambrosio MA, Garvin JL, et al. Connecting tubule glomerular feedback mediates acute tubuloglomerular feedback resetting. *Am J Physiol Renal Physiol*. 2012;302:F1300–F1304.
 141. Brenner BM, Troy JL. Postglomerular vascular protein concentration: evidence for a causal role in governing fluid reabsorption and glomerulotubular balance by the renal proximal tubule. *J Clin Invest*. 1971;50:336–349.
 142. Ichikawa I, Brenner BM. Importance of efferent arteriolar vascular tone in regulation of proximal tubule fluid reabsorption and glomerulotubular balance in the rat. *J Clin Invest*. 1980;65:1192–1201.
 143. Ichikawa I, Brenner BM. Mechanism of inhibition of proximal tubule fluid reabsorption after exposure of the rat kidney to the physical effects of expansion of extracellular fluid volume. *J Clin Invest*. 1979;64:1466–1474.
 144. Skorecki KL, Brenner BM. Body fluid homeostasis in congestive heart failure and cirrhosis with ascites. *Am J Med*. 1982;72:323–338.
 145. Imai M, Kokko JP. Effect of peritubular protein concentration on reabsorption of sodium and water in isolated perfused proximal tubules. *J Clin Invest*. 1972;51:314–325.
 146. Garcia NH, Ramsey CR, Knox FG. Understanding the role of paracellular transport in the proximal tubule. *News Physiol Sci*. 1998;13:38–43.
 147. Angelow S, Ahlstrom R, Yu AS. Biology of claudins. *Am J Physiol Renal Physiol*. 2008;295:F867–F876.
 148. Angelow S, Yu AS. Claudins and paracellular transport: an update. *Curr Opin Nephrol Hypertens*. 2007;16:459–464.
 149. Balkovetz DF. Tight junction claudins and the kidney in sickness and in health. *Biochim Biophys Acta*. 2009;1788:858–863.
 150. Kiuchi-Shaishin Y, et al. Differential expression patterns of claudins, tight junction membrane proteins, in mouse nephron segments. *J Am Soc Nephrol*. 2002;13:875–886.
 151. Enck AH, Berger UV, Yu AS. Claudin-2 is selectively expressed in proximal nephron in mouse kidney. *Am J Physiol Renal Physiol*. 2001;281:F966–F974.
 152. Gong Y, Hou J. Claudins in barrier and transport function—the kidney. *Pflugers Arch*. 2017;469:105–113.
 153. Romano G, Favret G, Damato R, et al. Proximal reabsorption with changing tubular fluid inflow in rat nephrons. *Exp Physiol*. 1998;83:35–48.
 154. Andreoli TE. An overview of salt absorption by the nephron. *J Nephrol*. 1999;12(suppl 2):S3–S15.
 155. Jamison RL, Sonnenberg H, Stein JH. Questions and replies: role of the collecting tubule in fluid, sodium, and potassium balance. *Am J Physiol*. 1979;237:F247–F261.
 156. Earley LE, Friedler RM. Changes in renal blood flow and possibly the intrarenal distribution of blood during the natriuresis accompanying saline loading in the dog. *J Clin Invest*. 1965;44:929–941.
 157. Friedler RM, Belleau LJ, Martino JA, et al. Hemodynamically induced natriuresis in the presence of sodium retention resulting from constriction of the thoracic inferior vena cava. *J Lab Clin Med*. 1967;69:565–583.
 158. Hall JE. The kidney, hypertension, and obesity. *Hypertension*. 2003;41:625–633.
 159. Hall JE, Guyton AC, Brands MW. Pressure-volume regulation in hypertension. *Kidney Int Suppl*. 1996;55:S35–S41.
 160. Cowley AW Jr. Role of the renal medulla in volume and arterial pressure regulation. *Am J Physiol*. 1997;273:R1–R15.
 161. Navar LG, Majid DS. Interactions between arterial pressure and sodium excretion. *Curr Opin Nephrol Hypertens*. 1996;5:64–71.
 162. Roman RJ, Zou AP. Influence of the renal medullary circulation on the control of sodium excretion. *Am J Physiol*. 1993;265:R963–R973.
 163. Cowley AW Jr, Mattson DL, Lu S, et al. The renal medulla and hypertension. *Hypertension*. 1995;25:663–673.
 164. Cowley AW Jr, Mori T, Mattson D, et al. Role of renal NO production in the regulation of medullary blood flow. *Am J Physiol Regul Integr Comp Physiol*. 2003;284:R1355–R1369.
 165. Mattson DL. Importance of the renal medullary circulation in the control of sodium excretion and blood pressure. *Am J Physiol Regul Integr Comp Physiol*. 2003;284:R13–R27.
 166. Granger JP. Pressure natriuresis. Role of renal interstitial hydrostatic pressure. *Hypertension*. 1992;19:19–117.

167. Granger JP, Alexander BT, Llinas M. Mechanisms of pressure natriuresis. *Curr Hypertens Rep.* 2002;4:152–159.
168. Evans RG, Majid DS, Eppel GA. Mechanisms mediating pressure natriuresis: what we know and what we need to find out. *Clin Exp Pharmacol Physiol.* 2005;32:400–409.
169. Dos Santos EA, Dahly-Vernon AJ, Hoagland KM, et al. Inhibition of the formation of EETs and 20-HETE with 1-aminobenzotriazole attenuates pressure natriuresis. *Am J Physiol Regul Integr Comp Physiol.* 2004;287:R58–R68.
170. Magyar CE, Zhang Y, Holstein-Rathlou NH, et al. Proximal tubule Na transporter responses are the same during acute and chronic hypertension. *Am J Physiol Renal Physiol.* 2000;279:F358–F369.
171. Majid DS, Navar LG. Blockade of distal nephron sodium transport attenuates pressure natriuresis in dogs. *Hypertension.* 1994;23:1040–1045.
172. Kline RL, Liu F. Modification of pressure natriuresis by long-term losartan in spontaneously hypertensive rats. *Hypertension.* 1994;24:467–473.
173. Hilliard LM, et al. Gender differences in pressure-natriuresis and renal autoregulation: role of the angiotensin type 2 receptor. *Hypertension.* 2011;57:275–282.
174. Jin C, et al. Effects of renal perfusion pressure on renal medullary hydrogen peroxide and nitric oxide production. *Hypertension.* 2009;53:1048–1053.
175. O'Connor PM, Cowley AW Jr. Modulation of pressure-natriuresis by renal medullary reactive oxygen species and nitric oxide. *Curr Hypertens Rep.* 2010;12:86–92.
176. Kompanowska-Jezierska E, et al. Renal nerves and nNOS: roles in natriuresis of acute isovolumetric sodium loading in conscious rats. *Am J Physiol Regul Integr Comp Physiol.* 2008;294:R1130–R1139.
177. Salom MG, Lahera V, Miranda-Guardiola F, et al. Blockade of pressure natriuresis induced by inhibition of renal synthesis of nitric oxide in dogs. *Am J Physiol.* 1992;262:F718–F722.
178. Patel AR, Granger JP, Kirchner KA. L-arginine improves transmission of perfusion pressure to the renal interstitium in Dahl salt-sensitive rats. *Am J Physiol.* 1994;266:R1730–R1735.
179. Majid DS, Godfrey M, Grisham MB, et al. Relation between pressure natriuresis and urinary excretion of nitrate/nitrite in anesthetized dogs. *Hypertension.* 1995;25:860–865.
180. Li N, Yi F, Dos Santos EA, et al. Role of renal medullary heme oxygenase in the regulation of pressure natriuresis and arterial blood pressure. *Hypertension.* 2007;49:148–154.
181. Sarkis A, Lopez B, Roman RJ. Role of 20-hydroxyecosatetraenoic acid and epoxyecosatrienoic acids in hypertension. *Curr Opin Nephrol Hypertens.* 2004;13:205–214.
182. Williams JM, et al. Elevations in renal interstitial hydrostatic pressure and 20-hydroxyecosatetraenoic acid contribute to pressure natriuresis. *Hypertension.* 2007;49:687–694.
183. Sipos A, et al. Connexin 30 deficiency impairs renal tubular ATP release and pressure natriuresis. *J Am Soc Nephrol.* 2009;20:1724–1732.
184. Reinhardt HW, Seeliger E. Toward an integrative concept of control of total body sodium. *News Physiol Sci.* 2000;15:319–325.
185. Rasmussen MS, Simonsen JA, Sandgaard NC, et al. Mechanisms of acute natriuresis in normal humans on low sodium diet. *J Physiol.* 2003;546:591–603.
186. Sandgaard NC, Andersen JL, Bie P. Hormonal regulation of renal sodium and water excretion during normotensive sodium loading in conscious dogs. *Am J Physiol Regul Integr Comp Physiol.* 2000;278:R11–R18.
187. Denton KM, Luff SE, Shweta A, et al. Differential neural control of glomerular ultrafiltration. *Clin Exp Pharmacol Physiol.* 2004;31:380–386.
188. Barajas L, Powers K. Monoaminergic innervation of the rat kidney: a quantitative study. *Am J Physiol.* 1990;259:F503–F511.
189. Dibona GF. Neural control of the kidney: functionally specific renal sympathetic nerve fibers. *Am J Physiol Regul Integr Comp Physiol.* 2000;279:R1517–R1524.
190. Eppel GA, Malpas SC, Denton KM, et al. Neural control of renal medullary perfusion. *Clin Exp Pharmacol Physiol.* 2004;31:387–396.
191. Dibona GF, Kopp UC. Neural control of renal function. *Physiol Rev.* 1997;77:175–197.
192. Jeffries WB, Pettinger WA. Adrenergic signal transduction in the kidney. *Miner Electrolyte Metab.* 1989;15:5–15.
193. Matsushima Y, Akabane S, Ito K. Characterization of alpha 1- and alpha 2-adrenoceptors directly associated with basolateral membranes from rat kidney proximal tubules. *Biochem Pharmacol.* 1986;35:2593–2600.
194. Summers RJ, Stephenson JA, Kuhar MJ. Localization of beta adrenoceptor subtypes in rat kidney by light microscopic autoradiography. *J Pharmacol Exp Ther.* 1985;232:561–569.
195. Dibona GF. Sympathetic nervous system and the kidney in hypertension. *Curr Opin Nephrol Hypertens.* 2002;11:197–200.
196. Friberg P, et al. Evidence for increased renal norepinephrine overflow during sodium restriction in humans. *Hypertension.* 1990;16:121–130.
197. McMurray JJ, Seidelin PH, Balfour DJ, et al. Physiological increases in circulating noradrenaline are antinatriuretic in man. *J Hypertens.* 1988;6:757–761.
198. Aperia A, Ibarra F, Svensson LB, et al. Calcineurin mediates alpha-adrenergic stimulation of Na⁺/K⁺(+)-ATPase activity in renal tubule cells. *Proc Natl Acad Sci USA.* 1992;89:7394–7397.
199. Nelson LD, Osborn JL. Role of intrarenal ANG II in reflex neural stimulation of plasma renin activity and renal sodium reabsorption. *Am J Physiol.* 1993;265:R392–R398.
200. Simon JK, Kasting NW, Ciriello J. Afferent renal nerve effects on plasma vasopressin and oxytocin in conscious rats. *Am J Physiol.* 1989;256:R1240–R1244.
201. Awazu M, et al. Renal sympathetic nerves modulate glomerular ANP receptors and filtration. *Am J Physiol.* 1991;261:F29–F35.
202. Brewster UC, Perazella MA. The renin-angiotensin-aldosterone system and the kidney: effects on kidney disease. *Am J Med.* 2004;116:263–272.
203. Schmieder RE. Mechanisms for the clinical benefits of angiotensin II receptor blockers. *Am J Hypertens.* 2005;18:720–730.
204. Santos RA, Ferreira AJ, Simões E, Silva AC. Recent advances in the angiotensin-converting enzyme 2-angiotensin(1-7)-Mas axis. *Exp Physiol.* 2008;93:519–527.
205. Schindler C, Bramlage P, Kirch W, et al. Role of the vasodilator peptide angiotensin-(1-7) in cardiovascular drug therapy. *Vasc Health Risk Manag.* 2007;3:125–137.
206. Kobori H, Nangaku M, Navar LG, et al. The intrarenal renin-angiotensin system: from physiology to the pathobiology of hypertension and kidney disease. *Pharmacol Rev.* 2007;59:251–287.
207. Seikaly MG, Arant BS Jr, Seney FD Jr. Endogenous angiotensin concentrations in specific intrarenal fluid compartments of the rat. *J Clin Invest.* 1990;86:1352–1357.
208. Matavelli LC, Siragy HM. AT2 receptor activities and pathophysiological implications. *J Cardiovasc Pharmacol.* 2015;65:226–232.
209. Fogo AB. Angiotensin receptors: beyond number one. *Curr Opin Nephrol Hypertens.* 2004;13:275–277.
210. Thomas WG, Mendelsohn FA. Angiotensin receptors: form and function and distribution. *Int J Biochem Cell Biol.* 2003;35:774–779.
211. Karnik SS, Khurajam D, Tirupula K, et al. Significance of Ang(1-7) coupling with MAS1 and other GPCRs to the Renin-Angiotensin system: IUPHAR review “X”. *Br J Pharmacol.* 2017.
212. Navar LG, et al. Paracrine regulation of the renal microcirculation. *Physiol Rev.* 1996;76:425–536.
213. Pagtalunan ME, Rasch R, Rennke HG, et al. Morphometric analysis of effects of angiotensin II on glomerular structure in rats. *Am J Physiol.* 1995;268:F82–F88.
214. Zhao D, Navar LG. Acute angiotensin II infusions elicit pressure natriuresis in mice and reduce distal fractional sodium reabsorption. *Hypertension.* 2008;52:137–142.
215. Satou R, Shao W, Navar LG. Role of stimulated intrarenal angiotensinogen in hypertension. *Ther Adv Cardiovasc Dis.* 2015;9:181–190.
216. Quan A, Baum M. Endogenous angiotensin II modulates rat proximal tubule transport with acute changes in extracellular volume. *Am J Physiol.* 1998;275:F74–F78.
217. Xu L, et al. Regulation of Na⁺/H⁺ exchanger-NHE3 by angiotensin-II in OKP cells. *Biochim Biophys Acta.* 2006;1758:519–526.
218. Zhou Y, Boron WF. Role of endogenously secreted angiotensin II in the CO₂-induced stimulation of HCO₃⁻ reabsorption by renal proximal tubules. *Am J Physiol Renal Physiol.* 2008;294:F245–F252.
219. Seldin DW, Preisig PA, Alpern RJ. Regulation of proximal reabsorption by effective arterial blood volume. *Semin Nephrol.* 1991;11:212–219.
220. Wang T, Giebisch G. Effects of angiotensin II on electrolyte transport in the early and late distal tubule in rat kidney. *Am J Physiol.* 1996;271:F143–F149.

221. Levine DZ, Iacovitti M, Buckman S, et al. Role of angiotensin II in dietary modulation of rat late distal tubule bicarbonate flux in vivo. *J Clin Invest*. 1996;97:120–125.
222. Loffing J, Korbmayer C. Regulated sodium transport in the renal connecting tubule (CNT) via the epithelial sodium channel (ENaC). *Pflugers Arch*. 2009;458:111–135.
223. Bader M. ACE2, angiotensin-(1-7), and Mas: the other side of the coin. *Pflugers Arch*. 2013;465:79–85.
224. Goodfriend TL. Aldosterone—a hormone of cardiovascular adaptation and maladaptation. *J Clin Hypertens (Greenwich)*. 2006;8:133–139.
225. Rad AK, Balment RJ, Ashton N. Rapid natriuretic action of aldosterone in the rat. *J Appl Physiol*. 2005;98:423–428.
226. Ashton N. Renal and vascular actions of urotensin II. *Kidney Int*. 2006;70:624–629.
227. Salyer SA, et al. Aldosterone regulates Na(+), K(+) ATPase activity in human renal proximal tubule cells through mineralocorticoid receptor. *Biochim Biophys Acta*. 2013;1833:2143–2152.
228. Williams GH. Aldosterone biosynthesis, regulation, and classical mechanism of action. *Heart Fail Rev*. 2005;10:7–13.
229. Pearce D, et al. Collecting duct principal cell transport processes and their regulation. *Clin J Am Soc Nephrol*. 2015;10:135–146.
230. Dooley R, Angibaud E, Yusef YR, et al. Aldosterone-induced ENaC and basal Na+/K+-ATPase trafficking via protein kinase D1-phosphatidylinositol 4-kinaseIIbeta trans Golgi signalling in M1 cortical collecting duct cells. *Mol Cell Endocrinol*. 2013;372:86–95.
231. Gunz ML, et al. The circadian clock protein Period 1 regulates expression of the renal epithelial sodium channel in mice. *J Clin Invest*. 2009.
232. Grimm PR, et al. SPAK isoforms and OSR1 regulate sodium-chloride co-transporters in a nephron-specific manner. *J Biol Chem*. 2012;287:37673–37690.
233. Hall JE, Brands MW, Henegar JR. Angiotensin II and long-term arterial pressure regulation: the overriding dominance of the kidney. *J Am Soc Nephrol*. 1999;10(suppl 12):S258–S265.
234. Hartupee J, Mann DL. Neurohormonal activation in heart failure with reduced ejection fraction. *Nat Rev Cardiol*. 2017;14:30–38.
235. Schiffrin EL. Effects of aldosterone on the vasculature. *Hypertension*. 2006;47:312–318.
236. Kortenoeven ML, Pedersen NB, Rosenbaek LL, et al. Vasopressin regulation of sodium transport in the distal nephron and collecting duct. *Am J Physiol Renal Physiol*. 2015;309:F280–F299.
237. Serradeil-Le GC, et al. Nonpeptide vasopressin receptor antagonists: development of selective and orally active V1a, V2 and V1b receptor ligands. *Prog Brain Res*. 2002;139:197–210.
238. Brown D, Hasler U, Nunes P, et al. Phosphorylation events and the modulation of aquaporin 2 cell surface expression. *Curr Opin Nephrol Hypertens*. 2008;17:491–498.
239. Bankir L. Antidiuretic action of vasopressin: quantitative aspects and interaction between V1a and V2 receptor-mediated effects. *Cardiovasc Res*. 2001;51:372–390.
240. Nielsen S, Kwon TH, Frokier J, et al. Regulation and dysregulation of aquaporins in water balance disorders. *J Intern Med*. 2007;261:53–64.
241. Esteve-Font C, Anderson MO, Verkman AS. Urea transporter proteins as targets for small-molecule diuretics. *Nat Rev Nephrol*. 2015;11:113–123.
242. Knepper MA, Kwon TH, Nielsen S. Molecular physiology of water balance. *N Engl J Med*. 2015;372:1349–1358.
243. Cowley AW Jr. Control of the renal medullary circulation by vasopressin V1 and V2 receptors in the rat. *Exp Physiol*. 2000;85(Spec No):223S–231S.
244. Aoyagi T, et al. Vasopressin regulates the renin-angiotensin-aldosterone system via V1a receptors in macula densa cells. *Am J Physiol Renal Physiol*. 2008;295:F100–F107.
245. Yasuoka Y, et al. Decreased expression of aquaporin 2 in the collecting duct of mice lacking the vasopressin V1a receptor. *Clin Exp Nephrol*. 2013;17:183–190.
246. Maybauer MO, Maybauer DM, Enkhbaatar P, et al. Physiology of the vasopressin receptors. *Best Pract Res Clin Anaesthesiol*. 2008;22:253–263.
247. Pelletier JS, Dicken B, Bigam D, et al. Cardiac effects of vasopressin. *J Cardiovasc Pharmacol*. 2014;64:100–107.
248. Olesen ET, Fenton RA. Is there a role for PGE2 in urinary concentration? *J Am Soc Nephrol*. 2013;24:169–178.
249. Nasrallah R, Clark J, Hebert RL. Prostaglandins in the kidney: developments since Y2K. *Clin Sci*. 2007;113:297–311.
250. Hao CM, Breyer MD. Physiological regulation of prostaglandins in the kidney. *Annu Rev Physiol*. 2008;70:357–377.
251. Simmons DL, Botting RM, Hla T. Cyclooxygenase isozymes: the biology of prostaglandin synthesis and inhibition. *Pharmacol Rev*. 2004;56:387–437.
252. Kramer BK, Kammerl MC, Komhoff M. Renal cyclooxygenase-2 (COX-2). Physiological, pathophysiological, and clinical implications. *Kidney Blood Press Res*. 2004;27:43–62.
253. Dunn MJ. Prostaglandin I2 and the kidney. *Arch Mal Coeur Vaiss*. 1989;82(Spec4):27–31.
254. Johns EJ, Kopp UC, Dibona GF. Neural control of renal function. *Compr Physiol*. 2011;1:731–767.
255. Scharschmidt LA, Douglas JG, Dunn MJ. Angiotensin II and eicosanoids in the control of glomerular size in the rat and human. *Am J Physiol*. 1986;250:F348–F356.
256. Laffi G, La VG, Pinzani M, et al. Arachidonic acid derivatives and renal function in liver cirrhosis. *Semin Nephrol*. 1997;17:530–548.
257. Sandhu GK, Heyneman CA. Nephrotoxic potential of selective cyclooxygenase-2 inhibitors. *Ann Pharmacother*. 2004;38:700–704.
258. Haas JA, Hammond TG, Granger JP, et al. Mechanism of natriuresis during intrarenal infusion of prostaglandins. *Am J Physiol*. 1984;247:F475–F479.
259. Bonvalet JP, Pradelles P, Farman N. Segmental synthesis and actions of prostaglandins along the nephron. *Am J Physiol*. 1987;253:F377–F387.
260. Rubinger D, Wald H, Scherzer P, et al. Renal sodium handling and stimulation of medullary Na-K-ATPase during blockade of prostaglandin synthesis. *Prostaglandins*. 1990;39:179–194.
261. Bonilla-Felix M. Development of water transport in the collecting duct. *Am J Physiol Renal Physiol*. 2004;287:F1093–F1101.
262. Breyer MD, Hao C, Qi Z. Cyclooxygenase-2 selective inhibitors and the kidney. *Curr Opin Crit Care*. 2001;7:393–400.
263. Qi Z, et al. Opposite effects of cyclooxygenase-1 and -2 activity on the pressor response to angiotensin II. *J Clin Invest*. 2002;110:61–69.
264. Volpe M, Carnovali M, Mastromarino V. The natriuretic peptides system in the pathophysiology of heart failure: from molecular basis to treatment. *Clin Sci*. 2016;130:57–77.
265. Kerkela R, Ulvila J, Magga J. Natriuretic peptides in the regulation of cardiovascular physiology and metabolic events. *J Am Heart Assoc*. 2015;4:e002423.
266. Kishimoto I, Tokudome T, Nakao K, et al. Natriuretic peptide system: an overview of studies using genetically engineered animal models. *FEBS J*. 2011;278:1830–1841.
267. Nishikimi T, Kuwahara K, Nakao K. Current biochemistry, molecular biology, and clinical relevance of natriuretic peptides. *J Cardiol*. 2011;57:131–140.
268. Kim SM, Kim SY, Kim SH, et al. Renal actions of dendroaspis natriuretic peptide in rabbits. *Peptides*. 2012;33:59–66.
269. Richards AM. Natriuretic peptides: update on peptide release, bioactivity, and clinical use. *Hypertension*. 2007;50:25–30.
270. Scotland RS, Ahluwalia A, Hobbs AJ. C-type natriuretic peptide in vascular physiology and disease. *Pharmacol Ther*. 2005;105:85–93.
271. Arora P, et al. Atrial natriuretic peptide is negatively regulated by microRNA-425. *J Clin Invest*. 2013;123:3378–3382.
272. Curry FR. Atrial natriuretic peptide: an essential physiological regulator of transvascular fluid, protein transport, and plasma volume. *J Clin Invest*. 2005;115:1458–1461.
273. Chen W, et al. Atrial natriuretic peptide enhances microvascular albumin permeability by the caveolae-mediated transcellular pathway. *Cardiovasc Res*. 2012;93:141–151.
274. Kuhn M. Endothelial actions of atrial and B-type natriuretic peptides. *Br J Pharmacol*. 2012;166:522–531.
275. Kuhn M, et al. The natriuretic peptide/guanylyl cyclase—a system functions as a stress-responsive regulator of angiogenesis in mice. *J Clin Invest*. 2009;119:2019–2030.
276. Roberts E, et al. The diagnostic accuracy of the natriuretic peptides in heart failure: systematic review and diagnostic meta-analysis in the acute care setting. *BMJ*. 2015;350:h910.
277. Rubattu S, et al. NPR-C: a component of the natriuretic peptide family with implications in human diseases. *J Mol Med*. 2010;88:889–897.
278. Zakeri R, Burnett JC Jr, Sangaralingham SJ. Urinary C-type natriuretic peptide: an emerging biomarker for heart failure and renal remodeling. *Clin Chim Acta*. 2015;443:108–113.
279. Davenport AP, et al. Endothelin. *Pharmacol Rev*. 2016;68:357–418.
280. Wypij DM, et al. Role of mast cell chymase in the extracellular processing of big-endothelin-1 to endothelin-1 in the perfused rat lung. *Biochem Pharmacol*. 1992;43:845–853.

281. Seyrantepe V, et al. Enzymatic activity of lysosomal carboxypeptidase (cathepsin) a is required for proper elastic fiber formation and inactivation of endothelin-1. *Circulation*. 2008;117:1973–1981.
282. Nakano D, Pollock D. New concepts in endothelin control of sodium balance. *Clin Exp Pharmacol Physiol*. 2012;39:104–110.
283. Kohan DE, Rossi NF, Inscho EW, et al. Regulation of blood pressure and salt homeostasis by endothelin. *Physiol Rev*. 2011;91:1–77.
284. Kopp UC. Endothelin in the control of renal sympathetic nerve activity. *Contrib Nephrol*. 2011;172:107–119.
285. Kohan DE. Biology of endothelin receptors in the collecting duct. *Kidney Int*. 2009;76:481–486.
286. Katoh T, Chang H, Uchida S, et al. Direct effects of endothelin in the rat kidney. *Am J Physiol*. 1990;258:F397–F402.
287. Wilkins FC Jr, Alberola A, Mizelle HL, et al. Systemic hemodynamics and renal function during long-term pathophysiological increases in circulating endothelin. *Am J Physiol*. 1995;268:R375–R381.
288. Gurbanov K, et al. Differential regulation of renal regional blood flow by endothelin-1. *Am J Physiol*. 1996;271:F1166–F1172.
289. Kon V, Yoshioka T, Fogo A, et al. Glomerular actions of endothelin in vivo. *J Clin Invest*. 1989;83:1762–1767.
290. Hoffman A, Abassi ZA, Brodsky S, et al. Mechanisms of big endothelin-1-induced diuresis and natriuresis: role of ET(B) receptors. *Hypertension*. 2000;35:732–739.
291. Pollock JS, Pollock DM. Endothelin and NOS1/nitric oxide signaling and regulation of sodium homeostasis. *Curr Opin Nephrol Hypertens*. 2008;17:70–75.
292. Gariepy CE, Ohuchi T, Williams SC, et al. Salt-sensitive hypertension in endothelin-B receptor-deficient rats. *J Clin Invest*. 2000;105:925–933.
293. Garvin JL, Herrera M, Ortiz PA. Regulation of renal NaCl transport by nitric oxide, endothelin, and ATP: clinical implications. *Annu Rev Physiol*. 2011;73:359–376.
294. Herrera M, Coffman TM. The kidney and hypertension: novel insights from transgenic models. *Curr Opin Nephrol Hypertens*. 2012;21:171–178.
295. Tennyson AG, Lippard SJ. Generation, translocation, and action of nitric oxide in living systems. *Chem Biol*. 2011;18:1211–1220.
296. Hyndman KA, et al. Renal collecting duct NOS1 maintains fluid-electrolyte homeostasis and blood pressure. *Hypertension*. 2013;62:91–98.
297. Carey RM, Jin X, Wang Z, et al. Nitric oxide: a physiological mediator of the type 2 (AT2) angiotensin receptor. *Acta Physiol Scand*. 2000;168:65–71.
298. Salazar FJ, Alberola A, Pinilla JM, et al. Salt-induced increase in arterial pressure during nitric oxide synthesis inhibition. *Hypertension*. 1993;22:49–55.
299. Ikeda Y, Saito K, Kim JI, et al. Nitric oxide synthase isoform activities in kidney of Dahl salt-sensitive rats. *Hypertension*. 1995;26:1030–1034.
300. Tolins JP, Shultz PJ. Endogenous nitric oxide synthesis determines sensitivity to the pressor effect of salt. *Kidney Int*. 1994;46:230–236.
301. Hu L, Manning RD Jr. Role of nitric oxide in regulation of long-term pressure-natriuresis relationship in Dahl rats. *Am J Physiol*. 1995;268:H2375–H2383.
302. Wilcox CS, Welch WJ. TGF and nitric oxide: effects of salt intake and salt-sensitive hypertension. *Kidney Int Suppl*. 1996;55:S9–S13.
303. Hoffman A, Haramati A, Dalal I, et al. Diuretic-natriuretic actions and pressor effects of big-endothelin (1-39) in phosphoramidon-treated rats. *Proc Soc Exp Biol Med*. 1994;205:168–173.
304. Pech V, et al. Nitric oxide reduces Cl(–) absorption in the mouse cortical collecting duct through an ENaC-dependent mechanism. *Am J Physiol Renal Physiol*. 2013;304:F1390–F1397.
305. Herrera M, Hong NJ, Ortiz PA, et al. Endothelin-1 inhibits thick ascending limb transport via Akt-stimulated nitric oxide production. *J Biol Chem*. 2009;284:1454–1460.
306. Monzon CM, Occhipinti R, Pignataro OP, et al. Nitric oxide reduces paracellular resistance in rat thick ascending limbs by increasing Na⁺ and Cl[–] permeabilities. *Am J Physiol Renal Physiol*. 2017;doi:10.1152/ajprenal.00671.2016.
307. Garcia NH, Pomposiello SI, Garvin JL. Nitric oxide inhibits ADH-stimulated osmotic water permeability in cortical collecting ducts. *Am J Physiol*. 1996;270:F206–F210.
308. Kakoki M, Smithies O. The kallikrein-kinin system in health and in diseases of the kidney. *Kidney Int*. 2009;75:1019–1030.
309. Rhaleb NE, Yang XP, Carretero OA. The kallikrein-kinin system as a regulator of cardiovascular and renal function. *Compr Physiol*. 2011;1:971–993.
310. Badzyska B, Sadowski J. Differential action of bradykinin on intrarenal regional perfusion in the rat: waning effect in the cortex and major impact in the medulla. *J Physiol*. 2009;587:3943–3953.
311. Taddei S, Bortolotto L. Unraveling the pivotal role of bradykinin in ACE inhibitor activity. *Am J Cardiovasc Drugs*. 2016;16:309–321.
312. Souza Dos Santos RA, Passaglio KT, Pesquero JB, et al. Interactions between angiotensin-(1-7), kinins, and angiotensin II in kidney and blood vessels. *Hypertension*. 2001;38:660–664.
313. Grilo A, et al. Identification of genetic factors associated with susceptibility to angiotensin-converting enzyme inhibitors-induced cough. *Pharmacogenet Genomics*. 2011;21:10–17.
314. Valenzuela-Sanchez F, Valenzuela-Mendez B, Rodriguez-Gutierrez JF, et al. New role of biomarkers: mid-regional pro-adrenomedullin, the biomarker of organ failure. *Ann Transl Med*. 2016;4:329.
315. Hirata Y, et al. Mechanisms of adrenomedullin-induced vasodilation in the rat kidney. *Hypertension*. 1995;25:790–795.
316. Schell DA, Vari RC, Samson WK. Adrenomedullin: a newly discovered hormone controlling fluid and electrolyte homeostasis. *Trends Endocrinol Metab*. 1996;7:7–13.
317. Kitamura K, Kangawa K, Eto T. Adrenomedullin and PAMP: discovery, structures, and cardiovascular functions. *Microsc Res Tech*. 2002;57:3–13.
318. Miura K, et al. Attenuation of adrenomedullin-induced renal vasodilation by NG-nitro L-arginine but not glibenclamide. *Br J Pharmacol*. 1995;115:917–924.
319. Majid DS, Kadowitz PJ, Coy DH, et al. Renal responses to intra-arterial administration of adrenomedullin in dogs. *Am J Physiol*. 1996;270:F200–F205.
320. Jougasaki M, Aarhus LL, Heublein DM, et al. Role of prostaglandins and renal nerves in the renal actions of adrenomedullin. *Am J Physiol*. 1997;272:F260–F266.
321. Lisy O, et al. Neutral endopeptidase inhibition potentiates the natriuretic actions of adrenomedullin. *Am J Physiol*. 1998;275:F410–F414.
322. Charles CJ, Rademaker MT, Richards AM. Hemodynamic, hormonal, and renal actions of adrenomedullin-5 in normal conscious sheep. *J Cardiovasc Pharmacol*. 2011;58:25–31.
323. Hong Y, Hay DL, Quirion R, et al. The pharmacology of adrenomedullin 2/intermedin. *Br J Pharmacol*. 2012;166:110–120.
324. Nishikimi T. Adrenomedullin in the kidney-renal physiological and pathophysiological roles. *Curr Med Chem*. 2007;14:1689–1699.
325. Langham RG, Kelly DJ. Urotensin II and the kidney. *Curr Opin Nephrol Hypertens*. 2013;22:107–112.
326. Charles CJ, Rademaker MT, Richards AM, et al. Urotensin II: evidence for cardiac, hepatic and renal production. *Peptides*. 2005;26:2211–2214.
327. Richards AM, Charles C. Urotensin II in the cardiovascular system. *Peptides*. 2004;25:1795–1802.
328. Abdel-Razik AE, Forty EJ, Balment RJ, et al. Renal haemodynamic and tubular actions of urotensin II in the rat. *J Endocrinol*. 2008;198:617–624.
329. Song W, et al. Urotensin II and renal function in the rat. *Kidney Int*. 2006;69:1360–1368.
330. Buckalew VM. Endogenous digitalis-like factors: an overview of the history. *Front Endocrinol (Lausanne)*. 2015;6:49.
331. Bagrov AY, Shapiro JI, Fedorova OV. Endogenous cardiotonic steroids: physiology, pharmacology, and novel therapeutic targets. *Pharmacol Rev*. 2009;61:9–38.
332. Hazelwood RL. The pancreatic polypeptide (PP-fold) family: gastrointestinal, vascular, and feeding behavioral implications. *Proc Soc Exp Biol Med*. 1993;202:44–63.
333. Winaver J, Abassi Z. Role of neuropeptide Y in the regulation of kallan function. *EXS*. 2006;123–132.
334. Callanan EY, et al. Renal and cardiac neuropeptide Y and NPY receptors in a rat model of congestive heart failure. *Am J Physiol Renal Physiol*. 2007;293:F1811–F1817.
335. O'Carroll AM, Lolait SJ, Harris LE, et al. The apelin receptor APJ: journey from an orphan to a multifaceted regulator of homeostasis. *J Endocrinol*. 2013;219:R13–R35.
336. Folino A, Montarolo PG, Samaja M, et al. Effects of apelin on the cardiovascular system. *Heart Fail Rev*. 2015;20:505–518.
337. Hus-Citharel A, et al. Apelin counteracts vasopressin-induced water reabsorption via cross talk between apelin and vasopressin receptor signaling pathways in the rat collecting duct. *Endocrinology*. 2014;155:4483–4493.
338. Hus-Citharel A, et al. Effect of apelin on glomerular hemodynamic function in the rat kidney. *Kidney Int*. 2008;74:486–494.

339. Skov J. Effects of GLP-1 in the kidney. *Rev Endocr Metab Disord*. 2014;15:197–207.
340. Skov J, et al. Glucagon-like peptide-1 (GLP-1): effect on kidney hemodynamics and renin-angiotensin-aldosterone system in healthy men. *J Clin Endocrinol Metab*. 2013;98:E664–E671.
341. Tokonami N, et al. alpha-Ketoglutarate regulates acid-base balance through an intrarenal paracrine mechanism. *J Clin Invest*. 2013;123:3166–3171.
342. Vallon V, Rieg T. Regulation of renal NaCl and water transport by the ATP/UTP/P2Y2 receptor system. *Am J Physiol Renal Physiol*. 2011;301:F463–F475.
343. Pavlov TS, et al. Deficiency of renal cortical EGF increases ENaC activity and contributes to salt-sensitive hypertension. *J Am Soc Nephrol*. 2013;24:1053–1062.
344. Lee IH, Song SH, Cook DI, et al. H-Ras mediates the inhibitory effect of epidermal growth factor on the epithelial Na⁺ channel. *PLoS ONE*. 2015;10:e0116938.
345. Kemp BA, Howell NL, Gildea JJ, et al. Intrarenal ghrelin receptors regulate ENaC-dependent sodium reabsorption by a cAMP-dependent pathway. *Kidney Int*. 2013;84:501–508.
346. Yang SY, et al. A low-salt diet increases the expression of renal sirtuin 1 through activation of the ghrelin receptor in rats. *Sci Rep*. 2016;6:32787.
347. Seki G. Unexpected effect of the appetite-stimulating hormone ghrelin on ENaC: hunger for sodium? *Kidney Int*. 2013;84:438–440.
348. Firsov D, Tokonami N, Bonny O. Role of the renal circadian timing system in maintaining water and electrolytes homeostasis. *Mol Cell Endocrinol*. 2012;349:51–55.
349. Nikolaeva S, et al. The circadian clock modulates renal sodium handling. *J Am Soc Nephrol*. 2012;23:1019–1026.
350. Rakova N, et al. Long-term space flight simulation reveals infradian rhythmicity in human Na(+) balance. *Cell Metab*. 2013;17:125–131.
351. Kirkman MA, Albert AF, Ibrahim A, et al. Hyponatremia and brain injury: historical and contemporary perspectives. *Neurocrit Care*. 2013;18:406–416.
352. Oh JY, Shin JI. Syndrome of inappropriate antidiuretic hormone secretion and cerebral/renal salt wasting syndrome: similarities and differences. *Front Pediatr*. 2014;2:146.
353. Leonard J, et al. Cerebral salt wasting after traumatic brain injury: a review of the literature. *Scand J Trauma Resusc Emerg Med*. 2015;23:98.
354. Hoorn EJ, Zietse R. Diagnosis and treatment of hyponatremia: compilation of the guidelines. *J Am Soc Nephrol*. 2017.
355. Madhusudan P, Tirupakuzhi Vijayaraghavan BK, Cove ME. Fluid resuscitation in sepsis: reexamining the paradigm. *Biomed Res Int*. 2014;2014:984082.
356. Siddall EC, Radhakrishnan J. The pathophysiology of edema formation in the nephrotic syndrome. *Kidney Int*. 2012;82:635–642.
357. Blantz RC. Pathophysiology of pre-renal azotemia. *Kidney Int*. 1998;53:512–523.
358. Rachoin JS, et al. The fallacy of the BUN:creatinine ratio in critically ill patients. *Nephrol Dial Transplant*. 2012;27:2248–2254.
359. Fujita H, Shinjoh M, Ishii T, et al. Utility of fractional excretion of urea in the differential diagnosis of acute kidney injury in children. *Pediatr Nephrol*. 2016;31:1349–1353.
360. Pahwa AK, Sperati CJ. Urinary fractional excretion indices in the evaluation of acute kidney injury. *J Hosp Med*. 2016;11:77–80.
361. Busse L, Davison DL, Junker C, et al. Hemodynamic monitoring in the critical care environment. *Adv Chronic Kidney Dis*. 2013;20:21–29.
362. Rajaram SS, et al. Pulmonary artery catheters for adult patients in intensive care. *Cochrane Database Syst Rev*. 2013;(2):CD003408.
363. Kumar L, Rajan S, Baalachandran R. Outcomes associated with stroke volume variation versus central venous pressure guided fluid replacements during major abdominal surgery. *J Anaesthesiol Clin Pharmacol*. 2016;32:182–186.
364. Sinkeler SJ, Damman K, van Veldhuisen DJ, et al. A re-appraisal of volume status and renal function impairment in chronic heart failure: combined effects of pre-renal failure and venous congestion on renal function. *Heart Fail Rev*. 2012;17:263–270.
365. Yunos NM, et al. Association between a chloride-liberal vs chloride-restrictive intravenous fluid administration strategy and kidney injury in critically ill adults. *JAMA*. 2012;308:1566–1572.
366. Lobo DN, Awad S. Should chloride-rich crystalloids remain the mainstay of fluid resuscitation to prevent 'pre-renal' acute kidney injury?: con. *Kidney Int*. 2014;86:1096–1105.
367. Myburgh JA. Fluid resuscitation in acute medicine: what is the current situation? *J Intern Med*. 2015;277:58–68.
368. Magrini F, Niarchos AP. Hemodynamic effects of massive peripheral edema. *Am Heart J*. 1983;105:90–97.
369. Intaglietta M, Zweifach BW. Microcirculatory basis of fluid exchange. *Adv Biol Med Phys*. 1974;15:111–159.
370. Messerli FH. Calcium antagonists in hypertension: from hemodynamics to outcomes. *Am J Hypertens*. 2002;15:94S–97S.
371. Kreimeier U. Pathophysiology of fluid imbalance. *Crit Care*. 2000;4(suppl 2):S3–S7.
372. Guyton AC, Taylor AE, Brace RA. A synthesis of interstitial fluid regulation and lymph formation. *Fed Proc*. 1976;35:1881–1885.
373. Singer DR, et al. Blood pressure and endocrine responses to changes in dietary sodium intake in cardiac transplant recipients. Implications for the control of sodium balance. *Circulation*. 1994;89:1153–1159.
374. Iwatsuki S, et al. Recovery from "hepatorenal syndrome" after orthotopic liver transplantation. *N Engl J Med*. 1973;289:1155–1159.
375. Schrier RW. Decreased effective blood volume in edematous disorders: what does this mean? *J Am Soc Nephrol*. 2007;18:2028–2031.
376. Schiller AM, Pellegrino PR, Zucker IH. The renal nerves in chronic heart failure: efferent and afferent mechanisms. *Front Physiol*. 2015;6:224.
377. Zucker IH, Patel KP, Schultz HD. Neurohumoral stimulation. *Heart Fail Clin*. 2012;8:87–99.
378. Goldsmith SR, Francis GS, Levine TB, et al. Regional blood flow response to orthostasis in patients with congestive heart failure. *J Am Coll Cardiol*. 1983;1:1391–1395.
379. Creager MA, Faxon DP, Rockwell SM, et al. The contribution of the renin-angiotensin system to limb vasoregulation in patients with heart failure: observations during orthostasis and alpha-adrenergic blockade. *Clin Sci*. 1985;68:659–667.
380. Zucker IH, et al. The origin of sympathetic outflow in heart failure: the roles of angiotensin II and nitric oxide. *Prog Biophys Mol Biol*. 2004;84:217–232.
381. Nishida Y, Ryan KL, Bishop VS. Angiotensin II modulates arterial baroreflex function via a central alpha 1-adrenoceptor mechanism in rabbits. *Am J Physiol*. 1995;269:R1009–R1016.
382. Wang WZ, Gao L, Pan YX, et al. AT1 receptors in the nucleus tractus solitarius mediate the interaction between the baroreflex and the cardiac sympathetic afferent reflex in anesthetized rats. *Am J Physiol Regul Integr Comp Physiol*. 2007;292:R1137–R1145.
383. Xiao L, Haack KK, Zucker IH. Angiotensin II regulates ACE and ACE2 in neurons through p38 mitogen-activated protein kinase and extracellular signal-regulated kinase 1/2 signaling. *Am J Physiol Cell Physiol*. 2013;304:C1073–C1079.
384. Xiao L, Gao L, Lazartigues E, et al. Brain-selective overexpression of angiotensin-converting enzyme 2 attenuates sympathetic nerve activity and enhances baroreflex function in chronic heart failure. *Hypertension*. 2011;58:1057–1065.
385. Damman K, Testani JM. The kidney in heart failure: an update. *Eur Heart J*. 2015;36:1437–1444.
386. Welling PA, Chang YP, Delpire E, et al. Multigene kinase network, kidney transport, and salt in essential hypertension. *Kidney Int*. 2010;77:1063–1069.
387. Winaver J, Hoffman A, Abassi Z, et al. Does the Heart's hormone, ANP, help in congestive heart failure? *News Physiol Sci*. 1995;10:247–253.
388. De Mello WC, Frohlich ED. On the local cardiac renin-angiotensin system. Basic and clinical implications. *Peptides*. 2011;32:1774–1779.
389. Pagliaro P, Penna C. Rethinking the renin-angiotensin system and its role in cardiovascular regulation. *Cardiovasc Drugs Ther*. 2005;19:77–87.
390. De Mello WC. Local renin-angiotensin-aldosterone systems and cardiovascular diseases. *Med Clin North Am*. 2017;101:117–127.
391. Delyani JA, Robinson EL, Rudolph AE. Effect of a selective aldosterone receptor antagonist in myocardial infarction. *Am J Physiol Heart Circ Physiol*. 2001;281:H647–H654.
392. Rudolph AE, Rocha R, McMahon EG. Aldosterone target organ protection by eplerenone. *Mol Cell Endocrinol*. 2004;217:229–238.
393. Pitt B, Pedro Ferreira J, Zannad F. Mineralocorticoid receptor antagonists in patients with heart failure: current experience and future perspectives. *Eur Heart J Cardiovasc Pharmacother*. 2017;3:48–57.
394. Ramkumar N, Kohan DE. Proximal tubule angiotensinogen modulation of arterial pressure. *Curr Opin Nephrol Hypertens*. 2013;22:32–36.
395. Bansal S, Lindenfeld J, Schrier RW. Sodium retention in heart failure and cirrhosis: potential role of natriuretic doses of mineralocorticoid antagonist? *Circ Heart Fail*. 2009;2:370–376.
396. Chaney E, Shaw A. Pathophysiology of fluid retention in heart failure. *Contrib Nephrol*. 2010;164:46–53.

397. Coble JP, Grobe JL, Johnson AK, et al. Mechanisms of brain renin angiotensin system-induced drinking and blood pressure: importance of the subfornical organ. *Am J Physiol Regul Integr Comp Physiol*. 2015;308:R238–R249.
398. Sobotka PA, Krum H, Bohm M, et al. The role of renal denervation in the treatment of heart failure. *Curr Cardiol Rep*. 2012;14:285–292.
399. Parati G, Esler M. The human sympathetic nervous system: its relevance in hypertension and heart failure. *Eur Heart J*. 2012;33:1058–1066.
400. Lieveise AG, et al. Renal and systemic hemodynamic effects of ibopamine in patients with mild to moderate congestive heart failure. *J Cardiovasc Pharmacol*. 1995;25:361–367.
401. Giamouzis G, Butler J, Triposkiadis F. Renal function in advanced heart failure. *Congest Heart Fail*. 2011;17:180–188.
402. Yusof AP, Yusoff NH, Suhaimi FW, et al. Role of supraspinal vasopressin neurones in the effects of atrial natriuretic peptide on sympathetic nerve activity. *Auton Neurosci*. 2009;148:50–54.
403. O'Tierney PF, Tse MY, Pang SC. Elevated renal norepinephrine in proANP gene-disrupted mice is associated with increased tyrosine hydroxylase expression in sympathetic ganglia. *Regul Pept*. 2007;143:90–96.
404. Villarreal D, Freeman RH, Johnson RA, et al. Effects of renal denervation on postprandial sodium excretion in experimental heart failure. *Am J Physiol*. 1994;266:R1599–R1604.
405. Feng QP, Hedner T, Hedner J, et al. Blunted renal response to atrial natriuretic peptide in congestive heart failure rats is reversed by the alpha 2-adrenergic agonist clonidine. *J Cardiovasc Pharmacol*. 1990;16:776–782.
406. Finley JJ, Konstam MA, Udelson JE. Arginine vasopressin antagonists for the treatment of heart failure and hyponatremia. *Circulation*. 2008;118:410–421.
407. Schrier RW, Sharma S, Shchekochikhin D. Hyponatraemia: more than just a marker of disease severity? *Nat Rev Nephrol*. 2013;9:37–50.
408. Balling L, Gustafsson F. Copeptin in heart failure. *Adv Clin Chem*. 2016;73:29–64.
409. Vukicevic T, Schulz M, Faust D, et al. The trafficking of the water channel aquaporin-2 in renal principal cells—a potential target for pharmacological intervention in cardiovascular diseases. *Front Pharmacol*. 2016;7:23.
410. Lechrich RW, Ortiz-Melo DI, Patel MB, et al. Role of vaptans in the management of hyponatremia. *Am J Kidney Dis*. 2013;62:364–376.
411. Grodin JL. Pharmacologic approaches to electrolyte abnormalities in heart failure. *Curr Heart Fail Rep*. 2016;13:181–189.
412. Peri A. Clinical review: the use of vaptans in clinical endocrinology. *J Clin Endocrinol Metab*. 2013;98:1321–1332.
413. Costello-Boerrigter LC, Boerrigter G, Burnett JC Jr. Pharmacology of vasopressin antagonists. *Heart Fail Rev*. 2009;14:75–82.
414. Ambrosy A, Goldsmith SR, Gheorghiade M. Tolvaptan for the treatment of heart failure: a review of the literature. *Expert Opin Pharmacother*. 2011;12:961–976.
415. Jankowska EA, et al. Identification of chronic heart failure patients with a high 12-month mortality risk using biomarkers including plasma C-terminal pro-endothelin-1. *PLoS ONE*. 2011;6:e14506.
416. Ertl G, Bauersachs J. Endothelin receptor antagonists in heart failure: current status and future directions. *Drugs*. 2004;64:1029–1040.
417. Ramirez GA. Endothelin ET receptor agonism as a new therapeutic strategy in pulmonary arterial hypertension and chronic heart failure. *Med Hypotheses*. 2013.
418. Kohan DE, Cleland JG, Rubin LJ, et al. Clinical trials with endothelin receptor antagonists: what went wrong and where can we improve? *Life Sci*. 2012;91:528–539.
419. Kazory A, Ross EA. Emerging therapies for heart failure: renal mechanisms and effects. *Heart Fail Rev*. 2012;17:1–16.
420. Abassi Z, et al. Regulation of intrarenal blood flow in experimental heart failure: role of endothelin and nitric oxide. *Am J Physiol*. 1998;274:F766–F774.
421. Liu C, et al. Glucocorticoids improve renal responsiveness to atrial natriuretic peptide by up-regulating natriuretic peptide receptor-A expression in the renal inner medullary collecting duct in decompensated heart failure. *J Pharmacol Exp Ther*. 2011;339:203–209.
422. Kobayashi D, Yamaguchi N, Takahashi O, et al. Human atrial natriuretic peptide treatment for acute heart failure: a systematic review of efficacy and mortality. *Can J Cardiol*. 2012;28:102–109.
423. Lohmeier TE, et al. Atrial natriuretic peptide and sodium homeostasis in compensated heart failure. *Am J Physiol*. 1996;271:R1353–R1363.
424. Rubattu S, Calvieri C, Pagliaro B, et al. Atrial natriuretic peptide and regulation of vascular function in hypertension and heart failure: implications for novel therapeutic strategies. *J Hypertens*. 2013;31:1061–1072.
425. Horii M, et al. Prognostic value of B-type natriuretic peptide and its amino-terminal proBNP fragment for cardiovascular events with stratification by renal function. *J Cardiol*. 2013;61:410–416.
426. Panagopoulou V, et al. NTproBNP: an important biomarker in cardiac diseases. *Curr Top Med Chem*. 2013;13:82–94.
427. van Veldhuisen DJ, et al. B-type natriuretic peptide and prognosis in heart failure patients with preserved and reduced ejection fraction. *J Am Coll Cardiol*. 2013;61:1498–1506.
428. Cleland JG, et al. Relationship between plasma concentrations of N-terminal pro brain natriuretic peptide and the characteristics and outcome of patients with a clinical diagnosis of diastolic heart failure: a report from the PEP-CHF study. *Eur J Heart Fail*. 2012;14:487–494.
429. McLellan J, et al. B-type natriuretic peptide-guided treatment for heart failure. *Cochrane Database Syst Rev*. 2016;(12):CD008966.
430. Li P, Luo Y, Chen YM. B-type natriuretic peptide-guided chronic heart failure therapy: a meta-analysis of 11 randomised controlled trials. *Heart Lung Circ*. 2013.
431. Noveanu M, et al. Direct comparison of serial B-type natriuretic peptide and NT-proBNP levels for prediction of short- and long-term outcome in acute decompensated heart failure. *Crit Care*. 2011;15:R1.
432. Manzano L, Escobar C, Cleland JG, et al. Diagnosis of elderly patients with heart failure. *Eur J Heart Fail*. 2012;14:1097–1103.
433. Heining L, Giesa C, Ewig S. MR-proANP, MR-proADM, and PCT in patients presenting with acute dyspnea in a medical emergency unit. *Lung*. 2016;194:185–191.
434. Del RS. C-type natriuretic peptide: a new cardiac mediator. *Peptides*. 2013;40:93–98.
435. Kalra PR, et al. C-type natriuretic peptide production by the human kidney is blunted in chronic heart failure. *Clin Sci*. 2010;118:71–77.
436. Potter LR. Regulation and therapeutic targeting of peptide-activated receptor guanylyl cyclases. *Pharmacol Ther*. 2011;130:71–82.
437. Egom EE, Feridooni T, Hotchkiss A, et al. Mechanisms of renal hyposensitiveness to BNP in heart failure. *Can J Physiol Pharmacol*. 2015;93:399–403.
438. Ghosh N, Haddad H. Atrial natriuretic peptides in heart failure: pathophysiological significance, diagnostic and prognostic value. *Can J Physiol Pharmacol*. 2011;89:587–591.
439. Mangiafico S, Costello-Boerrigter LC, Andersen IA, et al. Neutral endopeptidase inhibition and the natriuretic peptide system: an evolving strategy in cardiovascular therapeutics. *Eur Heart J*. 2013;34:886–893c.
440. Dickey DM, Potter LR. ProBNP(1-108) is resistant to degradation and activates guanylyl cyclase-A with reduced potency. *Clin Chem*. 2011;57:1272–1278.
441. Armaly Z, Assady S, Abassi Z. Corin: a new player in the regulation of salt-water balance and blood pressure. *Curr Opin Nephrol Hypertens*. 2013;22:713–722.
442. Andreassi MG, et al. Up-regulation of 'clearance' receptors in patients with chronic heart failure: a possible explanation for the resistance to biological effects of cardiac natriuretic hormones. *Eur J Heart Fail*. 2001;3:407–414.
443. Abassi Z, Haramati A, Hoffman A, et al. Effect of converting-enzyme inhibition on renal response to ANF in rats with experimental heart failure. *Am J Physiol*. 1990;259:R84–R89.
444. Pettersson A, Hedner J, Hedner T. Relationship between renal sympathetic activity and diuretic effects of atrial natriuretic peptide (ANP) in the rat. *Acta Physiol Scand*. 1989;135:323–333.
445. Gheorghiade M, et al. Soluble guanylate cyclase: a potential therapeutic target for heart failure. *Heart Fail Rev*. 2013;18:123–134.
446. Kemp-Harper BK, Horowitz JD, Ritchie RH. Therapeutic potential of nitroxyl (HNO) donors in the management of acute decompensated heart failure. *Drugs*. 2016;76:1337–1348.
447. Patel KP, Schultz HD. Angiotensin peptides and nitric oxide in cardiovascular disease. *Antioxid Redox Signal*. 2013;19:1121–1132.
448. Jonsson S, Agic MB, Narfstrom F, et al. Renal neurohormonal regulation in heart failure decompensation. *Am J Physiol Regul Integr Comp Physiol*. 2014;307:R493–R497.
449. Liu X, et al. Effect of asymmetric dimethylarginine (ADMA) on heart failure development. *Nitric Oxide*. 2016;54:73–81.
450. Harirforoosh S, Asghar W, Jamali F. Adverse effects of nonsteroidal antiinflammatory drugs: an update of gastrointestinal, cardiovascular and renal complications. *J Pharm Pharm Sci*. 2013;16:821–847.

451. Winkelmayer WC, Waikar SS, Mogun H, et al. Nonselective and cyclooxygenase-2-selective NSAIDs and acute kidney injury. *Am J Med.* 2008;121:1092–1098.
452. Dzau VJ, et al. Prostaglandins in severe congestive heart failure. Relation to activation of the renin–angiotensin system and hyponatremia. *N Engl J Med.* 1984;310:347–352.
453. Castellani S, et al. Increased renal formation of thromboxane A2 and prostaglandin F2 alpha in heart failure. *Am Heart J.* 1997;133:94–100.
454. Riegger GA, Elsner D, Hildenbrand J, et al. Prostaglandins, renin and atrial natriuretic peptide in the control of the circulation and renal function in heart failure in the dog. *Prog Clin Biol Res.* 1989;301:455–458.
455. Townsend JN, Doran J, Lote CJ, et al. Peripheral haemodynamic effects of inhibition of prostaglandin synthesis in congestive heart failure and interactions with captopril. *Br Heart J.* 1995;73:434–441.
456. Gislason GH, et al. Increased mortality and cardiovascular morbidity associated with use of nonsteroidal anti-inflammatory drugs in chronic heart failure. *Arch Intern Med.* 2009;169:141–149.
457. Grosser T, Yu Y, Fitzgerald GA. Emotion recollected in tranquility: lessons learned from the COX-2 saga. *Annu Rev Med.* 2010;61:17–33.
458. Kose F, et al. Effects of selective Cox-2 inhibitor, rofecoxib, alone or combination with furosemide on renal functions and renal Cox-2 expression in rats. *Clin Exp Nephrol.* 2010;14:22–27.
459. Akinbamowo AO, Salzberg DJ, Weir MR. Renal consequences of prostaglandin inhibition in heart failure. *Heart Fail Clin.* 2008;4:505–510.
460. Nishikimi T, Matsuoka H. Cardiac adrenomedullin: its role in cardiac hypertrophy and heart failure. *Curr Med Chem Cardiovasc Hematol Agents.* 2005;3:231–242.
461. Rademaker MT, et al. Adrenomedullin and heart failure. *Regul Pept.* 2003;112:51–60.
462. Shah RV, et al. Mid-regional pro-atrial natriuretic peptide and pro-adrenomedullin testing for the diagnostic and prognostic evaluation of patients with acute dyspnoea. *Eur Heart J.* 2012;33:2197–2205.
463. Alehagen U, Dahlstrom U, Rehfeld JF, et al. Pro-A-type natriuretic peptide, proadrenomedullin, and N-terminal pro-B-type natriuretic peptide used in a multimarker strategy in primary health care in risk assessment of patients with symptoms of heart failure. *J Card Fail.* 2013;19:31–39.
464. Cinar O, et al. Evaluation of mid-regional pro-atrial natriuretic peptide, procalcitonin, and mid-regional pro-adrenomedullin for the diagnosis and risk stratification of dyspneic ED patients. *Am J Emerg Med.* 2012;30:1915–1920.
465. Hirose T, et al. Increased expression of adrenomedullin 2/intermedin in rat hearts with congestive heart failure. *Eur J Heart Fail.* 2008;10:840–849.
466. Totsune K, et al. Increased gene expression of adrenomedullin and adrenomedullin-receptor complexes, receptor-activity modifying protein (RAMP)2 and calcitonin-receptor-like receptor (CRLR) in the hearts of rats with congestive heart failure. *Clin Sci.* 2000;99:541–546.
467. Jougasaki M, Heublein DM, Sandberg SM, et al. Attenuated natriuretic response to adrenomedullin in experimental heart failure. *J Card Fail.* 2001;7:75–83.
468. Stenberg TA, Kildal AB, How OJ, et al. Adrenomedullin-epinephrine cotreatment enhances cardiac output and left ventricular function by energetically neutral mechanisms. *Am J Physiol Heart Circ Physiol.* 2012;302:H1584–H1590.
469. Nishikimi T, et al. Effects of long-term intravenous administration of adrenomedullin (AM) plus hANP therapy in acute decompensated heart failure. *Circ J.* 2009;73:892–898.
470. Ross B, McKendry K, Gaiad A. Role of urotensin II in health and disease. *Am J Physiol Regul Integr Comp Physiol.* 2010;298:R1156–R1172.
471. Ejaz A, LoGerfo FW, Pradhan L. Diabetic neuropathy and heart failure: role of neuropeptides. *Expert Rev Mol Med.* 2011;13:e26.
472. Galanth C, Hus-Citharel A, Li B, et al. Apelin in the control of body fluid homeostasis and cardiovascular functions. *Curr Pharm Des.* 2012;18:789–798.
473. Yu XH, et al. Apelin and its receptor APJ in cardiovascular diseases. *Clin Chim Acta.* 2013.
474. Dalzell JR, Jackson CE, Chong KS, et al. Do plasma concentrations of apelin predict prognosis in patients with advanced heart failure? *Biomark Med.* 2014;8:807–813.
475. Dalzell JR, et al. The emerging potential of the Apelin-APJ system in heart failure. *J Card Fail.* 2015;21:489–498.
476. Gerbier R, et al. Development of original metabolically stable apelin-17 analogs with diuretic and cardiovascular effects. *FASEB J.* 2017;31:687–700.
477. Sarma S. Use of clinically available PPAR agonists for heart failure; do the risks outweigh the potential benefits? *Curr Mol Pharmacol.* 2012;5:255–263.
478. Gilbert RE, Krum H. Heart failure in diabetes: effects of anti-hyperglycaemic drug therapy. *Lancet.* 2015;385:2107–2117.
479. Beltowski J, Rachanczyk J, Wlodarczyk M. Thiazolidinedione-induced fluid retention: recent insights into the molecular mechanisms. *PPAR Res.* 2013;2013:628628.
480. Tahrani AA, Barnett AH, Bailey CJ. Pharmacology and therapeutic implications of current drugs for type 2 diabetes mellitus. *Nat Rev Endocrinol.* 2016;12:566–592.
481. Wong F. Recent advances in our understanding of hepatorenal syndrome. *Nat Rev Gastroenterol Hepatol.* 2012;9:382–391.
482. Rahimi RS, Rockey DC. End-stage liver disease complications. *Curr Opin Gastroenterol.* 2013;29:257–263.
483. Acevedo JG, Cramp ME. Hepatorenal syndrome: update on diagnosis and therapy. *World J Hepatol.* 2017;9:293–299.
484. Moller S, Bernardi M. Interactions of the heart and the liver. *Eur Heart J.* 2013;34:2804–2811.
485. Fagundes C, Gines P. Hepatorenal syndrome: a severe, but treatable, cause of kidney failure in cirrhosis. *Am J Kidney Dis.* 2012;59:874–885.
486. Bellot P, Frances R, Such J. Pathological bacterial translocation in cirrhosis: pathophysiology, diagnosis and clinical implications. *Liver Int.* 2013;33:31–39.
487. Huang HC, et al. Cannabinoid receptor 2 agonist ameliorates mesenteric angiogenesis and portosystemic collaterals in cirrhotic rats. *Hepatology.* 2012;56:248–258.
488. Iwakiri Y, Groszmann RJ. The hyperdynamic circulation of chronic liver diseases: from the patient to the molecule. *Hepatology.* 2006;43:S121–S131.
489. Machicao VI, Balakrishnan M, Fallon MB. Pulmonary complications in chronic liver disease. *Hepatology.* 2013.
490. Hu LS, George J, Wang JH. Current concepts on the role of nitric oxide in portal hypertension. *World J Gastroenterol.* 2013;19:1707–1717.
491. Di Pascoli M, Sacerdoti D, Pontisso P, et al. Molecular mechanisms leading to splanchnic vasodilation in liver cirrhosis. *J Vasc Res.* 2017;54:92–99.
492. Kato Y, et al. Inhibition of nitric oxide synthase in hyperdynamic circulation of rats with early or late cirrhosis secondary to common bile duct ligation. *J Nippon Med Sch.* 2011;78:146–155.
493. La VG, et al. Hemodynamic, renal, and endocrine effects of acute inhibition of nitric oxide synthase in compensated cirrhosis. *Hepatology.* 2001;34:19–27.
494. Campillo B, et al. Inhibition of nitric oxide synthesis in the forearm arterial bed of patients with advanced cirrhosis. *Hepatology.* 1995;22:1423–1429.
495. Thiessen HC, Skott O, Jespersen B, et al. Nitric oxide synthase inhibition does not improve renal function in cirrhotic patients with ascites. *Am J Gastroenterol.* 2003;98:180–186.
496. Iwakiri Y, Kim MY. Nitric oxide in liver diseases. *Trends Pharmacol Sci.* 2015;36:524–536.
497. Wiest R, Groszmann RJ. The paradox of nitric oxide in cirrhosis and portal hypertension: too much, not enough. *Hepatology.* 2002;35:478–491.
498. Mookerjee RP, Vairappan B, Jalan R. The puzzle of endothelial nitric oxide synthase dysfunction in portal hypertension: the missing piece? *Hepatology.* 2007;46:943–946.
499. Mookerjee RP, et al. Increasing dimethylarginine levels are associated with adverse clinical outcome in severe alcoholic hepatitis. *Hepatology.* 2007;45:62–71.
500. Leiper J, et al. Disruption of methylarginine metabolism impairs vascular homeostasis. *Nat Med.* 2007;13:198–203.
501. Lluch P, Segarra G, Medina P. Asymmetric dimethylarginine as a mediator of vascular dysfunction in cirrhosis. *World J Gastroenterol.* 2015;21:9466–9475.
502. Baldassarre M, et al. The endocannabinoid system in advanced liver cirrhosis: pathophysiological implication and future perspectives. *Liver Int.* 2013;33:1298–1308.
503. Kostreva DR, Castaner A, Kampine JP. Reflex effects of hepatic baroreceptors on renal and cardiac sympathetic nerve activity. *Am J Physiol.* 1980;238:R390–R394.
504. Burnstock G, Vaughn B, Robson SC. Purinergic signalling in the liver in health and disease. *Purinergic Signal.* 2014;10:51–70.

505. Aller MA, et al. The interstitial lymphatic peritoneal mesothelium axis in portal hypertensive ascites: when in danger, go back to the sea. *Int J Inflam*. 2010;2010:148689.
506. Parasher VK, et al. Observation of thoracic duct morphology in portal hypertension by endoscopic ultrasound. *Gastrointest Endosc*. 1998;48:588–592.
507. Facciorusso A, et al. The use of human albumin for the treatment of ascites in patients with liver cirrhosis: item of safety, facts, controversies and perspectives. *Curr Drug Saf*. 2011;6:267–274.
508. Sikuler E, Kravetz D, Groszmann RJ. Evolution of portal hypertension and mechanisms involved in its maintenance in a rat model. *Am J Physiol*. 1985;248:G618–G625.
509. Atucha NM, et al. Role of vascular nitric oxide in experimental liver cirrhosis. *Curr Vasc Pharmacol*. 2005;3:81–85.
510. Better OS, Guckian V, Giebisch G, et al. The effect of sodium taurocholate on proximal tubular reabsorption in the rat kidney. *Clin Sci*. 1987;72:139–141.
511. Green J, Better OS. Systemic hypotension and renal failure in obstructive jaundice-mechanistic and therapeutic aspects. *J Am Soc Nephrol*. 1995;5:1853–1871.
512. Gattinoni L, Carlesso E. Supporting hemodynamics: what should we target? What treatments should we use? *Crit Care*. 2013;17(suppl 1):S4.
513. Bernardi M, et al. Systemic and regional hemodynamics in pre-ascitic cirrhosis: effects of posture. *J Hepatol*. 2003;39:502–508.
514. Wong F, Liu P, Blendis L. The mechanism of improved sodium homeostasis of low-dose losartan in preascitic cirrhosis. *Hepatology*. 2002;35:1449–1458.
515. Ubeda M, et al. Renin and angiotensinogen mRNA expression in the kidneys of rats subjected to long-term bile duct ligation. *Hepatology*. 1994;19:1431–1436.
516. Schneider AW, Kalk JF, Klein CP. Effect of losartan, an angiotensin II receptor antagonist, on portal pressure in cirrhosis. *Hepatology*. 1999;29:334–339.
517. Gentilini P, et al. Effects of low-dose captopril on renal hemodynamics and function in patients with cirrhosis of the liver. *Gastroenterology*. 1993;104:588–594.
518. Fialla AD, Thieson HC, Bie P, et al. Internal dysregulation of the renin system in patients with stable liver cirrhosis. *Scand J Clin Lab Invest*. 2017;1–15.
519. Wadei HM, Mai ML, Ahsan N, et al. Hepatorenal syndrome: pathophysiology and management. *Clin J Am Soc Nephrol*. 2006;1:1066–1079.
520. Grace JA, et al. Activation of the MAS receptor by Angiotensin-(1-7) in the Renin-Angiotensin system mediates mesenteric vasodilatation in cirrhosis. *Gastroenterology*. 2013;145:874–884.
521. Herath CB, Grace JA, Angus PW. Therapeutic potential of targeting the renin angiotensin system in portal hypertension. *World J Gastrointest Pathophysiol*. 2013;4:1–11.
522. Klein S, et al. Hemodynamic effects of the non-peptidic angiotensin-(1-7) agonist AVE0991 in liver cirrhosis. *PLoS ONE*. 2015;10:e0138732.
523. Sola E, Gines P. Challenges and management of liver cirrhosis: pathophysiology of renal dysfunction in cirrhosis. *Dig Dis*. 2015;33:534–538.
524. Martell M, Coll M, Ezkurdia N, et al. Pathophysiology of splanchnic vasodilation in portal hypertension. *World J Hepatol*. 2010;2:208–220.
525. Genovesi S, et al. Baroreceptor sensitivity and baroreceptor effectiveness index in cirrhosis: the relevance of hepatic venous pressure gradient. *Liver Int*. 2010;30:232–239.
526. Rodriguez-Martinez M, Sawin LL, Dibona GF. Arterial and cardiopulmonary baroreflex control of renal nerve activity in cirrhosis. *Am J Physiol*. 1995;268:R117–R129.
527. Dibona GF, Sawin LL, Jones SY. Characteristics of renal sympathetic nerve activity in sodium-retaining disorders. *Am J Physiol*. 1996;271:R295–R302.
528. Bomzon A, et al. Systemic hypotension and decreased pressor response in dogs with chronic bile duct ligation. *Hepatology*. 1986;6:595–600.
529. Ryan J, Sudhir K, Jennings G, et al. Impaired reactivity of the peripheral vasculature to pressor agents in alcoholic cirrhosis. *Gastroenterology*. 1993;105:1167–1172.
530. Stadlbauer V, et al. Relationship between activation of the sympathetic nervous system and renal blood flow autoregulation in cirrhosis. *Gastroenterology*. 2008;134:111–119.
531. Hamza SM, Kaufman S. Splenic neurohormonal modulation of mesenteric vascular tone. *Exp Physiol*. 2012;97:1054–1064.
532. Hamza SM, Kaufman S. Role of spleen in integrated control of splanchnic vascular tone: physiology and pathophysiology. *Can J Physiol Pharmacol*. 2009;87:1–7.
533. Yang YY, et al. Association of the G-protein and alpha2-adrenergic receptor gene and plasma norepinephrine level with clonidine improvement of the effects of diuretics in patients with cirrhosis with refractory ascites: a randomised clinical trial. *Gut*. 2010;59:1545–1553.
534. Sansoe G, Aragno M, Mastrocola R, et al. Alpha-2A adrenoceptor agonist guanfacine restores diuretic efficiency in experimental cirrhotic ascites: comparison with clonidine. *PLoS ONE*. 2016;11:e0158486.
535. de Mattos AZ, de Mattos AA, Mendez-Sanchez N. Hepatorenal syndrome: current concepts related to diagnosis and management. *Ann Hepatol*. 2016;15:474–481.
536. Sinha VK, Ko B. Hyponatremia in cirrhosis—pathogenesis, treatment, and prognostic significance. *Adv Chronic Kidney Dis*. 2015;22:361–367.
537. Kim JK, Summer SN, Howard RL, et al. Vasopressin gene expression in rats with experimental cirrhosis. *Hepatology*. 1993;17:143–147.
538. Huang YY, Sun JY, Wang JY, et al. Terlipressin resolves ascites of cirrhotic rats through downregulation of aquaporin 2. *J Int Med Res*. 2012;40:1735–1744.
539. Koshimizu TA, et al. Vasopressin V1a and V1b receptors: from molecules to physiological systems. *Physiol Rev*. 2012;92:1813–1864.
540. Arroyo V, Claria J, Salo J, et al. Antidiuretic hormone and the pathogenesis of water retention in cirrhosis with ascites. *Semin Liver Dis*. 1994;14:44–58.
541. Martinet JP, et al. Changes in plasma endothelin-1 and Big endothelin-1 induced by transjugular intrahepatic portosystemic shunts in patients with cirrhosis and refractory ascites. *J Hepatol*. 1996;25:700–706.
542. Angus PW. Role of endothelin in systemic and portal resistance in cirrhosis. *Gut*. 2006;55:1230–1232.
543. Tsiakalos A, et al. Portopulmonary hypertension and serum endothelin levels in hospitalized patients with cirrhosis. *Hepatobiliary Pancreat Dis Int*. 2011;10:393–398.
544. Cavaasin MA, et al. Acute effects of endothelin receptor antagonists on hepatic hemodynamics of cirrhotic and noncirrhotic rats. *Can J Physiol Pharmacol*. 2010;88:636–643.
545. Ling L, et al. Comparison of endothelin receptors in normal versus cirrhotic human liver and in the liver from endothelial cell-specific ETB knockout mice. *Life Sci*. 2012;91:716–722.
546. Lebrech D, et al. Hemodynamics and pharmacokinetics of tezosentan, a dual endothelin receptor antagonist, in patients with cirrhosis. *Eur J Clin Pharmacol*. 2012;68:533–541.
547. Vaughan RB, Angus PW, Chin-Dusting JP. Evidence for altered vascular responses to exogenous endothelin-1 in patients with advanced cirrhosis with restoration of the normal vasoconstrictor response following successful liver transplantation. *Gut*. 2003;52:1505–1510.
548. Kapoor D, Redhead DN, Hayes PC, et al. Systemic and regional changes in plasma endothelin following transient increase in portal pressure. *Liver Transpl*. 2003;9:32–39.
549. Anand R, et al. Endothelin is an important determinant of renal function in a rat model of acute liver and renal failure. *Gut*. 2002;50:111–117.
550. Takashimizu S, et al. Effect of endothelin A receptor antagonist on hepatic hemodynamics in cirrhotic rats. Implications for endothelin-1 in portal hypertension. *Tokai J Exp Clin Med*. 2011;36:37–43.
551. Arroyo V, Fernandez J. Management of hepatorenal syndrome in patients with cirrhosis. *Nat Rev Nephrol*. 2011;7:517–526.
552. Principe A, et al. The hepatic apelin system: a new therapeutic target for liver disease. *Hepatology*. 2008;48:1193–1201.
553. Yokomori H, et al. Overexpression of apelin receptor (APJ/AGTRL1) on hepatic stellate cells and sinusoidal angiogenesis in human cirrhotic liver. *J Gastroenterol*. 2011;46:222–231.
554. Yokomori H, Oda M, Yoshimura K, et al. Enhanced expressions of apelin on proliferative hepatic arterial capillaries in human cirrhotic liver. *Hepatol Res*. 2012;42:508–514.
555. Wong F, Blendis L. Pathophysiology of sodium retention and ascites formation in cirrhosis: role of atrial natriuretic factor. *Semin Liver Dis*. 1994;14:59–70.
556. Levy M. Atrial natriuretic peptide: renal effects in cirrhosis of the liver. *Semin Nephrol*. 1997;17:520–529.
557. Poulos JE, Gower WR, Fontanet HL, et al. Cirrhosis with ascites: increased atrial natriuretic peptide messenger RNA expression in rat ventricle. *Gastroenterology*. 1995;108:1496–1503.

558. Rector WG Jr, Adair O, Hossack KF, et al. Atrial volume in cirrhosis: relationship to blood volume and plasma concentration of atrial natriuretic factor. *Gastroenterology*. 1990;99:766–770.
559. Wong F, Liu P, Tobe S, et al. Central blood volume in cirrhosis: measurement with radionuclide angiography. *Hepatology*. 1994;19:312–321.
560. Wong F, Liu P, Blendis L. Sodium homeostasis with chronic sodium loading in preascitic cirrhosis. *Gut*. 2001;49:847–851.
561. Tobe SW, et al. Angiotensin II modulates atrial natriuretic factor-induced natriuresis in cirrhosis with ascites. *Am J Kidney Dis*. 1993;21:472–479.
562. Morali GA, Tobe SW, Skorecki KL, et al. Refractory ascites: modulation of atrial natriuretic factor unresponsiveness by mannitol. *Hepatology*. 1992;16:42–48.
563. Abraham WT, Lauwaars ME, Kim JK, et al. Reversal of atrial natriuretic peptide resistance by increasing distal tubular sodium delivery in patients with decompensated cirrhosis. *Hepatology*. 1995;22:737–743.
564. MacGilchrist A, Craig KJ, Hayes PC, et al. Effect of the serine protease inhibitor, aprotinin, on systemic haemodynamics and renal function in patients with hepatic cirrhosis and ascites. *Clin Sci*. 1994;87:329–335.
565. Koepke JP, Jones S, Dibona GF. Renal nerves mediate blunted natriuresis to atrial natriuretic peptide in cirrhotic rats. *Am J Physiol*. 1987;252:R1019–R1023.
566. Tobe SW, Morali GA, Greig PD, et al. Peritoneovenous shunting restores atrial natriuretic factor responsiveness in refractory hepatic ascites. *Gastroenterology*. 1993;105:202–207.
567. Piccinni P, et al. Human natriuretic factor in cirrhotic patients undergoing orthotopic liver transplantation. *Transpl Int*. 1995;8:51–54.
568. Warner L, Skorecki K, Blendis LM, et al. Atrial natriuretic factor and liver disease. *Hepatology*. 1993;17:500–513.
569. La VG, et al. Blunted natriuretic response to low-dose brain natriuretic peptide infusion in nonazotemic cirrhotic patients with ascites and avid sodium retention. *Hepatology*. 1995;22:1745–1750.
570. Henriksen JH, et al. Increased circulating pro-brain natriuretic peptide (proBNP) and brain natriuretic peptide (BNP) in patients with cirrhosis: relation to cardiovascular dysfunction and severity of disease. *Gut*. 2003;52:1511–1517.
571. Yildiz R, Yildirim B, Karıncaoglu M, et al. Brain natriuretic peptide and severity of disease in non-alcoholic cirrhotic patients. *J Gastroenterol Hepatol*. 2005;20:1115–1120.
572. Radvan M, Svoboda P, Radvanova J, et al. Brain natriuretic peptide in decompensation of liver cirrhosis in non-cardiac patients. *Hepatogastroenterology*. 2009;56:181–185.
573. Pimenta J, et al. B-type natriuretic peptide is related to cardiac function and prognosis in hospitalized patients with decompensated cirrhosis. *Liver Int*. 2010;30:1059–1066.
574. Gulberg V, Moller S, Henriksen JH, et al. Increased renal production of C-type natriuretic peptide (CNP) in patients with cirrhosis and functional renal failure. *Gut*. 2000;47:852–857.
575. Henriksen JH, Gulberg V, Gerbes AL, et al. Increased arterial compliance in cirrhosis is related to decreased arterial C-type natriuretic peptide, but not to atrial natriuretic peptide. *Scand J Gastroenterol*. 2003;38:559–564.
576. Fabrega E, et al. Dendroaspis natriuretic peptide in hepatic cirrhosis. *Am J Gastroenterol*. 2001;96:2724–2729.
577. Claria J, Arroyo V. Prostaglandins and other cyclooxygenase-dependent arachidonic acid metabolites and the kidney in liver disease. *Prostaglandins Other Lipid Mediat*. 2003;72:19–33.
578. Claria J, et al. Effects of celecoxib and naproxen on renal function in nonazotemic patients with cirrhosis and ascites. *Hepatology*. 2005;41:579–587.
579. Sansoe G, et al. Calcium-dependent diuretic system in preascitic liver cirrhosis. *J Hepatol*. 2010;53:856–862.
580. Gao JH, et al. Celecoxib ameliorates portal hypertension of the cirrhotic rats through the dual inhibitory effects on the intrahepatic fibrosis and angiogenesis. *PLoS ONE*. 2013;8:e69309.
581. Govindarajan S, et al. Immunohistochemical distribution of renal prostaglandin endoperoxide synthase and prostacyclin synthase: diminished endoperoxide synthase in the hepatorenal syndrome. *Hepatology*. 1987;7:654–659.
582. Gines A, et al. Oral misoprostol or intravenous prostaglandin E2 do not improve renal function in patients with cirrhosis and ascites with hyponatremia or renal failure. *J Hepatol*. 1993;17:220–226.
583. Chang AM, Maisel AS, Hollander JE. Diagnosis of heart failure. *Heart Fail Clin*. 2009;5:25–35, vi.
584. Chow SL, et al. Role of biomarkers for the prevention, assessment, and management of heart failure: a scientific statement from the American Heart Association. *Circulation*. 2017.
585. Parrinello G, et al. Troponin I release after intravenous treatment with high furosemide doses plus hypertonic saline solution in decompensated heart failure trial (Tra-HSS-Fur). *Am Heart J*. 2012;164:351–357.
586. Westermann D, Neumann JT, Sorensen NA, et al. High-sensitivity assays for troponin in patients with cardiac disease. *Nat Rev Cardiol*. 2017.
587. Aliti GB, et al. Aggressive fluid and sodium restriction in acute decompensated heart failure: a randomized clinical trial. *JAMA Intern Med*. 2013;173:1058–1064.
588. DiNicolantonio JJ, Chatterjee S, O'Keefe JH. Dietary salt restriction in heart failure: where is the evidence? *Prog Cardiovasc Dis*. 2016;58:401–406.
589. Bikdeli B, et al. Intravenous fluids in acute decompensated heart failure. *JACC Heart Fail*. 2015;3:127–133.
590. Paterna S, et al. Short-term effects of hypertonic saline solution in acute heart failure and long-term effects of a moderate sodium restriction in patients with compensated heart failure with New York Heart Association class III (class C) (SMAC-HF study). *Am J Med Sci*. 2011;342:27–37.
591. Paterna S, et al. Hypertonic saline in conjunction with high-dose furosemide improves dose-response curves in worsening refractory congestive heart failure. *Adv Ther*. 2015;32:971–982.
592. Bernstein PL, Ellison DH. Diuretics and salt transport along the nephron. *Semin Nephrol*. 2011;31:475–482.
593. Winaver J, Teredesai P, Anast C, et al. Investigations into the mechanism of the phosphaturia induced by chlorothiazide. *J Pharmacol Exp Ther*. 1981;218:46–54.
594. Moranville MP, Choi S, Hogg J, et al. Comparison of metolazone versus chlorothiazide in acute decompensated heart failure with diuretic resistance. *Cardiovasc Ther*. 2015;33:42–49.
595. Chiandussi L, Bartoli E, Arras S. Reabsorption of sodium in the proximal renal tubule in cirrhosis of the liver. *Gut*. 1978;19:497–503.
596. Turagam MK, et al. Outcomes of furosemide-mannitol infusion in hospitalized patients with heart failure: an observational single-center cohort study of 122 patients. *Int J Cardiol*. 2011;151:232–234.
597. Grieff M, Bushinsky DA. Diuretics and disorders of calcium homeostasis. *Semin Nephrol*. 2011;31:535–541.
598. Palmer BF. Metabolic complications associated with use of diuretics. *Semin Nephrol*. 2011;31:542–552.
599. Hardin EA, Grodin JL. Diuretic strategies in acute decompensated heart failure. *Curr Heart Fail Rep*. 2017;14:127–133.
600. Greenberg A. Diuretic complications. *Am J Med Sci*. 2000;319:10–24.
601. Courivaud C, et al. Peritoneal dialysis reduces the number of hospitalization days in heart failure patients refractory to diuretics. *Perit Dial Int*. 2013.
602. Bart BA, et al. Ultrafiltration in decompensated heart failure with cardiorenal syndrome. *N Engl J Med*. 2012;367:2296–2304.
603. Mentz RJ, et al. The past, present and future of renin-angiotensin aldosterone system inhibition. *Int J Cardiol*. 2013;167:1677–1687.
604. Orsborne C, Chaggar PS, Shaw SM, et al. The renin-angiotensin-aldosterone system in heart failure for the non-specialist: the past, the present and the future. *Postgrad Med J*. 2017;93:29–37.
605. Miura M, et al. Prognostic impact of loop diuretics in patients with chronic heart failure- effects of addition of renin-angiotensin-aldosterone system inhibitors and beta-Blockers. *Circ J*. 2016;80:1396–1403.
606. Kotecha D, et al. Heart rate, heart rhythm, and prognostic benefits of beta-blockers in heart failure: individual patient-data meta-analysis. *J Am Coll Cardiol*. 2017.
607. Kraehling JR, Sessa WC. Contemporary approaches to modulating the Nitric Oxide-cGMP pathway in cardiovascular disease. *Circ Res*. 2017;120:1174–1182.
608. Cole RT, et al. Hydralazine and isosorbide dinitrate in heart failure: historical perspective, mechanisms, and future directions. *Circulation*. 2011;123:2414–2422.
609. Ghosh RK, et al. Serelaxin in acute heart failure: most recent update on clinical and preclinical evidence. *Cardiovasc Ther*. 2017;35:55–63.
610. Tei C, et al. Waon therapy for managing chronic heart failure- results from a multicenter prospective randomized WAON-CHF study. *Circ J*. 2016;80:827–834.

611. Ichiki T, et al. Neurohumoral modulation during waon therapy in chronic heart failure- subanalysis of Waon-CHF study. *Circ J*. 2017;81:709–716.
612. Zheng H, Liu X, Patel KP. Angiotensin-converting enzyme 2 overexpression improves central nitric oxide-mediated sympathetic outflow in chronic heart failure. *Am J Physiol Heart Circ Physiol*. 2011;301:H2402–H2412.
613. Basu R, et al. Roles of angiotensin peptides and recombinant human ACE2 in heart failure. *J Am Coll Cardiol*. 2017;69:805–819.
614. Packer M, et al. Long-term effect of endothelin receptor antagonism with bosentan on the morbidity and mortality of patients with severe chronic heart failure: primary results of the ENABLE trials. *JACC Heart Fail*. 2017;5:317–326.
615. Valero-Munoz M, et al. Dual endothelin-A/endothelin-B receptor blockade and cardiac remodeling in heart failure with preserved ejection fraction. *Circ Heart Fail*. 2016;9.
616. Krupicka J, Janota T, Kasalova Z, et al. Natriuretic peptides - physiology, pathophysiology and clinical use in heart failure. *Physiol Res*. 2009;58:171–177.
617. Korinek J, Boerrigter G, Mohammed SF, et al. Insights into natriuretic peptides in heart failure: an update. *Curr Heart Fail Rep*. 2008;5:97–104.
618. Colucci WS. Nesiritide for the treatment of decompensated heart failure. *J Card Fail*. 2001;7:92–100.
619. Chen HH, et al. Low-dose dopamine or low-dose nesiritide in acute heart failure with renal dysfunction: the ROSE acute heart failure randomized trial. *JAMA*. 2013.
620. Del Ry S, Cabiati M, Clerico A. Natriuretic peptide system and the heart. *Front Horm Res*. 2014;43:134–143.
621. McDowell G, Nicholls DP. The endopeptidase inhibitor, candosatriel, and its therapeutic potential in the treatment of chronic cardiac failure in man. *Expert Opin Investig Drugs*. 1999;8:79–84.
622. Chen HH, Lainchbury JG, Harty GJ, et al. Maximizing the natriuretic peptide system in experimental heart failure: subcutaneous brain natriuretic peptide and acute vasopeptidase inhibition. *Circulation*. 2002;105:999–1003.
623. Solomon SD, Claggett B, McMurray JJ, et al. Combined neprilysin and renin-angiotensin system inhibition in heart failure with reduced ejection fraction: a meta-analysis. *Eur J Heart Fail*. 2016;18:1238–1243.
624. Rodgers JE. Sacubitril/Valsartan: the newest addition to the toolbox for guideline-directed medical therapy of heart failure. *Am J Med*. 2017.
625. Lehrich RW, Greenberg A. Hyponatremia and the use of vasopressin receptor antagonists in critically ill patients. *J Intensive Care Med*. 2012;27:207–218.
626. Costello-Boerrigter LC, et al. Vasopressin-2-receptor antagonism augments water excretion without changes in renal hemodynamics or sodium and potassium excretion in human heart failure. *Am J Physiol Renal Physiol*. 2006;290:F273–F278.
627. Konstam MA, et al. Effects of oral tolvaptan in patients hospitalized for worsening heart failure: the EVEREST Outcome Trial. *JAMA*. 2007;297:1319–1331.
628. Udelsion JE, et al. Acute hemodynamic effects of conivaptan, a dual V(1A) and V(2) vasopressin receptor antagonist, in patients with advanced heart failure. *Circulation*. 2001;104:2417–2423.
629. Goldsmith SR, Elkayam U, Haught WH, et al. Efficacy and safety of the vasopressin V1A/V2-receptor antagonist conivaptan in acute decompensated heart failure: a dose-ranging pilot study. *J Card Fail*. 2008;14:641–647.
630. Rossi J, et al. Improvement in hyponatremia during hospitalization for worsening heart failure is associated with improved outcomes: insights from the Acute and Chronic Therapeutic Impact of a Vasopressin Antagonist in Chronic Heart Failure (ACTIV in CHF) trial. *Acute Card Care*. 2007;9:82–86.
631. Ghali JK, Orlandi C, Abraham WT. The efficacy and safety of lixivaptan in outpatients with heart failure and volume overload: results of a multicentre, randomized, double-blind, placebo-controlled, parallel-group study. *Eur J Heart Fail*. 2012;14:642–651.
632. Borne RT, Krantz MJ. Lixivaptan for hyponatremia—the numbers game. *JAMA*. 2012;308:2345–2346.
633. Udelsion JE, et al. Multicenter, randomized, double-blind, placebo-controlled study on the effect of oral tolvaptan on left ventricular dilation and function in patients with heart failure and systolic dysfunction. *J Am Coll Cardiol*. 2007;49:2151–2159.
634. Eguchi A, et al. Long-term administration of tolvaptan increases myocardial remodeling and mortality via exacerbation of congestion in mice heart failure model after myocardial infarction. *Int J Cardiol*. 2016;221:302–309.
635. Matsue Y, et al. Tolvaptan reduces the risk of worsening renal function in patients with acute decompensated heart failure in high-risk population. *J Cardiol*. 2013;61:169–174.
636. Jujo K, et al. Randomized pilot trial comparing tolvaptan with furosemide on renal and neurohumoral effects in acute heart failure. *ESC Heart Fail*. 2016;3:177–188.
637. Udelsion JE, et al. A multicenter, randomized, double-blind, placebo-controlled study of tolvaptan monotherapy compared to furosemide and the combination of tolvaptan and furosemide in patients with heart failure and systolic dysfunction. *J Card Fail*. 2011;17:973–981.
638. Matsue Y, et al. Clinical effectiveness of tolvaptan in patients with acute decompensated heart failure and renal failure: design and rationale of the AQUAMARINE study. *Cardiovasc Drugs Ther*. 2013.
639. Matsue Y, et al. Clinical effectiveness of tolvaptan in patients with acute decompensated heart failure and renal failure: design and rationale of the AQUAMARINE study. *Cardiovasc Drugs Ther*. 2014;28:73–77.
640. Alskaf E, Tridante A, Al-Mohammad A. Tolvaptan for heart failure, systematic review and meta-analysis of trials. *J Cardiovasc Pharmacol*. 2016;68:196–203.
641. Martens P, Mathieu C, Verbrugge FH. Promise of SGLT2 inhibitors in heart failure: diabetes and beyond. *Curr Treat Options Cardiovasc Med*. 2017;19:23.
642. Weber ML, Ibrahim HN, Lake JR. Renal dysfunction in liver transplant recipients: evaluation of the critical issues. *Liver Transpl*. 2012;18:1290–1301.
643. Sanyal AJ, et al. Reversal of hepatorenal syndrome type 1 with terlipressin plus albumin vs. placebo plus albumin in a pooled analysis of the OT-0401 and REVERSE randomised clinical studies. *Aliment Pharmacol Ther*. 2017;45:1390–1402.
644. Sanyal AJ, et al. A randomized, prospective, double-blind, placebo-controlled trial of terlipressin for type 1 hepatorenal syndrome. *Gastroenterology*. 2008;134:1360–1368.
645. Martin-Llahi M, et al. Terlipressin and albumin vs albumin in patients with cirrhosis and hepatorenal syndrome: a randomized study. *Gastroenterology*. 2008;134:1352–1359.
646. Boyer TD, et al. Terlipressin plus albumin is more effective than albumin alone in improving renal function in patients with cirrhosis and hepatorenal syndrome type 1. *Gastroenterology*. 2016;150:1579–1589, e1572.
647. Wong F, et al. Terlipressin improves renal function and reverses hepatorenal syndrome in patients with systemic inflammatory response syndrome. *Clin Gastroenterol Hepatol*. 2017;15:266–272, e261.
648. Cavallin M, et al. Terlipressin given by continuous intravenous infusion versus intravenous boluses in the treatment of hepatorenal syndrome: a randomized controlled study. *Hepatology*. 2016;63:983–992.
649. Maddukuri G, Cai CX, Munigala S, et al. Targeting an early and substantial increase in mean arterial pressure is critical in the management of type 1 hepatorenal syndrome: a combined retrospective and pilot study. *Dig Dis Sci*. 2013.
650. Sola E, Cardenas A, Gines P. Results of pretransplant treatment of hepatorenal syndrome with terlipressin. *Curr Opin Organ Transplant*. 2013;18:265–270.
651. Fayed N, Refaat EK, Yassein TE, et al. Effect of perioperative terlipressin infusion on systemic, hepatic, and renal hemodynamics during living donor liver transplantation. *J Crit Care*. 2013;28:775–782.
652. Mukhtar A, et al. Intraoperative terlipressin therapy reduces the incidence of postoperative acute kidney injury after living donor liver transplantation. *J Cardiothorac Vasc Anesth*. 2015;29:678–683.
653. Esrailian E, Pantangco ER, Kyulo NL, et al. Octreotide/Midodrine therapy significantly improves renal function and 30-day survival in patients with type 1 hepatorenal syndrome. *Dig Dis Sci*. 2007;52:742–748.
654. Karwa R, Woodis CB. Midodrine and octreotide in treatment of cirrhosis-related hemodynamic complications. *Ann Pharmacother*. 2009;43:692–699.
655. Skagen C, Einstein M, Lucey MR, et al. Combination treatment with octreotide, midodrine, and albumin improves survival in patients with type 1 and type 2 hepatorenal syndrome. *J Clin Gastroenterol*. 2009;43:680–685.
656. Rice JP, Skagen C, Said A. Liver transplant outcomes for patients with hepatorenal syndrome treated with pretransplant vasoconstrictors and albumin. *Transplantation*. 2011;91:1141–1147.

657. Alessandria C, et al. Midodrine in the prevention of hepatorenal syndrome type 2 recurrence: a case-control study. *Dig Liver Dis.* 2009;41:298–302.
658. Cavallin M, et al. Terlipressin plus albumin versus midodrine and octreotide plus albumin in the treatment of hepatorenal syndrome: a randomized trial. *Hepatology.* 2015;62:567–574.
659. Yan L, et al. The treatment of vasopressin V2-receptor antagonists in cirrhosis patients with ascites: a meta-analysis of randomized controlled trials. *BMC Gastroenterol.* 2015;15:65.
660. Fukui H. Do vasopressin V2 receptor antagonists benefit cirrhotics with refractory ascites? *World J Gastroenterol.* 2015;21:11584–11596.
661. Pose E, et al. Limited efficacy of tolvaptan in patients with cirrhosis and severe hyponatremia: real-life experience. *Am J Med.* 2017;130:372–375.
662. Wong F, et al. Satavaptan for the management of ascites in cirrhosis: efficacy and safety across the spectrum of ascites severity. *Gut.* 2012;61:108–116.
663. Bai M, et al. TIPS improves liver transplantation-free survival in cirrhotic patients with refractory ascites: an updated meta-analysis. *World J Gastroenterol.* 2014;20:2704–2714.
664. Ortega R, et al. Terlipressin therapy with and without albumin for patients with hepatorenal syndrome: results of a prospective, nonrandomized study. *Hepatology.* 2002;36:941–948.
665. Brensing KA, et al. Long term outcome after transjugular intrahepatic portosystemic stent-shunt in non-transplant cirrhotics with hepatorenal syndrome: a phase II study. *Gut.* 2000;47:288–295.
666. Mumtaz K, et al. Impact of transjugular intrahepatic porto-systemic shunt on post liver transplantation outcomes: Study based on the United Network for Organ Sharing database. *World J Hepatol.* 2017;9:99–105.
667. Fagioli S, et al. Consensus conference on TIPS management: techniques, indications, contraindications. *Dig Liver Dis.* 2017;49:121–137.
668. Sourianarayanan A, Raina R, Garg G, et al. Management and outcome in hepatorenal syndrome: need for renal replacement therapy in non-transplanted patients. *Int Urol Nephrol.* 2013.
669. Gonwa TA, Wadei HM. The challenges of providing renal replacement therapy in decompensated liver cirrhosis. *Blood Purif.* 2012;33:144–148.
670. Donati G, et al. Acute systemic, splanchnic and renal haemodynamic changes induced by molecular adsorbent recirculating system (MARS) treatment in patients with end-stage cirrhosis. *Aliment Pharmacol Ther.* 2007;26:717–726.
671. Mitzner SR, et al. Improvement of hepatorenal syndrome with extracorporeal albumin dialysis MARS: results of a prospective, randomized, controlled clinical trial. *Liver Transpl.* 2000;6:277–286.
672. Tsiptotis E, Shuja A, Jaber BL. Albumin dialysis for liver failure: a systematic review. *Adv Chronic Kidney Dis.* 2015;22:382–390.
673. Lata J. Hepatorenal syndrome. *World J Gastroenterol.* 2012;18:4978–4984.
674. Fabrizi F, Messa P. Challenges in renal failure treatment before liver transplant. *Clin Liver Dis.* 2017;21:303–319.
675. Marik PE, Wood K, Starzl TE. The course of type 1 hepato-renal syndrome post liver transplantation. *Nephrol Dial Transplant.* 2006;21:478–482.
676. Xing T, Zhong L, Chen D, et al. Experience of combined liver-kidney transplantation for acute-on-chronic liver failure patients with renal dysfunction. *Transplant Proc.* 2013;45:2307–2313.
677. Pham PT, Lunsford KE, Bunnapradist S, et al. Simultaneous liver-kidney transplantation or liver transplantation alone for patients in need of liver transplantation with renal dysfunction. *Curr Opin Organ Transplant.* 2016;21:194–200.
678. Angeli P, Gines P. Hepatorenal syndrome, MELD score and liver transplantation: an evolving issue with relevant implications for clinical practice. *J Hepatol.* 2012;57:1135–1140.
679. Kalra A, Wedd JP, Biggins SW. Changing prioritization for transplantation: MELD-Na, hepatocellular carcinoma exceptions, and more. *Curr Opin Organ Transplant.* 2016;21:120–126.
680. Kwong AJ, Fix OK. Update on the management of the liver transplant patient. *Curr Opin Gastroenterol.* 2015;31:224–232.